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GEG Geotechnical Engineering Group University of Salerno

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22 July

11:00	120 min		**Registration**
13:30	120 min		**ICGdR Annual Meeting**
15:30	60 min		**Registration**
16:30	end		

23 July

8:00	60 min		**Registration**
9:00	50 min		**Opening of the 22ISGdR and ICGdR Awards**
9:50	40 min		**KL1: Cascini L. (LARAM School)**
10:30	20 min		Coffee Break

10:50	90 min		**Session 1 (room A)**
		ID	Chair: Cuomo S. (Salerno, Italy)
	30 min	007	IL: Marinos V. (IAEG)
	15 min	155	IP: Di Vito (INGV, Italy)
	15 min	111	IP: Duro A. (Civil Protection, Italy)
	15 min	021	IP: Spizzichino (Geol. Survey, Italy)
	15 min	022	IP: Cagelli (Net.pro, Italy)

10:50	90 min		**Session 1 (room B)**
		ID	Chair: Valentino R. (Parma, Italy)
	15 min	026	Nicodemo G. (Salerno, Italy)
	15 min	039	Gong B. (London, UK)
	15 min	040	Gu K.(Nanjing, China)
	15 min	050	Li P. (Sanhe, China)
	15 min	037	Durante M.G. (Cosenza, Italy)
	15 min	029	Casablanca O. (Messina, Italy)

12:20	10 min		**Group Photo**
12:30	60 min		Lunch Time
13:30	30 min		**KL2: Fathani T.F. (Java, Indonesia)**

14:00	90 min		**Session 2 (room A)**
		ID	Chair: Wang F. (Shanghai, China)
	25 min	062	IL: Miyajima M. (Kanazawa, Japan)
	25 min	011	IL: Choi C. (Hong Kong, China)
	15 min	106	IP: Petrosino P. (Napoli, Italy)
	15 min	094	Zhang S. (Beijing, China)
	15 min	074	Tian Z. Y. (Qingdao, China)

14:00	90 min		**Session 2 (room B)**
		ID	Chair: Nicodemo (Salerno, Italy)
	15 min	032	Chen Y. (Sanhe, China)
	15 min	038	Feng Y. (Shanghai, Italy)
	15 min	079	Wang X. (Chengdu, China)
	15 min	100	Zimmaro P. (Cosenza, Italy)
	15 min	035	Dai G.L. (Nanjing, China)
	15 min	025	Sarno L. (Salerno, Itay)

15:30	30 min		Coffee Break
16:00	30 min		**KL3: Yaboyedoff M. (Lausanne, Switzerland)**

16:20	100 min		**Session 3 (room A)**
		ID	Chair: Ferlisi S. (Salerno, Italy)
	15 min	102	IP: Corbelli V. (River Basin Auth.DAM, Italy)
	15 min	103	IP: Capasso G. (River Basin Auth.DAM, Italy)
	15 min	105	IP: Vitone C. (Bari, Italy)
	15 min	104	Cuomo S. (Salerno, Itay)
	15 min	113	Wang F. (Shanghai, China)

16:20	100 min		**Session 3 (room B)**
		ID	Chair: QI S. (Beijing, China)
	25 min	017	IL: Peranic J. (Rijeka, Croatia)
	15 min	031	Tursi M. (Napoli, Italy)
	15 min	030	Di Leo A. (Salerno, Italy)
	15 min	055	Liu X.L. (Qingdao, China)
	15 min	057	Lyu Q. (Hangzhou, China)
	15 min	058	Ma H. (Shanghai, China)

18:00	end		
19:30			**Social Dinner**

24 July

8:30	30 min		**Registration**
9:00	30 min		**KL4: He M.C. (Beijing, China)**

9:30	70 min		**Session 4 (room A)**
		ID	Chair: Esposito (CNR, Italy)
	25 min	012	IL: Catani F. (Padova, Italy)
	15 min	019	IP: Matano F. (CNR, Italy)
	15 min	087	Wu S.W. (Taichung, Taiwan)
	15 min	028	Fraccica A. (Geol. Survey, Italy)

9:30	70 min		**Session 4 (room B)**
		ID	Chair: Pirulli M. (Torino, Italy)
	25 min	013	IL: Tofani V. (Firenze, Italy)
	15 min	027	Pecoraro G. (Salerno, Italy)
	15 min	098	Zheng B.W. (Beijing, China)
	15 min	092	Yue P. (Kobe, Japan)

10:40	20 min		Coffee Break
11:00	30 min		**KL5: Yoshida M. (Fukui, Japan)**

11:30	70 min		**Session 5 (room A)**
		ID	Chair: Matano F. (CNR, Italy)
	25 min	009	IL: Pirulli M. (Torino, Italy)

11:30	70 min		**Session 5 (room B)**
		ID	Chair: Pecoraro (Salerno, Italy)
	25 min	015	IL: Martinelli M. (Delft, Netherlands)

	15 min	080	Wang T. (Beijing, China)		15 min	047	Li B. (Shanghai, Cina)
	15 min	065	Nian T.K. (Dalian, China)		15 min	086	Wu H. (Nanjing, China)
	15 min	069	Song K. (Yichang, China)		15 min	075	Valentino R. (Parma, Italy)
12:40	80 min		Lunch Time				
14:00	30 min		**KL6: Winter M. (Kirknewton, UK)**				
14:30	70 min		**Session 6 (room A)**	**14:30**	70 min		**Session 6 (room B)**
		ID	Chair: Tofani (Florence, Italy)			ID	Chair: Peranic J. (Rijeka, Croatia)
	25 min	036	IL: Dai F.C. (Beijing, China)		25 min	014	IL: QI S. (Beijing, China)
	15 min	020	IP: Esposito G. (CNR, Italy)		15 min	076	IP: Viccione G. (Salerno, Italy)
	15 min	024	IP: Petti L. (Salerno, Italy)		15 min	041	Guo F. (Yichang, China)
	15 min	073	Tian Y.Y. (Beijing, China)		15 min	108	Peng M. (Shanghai, China)
15:40	20 min		Coffee Break				
16:00	60 min		**Session 7 (room A)**	**16:00**	60 min		**Session 7 (room B)**
		ID	Chair: Moscariello (Salerno, Italy)				Chair: Iannuzzi E. (Salerno, Italy)
	15 min	048	Li X. B. (Sanhe, China)			ID	Short Presentations
			Short Presentations			033	Cheng Q. (Nanjing, China)
		018	Bhowmik R. (Jammu, India)			034	Dai G.W. (Shanghai, China)
		156	Ciaglia R. (Salerno, Italy)			043	Hu Y.X. (Shanghai, Cina)
		116	Guo Y. (Lanzhou, China)			049	Li R. (Beijing, China)
		046	Li J.B. (Lanzhou, China)			064	NI A. (Beijing, China)
		054	Liu J.M. (Beijing, China)			115	Shan W. (Lanzhou, China)
		059	Mahmoudi E. (Delft, Netherlands)			068	Shao X.H. (Shanghai, Cina)
		060	Meng H.J. (Beijing, China)			110	Valletta A. (Parma, Italy)
		063	Murata T. (Tokyo, Japan)			084	Wei S. C. (Taichung, Taiwan)
		070	Sun X.L. (Xi'an, China)			109	Yang S. (Wuhan, China)
		083	Wei L.D. (Xi'an, China)			117	Zhang C. (Lanzhou, China)
		088	Xia C.Z. (Shanghai, Cina)			096	Zhang H. (Dalian, China)
		099	Zhu M.Y. (Shanghai, Cina)			095	Zhang W.L. (Shanghai, Cina)
						101	Zou Y. (Beijing, China)
17:00	30 min		**Closing Ceremony**				
17:30	30 min		Walking to Zaha Hadid Maritime Station				
18:00	30 min		Guided Visit to Zaha Hadid Maritime Station				
19:00	120 min		**Live Concert and Farewall party**				
21:00	end						

25 July	**07:30**	**18:30**	**Technical Tour**

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- AEG (International Association for Engineering Geology and the Environment), **https://iaeg.info/**
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- INGV (Istituto Nazionale di Geofisica e Vulcanologia), **www.ingv.it**
- IAEG-Italy (Italian Chapter of IAEG), **https://www.iaeg.it/**
- IGS-Italy (Italian Chapter of IGS), **link**
- AGI (Associazione Geotecnica Italiana), **www.associazionegeotecnica.it**
- CNR-IRPI (National Research Council, Institute for Geo-Hydrological Protection), **link**
- CNR-ISMAR (National Research Council, Institute of Marine Sciences), **link**
- CUGRI (InterUniversity Centre for the Prediction and Prevention of Major Hazards), **www.cugri.it**
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Analysis and management of geo-hazards through integrated approaches

Leonardo Cascini[1]

Abstract

The paper proposes different approaches to overcome the limits of current landside research, usually based on a monodisciplinary vision that limits its diffusion. The proposed approaches essentially follow two different paths, respectively defined as the horizontal and the vertical path, both aimed at the generalization of knowledge. The first analyzes landslides in their current spatial and temporal condition while the second one connects, through different topographical scales, the contributions of various disciplines that study landslides from different points of view. For all the approaches, the paper provides the results obtained and the period of time necessary to achieve them, strictly linked to the path followed.

Keywords

landslides, analysis, inter-disciplinary, multi-scale

Introduction

Several geo-hazards threaten the urbanized areas all over the world. The documents of the International Consortium on Geo-disaster Reduction (ICGdR, www.icgdr.com) mentions earthquakes, volcanic activity, landslides, tsunamis, intensive erosion, and hydro-meteorological events, encouraging the implementation of multidisciplinary approaches to improve their understanding and the related risk management.

However, these multidisciplinary approaches as well as a full sharing of knowledge are far from being systematically applied to landslides, which are often linked to many other geo-hazards and, nevertheless, usually analyzed through single disciplines (e.g., Geology, Geomorphology, Geography, Geotechnics, Geomechanics). Then the acquired knowledge often remains confined for a long time in well-defined research areas, without the diffusion it deserves. Indeed, there is a need for a paradigm change.

Aware of the progress recorded in recent decades - thanks to the efforts devoted in many disciplinary sectors - and following the suggestions of the ICGdR Consortium, the question is: how can we pursue a broader generalization of the available knowledge?

An answer to this complex question is today pursued essentially through the Artificial Intelligence (AI) which implicitly refers to the theory of "Complex Systems" as defined in Physics (Parisi,1999). From this perspective, researchers are paying an increasing attention to big data and their correlations to solve many issues of great practical interest.

Although this field of research is very promising, the presentation focuses on a completely opposite approach, based on the chain of "cause and effect" which is typical of the physical sciences. From this perspective, it is preliminarily necessary to understand whether the poor generalization of knowledge is due to the need for greater understanding of many crucial issues related to landslides or other reasons.

It is my belief that this is essentially due to the limited application of the multidisciplinary approach invoked by the ICGdR Consortium which invites us to break down the barriers and to share the knowledge. Starting from this assumption, the basic principles of the developed approaches and explanatory examples of their application to different types of landslides are provided in the following sections.

Basic principles of the proposed approaches

Classification systems introduce and describe a variety of landslides that differ in shape, mechanisms, materials involved and many other factors. Having no specific references in the literature, when the author decided to investigate possible common features between so different phenomena, the first problem to solve was the possibility of grouping landslides into a more limited number of classes. This issue was addressed observing that landslides described in all

the classification systems fall into two main classes called "existing landslides" and "first-failure landslides", respectively.

This observation led to investigating the possible differences between the two classes of phenomena for which the only reference available in the literature (Leroueil et al., 1996) indicated, on a phenomenological basis, totally different evolution stages. The suggestion of these authors was considered worthy of further investigation for several reasons among which the residual shear strength usually mobilized along the slip surface of an existing landslide and a significant higher value of the shear strength in a first-failure landslide such as those caused in quarries. The conclusion from these and few other considerations was that *i)* existing landslides are the result of the long geological history underlying their origin and evolution stages while *ii)* first-failure phenomena are comparable to the initial stage of an existing landslide even if the evolution in time and space is quite different where the anthropic triggering factors prevail.

This gave rise to two different research activities. The first aimed to identify common kinematic features for deep-seated existing landslides and first-failure phenomena. The second one dedicated to connecting the geological history of a site with the landslides present inside to understand their spatial location and evolution in the near future.

The first line of research follows an horizontal path focusing the attention on landslides that can be very different from each other but all in their current spatial and temporal configuration. This research can be considered concluded as the theoretical framework has been completely defined and the results obtained are particularly promising as illustrated below.

The second line of research is, vice-versa, vertical in nature as it analyzes the evolution of the territory over time through a multi-scale approach which can be developed following a bottom-up, top-down or a mixed path as defined in Cascini (2015). All are markedly multidisciplinary approaches and blend together the contributions usually provided by Geology, Geotechnics and other disciplines involved in studying the landslides.

The experience acquired so far clearly highlights that the best results, in terms of quality and time, are provided by the top-down approach especially if the geological maps developed at the smallest (territorial) scales are translated into thematic maps capable of correlating the geological evolution of a territory with the current state of landslides and their possible future evolution. The explanatory examples discussed in the following also highlight other potentialities of this approach which can correlate different geo-hazards together highlighting the role of each of them on the global risk that affects an area.

Common kinematic features of deep-seated landslides

As summarized in Babilio et al. (2021), understanding and quantifying the time evolution of landslides has always engaged researchers for the consequences of such phenomena on the stability of buildings and infrastructure and the loss of life, as in the case of the Vajont landslide which caused 1,917 victims.

The scientific literature reports both mechanical and phenomenological approaches to forecast a landslide evolution. Mechanical approaches develop limit equilibrium, coupled or uncoupled stress-strain analyses and other sophisticated analyses, depending on the objectives to be pursued and the available data. They are usually applied to a single landslide case study and the results obtained cannot be generalized. On the other end, phenomenological approaches implemented as black box models often fail in the prediction of the landslide collapse time. Indeed, there is a lack of criteria to forecast the landslide evolution on a consistent basis.

Fifteen years ago, the University of Salerno started a research activity to bridge this gap through the analysis of the landslide dynamic equilibrium based on the inverse approximation of the cumulative displacement vs. time relationship.

The first research task consisted in collecting well documented landslides in the scientific literature. Then, the displacement trend of each landslide (e.g., Figure 1a) was divided into a sequence of activity phases distinguished with the aid of the landslide triggering factors (Figure 1b).

This preliminary and fundamental step allowed the recognition, in the dimensionless space, of common growth properties of different landslide displacements, independently from the volume of the landslide mass, the involved geomaterials, etc. (Figure 1c).

The third step implemented a power-law regularization of the experimental data to compute the time derivatives of the dimensionless cumulative displacements up to the third order (velocity, acceleration and jerk).

This procedure allowed us to understand and quantify the behaviour of all the monitored landslides through a stability chart which relates a different stability regime to each of the dimensionless displacement trends in Figure 1c (Babilio et al., 2021).

Once calibrated and validated, the model was used to test its ability to predict the evolution of all the landslides in the dataset.

The results obtained demonstrate that this objective, unimaginable until a few years ago, can now be achieved when an adequate characterization of the landslide and reliable measurements of its movements are available.

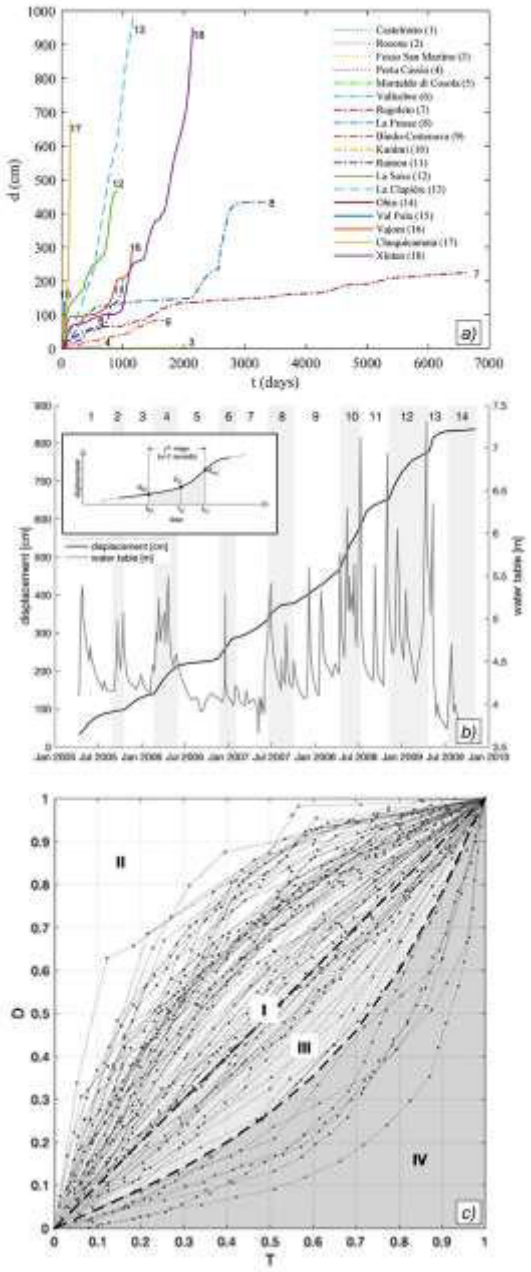


Figure 1. a) Dimensional displacement of the landslides collected in the dataset, b) isolating the activity stages through the shape of the displacement trend and the triggering factors for the case study of Bindo Cortenova, c) from dimensional to dimensionless displacement trends.

The multi-scale approach

Another possibility to enhance the landslide analysis and related risk management is provided by the multi-scale approach which is able to connect each other the skills of the many disciplines that, from different points of view, study the landslide.

Greatly simplifying the problem, we can say that Geological sciences usually analyse through heuristic methods large areas to investigate the predisposing factors of the landsliding while the engineering approaches are much more interested in quantifying the role played by triggering factors on a single slope. Indeed, merging the two different approaches allow passing from large areas to small ones or vice-versa. This can be done through the paths suggested in Cascini (2015), respectively named bottom-up and top-down approaches (Figure 2). Explanatory examples of these paths are provided in the following sections.

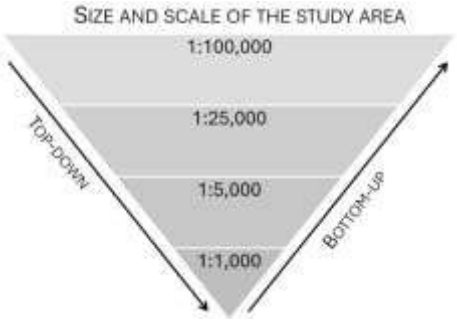


Figure 2 Bottom-up and top-down paths.

Example of the bottom-up approach

The bottom-up approach starts from studies developed at large/detailed scale on landslides representative of a well-defined geological context and the results achieved are progressively generalized to broader contexts than those analysed in detail.

These studies are developed through in situ and laboratory investigations and engineering models implemented to quantify the stability of few slopes and their evolution over time.

In this perspective, a proper analysis of the effects caused by the landslide triggering factors is of fundamental relevance since the displacements for an existing landslide and the mobilized volumes and runout for the first-failure phenomena strictly depend on these factors.

The applicability of the results achieved in areas much broader than those analysed in detail is obtained through statistical and/or heuristic methods which must be calibrated on the case studies analysed in depth. This process allows us to frame all the landslides in a geological context with a considerably less effort than would have been necessary through detailed investigations of every single landslide present in the area.

The presentation provides an explanatory example of this path by referring to a complex geological unit of weathered gneisses whose heterogeneity varies from site to site and, sometimes, to a marked extent in

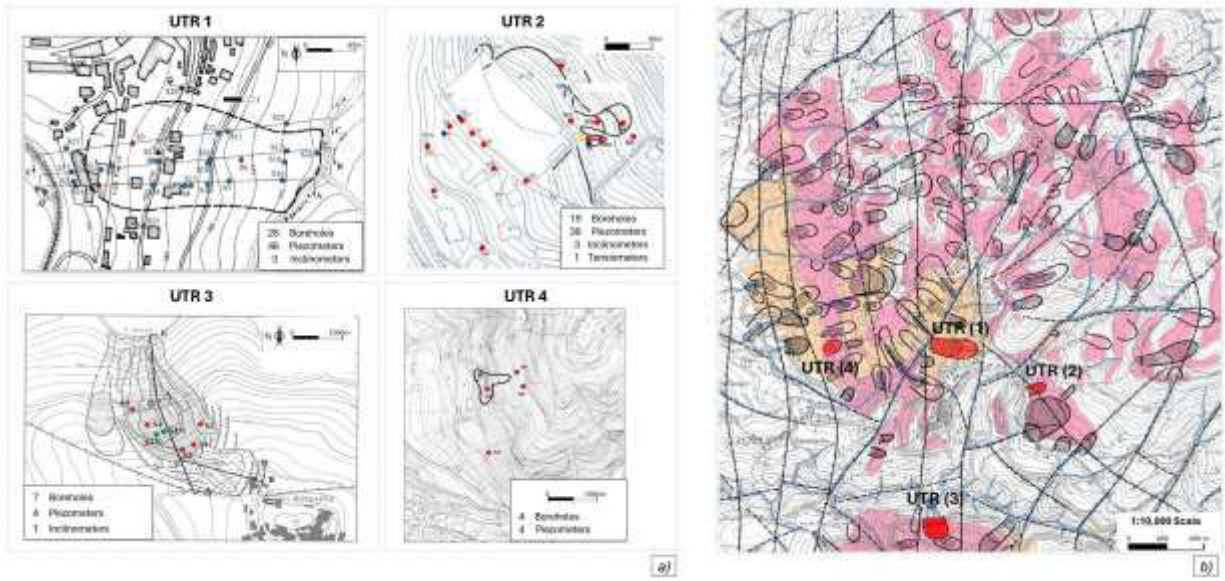


Figure 3 a) In-situ investigation of landslides studied in detail and b) identification of the landslide predisposing factors in a geologically homogeneous area.

the same site and along the same vertical. Figure 3a indicates the in-situ investigations developed in the first step of the research activity when four different landslides were analysed in detail.

Then, the obtained results were generalized over an area of about 10 km^2, Figure 3b, by implementing the geological model based on the landslides studied in detail (Cascini, 2015).

The time required to start and conclude the research activity was approximately 15 years.

Example of the top-down approach

Rainfall-induced shallow landslides involve several geo-environmental contexts and different types of soils. In clayey soils, they affect the most superficial layer, which is generally constituted by physically weathered soils characterised by a diffuse pattern of cracks. These landslides, which usually develop over large areas through a multiple sequence of phenomena, frequently cause significant consequences in terms of environmental and economic damage. Indeed, landslide susceptibility zoning is a relevant issue for land use planning and/or design purposes.

The report proposes a multi-scale analysis implemented through the top-down approach to reach this goal. The reliability of the approach is tested and validated over an area in southern Italy affected by widespread shallow landslide that can be classified as earth-slides and earth-slide-flows (Cascini et al., 2015). Specifically, by moving from a small scale (1:100,000) to a medium scale (1:25,000), with the aid of heuristic and statistical methods, the approach identifies the main factors leading to the landslide

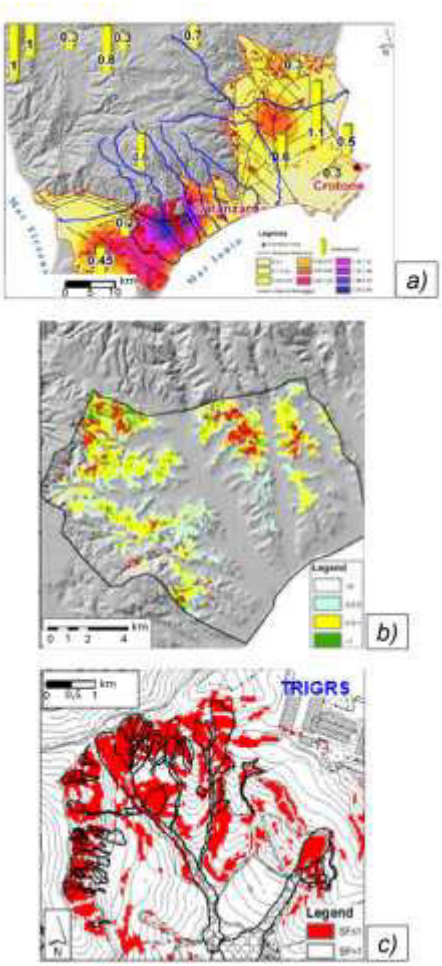


Figure 4 Identification of a) areas and b) slopes affected by shallow landslides, c) geotechnical analyses to quantify the landslide evolution.

spatial occurrence (Figure 4a), and effectively detects the areas potentially affected (Figure 4b).

Finally, at large scale (1:5,000) physically based models (TRIGRS and TRIGRS-unsaturated) allow the quantitative landslide susceptibility assessment in sample areas representative of those that are affected by shallow landslides (Figure 4c).

Considering the reliability of the results obtained and the time of 3 years required to develop the research activity, the proposed approach seems useful for analysing other case studies in similar or different geological contexts.

A new frontier for the geo-hazard analysis and management

The reliability and performance of the top-down approach led to formally defining the steps to follow moving from small to large scale, recently tested in the emergency management of the Casamicciola Terme area (island of Ischia, southern Italy), severely damaged, in November 2022, by a large number of landslides (Figure 5a-b). In this case study the territory was preliminarily divided into macro-sectors selected on the basis of their geological history (Figure 5c).

Then, each macro-sector was divided into sectors and territorial units, referring to their evolution caused – over the last hundreds of years – by landslides, floods, etc. This type of zoning, adequately supported by geotechnical analyses, has made it possible to quantitatively assess the landslide risk and to outline the links between volcanic, seismic, landslide and alluvial hazards affecting a large area (Figure 5d). This objective was achieved after just 2 months by a highly multidisciplinary study group made up of 20 scientists and technicians.

This procedure is currently being applied to on-going research on a much larger area, namely southern Italy which extends for 68,000 km^2 in the centre of the Mediterranean Sea (Figure 6a). Although the research is only at the beginning, the results achieved so far appear promising as it concerns the relationships between geology and geo-hazards as well as the relationships that exist between many of them (Figure 6b-d). The presentation will provide every useful detail to understand the major points and results achieved so far.

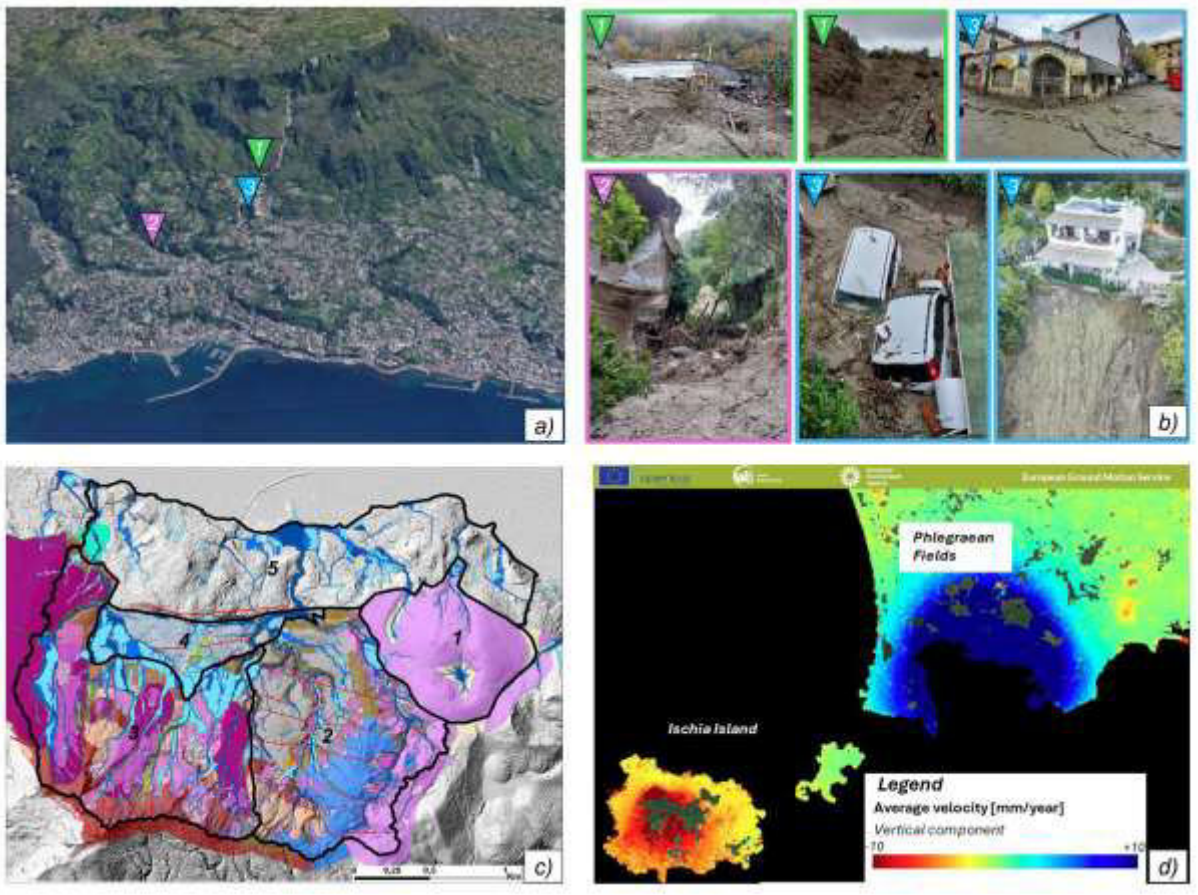


Figure 5 a) The territory of Casamicciola Terme (island of Ischia, NA – Italy) affected by the landslide event dated November 2022, b) examples of consequences caused by the landslides, c) territory zoning based on geological criteria, d) relationships between the geo-hazards threatening the territory (average velocity of the vertical component displacement provided by the satellite data (https://land.copernicus.eu/).

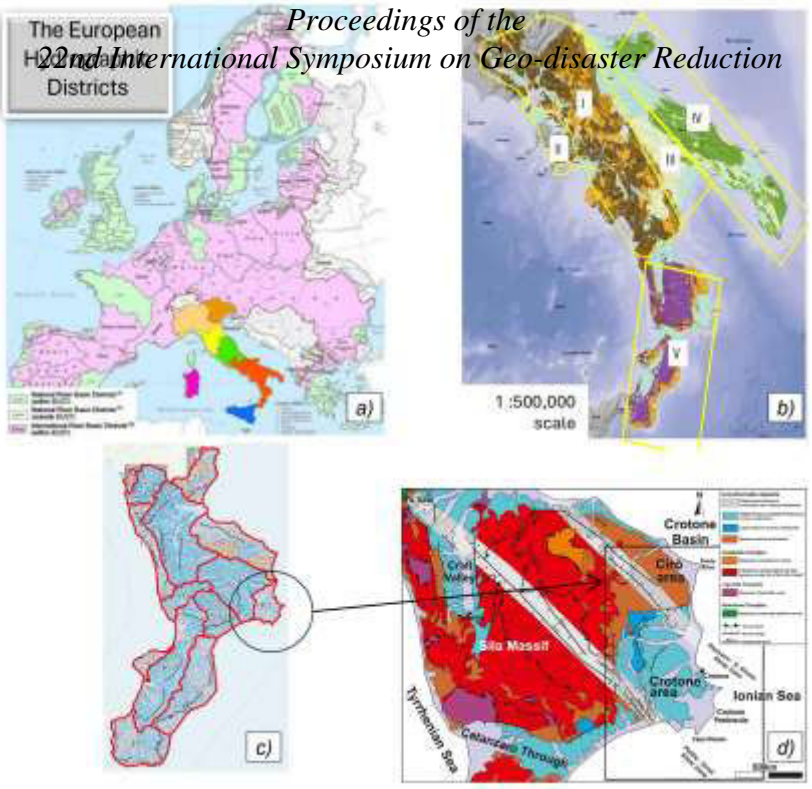


Figure 6 a) Hydrographic District of the Southern Italian Apennine in red; b) geodynamic macrozonation of the District territory: (I) Apennine chain, (II) Tyrrhenian area, (III) Bradanic foredeep, (IV) Apulia foreland, (V) Calabro Arc; c) zoning the macro-sectors of the Calabria region; d) the Crotone basin affected by a megalandslide according to the literature.

Concluding remarks

The case studies discussed in the presentation and summarized in this report highlight that sharing knowledge and linking geology and geotechnics are among the most promising ways to improve landslide analysis and related risk management.

Among these, the analyses introduced in section 3 and the top-down approach dealt with in section 4.2 appear particularly promising, especially if the latter is based on a territory zoning developed with the aid of geological sciences and deepened in few representative areas through the geotechnical models.

The author hopes for a broad discussion during the symposium to share this opinion and to have the opportunity of testing the described approaches in many other geological areas of the world.

Acknowledgments

The author is deeply grateful to all those who, over the decades, have contributed to deepen the themes summarized in the abstract and discussed in detail in the Symposium.

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Leonardo Cascini

[1] Honorary President of LARAM School, Salerno, Italy.E-mail: cascinileonardo@gmail.com

2018 Central Sulawesi Earthquake and Tsunami: Toward the Reconstruction and Recovery

Teuku Faisal Fathani[1,*]

[1] Center for Disaster Mitigation and Technological Innovation (GAMA-InaTEK), Faculty of Engineering, Universitas Gadjah Mada, Yogyakarta 55281, Indonesia

[*] Corresponding author. Tel: +62-274545675; E-mail: tfathani@ugm.ac.id

Abstract

On 28 September 2018, an earthquake followed by tsunami in Central Sulawesi caused severe damage and resulted in a death toll of more than 2,000 people. The epicenter of the M7.4 earthquake was northeast of Palu City, at a relatively shallow depth of 10 kilometers. The earthquake occurred at 18:02:45 Central Indonesian Time. Seven aftershocks were recorded until 18:21 with magnitudes of M4.7 to M6.3. The M7.4 earthquake was followed by a series of tsunami waves as high as 6 m. The earthquake caused liquefaction and earth flows, which affected a 380-hectare area in Sigi District and Palu City.

Sulawesi Island has several geological structures, one of which is the Palu Koro Fault. The fault is 240 km long north to south, crossing Palu City to Bone Bay. The Palu Koro Fault is an active sinistral fault which moves north 25-30 mm/year. As a result, earthquakes often occur in the area. One of the important historical earthquakes is the 1927 M6.3 earthquake which was followed by a 15 m high tsunami wave causing 2,500 casualties. The 2018 earthquake was also caused by the Palu Koro Fault. Other notable phenomena caused by the 2018 earthquake were tsunami, liquefaction, and earth flows. The expert opinion is that the tsunami was most likely caused by a submarine landslide. The landslide occurred in Palu Bay between 200 and 500 m deep. Meanwhile, the liquefaction that occurred in several areas was then followed by earth flows. The liquefaction was caused by earthquake induced shaking in a soil layer dominated by saturated fluvial and alluvial sediments.

Since 2014, Gadjah Mada University has carried out research related to seismic activity in Palu, Central Sulawesi. The research on sediment thickness using microtremors indicates that the mountainous area has a low seismic hazard index while the coastal plain has a high seismic hazard index. The sediment thickness in Palu ranges from >125 m to <25 m. The sediment thickness can influence the amplification of seismic waves; thicker sediment in an area means higher amplification and the damaging effects of the earthquake are more significant as a result.

This research can be used to guide the recovery and future development plan for Palu City. Changing the mindset from a culture of 'being prepared to respond' to being a 'resilient city' is a long process and can take generational change. Palu is making good progress on this journey, however, given the damage and the high death toll from the 28 September 2018 earthquake, continued mitigation efforts from relevant stakeholders and the community is still very much required.

Accurate Prediction of Landslide

Manchao He

Accurate prediction of landslide is the key to disaster prevention and mitigation. At present, the success rate of landslide early warning is relatively low and the early-warning time is lagging. In response to these problems, a new monitoring and prediction method based on the academic idea of "the sufficient and necessary condition for landslide occurring is the change of Newton force" has been proposed. First, the Newton force change law of rock mechanics was proposed for the first time in the field of rock mechanics, and a double-block mechanics model based on Newton force change measurement was constructed, and a complete set of landslide Newton force measurement theory was formed. Then the monitoring and prediction system and the NPR anchor cable suitable for landslide monitoring with high constant resistance and large deformation were independently developed. The warning mode and warning level of landslide with Newton force was proposed and an integrated landslide control technology was formed. Finally, the system was implemented in 723 monitoring points in 26 demonstration areas in China. All 14 landslide disasters within the scope of the deployment were successfully predicted, more than 100 lives and hundreds of millions of equipment and property was saved.

From susceptibility to risk assessment of geohazard

Michel Jaboyedoff[1,*]

[1] Risk-group -ISTE - Institute of Earth Sciences - GEOPOLIS – 3793; University of Lausanne; CH-1015 Lausanne – Switzerland

[*] Corresponding author. Tel: +41-797523515; E-mail: Michel.Jaboyedoff@unil.ch

Abstract

For the United Nations Office for Disaster Risk Reduction (UNDRR), the word appears in the definition or vulnerability: "*The conditions determined by physical, social, economic and environmental factors or processes which increase the susceptibility of an individual, a community, assets or systems to the impacts of hazards.*", in susceptibility risk assessment, for soil liquefaction. But as the word has a broad meaning, it can be used for several other risks such as the susceptibility of forests to hurricane damage (Imbert et al., 1996; McNulty, 2002), forest fires (Ghorbanzadeh et al., 2019), floods (Zhao et al., 2018), seismic amplification (Jena et al., 2020) or for multi-hazard purposes (Karakas et al., 2023). But many susceptibility scales today are derived from the machine learning approach, which often creates relative scales based on various indicators.

For example, the use of susceptibility in landslide studies is essentially applied to hazard, it is based on an empirical scale of the probability of a hazard at a given location based on the assessment of predisposing factors and summarizing them in a relative hazard scale (Ferrari et al., 2016; Fell et al., 2008; Corominas et al. 2023). It is often used to qualify the failure hazard or reactivation on the basis of various choice indicators.

The degree of destruction depends on the types of objects and their sensitivity to the level of intensity of the dangerous phenomenon. This sensitivity can be cross-referenced with the relative frequency of the intensity of the hazard, which makes it possible to qualify the level of risk. In fact, risk matrix approaches are based on the above statement, which uses both susceptibilities to qualify the hazard and vulnerability or the level of damage.

These empirical or expert approaches are so common that certain rules need to be developed to avoid problems, such as classes definition. In addition, as can be seen from the comparison with observations, uncertainty can appear to be significant (Cardinali et al., 2002). This is why it is necessary to develop different ways of introducing uncertainty into these approaches. This can be done by weighting the data or by asking to assess the possibility of belonging to each class of damage and risk, or by using methods such as the Bayesian approach to distribute weights over neighboring classes. Examples of these approaches will be given. Examples of these approaches will be given.

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Debris Flow and Road Infrastructure Vulnerability, Risk and Resilience

Mike G Winter[1]

Abstract

The provision of safe, reliable and resilient networks is a fundamental requirement placed by society on those responsible for our transport systems. Resilience can be viewed at a number of levels ranging from asset, through link, node, and route, to network; each must be resilient in order to achieve the goal of a resilient road network. In this paper a personal view of resilience, the effects of climate change thereon and upon our assets is given. Risk and vulnerability are considered and applied examples of quantitative risk assessment described along with their application. Two case studies are presented to illustrate the potential challenges of both asset redundancy (or diversion) and recovery, the latter in terms of potentially complex governance.

Keywords

Debris flow, road, infrastructure, vulnerability, risk, resilience

Introduction

Road and highway infrastructure perform a fundamental role in ensuring the smooth and effective running of society. They provide a means to move goods in a timely manner and access to education, employment, health opportunities, and to social and leisure activities. Industries that tend to be particularly dependent on transport generally, and road transport specifically, include forestry, fishing, agriculture and tourism as well as manufacturing.

The provision of a reliable and resilient network is thus a key objective for governments and their transport and road authorities. Natural hazards such as inter alia landslides, floods and wildfires present major challenges to the achievement of this objective and these challenges are significantly compounded in the light of ongoing climate change. In this context resilience can be defined as "… the ability of assets, networks and systems to anticipate, absorb, adapt to, and/or rapidly recover from a disruptive event" (Anon. 2011). Reeves et al. (2019a, 2019b) delivered guides to enhancing resilience for road and rail and illustrated the components of resilience as the four Rs (Fig 1):

- Redundancy (spare capacity or diversion routes).
- Reliability (ability to operate under a variety of conditions).
- Resistance (physical robustness).
- Recovery (respond and recover from disruption).

Typically, the focus of engineers is on assets, and while this often considers how different assets interact at either a nodal or route level, the bigger picture is represented by the route or the network (Fiddes et al. 2024).

In this paper a short overview of some of the key challenges that face geotechnical engineers is given using case studies to illustrate key points.



Fig. 1 The components of resilience (after Reeves et al. 2019a, b).

Effects of climate flux

The effects of climate change in terms of changing rainfall patterns, increased temperatures and other key impacts are well known at a conceptual level. However, the precise magnitude of these changes remains the subject of significant uncertainty, and in addition the weather patterns associated with climate are inexorably linked to variability and extremes (Winter et al. 2010a). Winter et al. (2010b) articulated

the uncertainties inherent in climate flux within the Rumsfeldian framework of:

- Known knowns: These include historic and recent climate trends, their relation to current patterns and the fact of climate change (the sequence of greenhouse gas emissions, global warming, and climate change and instability).
- Known unknowns: The precise degree and nature of climate change and some of its impacts, particularly in the light of the variability in climate change forecasts and likely instability in year-on-year climate patterns. These impacts might, for example, include the reaction of vulnerable human populations to both climate change and instability.
- Unknown unknowns: The nature of some other impacts of climate change, although as these are genuinely unknown unknowns these will really have to wait until our knowledge is more complete – that these are unknowns is after all the point. Possibly the real value of this element of the framework is as a reminder that there will always be issues that arrive unexpectedly out of left field.

Such climate change has a fundamental effect on the resilience of a network and Winter (2024) demonstrates the transition from a low frequency base in which resilience can be increased, at least in principle, through a scenario in which the recurrence interval matches the recovery time and resilience can barely be maintained. This culminates in a scenario in which the recurrence interval reduces (frequency increase) so as to be greater than the time required to recover, and resilience inevitably declines as full recovery is not possible between events (Fig. 2).

Risk and vulnerability

Typically, we consider the risk to a person or object as the product of the hazard (H), the elements at risk (E, the person or object), and the vulnerability of the elements at risk (V), or:

$R = H \times E \times V$

Clearly, there is a need to evaluate each component of this equation in order to properly assess risk assessment. Despite the tendency of many geoscientists and engineers to focus on the hazards it is important to note that the elements at risk and vulnerability are equally important in a risk assessment. The proportion of resources allocated to each element is situationally dependent and Winter et al. (2024) presented a case in which the lack of data on the elements at risk led to the greater share of resources being allocated to that part of the assessment.

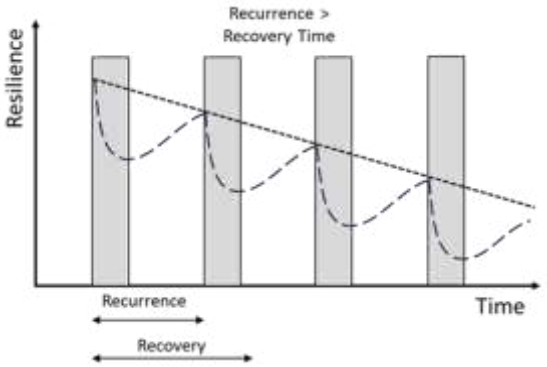


Fig. 2 A simplified, conceptual illustration of the effects of repeated hazard events on infrastructure system resilience when the recovery period is greater than the recurrence period thus creating a decrease in system resilience over time. As adapted from the original by Dijkstra & Dixon (2010), representing vulnerability, and sequentially adapted by Milne et al. (2016), Reeves et al. (2019a, 2019b) and added to by Winter (2024).

Until relatively recently the assessment of vulnerability for landslides was an area that was relatively poorly understood and applied. However, this has changed substantially in the last decade or so as the concept of fragility curves has been adopted from seismic engineering (Fig. 3).

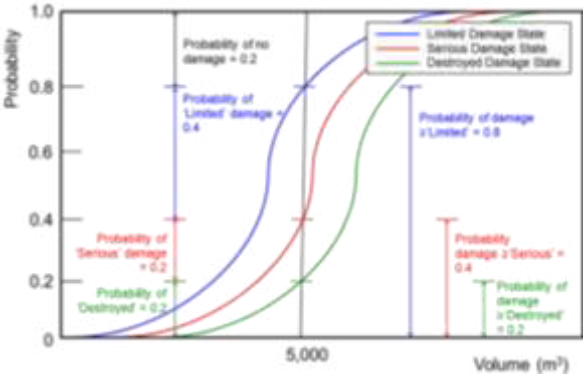


Fig. 3 Hypothetical fragility curve illustrating the probabilities of a given damage state being exceeded and the conditional probabilities for each damage state for a 5,000m^3 event (adapted from Winter et al. 2014).

The vulnerability to debris flow for impacted buildings has been expressed using fragility curves and/or probabilities of exceedance of damage states (Haugen & Kaynia 2008; Jakob et al. 2012; Quan Luna et al. 2011; Papathoma-Khöle et al. 2012), while Winter et al. (2014) developed fragility curves for the effects of debris flow on road infrastructure and Mavrouli et al. (2014) developed such curves for building as part of the same project.

The concept of fragility also lends itself to being based on the results of modelling and in that sense has been widely applied including to damage related to both highway and railway embankments and cut slopes (Argyroudis & Kaynia 2015), to cantilever retaining walls (Argyroudis et al. 2013), to the settlement of bridges (Peduto et al. 2018) and to moisture ingress and scour of embankments (McKenna et al. 2021).

As noted in the introduction the resilience of road infrastructure is heavily dependent upon the resilience of the individual assets (Fiddes et al. 2024) but the overall performance is dependent upon the resilience of the individual assets that exist on the links between nodes, the route and ultimately the network. To this end Argyroudis et al. (2019) developed the concept of Systems of Assets – a combination of interdependent assets exposed not to one, but to multiple, hazards depending on the environment within which these reside (Fig. 4). This multi-element, multi-hazard approach presents a far more realistic, real-world, approach to understanding and assessing the behaviour and the fragility of assets and Systems of Assets subjected to one or a sequence of similar or disparate hazards and may be expected to come to the fore in future studies.



Fig. 4 Transport infrastructure in diverse ecosystems exposed to multiple hazards: High capacity and speed roads (e.g. motorways) in mountainous areas.

Quantitative risk assessment

The use of fragility curves to express vulnerability as part of quantitative risk analysis is a potentially powerful tool. While such studies are as yet rare, recent quantitative risk assessment work has at least been informed by the vulnerability of people to debris flow hazards as considered through the means of fragility curves (Winter & Wong 2020; Winter et al. 2024).

Such QRA can be a powerful tool to understand both the absolute risk at a given site but also the relative risk between different sites. Work on three sites in Scotland is illustrated in Fig. 5 and can be used to aid the prioritization of budgets between sites and routes (Winter & Waaser, 2024a). The QRA results presented in Fig. 5 include the A83 Rest and be Thankful, for which the risk is presented both without and with mitigation, the A85 Glen Ogle and the A82 Glen Coe. Each of the QRAs considers the risk to road users in moving vehicles while the A82 Glen Coe QRA additionally considers the risk to road users who stop in car parks that are built on material from past debris flows.

Case studies

Luib debris flow Isle of Skye, Scotland

The Luib debris flow on the Isle of Skye in Scotland occurred on 30 September 2022 in response to significant rainfall of around 120mm, with a peak 15-minute rainfall of 7.8mm (i.e. an intensity of 31.2mm/hr). The area is subject to high rainfall with a mean for the years 2002 to 2022 of 3,918mm (minima 2,732mm; maxima 4,983mm). Details of the event and a detailed rainfall analysis are presented by Winter & Waaser (2024b).

The debris flow blocked the road for slightly in excess of two hours after which the eastbound (downslope) carriageway was reopened to alternating one-way traffic controlled by traffic lights. The response to the event is an exemplar of recovery (see Fig. 1) but it does raise a number of issues regarding redundancy.

The section of the A87 trunk (strategic) road in question forms a single link between the village of Broadford to the southeast and the junction with the A863 at Sligachan to the northwest, a distance of approximately 26km. While there is no suggestion that a larger single event, or series of events, that may close the road for a longer period is likely, without a road-based, on-island diversion the question as to what would happen in such a case is relevant.

The diversion would involve both road and ferry route via the Outer Hebrides, with between 142km and 303km by road and between two and three ferries with a sailing time of between five and six hours (plus waiting times). It seems clear that any longer-term closure in this area would cause major access issues for the largest settlement on the Isle of Skye, Portree, which is located to the north and west.

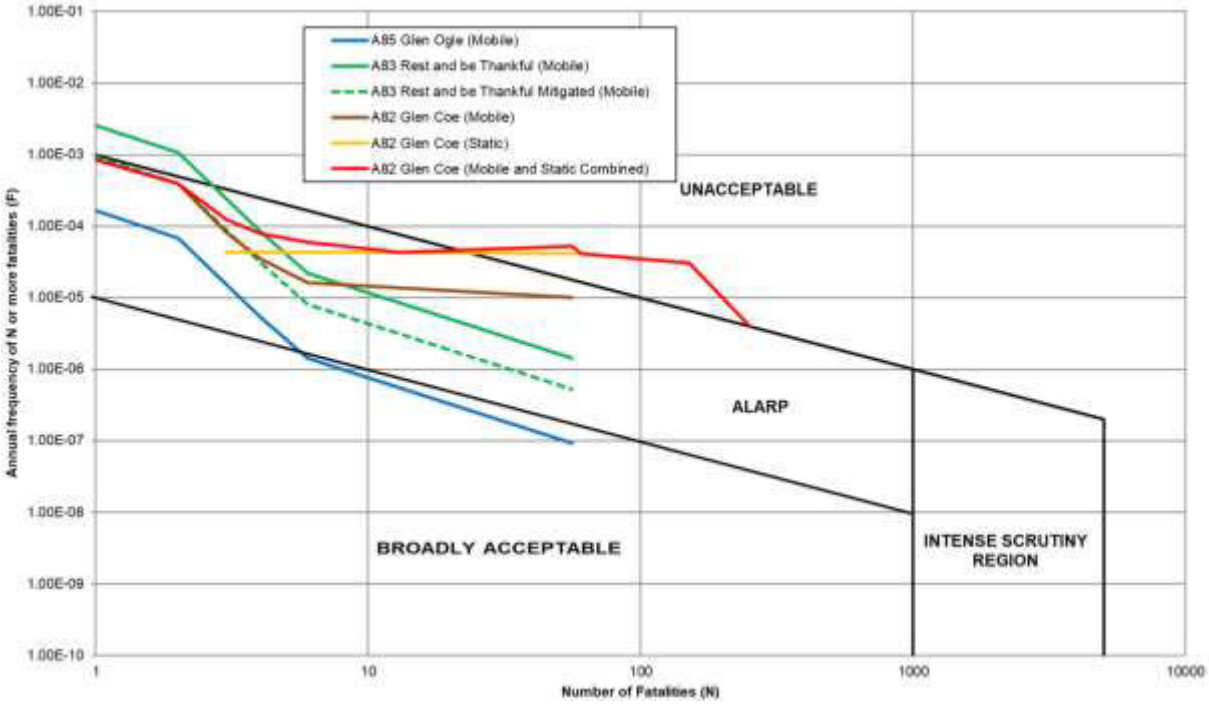


Fig. 5 F-N curves based on the Wong et al. (2004) approach for the A83 Rest and be Thankful, before and after mitigation measures, A85 Glen Ogle and A82 Glen Coe, for both mobile traffic/road users and static elements at risk (parked vehicles and people) and those combined (after Winter et al. 2024).



Fig. 6 The Luib debris flow. The A87 road is at the base of the slope.

Kosova landslide, Bosnia & Herzegovina

The Kosova landslide occurred on 19 May 2014 following a prolonged period of heavy rain and flooding that caused significant disruption in Bosnia and Herzegovina and other parts of the Balkans. As of 20 May, an estimated 24 lives had been lost in Bosnia and Herzegovina as a direct result of the floods with a further seven people still missing. The rainfall was the highest in 120 years of recorded measurements.

The landslide is located on the west bank of the Bosna River, which forms the boundary between the Federation of Bosnia and Herzegovina and Republika Srpska. These two entities, along with the District of Brčko, form the country of Bosnia and Herzegovina.

The landslide is described by Winter et al. (2020) including damage and morphology, geology, future impacts, investigation and monitoring, and governance.

The Kosova landslide is a translational slide movement over a significant area. It has already wreaked considerable damage to the village and rendered a significant number of properties uninhabitable; it would not be inappropriate to describe these properties as having been completely destroyed. It has also caused significant damage to the main road that runs in close proximity to the postulated location of the toe. The Kosova landslide clearly presents a major operational risk for the M17 (E73) road and thus to the road authority.



Fig. 7 View of the northern lateral margins of the Kosova landslide. Scarp features and damaged buildings are evident.

Given the attendant risk to houses on the flood plain there is a case to be made that the Municipality and/or the Canton should be involved. The potential for damming the Bosna River should there be a substantial downslope movement, also brings both the River Basin Agency and the Civil Protection Agency into that forum. However, given that the Bosna River forms the border between the Federation of Bosnia and Herzegovina and the Republika Srpska at this location, any flooding (both upstream and downstream) would affect both the Federation and the Republika; it thus seems appropriate that the relevant agencies from the latter should also be involved. In this context consideration ought to be given to a national, Bosnia and Herzegovina, effort to monitor the landslide and to effect remediation risk reduction where necessary.
The damage to date is relatively slight compared to the potential for future damage. This clearly brings into play a need for a joined-up multi-entity and/or national approach to the investigation, monitoring and remediation of the Kosova landslide.

Closing Remarks

The successful operation of a safe, reliable and resilient road network is challenging. Such challenges are significantly magnified by the ongoing process of climate change and its associated uncertainties. The increased frequency (and magnitude) of events such as landslides, debris flows and floods can rapidly lead to a situation in which the time required to recover from an event is greater than the interval between successive incidents leading to an inevitable reduction in resilience.
In order to achieve resilience a view that encompasses not only the asset but the links and nodes, the route and ultimately the asset must prevail.
Risk assessment can inform decision-making including the allocation of budgets but also be an important tool for understanding and articulating issues surrounding resilience and the lack thereof.
As climate change progresses, we will see some extreme examples and although largely hypothetical in terms of its likelihood, the Isle of Skye case study articulates a ‘worst case’ scenario in terms of achieving resilience through redundancy. The example of the Kosova landslide in Bosnia and Herzegovina illustrates the challenges associated not only with the impacts of a potentially significant event but also the need for effective and appropriate governance.
There is an exciting and challenging road ahead.

Acknowledgments

The views presented here are the views of the author and are not intended to reflect those of any other individual or organisation.

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Mike Winter (✉)
[1] Winter Associates Limited, Kirknewton, EH27 8AF, United Kingdom
E-mail: mwinter@winterassociates.co.uk

Contribution to Global Warming Countermeasures via Use of Wood for Ground Reinforcement

Masaho Yoshida[1*]

[1] Department of Civil Engineering, National Institute of Technology, Fukui College, Sabae, Fukui 916-8507, Japan

[*] Corresponding author. Tel: +81-778-62-8305; E-mail: masaho@fukui-nct.ac.jp

Abstract

Unequal settlement owing to soil liquefaction is a type of earthquake damage to structures such as detached houses. If the inclination of the house is large, then reconstruction or repair work is required. Even if the inclination is minor, cracks in the foundation and outer walls as well as problems in opening and closing doors may occur over time.

Hence, we propose a liquefaction countermeasure to increase the density of the ground such that it is equivalent to the volume of logs using logs as an improvement material in loose saturated sandy soil under and around the structure at depths below the groundwater level as shown in Fig.1. (Yoshida et al., 2009). This method contributes to global warming through carbon storage because carbon-containing wood is stored underground for a long time as a ground-reinforcement material. This lecture introduces the background in the development of this method and presents some recent studies.

Examples of these studies are as follows: This method is based on the principle of densification and does not consider the effect of the presence of wood, which is stiffer than soil, in soil, i.e., an increase in the liquefaction strength due to the shear resistance of the composite soil and logs. Simple shear tests were conducted using a model ground to clarify the effect of log stiffness on liquefaction resistance in saturated sandy soil. Consequently, the shear resistance increased by installing the column in the ground, and a greater column stiffness increased the shear modulus of the composite ground.

Shaking table tests under a 1 g gravity field were conducted using saturated sandy soil with the same relative density using the same diameter and length of the log but different log-replacement ratios. Consequently, the shear stiffness of the soil and liquefaction resistance increased with the replacement ratio of the logs, although the relative density remained the same.

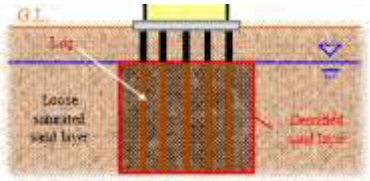


Fig. 1 Principle of proposed method

Acknowledgments

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Employing comprehensive engineering geological models in assessing rock slope behavior modes and geotechnical engineering design

Vassilis Marinos[1,*]

[1] School of Civil Engineering, National Technical University of Athens, Athens 15780, Greece
[*] Corresponding author. Tel: +30-2107723749; E-mail: marinosv@civil.ntua.gr

Abstract

Rock masses are often connected with severe deep-ground sliding, rotational and translational sliding, mudflows but also with systematic structural instabilities and rockfalls. Responsible for these variant sliding mechanisms are the different geological models that have developed after their genesis, tectonism, metamorphism, alteration, weathering or also recent tectonic activity. Geological model of sedimentary varied and heavily sheared rock masses like flysch are examined. Geological models characterized by severe alteration, like in ophiolitic melanges, have their own rubrics and can present different sliding mechanisms. In an environment of distinctive and /or deep weathering profiles, like in crystallic rocks of gneiss or granites or other, we can identify structural failures, multiplanar slides or classic rotational isotropic failures. These various failure modes are directly linked to rock mass types composed within these diverse geological models. The geotechnical design-analysis can then be relied on a comprehensive geological model using realistic parameters and an acknowledged failure mechanism.

For example, flysch series that is composed of varying alternations of competent sandstones and weak, of low generally strength, pelitic sediments, and is associated with orogenesis, deformed from light fracturing and/or folding to severe thrusting and shearing. The flysch has thus suffered from large com-pression, and very weak rock masses have been produced. In these cases, the original structure is no longer recognizable, and blockiness is lost. Additionally, siltstone members (or clayey rocks as clayshales and claystones) are very vulnerable to weathering, and fissility may be developed parallel to the bedding when these rocks are exposed to the surface or are very close to it from wetting – drying processes. Therefore, the special weathering profiles and groundwater conditions but also the paleogeographic evolution that can be met in a flysch environment complete the possible geological model. Consequently, numerous lithotypes and rock mass entities can be produced in a flysch environment. As a result, slope failure mechanisms in flysch formations are different according to the siltstone-sandstone participation, the intensity of tectonic disturbance and the weathering degree. Several conceptual geological models, related to numerous landslide case studies, are illustrated and analyzed in accordance to the different sliding mechanisms.

Finally, since the evaluation of the principal anticipated failure mechanism is essential for the selection of the appropriate

geotechnical parameters, be it the overall rock mass strength, intact rock strength and joint strength, used for the stability analysis, a standardization of the qualitative engineering geological characteristics and the assessment of the behavior in slope stability for flysch rock masses is analyzed.

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Multiscale modeling of multiphase flow interaction with mitigation structures

Marina Pirulli[1], Andrea Pasqua[1,2], Alessandro Leonardi[2]

Abstract

Debris flows are multiphase flows consisting of water, fine-grained, and coarse-grained sediments, whose complex behavior poses challenges for understanding the process dynamics, and mitigating the effects on the territory. Standards for designing effective protection measures are lacking, and are unlikely to become available soon. A variety of simulation methods exist, ranging from microscopic to macroscopic scales, offering tools to model multiphase flows and address the above challenges. Discrete methods represent flows as assemblies of colliding particles. They are useful for studying particle clogging, but appear inadequate for capturing the entirety of events due to computational demands. Continuum-based numerical methods, such as depth-averaged and three-dimensional models, offer more comprehensive simulations. Depth-averaged models efficiently simulate the entire process, but struggle with flow-structure interactions. In contrast, three-dimensional models, while more complex, excel at studying specific aspects like the impact of debris flow against mitigation structures. This study focuses on the Gran Valley catchment, leveraging historical data and existing mitigation structures. It explores how different modeling approaches contribute to understanding the dynamics of debris flows and their interaction with protective measures. By examining both the process dynamics and structural interactions, the research aims to provide insights into effective mitigation strategies

Keywords

Debris flows, numerical modelling, mitigation structures

Introduction

Debris flows represent a significant natural hazard, characterized by the rapid movement of a mixture of water, fine-grained, and coarse-grained sediments down steep slopes. These events are often triggered by intense rainfall, snowmelt, or seismic activity, posing threats to communities, infrastructure, and the environment in mountainous regions worldwide. The heterogeneous composition and complex behavior of debris flows make them challenging to model accurately. Advanced numerical techniques and extensive field observations are then required to understand their dynamics and mitigate their impacts.

Effective protection measures against debris flows rely on a comprehensive understanding of their behavior and interaction with mitigation structures. While physical barriers, such as slit-check dams and retention basins, can help reduce the risk posed by these events, their design and implementation require a careful consideration of local topography, flow dynamics, and sediment characteristics. Numerical modeling plays a crucial role in assessing the effectiveness of mitigation measures, providing insights into flow patterns, deposition zones, and potential failure scenarios.

In this paper, we explore the contributions of discrete and continuum-based numerical techniques to the understanding of debris flow dynamics and their interaction with mitigation structures. Drawing on data from the Gran Valey catchment - a region with a history of debris flow events and with installed mitigation measures - we conduct a comparative analysis of different modeling approaches. By examining their strengths, limitations, and potential synergies, we aim to develop a multiscale framework, with the final goal of increasing our ability to predict and mitigate the impacts of debris flows in vulnerable areas.

Numerical modeling

Discontinuum-based approach

Discrete numerical modeling approaches represent debris flows as interacting particles (Fig. 1), allowing for detailed analysis of particle motion, collision, and aggregation (e.g., Cundall and Strack, 1979; Galindo-Torres, 2013). These models simulate individual particle behaviour, considering factors like size, shape,

density, and surface roughness (e.g., Zhou *et al.*, 2016). While highly accurate at capturing particle-level interactions, they are computationally demanding, leading to a struggle in simulating large-scale debris flow processes (e.g., Leonardi, 2015).

A key strength of discrete modeling is its ability to simulate particle clogging in mitigation structures, such as barriers and check dams (e.g., Shen *et al.*, 2018), by tracking individual particle movements and interactions. This helps assess barrier effectiveness and identify potential failure modes. Discrete models also investigate flow stratification, sediment segregation, and the formation of levees and lobes during debris flow propagation (e.g., Zhou *et al.*, 2016).

However, scaling up simulations for large-scale events is challenging due to the high computational cost of simulating numerous, limiting the model spatial and temporal resolution. This can hinder the accurate representation of long-distance propagation, channel avulsion, and the interaction of multiple debris flows merging downstream (e.g., La Porta *et al.*, 2024). Despite these limitations, discrete modeling is valuable for understanding particle-scale processes and evaluating the performance of mitigation structures at the local level.

Continuum-based approach

Continuum-based numerical models represent debris flows as continuous fluid-solid mixtures (Fig. 1), using balance equations for mass, momentum, and energy (e.g., Pudasaini, 2012). These models simulate flow dynamics over large spatial and temporal scales (e.g., Pirulli, 2010) and can be divided into depth-averaged models and three-dimensional models.

Depth-averaged models, or two-dimensional models, simplify debris flow dynamics by averaging flow properties over the vertical dimension (e.g., Savage & Hutter, 1989). They are computationally efficient and can simulate the entire debris flow process from initiation to deposition (e.g., Pirulli et al., 2011), making them suitable for large-scale hazard assessment (e.g., La Porta *et al.*, 2024). However, they cannot capture three-dimensional flow features or flow-structure interactions, potentially oversimplifying flow dynamics.

In contrast, three-dimensional models provide a detailed representation of debris flow dynamics, accounting for spatial variations and topographic features (e.g., Leonardi, 2015), by solving the full three-dimensional Navier-Stokes equations. These models can simulate complex flow phenomena (e.g., Gray and Tang, 2018) and flow-structure interactions, providing insights into energy dissipation, pressure distribution, and structural loading (e.g., Pasqua *et al.*, 2022). However, they are computationally intensive and require high-resolution topographic data (e.g., Pastor *et al.*, 2011), making them challenging to apply over large spatial domains. These models also require a more detailed parameterization of turbulence, sediment transport, and bed roughness, which may introduce uncertainty and complexity into the modeling process (e.g., Mangeney *et al.*, 2010). Despite these challenges, three-dimensional modeling offers valuable insights into specific aspects of debris flow behavior, such as flow-structure interaction and sediment deposition patterns.

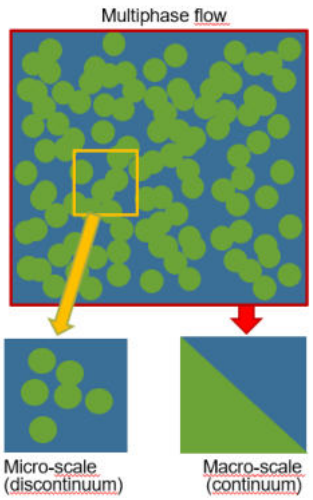


Fig. 1 Discrete (micro) vs. continuum (macro) representation of a multiphase flow.

Comparative analysis

In this section, we conduct a comparative analysis of discrete and continuum-based numerical modeling approaches applied to debris flows, using data from the Gran Valey catchment as a case study. We examine their performance in replicating past debris flow events, predicting the effectiveness of mitigation structures, and capturing the overall dynamics of debris flow propagation. By comparing the strengths and limitations of each modeling approach, we aim to identify opportunities for improving hazard assessment and risk management in vulnerable areas.

The Gran Valley catchment serves as a valuable testbed for evaluating debris flow models and mitigation strategies due to its history of past events and installed mitigation measures. In this section, we provide an overview of the catchment characteristics, past debris flow events, and implemented mitigation

measures. We then conduct a detailed analysis of debris flow dynamics and interaction with mitigation structures using both discrete- and continuum-based numerical models.

Gran Valey Catchment

The town of Saint-Vincent, nestled at the base of Mount Zerbion in Aosta Valley Region (Italy), is prone to rapid landslides due to the steep slopes and fractured rock that characterize the upper Grand Valey basin. Historical records detail frequent debris flows often triggered by intense rainfall in spring or summer, involving thousands of cubic Glendameters of material (Leonardi & Pirulli, 2020). To safeguard Saint-Vincent and surrounding areas from such events, defense structures were constructed along the Grand Valey channel. These include two check dams near the Pèrriere hamlet (Figs. 2 and 3), with a couple of steel net barriers and a terminal check dam further downstream.



Fig. 2 Grand Valey catchment. A: main release channel; B: secondary release channel; Area 1, Area 2: triggering areas; (1) the two check dams located close to Perrière. The arrow indicates the confluence of the two channels (modified after Leonardi and Pirulli, 2020).

To assess the impact forces of debris flows on defense structures, the Aosta Valley Region government, in collaboration with Politecnico di Torino, implemented a monitoring system. This system equips the upper check dam with sensors, converting its 18 vertical steel beams into transducers capable of measuring horizontal loads from the flowing mass. These sensors monitor both the impacts and intervals between them, recording strain at the base of the steel beams at preset intervals. Upon detecting a sudden strain change, the system captures continuous high-speed data streams, enabling a time-history of force measurements even during rapid events such as debris flows (Leonardi & Pirulli, 2020).

Two debris flow events recorded by the above described monitoring system are used here: on June 9th and July 11th 2016 (Fig. 3), respectively.

The mass mobilized in both events was retained almost completely by the two filter barriers at Perrière. For this reason, the estimation of the involved volumes was made on the basis of the material removed from the basins after each event: 1875 m^3 in the June event and 4420 m^3 in the July event (Leonardi *et al.*, 2021). The available data was used for comparison with the different numerical approaches proposed in the following section.



Fig. 3 The check dams in Perrière filled afted the July 11th 2016 debris flow event.

Numerical results

Depending on the aspect to investigate, the most appropriate numerical approach has to be selected to obtain useful and meaningful results.With this in mind, different numerical methods have been applied to analyse the dynamic behavior of flow movements occurring along the Grand Valey catchment.

The first approach was devoted to an overall analysis of the process dynamics. As detailed in the numerical section, depth-averaged continuum models are well suited to these purposes. In the last decades, these methods have been widely validated and it has been proven their capability to reproduce the entire debris flow process, despite the loss of in-depth information concerning the velocity profile during movement. The depth-averaged RASH3D code

(Pirulli, 2010) has been then used to reproduce the Grand Valey event that occurred on July 11th 2016. This event caused the filling of the two basins existing upstream of the 2 check dams located near Pèrriere. It was also characterized by the release of material from both branches in which the Grand Valey splits in the upper part of the slope (Area 1 and Area 2, Fig. 2). Branch B activates only during the most intensive rainfall events.

An intriguing aspect that emerges is that, despite the material being released simultaneously from Area 1 and Area 2, the flow from Area 2 is delayed compared to Area 1 (Leonardi *et al.*, 2021). This behavior aligns with the site survey interpretations. The initial surge from Area 1 likely caused clogging of the first barrier. Approximately 300 seconds later, a subsequent surge carrying woody debris arrived from Area 2, flowing over the deposit in the retention basin soon after the initial event, and clogging the second barrier (Fig. 4).

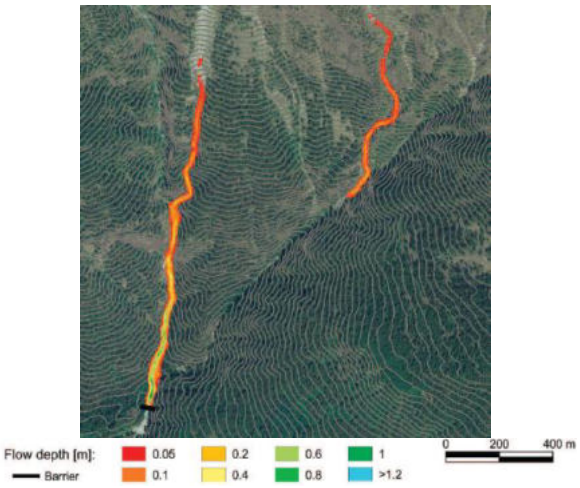


Fig. 4 Modeling of the event on July 11 with a Voellmy rheology (modified after Leonardi *et al.*, 2021)

Moving to the analysis of the flow-structure interaction, especially with reference to the check dam equipped with the monitoring system. It emerged the need of moving to a fully three-dimensional approach that guarantees a detailed description of the mass in-depth velocity distribution. To retain the capability of simulating the entire debris flow process, a coupling between a depth-averaged and a fully three-dimensional model was proposed (Pasqua *et al.*, 2022). This allows to have a computationally efficient tool and an appropriate level of accuracy when the flow impact the structure. A comparison between monitoring recordings and numerical results is given in Figure 5 to highlight the good match between measured and computed forces acting on the central elements (07, 08 and 10) of the filter impacted by the flow in Pèrriere.

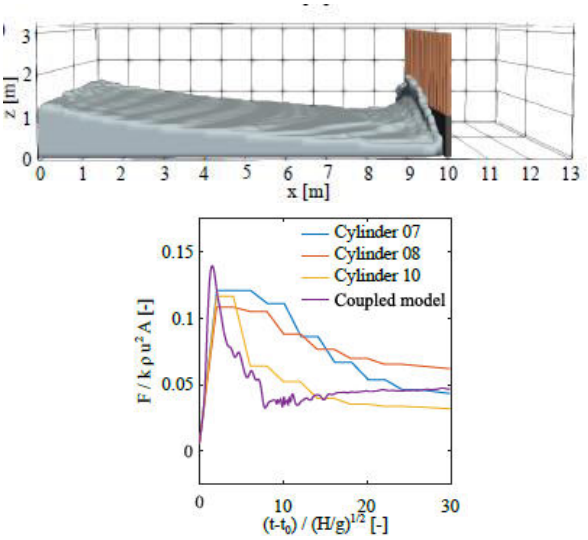


Fig. 5 Numerical results obtained with the coupled depth-averaged and three-dimensional continuum-based numerical model for the July 11 debris flow event. Comparison between measured and simulated impact force F against the Perrière monitored check dam. The forces are scaled to the dynamic load applied to barrier cross-section, while the time is scaled with the time of first impact (t_0) (modified after Pasqua, 2023).

Focusing on the flow-structure interaction, some aspects are however uncatchable by a continuum model, even if three-dimensional. This mainly concerns the possible clogging of the barrier filter and its sizing on the basis of the granulometric composition of the flowing mass. This analysis requires that the numerical model is able to describe particles that are part of the movement and their behavior. For this reason, a discrete-based numerical model is necessary. This allows to investigate the possible clogging of the filter and its development over time, in reason of the granulometric characteristics of the moving mass.

In order to give an interpretation to the signals recorded on St. Vincent site, the discrete element method (DEM) is here proposed. The type of load exerted on the barrier is evaluated by assuming the barrier itself is hit by a single surge of grains (Fig. 6).

The output is plugged into a finite-element model (FEM) of the barrier as a time-history of external actions. A dynamic analysis is then performed,

studying how the bending moments at the base of the beams evolve. The strain at the base is then compared to the site recordings. the main factor that determines amplitude and sign of ε_{zz} is M_x, rather than M_y. This is confirmed by the FEM computations of ε_{zz}. Notwithstanding the many simplifications adopted in the procedure, the model is able to capture the order of magnitude of the recorded strains (Fig. 7).

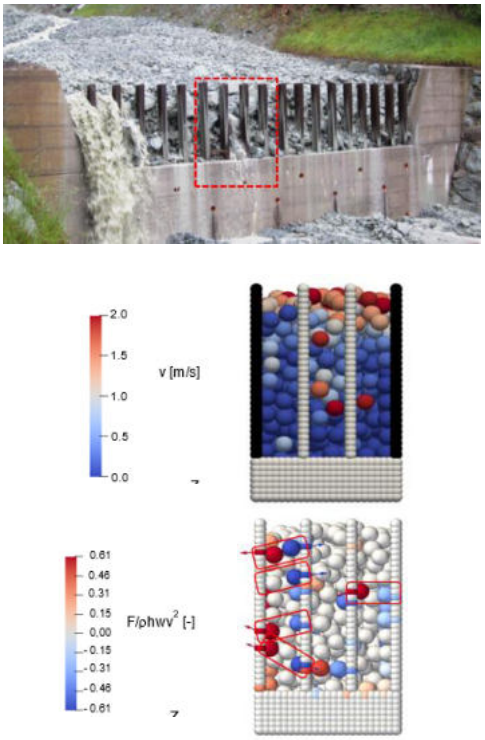


Fig. 6 Schematic of the mechanism of force transmission to be barrier, and corresponding strain field on the beams. The strongest granular arches are highlighted with red boxes (modified after Leonardi *et al.*, 2019).

Conclusions

Debris flows are complex natural phenomena that pose significant risks to human life, property, and infrastructure in mountainous regions worldwide. Effective mitigation of these hazards requires a thorough understanding of their dynamics and interaction with mitigation structures. In this paper, we have provided a comprehensive analysis of discrete and continuum-based numerical modeling approaches applied to debris flows, using data from the Grand Valey catchment as a case study.

Our analysis highlights the strengths and limitations of each modeling approach, as well as their potential synergies in providing comprehensive insights into debris flow dynamics and mitigation strategies. While discrete models excel in capturing particle-level interactions and clogging phenomena, continuum-based models offer a broader understanding of flow dynamics and can simulate the entire debris flow process. By integrating these approaches into comprehensive analyses, we can enhance our ability to predict and mitigate the impacts of debris flows in vulnerable areas.

Moving forward, future research should focus on refining numerical models, integrating data-driven approaches, validating models with field data, improving mitigation structure design, and increasing public awareness and community engagement. By addressing these challenges and embracing emerging technologies and methodologies, researchers can contribute to the development of more effective debris flow mitigation strategies, ultimately saving lives and protecting communities from the devastating effects of debris flows.

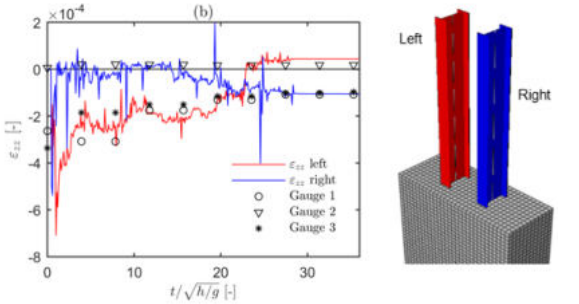


Fig. 7 Time-history of the bending moment registered at the location where the strain gauges are installed, and comparison with the results of the DEM numerical simulations (modified after Leonardi *et al.*, 2019).

Acknowledgments

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Marina Pirulli (✉)
[1] Department of Structural Geotechnical and Building Engineering, Politecnico di Torino, Torino, Italy
E-mail: marina.pirulli@polito.it

Andrea Pasqua
[1] Department of Structural Geotechnical and Building Engineering, Politecnico di Torino, Torino, Italy
[2] Department of Civil and Structural Engineering, University of Sheffield, Sheffield, UK
E-mail: andrea.pasqua@polito.it; a.pasqua@sheffield.ac.uk

Alessandro Leonardi
[2] Department of Civil and Structural Engineering, University of Sheffield, Sheffield, UK
E-mail: a.leonardi@sheffield.ac.uk

Simulating the Ability of Forests as Nature-Based Solutions for Geohazards: MPM-LSDEM

Clarence Edward Choi[1,*], Zhengyu Liang[1]

[1] Department of Civil Engineering, The University of Hong Kong, Pokfulam Road, Hong Kong SAR.
[*] Corresponding author. Tel: +852 98695544; E-mail: cechoi@hku.hk

Abstract

Forests are often considered environmentally friendly barriers against granular geophysical flows (Liang et al. 2023; Perzl et al. 2021). To harness these barrier effects, it is essential to model the interactions between flows and forests in detail. However, the complexity of flow-forest interactions poses significant challenges for traditional physical experiments and numerical simulations. Key challenges include simulating flow-tree contact and modeling the biomechanical behaviors of trees. To address these issues, we propose a novel numerical simulator based on the Material Point Method (MPM) and the Level-Set Discrete Element Method (LSDEM). This simulator utilizes the GJK algorithm (Bergen 1999) for contact detection and employs a sampling point level-set strategy to create a realistic contact manifold. Validation of the simulator is achieved through comparison with experimental results and analytical solutions. Findings from numerical parametric studies indicate that the biomechanical behaviors of trees significantly influence the mobility of dry granular flows. In forests with squared tree patterns, shock-shock interactions can enhance flow mobility. Additionally, tree uprooting dissipates flow kinetic energy and reduces flow momentum, though it also produces wood-laden flows, which can intensify the destructive potential of these flows. The results of this study provide new insights into the ability of cultivated or natural forests to mitigate granular geophysical flows.

Acknowledgments

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Lifting the curse of data contamination in landslide hazard models: towards self-improving inventories

Filippo Catani[1,*], Kushanav Bhuyan[1], Sansar Raj Meena[1], Lorenzo Nava[1]

[1] Machine Intelligence and Slope Stability Lab, Department of Geosciences, University of Padova, Padova 35131, Italy

[*] Corresponding author. Tel: +39-0498279193; E-mail: filippo.catani@unipd.it

Abstract

Maps and inventories of geo-hazards are the key data in building reliable predictive models, in particular when using data-driven approaches and machine learning. Errors and omissions in the datasets propagate easily and with unpredictable effects on the final results. For landslide hazard data contamination may often include unmapped occurrences, misclassification, lack of information on material, type, and age, wrong or absent separation of kinematic zones. Studies and recommendations exist that precisely state what should be desirably included in a complete and useful landslide inventory but, despite this, to go from theory to practice remains extremely challenging. Hopefully, some solutions to this problem are suggesting themselves from the wide and flourishing realm of artificial intelligence. Among them, we will briefly summarize here examples of new AI methods and applications capable to improve the automated mapping of landslides in event-based circumstances, the generation of multi-temporal inventories from existing archives of EO data, the classification of landslide typology based on 3D topology, the delimitation of kinematic zones into detachment, runout, and deposition areas where such information is missing.

These techniques, when carefully adopted along with more traditional approaches, may constitute a set of tools able to support a semi-automated and self-improving operational pipeline which, starting from available maps and datasets, might be routinely used over large areas to produce missing information, reduce errors and omissions, and extract model calibration parameters for hazard prediction and prevention purposes.

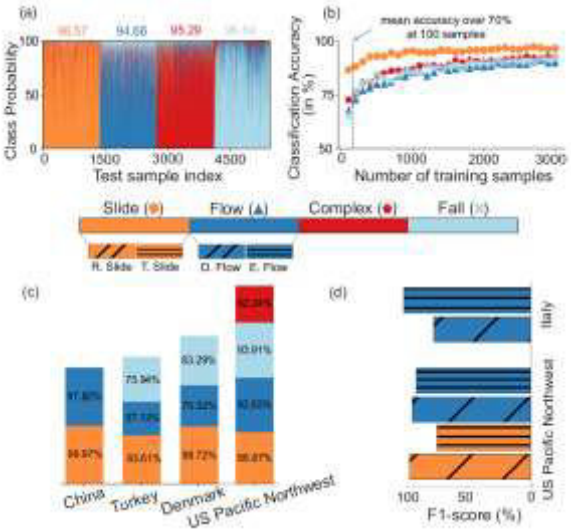


Fig. 1 Accuracy in automated classification (From Bhuyan et al., 2024, modified).

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Modelling the occurrence of landslides at regional scale under a changing climate

Veronica Tofani[1,*], Elena Benedetta Masi[1], Guglielmo Rossi[1], Pierluigi Confuorto[1], Francesco Caleca[1],Nicolò Brilli[1]

[1] Department of Earth Sciences, University of Florence, Florence, Italy
[*] Corresponding author: E-mail: veronica.tofani@unifi.it

Abstract

Landslides triggered by extreme rainfall events pose a significant threat to communities in many regions of the world. These events occur when heavy rainfall exceeds the drainage capacity of soils, leading to unstable conditions. As a result, mass movements are mainly triggered by an increase in pore pressure and an associated decrease in soil strength.

Shallow landslides are one of the most dangerous categories of mass movements, mainly because they often develop into rapid mass movements that take on characteristics of debris avalanches and flows. As climate change continues to influence global weather patterns, the frequency and intensity of these events is expected to increase, increasing the likelihood of landslides occurring. This poses an increasing threat to communities living in landslide-prone areas.

Due to the large number of areas prone to shallow landslides, regional modelling of this type of slope instability is crucial for assisting administrations and disaster management agencies in risk mitigation, especially in hazard assessment for land management decisions and in predicting the temporal and spatial distribution of events for early warning systems.

In this work we present the application of shallow landslide modelling at a regional scale using a physically based slope stability model, namely the HIRESSS model (Rossi et al., 2013, Tofani et al., 2017).

The model is applied in two different study areas in Italy: in the Valle d'Aosta region in northern Italy and in the Marche region in central Italy. These areas have been hit by intense rainfall, which has led to numerous shallow landslides. During the Marche event, on September 15, 2022, the region was hit by an unprecedented rainfall event with a cumulative rainfall of 204 mm in twelve hours, triggering 700 shallow landslides that caused major damage and cost lives.

The work deals with the simulation of past events for the two study areas using the HIRESSS model and focuses on both data preparation and validation of the results.

The results are critically analyzed to determine the advantages and disadvantages of the model for real-time prediction of shallow landslide prediction and the early warning system.

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Advances in Geotechnical Characterization and Failure Analysis through MPM Simulations

Mario Martinelli[1,2*]

[1] Geo-Unit Engineering, Deltares, The Netherlands

[2] Adjunct Research Professor, Department of Civil and Environmental Engineering at Carleton University in Ottawa, Canada.

[*] Corresponding author. E-mail: mario.martinelli@deltares.nl

Abstract

This lecture focuses on recent advancements in geotechnical characterization and failure analysis utilizing numerical simulations based on the Material Point Method. It addresses two specific areas: site characterization and failure analysis.

In terms of site characterization, the presentation explores the modeling of Cone Penetration Tests (CPTs), including validation examples in clay and sand, discussions on employing state-dependent models for CPT validation across (Fig. 1) various drainage conditions, modeling experiences in layered soils with thin layers of soft clay, and the determination of state parameters via multivariate regression based on MPM CPT simulations.

For failure analysis, it examines the application of MPM in simulating landslides and tailing dam failures. Case studies include a lab-scale retrogressive landslide in liquefiable soil and an analysis of a tailing dam failure, illustrating failure modes and contributing factors.

The lecture showcases recent research findings, particularly highlighting the predictive capabilities of integrating critical state models with the Material Point Method. It offers valuable insights for researchers and practitioners in geotechnical engineering, providing an overview of recent advancements and challenges in both site characterization and failure analysis.

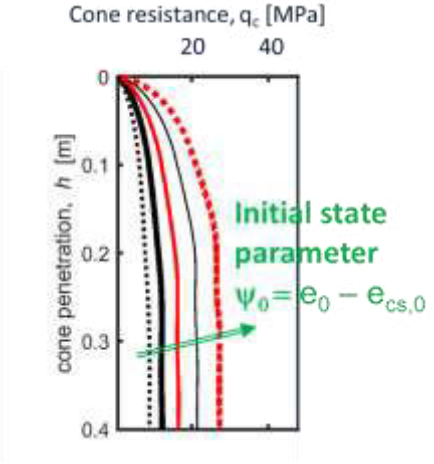


Fig. MPM simulation of CPTs in sand: effect of the initial state parameter.

Acknowledgments

The examples provided in this presentation summarize ongoing international research activities. Sincere gratitude goes to all contributors from Deltares, VirginiaTech, University of Salerno, University of Western Australia, and Carleton University, who will be named and acknowledged within the presentation.

Landslide-induced tsunamis – global overview and expert-based hazard mapping in Indonesia

Katrin Dohmen[1], Anika Braun[1,*], Hendy Setiawan[2], Fikri Faris[3], Tomas Manuel Fernandez-Steeger[1]

[1] Department of Engineering Geology, Technische Universität Berlin, 10587 Berlin, Germany
[2] Department of Geological Engineering, Universitas Gadjah Mada, Yogyakarta, Indonesia
[3] Department of Civil and Environmental Engineering, Universitas Gadjah Mada, Yogyakarta, Indonesia
[*] Corresponding author. Tel: +49 30 314-72870; E-mail: anika.braun@tu-berlin.de

Abstract

„Atypical“ or „silent“ tsunamis refer to tsunami waves that are not primarily induced by earthquakes, but by other phenomena, such as volcanic flank collapses or landslides. Submarine and coastal l andslides can induce tsunami waves that reach extremely high heights of up to several tens of meters. These tsunamis typically impact areas within approximately 100 km of the landslide source. Due to the short time between wave initiation and arrival at the coast in addition to the challenges in detecting the triggering landslide, early warning is extremely difficult. One first step towards reducing landslide-induced tsunami risk is to understand which parts of the coastline are prone to this type of hazard so the affected communities can be made aware and further measures can be taken by local authorities.

In this study, we first conducted a systematic global review of historic submarine and coastal landslide-induced tsunami cases documented in the literature to get a better understanding of this phenomenon. We collected data on parameters characterizing the triggering landslides and the resulting tsunami waves and we categorized the cases regarding the landslide trigger and type of water body.

To get a more detailed understanding of where tsunami-inducing landslides can occur, we opted for an expert-based framework. In two online workshops and one survey, we identified 37 parameters and together with a group of international experts on landslides and tsunamis rated them regarding their importance for submarine and coastal landslide susceptibility and wave generation.

The outcomes are currently being translated into a GIS-based heuristic mapping approach for a pilot area in Indonesia, a country highly prone to different types of tsunamis due to the long coastline and high seismic and volcanic activity. We are working with publicly available data, so the mapping approach can be extended to further areas in Indonesia and beyond.

Acknowledgments

This work was sponsored by the German Federal Ministry of Education and Research under Grant No. 03G0906E. We also want to express our sincere thanks to the experts who participated in two online workshops and one survey. Their time and expertise are highly appreciated.

Hydraulic characterization of soil through physical slope modelling tests

Josip Peranić[1,*]

[1] University of Rijeka, Faculty of Civil Engineering, Rijeka, Radmile Matejčić 3, Croatia
[*] Corresponding author. Tel: +385-511 265 943; E-mail: josip.peranic@uniri.hr

Abstract

Despite the pivotal role that the soil water characteristic curve plays in unsaturated soil mechanics, the experimental procedures and measurement techniques commonly used for their determination still have some limitations in deriving representative SWCC for the soil and processes under consideration. These include, among others, loss of hydraulic contact, reduced hydraulic conductivity, unfeasible equilibration periods, artificial testing conditions that differ from those in the field, hydraulic hysteresis, and limitations in the measurement range. On the other hand, field data obtained from hydraulic monitoring of the shallow unsaturated zone of slopes indicate possible deviations from the hydraulic responses predicted by the commonly applied procedures. Therefore, further efforts should be made to improve the understanding of different rainfall infiltration related phenomena and their effects on the stability of slopes, especially considering the change in boundary conditions forced by the ongoing climate change.

This contribution presents some of the results on determining the SWCC of soils by physical slope modelling tests. The study encompasses uniformly graded fine sand – the type of soil that has been used extensively in small-scale landslide modelling, but for which high nonlinearity and steepness in the shape of SWCC typically makes hydraulic characterization difficult. A 30 cm high slope inclined at 35 degrees and instrumented with soil moisture and pore water pressure sensors was subjected to a range of rainfall intensities, to obtain steady-state conditions on soil moisture and soil suction. Initial experience showed that the approach is useful not only for determining SWCC, but also for investigating other phenomena associated with rainfall-induced landslides.

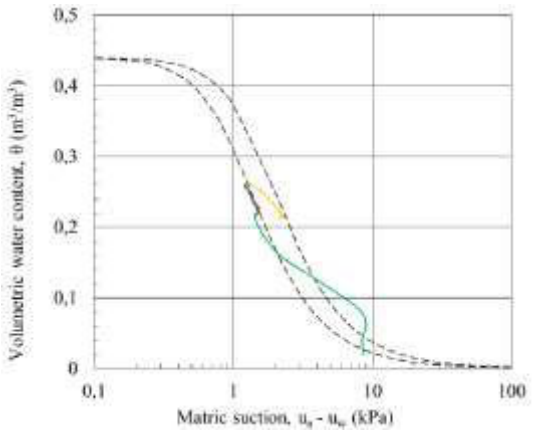


Fig. 1 The reconstructed hydraulic paths.

Acknowledgments

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Liquefied Ground Flow in Uchinada Town induced by the 2024 Noto Peninsula Earthquake, Japan

Yuko Serikawa[1*], Midori Asai[1], Masakatsu Miyajima[2]

[1] Department of Civil Engineering, National Institute of Technology, Fukui college, Sabae 916-8507, Japan
[2] Lifeline Disaster Mitigation Lab., Kanazawa 920-0965, Japan
[*] Corresponding author. Tel: +81-9056840894; E-mail: masa.42.1950@gmail.com

Abstract

This presentation is a report on the liquefied ground flow in Uchinada Town, Ishikawa Prefecture Japan caused by the 2024 Noto Peninsula earthquake. At 16:10 JST (07:10 UTC) on 1 January 2024, a MJ7.6 (Mw 7.5) earthquake struck 7 km north-northwest of Suzu city located on the Noto Peninsula of Ishikawa Prefecture, Japan. The reverse-faulting shock achieved a maximum JMA seismic intensity of 7 and Modified Mercalli intensity of X–XI (Extreme). The shaking and accompanying tsunami caused widespread destruction on the Noto Peninsula, particularly in the towns of Suzu city, Wajima city, Noto town and Anamizu town, with the neighboring prefectures of Toyama and Niigata also suffering significant damage. About 250 people were killed by the earthquake even in the depopulated and aging area. The power and water outages continued for a long time in the Noto Peninsula. Extensive liquefied ground flow was occurred in Uchinada town, more than 100 km away from the epicenter. This was the boundary between the reclaimed land and the inland side of Uchinada Dune, and was an area with a high risk of liquefaction. As the roads became undulating, many houses became sloping and displaced, making them uninhabitable. In this study, we clarified the effect on the amount of ground flow by comparing the old version of the topography with the current topography, and examined the relationship of the amount of ground flow with slope of the house.



Fig. 1 Inclination of house by the liquefied ground flow

Acknowledgments

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Dynamic processes of a flow-like loess landslide reactivated on October 5, 2021, in Gansu Province, Northwest China

Ping Sun[1,2,*], Haojie Wang[1,2], Shuai Zhang[1,2]

[1] Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing 100081, China.

[2] Observation and Research Station of Geological Disaster in Baoji, Shaanxi Province, Ministry of Natural Resources, Baoji 721001, China.

[*] Corresponding author. Tel: +86-15810865191; E-mail: sunpingcgs@foxmail.com

Abstract

On October 5, 2021, a small-scale catastrophic loess landslide occurred in Niangniangba town, Tianshui City, Gansu Province in China. The landslide with a volume of approximately 15,360 m3 buried 2 houses and damaged another 2 houses. On the basis of field investigation, unmanned aerial vehicle photography, and numerical simulation, the characteristics and dynamic processes of the Zhongzhai landslide are analyzed, and the potential destructive zone of the unstable loess mass is preliminarily assessed. The Zhongzhai landslide is a typical rainfall–reactivated flow-type loess landslide, and its failure mode is creep-tensile type. The upper sliding mass is the typical retrogressive failure, and the lower sliding mass shows thrust load caused failure. The runout simulation results show that the entire sliding process of Zhongzhai landslide lasted for approximately 35 s, with a maximum velocity of 12 m/s. The maximum travel distance of the sliding mass is 130 m. In addition, the presence of masonry buildings has a great influence on the sliding distance. Simulation results of potential destructive zone of the unstable loess mass show that the travel distance of the unstable loess mass is approximately 250–295 m, the thickness of the deposited mass is approximately 2.7–3.3 m in the potential destructive zone, and the maximum velocity is 28 m/s. This work contributes to improving the understanding of evolution law, dynamic processes, and risk prevention of this kind of landslide in loess plateau areas.



Fig. 1 Photographs showing details of the Zhongzhai landslide.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (grant number 42130720) and the China Geological Survey (grant numbers: DD20221738 and DD20190717).

Strength behavior of red clay in the Qinghai-Tibetan Plateau considering climate warming

Zhihong Zhang[1], Fuchu Dai[2], Yanhai Yu[3]

Abstract

Under the background of global warming, the degradation of permafrost in the Qinghai-Tibetan Plateau (QTP) has led to frequent freeze-thaw geo-disasters. It is of great significance to investigate the effect of freeze-thaw on damage of soils under climate warming conditions. However, fixed freezing and thawing temperatures are set in most experiments, the impact of climate warming on the mechanical behavior of soils is ignored. Employing increased freezing temperature to reflect the climate warming, a series of freeze-thaw cycling tests were conducted on red clay specimens taken from the QTP under fixed and variable temperature paths. Triaxial tests and nuclear magnetic resonance (NMR) tests were conducted to investigate the freeze-thaw cycling effect on the shear strength and the corresponding micro-mechanism. Triaxial test results indicate that the measured shear strength of specimens under fixed and variable freezing temperature paths significantly decreases with the increase of freeze-thaw cycling numbers, yet, different descent trends have been found, S-type descent is for variable temperature path and exponential type is for fixed temperature path. NMR test results indicate that the red clay exhibits a bimodal pore size distribution. As the freeze-thaw cycling number increases, the micro-porosity gradually decreases, while the macro-porosity gradually increases. Compared to the pore size distribution for specimens under fixed freezing temperature path, the macro-porosity for specimens under variable freezing temperature path first decreases and then increases with the increase in the freezing temperature. It means that the structural damage of specimens under variable freezing temperature path is more severe, which verified the variation of shear strength from a microscale perspective.

Keywords

freeze-thaw cycling; climate warming trend; red clay; shear strength; microstructure

Introduction

The Qinghai-Tibet Plateau (QTP) is known as the third pole of the world, which is distributed with the largest area of frozen soil in the middle and low latitudes (Zhao et al., 2004; Luo et al., 2020; Li et al., 2023; Yan et al., 2023). With global warming, the climate of the QTP has become warmer (as shown in Fig. 1), which leads to an increase in the thickness of active layer (Guo et al., 2013; Kuang et al., 2016; Li et al., 2019; Zhang et al., 2019; Biskaborn et al., 2020; Lin et al., 2023; Zhang et al., 2023). The strength of soil in the active layer will weaken under the influence of freeze-thaw cycles (Ye et al., 2019; Lu et al., 2019; Wang et al., 2020; Lu et al., 2023). Red clay is widely used as the natural engineering material in the QTP due to its low compressibility, low permeability, and high strength (Sani et al., 2019; Lang et al., 2021; Zeng et al., 2022). Therefore, it is necessary to study the strength of red clay after freeze-thaw cycling (FTC) under the trend of climate warming in QTP.

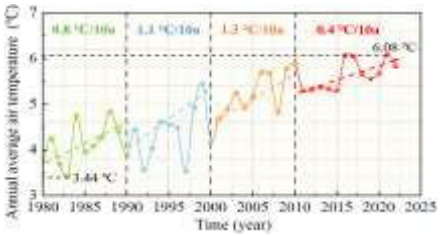


Fig. 1. Changes of annual average air temperature in the QTP.

At present, extensive experimental studies on FTC effect on mechanical behavior of soils have been conducted, and significant results have been achieved. Xie et al. (2015) conducted multiple FTC tests on soil specimens at a freezing temperature of -40°C and a thawing temperature of 20°C. As the number of FTC increases, the uniaxial compression strength decreased significantly within 6 FTCs. Wang et al. (2018) carried out multiple FTCs and triaxial tests on clay at a freezing temperature of -25°C and a thawing temperature of 10°C. The results showed that the strength and cohesion decreased exponentially with the increase in FTCs, while the internal friction angle increased exponentially. Tang et al. (2018) and Kong

et al. (2023) performed FTC and triaxial tests on soil specimens at a freezing temperature of -10°C and a thawing temperature of 10°C. They found that the strength of expansive soils and loess decreased significantly with increasing FTC numbers. Zhang et al. (2019) carried out multiple FTCs and triaxial tests on loess at a freezing temperature of -15°C and a thawing temperature of 20°C. The strength of lime-stabilized loess decreased by about 15% to 25% from the initial state after 3 FTCs. It is also found that most of the FTC tests were conducted within a temperature range between -20°C and 20°C (Xu et al., 2018; Zou et al., 2019; Zheng et al., 2021; Xu et al., 2022; Chen et al., 2024).

The freezing temperature is one of the most important factors affecting the influence of FTC on soil behavior, which has attracted widespread attention. Liu et al. (2009) found that the failure stress increased with decreasing freezing temperature. Christ et al. (2009) found that the strength of sand increased with the decrease in freezing temperature. Wang et al. (2017) conducted triaxial tests on specimens frozen at different freezing temperatures (-10°C, -5°C, and -2°C). The measured shear strength decreased linearly as the freezing temperature raised from -10°C to -2°C. However, it was also found that the freezing temperature had a weak effect on the strength of silty sand in the QTP (Liu et al., 2016). Zhou et al. (2018) conducted several FTC on loess at different freezing temperatures, and they found that the freezing low temperatures from -6 °C to -12 °C did not have a significant effect on the test results at the same FTC number.

All of the above FTC tests are carried out at a fixed freezing and thawing temperature. In fact, the freezing temperature gradually increases due to the climate warming. Therefore, it is of great significance to carry out the FTC tests under the variable temperature path (VTP) to investigate the influence of climate warming trend on the shear strength of red clay.

In this study, FTC tests were carried out on a red clay under variable temperature path (VTP) and fixed temperature path (FTP). A series of triaxial shear and NMR tests were carried out on the specimens after FTC. The test results were analyzed to investigate the influence of FTC on the shear strength of red clay under the trend of climate warming. The research results can provide a theoretical reference for the engineering construction and ecological environment management in red clay areas of the QTP.

Materials and Methods

Materials

The red clay used in this study was taken from a slope near Nangqian County, Yushu Prefecture, Qinghai Province, in the eastern part of QTP. The location where the soil was taken is shown in Fig. 2.

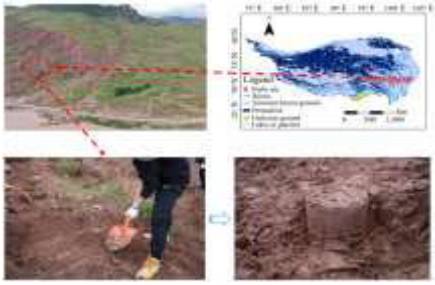


Fig. 2. Sampling location.

The retrieved red clay was dried naturally, crushed and sieved through a 2 mm sieve. The coefficient of uniformity Cu and coefficient of curvature Cc are 4.64 and 0.54, respectively. The basic physical characteristics of red clay were shown in Table 1.

Table 1. Basic physical properties of red clay.

Natural moisture content (%)	Specific gravity	Liquid limit (%)	Plastic limit (%)
24.3	2.78	45.5	23.9
Dry density (g/cm^3)	Optimal moisture content (%)	Maximum Dry density (g/cm^3)	
1.63	18.9	1.685	

Specimen preparation

The soil particles were mixed with distilled water to achieve a water content of 24%. The soil mixture was wrapped in a plastic wrap for 24 hours to ensure that the moisture was evenly distributed throughout the soil. Then, the soil specimen was compacted in five layers. The surface was scraped after each layer was compacted to ensure even compaction of specimens. The specimens were cylindrical with a diameter of 50 mm and a height of 100 mm.

Testing procedure

Freeze-thaw cycling tests

The FTC tests were carried out in two groups. One group was the freeze-thaw cycling (FTC) test under FTP, with the freezing temperature at -17°C and the thawing temperature at 20°C. The other group is the FTC test under VTP, the freezing temperature gradually increased during the FTC, the detail of which

was shown in Fig. 3. The freezing and thawing durations of both groups were set to 12 hours to ensure that the specimens were completely frozen and melted, which was defined as one FTC. The number of FTC was set to 0, 1, 2, 3, 4, 5, 6, and 7. The FTC test was carried out in a temperature-controlled test chamber, which had a temperature range of -20℃ to 25℃ with a precision of 0.1℃

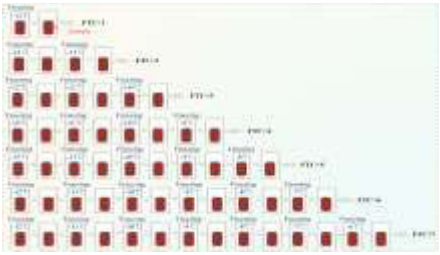


Fig. 3. Flow chart of FTC test under VTP.

Triaxial shear tests

A BS01015 fully automated triaxial instrument was used for the triaxial test. The initial confining pressures were 50 kPa, 100 kPa, 200 kPa, and 400 kPa. The triaxial test was conducted with unconsolidated and undrained triaxial shear (UU). The axial shear rate was 0.4 mm /min. As the axial strain reached 20%, the triaxial test was ceased.

Nuclear magnetic resonance tests

NMR technique was used to measure the pore size distribution of the soil specimen (Li et al., 2021). The instrument used for the NMR tests was a MesoMR14-060H-1 low-field nuclear magnetic resonance machine. The tests were performed using a Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence. The parameters of the pulse sequence are shown in Table 2. A pin-spin relaxation time (T2) signal curve can be obtained after the NMR test. T2 is in direct proportion to the pore radius r (Song et al., 2023)

$$\frac{1}{T_2} \approx \rho_2 \frac{F}{r} \tag{1}$$

where ρ_2 is the surface relaxivity, and F is the shape factor, assumed to be a constant in practice (Tian et al., 2019). According to Eq. (1), the pore size distribution of the tested soil sample can be reflected by the T2 signal curve. The NMR test was carried out with the same specimen after FTC for 0, 1, 3, 5, and 7 times. Before the NMR test, the specimen was saturated by vacuum to ensure that the specimen was in a saturated state.

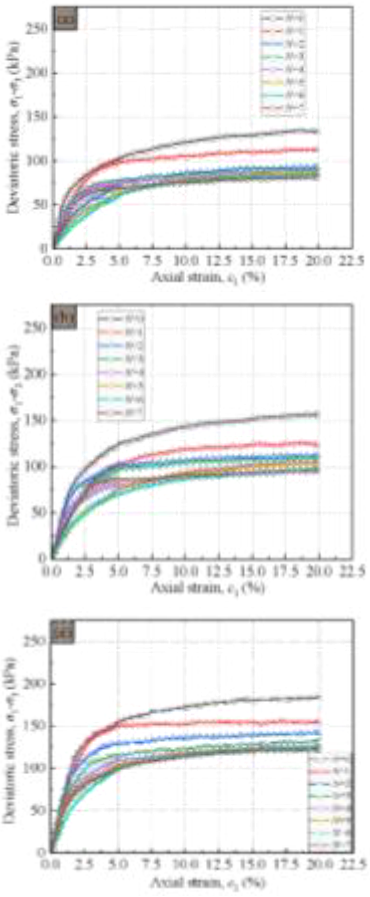
Table 2. Parameters of the CPMG pulse sequence.

Sampling frequency (kHz)	Main frequency (MHz)	Pulse 90° (μs μs)	Pulse 180° (μs μs)
250	12	13.52	27.04
Regulate first data (msms)	Regulate analog gain (dB)	Regulate digital gain	Echo time (msms)
0.002	10	3	0.2

Results and Discussions

Triaxial test results

Figs. 4 and 5 display the stress-strain curves of soil specimens after FTP-FTC and VTP-FTC, respectively. The deviatoric stress increases with the increase of the confining pressure, while decreases as the number of FTC increases. The deviatoric stress decreases significantly after the first FTC, then gradually stabilizes after 5 FTCs. In addition, the variation of FTC and freezing temperature have no obvious influence on the type of stress-strain curves. The deviatoric stress under different initial confining pressures increases with the increase of axial strain and then stabilizes, which exhibits a strain-hardening type.



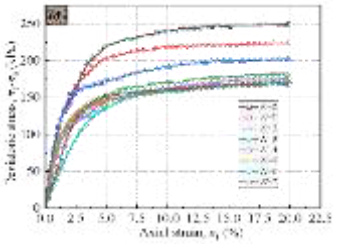


Fig.4. Stress-strain curves of specimens after FTP-FTC under confining pressures of: (a) 50 kPa; (b) 100 kPa; (c) 200 kPa; and (d) 400 kPa.

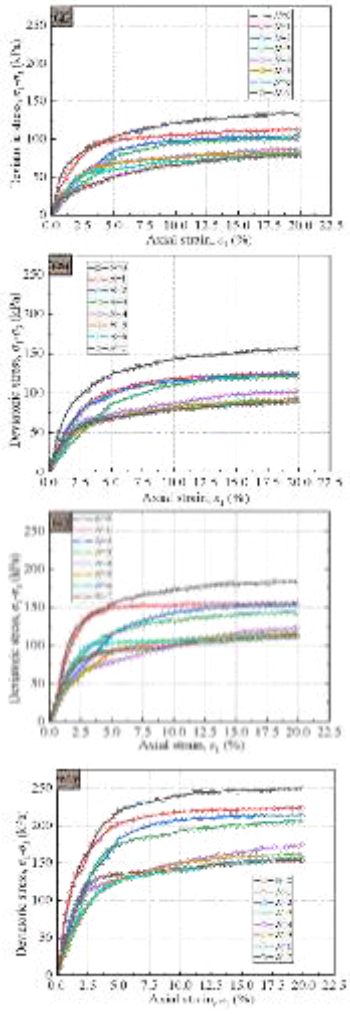


Fig.5. Stress-strain curves of specimens after VTP-FTC under confining pressures of: (a) 50 kPa; (b) 100 kPa; (c) 200 kPa; and (d) 400 kPa.

The deviatoric stresses of the strain-hardening type curves at 15% axial strain are taken as the shear strength. Fig. 6(a) shows the variation of the shear strength of specimens after FTP-FTC and VTP-FTC. The shear strength of the specimens gradually decreases with an increase in the FTC number. The magnitude of strength reduction increases with an increase of the initial confining pressure. The decline trend of the shear strength after FTP-FTC is exponential. It can be roughly divided into three phases: rapid descent (FTCs less than 3 times), slow descent (FTCs less than 5 times) and stabilization (FTCs more than 5 times). However, the descent trend of the shear strength of the specimens under VTP exhibits a S-type. The shear strength decreases more rapidly after the first and fourth FTC; while for the remaining cycles, the shear strength shows a slower decreasing trend. Unlike VTP, the descent trend of the shear strength of the specimens under FTP exhibits an exponential type.

Comparison between the test results after FTP-FTC and VTP-FTC shows that the shear strength after the first three FTCs is slightly higher under the VTP than under the FTP. However, from the fourth FTCs onwards, the shear strength of the specimens subjected to VTP-FTC is slightly lower than that of the specimens subjected to FTP-FTC. The shear strength difference between the two test groups increases with the initial confining pressure.

Cohesion and internal friction angle are two important indicators to evaluate the shear strength of soils. Fig. 6(b) illustrates the measured cohesion and internal friction after FTP-FTC and VTP-FTC. As the number of FTC increases, the cohesion of the specimens gradually decreases and tends to stabilize. Compared with the initial state, the cohesion

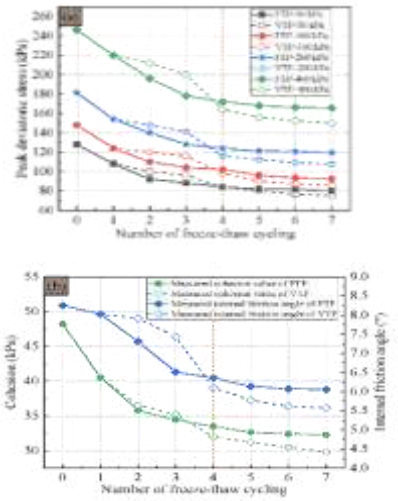


Fig.6. Variation curves of (a) peak deviatoric stress, (b) cohesion and internal friction angle of red clay with the number of FTP-FTC and VTP-FTC.

decreases by 37.2% and 38.2% after seven FTP-FTCs and VTP-FTCs, respectively. The internal friction angle also exhibits a decreasing trend, although the change is not noticeable. The variation of cohesion and internal friction angle with the number of FTC was curve-fitted, the fitted equations are as follows:

$$c_{FTP} = 31.5 + 16.7 / \left(1 + 0.9N^{1.6}\right) \quad (2)$$

$$\varphi_{FTP} = 6 + 2.2 / \left(1 + 0.1N^{3.2}\right) \quad (3)$$

$$c_{VTP} = 22.1 + 26.1 / \left(1 + 0.42N^{0.9}\right) \quad (4)$$

$$\varphi_{VTP} = 5.6 + 2.49 / \left(1 + 0.00007N^{7.8}\right) \quad (5)$$

where c is cohesion; φ is internal friction angle; and N is the number of FTC.

Comparison between the test results after FTP-FTC and VTP-FTC shows that most of the reduction of the cohesion occurs after the first FTC. During the first three FTCs, the cohesion under VTP is slightly higher than that under FTP; afterwards, it shows to be lower. The internal friction angle of specimens exhibits a similar trend. Hu et al. (2017) found that there exists a most unfavorable freezing temperature during the freezing process of the soil, which makes the worst mechanical properties of the soil after experiencing FTC at this temperature. For the tested red clay, the most unfavorable freezing temperature range is between -13°C and -11°C, and the FTC at this freezing temperature range causes the greatest attenuation of shear strength, cohesion, and internal friction angle.

***Nuclear magnetic resonance* results**

Fig. 7 presents the T_2 spectrum curves of red clay after FTP-FTC and VTP-FTC. The T_2 spectrum curve exhibits a distinct bimodal distribution characteristic. According to Eq. (1), it reflects the bimodal pore size distribution of the soil specimen. Before FTC, the T_2 spectrum area of peak 1 accounts for approximately 95% of the total area, primarily between 0.01 and 4.82 ms, while the T_2 spectrum area of peak 2 accounts for approximately 5% of the total area, primarily between 7.31 and 72.32 ms. Based on the relationship of T_2 and pore radius r, peak 1 corresponds to micropores, while peak 2 corresponds to macropores. After the FTCs, the signal intensity of the T_2 spectra corresponding to peak 1 gradually decreases, while the signal intensity of the T_2 spectra corresponding to peak 2 increases. This means an increase in the macro-porosity of the red soil, i.e., an enhancement of the structural damage.

Fig. 8(a) shows the relative change in the area below T_2 spectral curves with the number of FTC under FTP and VTP. During the first five FTCs, the area below T_2 spectral corresponding to peak 1 gradually decreases while the area below T_2 spectral corresponding to peak 2 increases. After five FTCs, the change of the area below T_2 spectral curve is weak, indicating that the pore structure of the red clay reaches a new stable state. Fig. 8(b) shows the comparison of the relative change in T_2 spectrum area corresponding to the macropore after FTP-FTC and VTP-FTC. It can be seen that the area for specimen after VTP-FTC is higher than that after FTP-FTC. The results indicate that the macro-porosity of specimens subjected to VTP-FTC is larger than that subjected to FTP-FTC, which verified the shear strength behavior of the red clay at the microscopic scale.

The above results demonstrate that the structural damage of specimens caused by FTC is mainly caused by pores redistribution. Therefore, a damage factor based on the area below T_2 spectrum is introduced. The damage factor is defined as η, which can represent the change of T_2 spectral area after FTC.

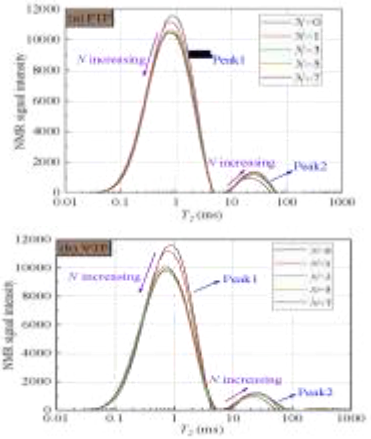


Fig. 7. (a) and (b) T_2 spectrum curves of red clay after FTP-FTC and VTP-FTC.

The greater the value of η, the greater the deterioration to soil structure. The formula of η is

$$\eta = \frac{\sum I_N / \sum I_{Micropore,N} - \sum I_0 / \sum I_{Micropore,0}}{\sum I_0 / \sum I_{Micropore,0}} \quad (6)$$

where I_0 is the signal amplitude corresponding to the T_2 value of specimen without FTC, and I_N is the signal amplitude corresponding to the T_2 value of specimen after N FTCs. $I_{Micropore,0}$ the signal amplitude corresponding to the T_2 value of peak 1 without FTC, and $I_{Micropore,N}$ is the signal amplitude corresponding to the T_2 value of peak 1 after N FTCs.

The variation of η with the increase of FTC under FTP and VTP can be fitted as Eq. (7) and Eq. (8), the result is illustrated in Fig. 8 (c) and (d).

$$\eta = 3\left(1 - e^{-0.4N}\right) \quad (7)$$

$$\eta = 3.4\left(1 - e^{-0.42N}\right) \tag{8}$$

Fig. 8(c) and (d) show that as the FTC increases, the damage factor gradually increases and then reaches a stable state. This suggests that pores redistribution caused by FTC significantly affect the shear strength of red clay. Therefore, special attention of pores redistribution should be paid in engineering practice.

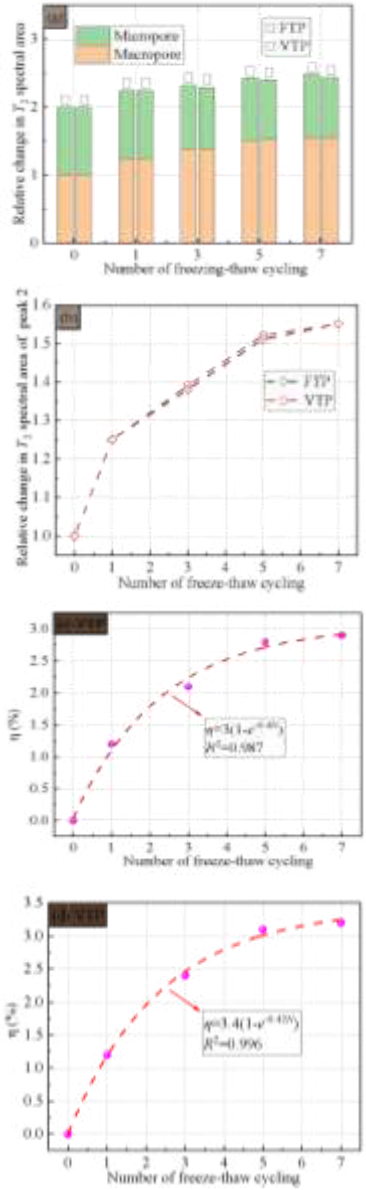


Fig.8. (a) and (b) Relative change in area below T_2 spectral curves of red clay after FTP-FTC and VTP-FTC; (c) and (d) Variation of η with the number of FTP-FTC and VTP-FTC.

Conclusions

In this paper, the deterioration mechanism of red clay's strength under climate warming trend was analyzed from the macro and micro scales through triaxial shear and NMR tests. The conclusions are summarized as follows.

(1) As the FTC increases, the shear strength of red clay declines significantly. The decline trend of the shear strength after FTP-FTC is exponential. It can be roughly classified as three phases: rapid reduction, slow reduction and stabilization. For the specimens after VTP-FTC, the strength declines rapidly after the first and fourth FTC, which exhibits a S-type descent trend.

(2) With the increase in the number of FTC, the cohesion of the post-freeze-thaw red clay under FTP and VTP decreases significantly. While the angle of internal friction decreases slightly with insignificant changes. Both trends can be fitted by Logistic model.

(3) Comparative analysis of the results of specimens under FTP and VTP revealed that there exists a most unfavorable freezing temperature between -13℃~-11℃, at which the strength of red clay decreases the most after FTC. In engineering practice, attention should be paid to avoid FTC at this temperature.

(4) Under the action of FTC, some of the micropores are developed into macropores, leading to the loosening structure and the attenuation of the shear strength. With climate warming, the freeze-thaw effect is more significant, leading to a larger volume of macropores inside the red clay and a greater degree of shear strength attenuation.

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Zhihong Zhang

1 Key Laboratory of Urban Security and Disaster Engineering, Ministry of Education, Beijing University of Technology, Beijing 100124, China.

Email: zhangzh2002@126.com

Fuchu Dai

2 Key Laboratory of Urban Security and Disaster Engineering, Ministry of Education, Beijing University of Technology, Beijing 100124, China.

Email: daifc@bjut.edu.cn

Yanhai Yu

3 Key Laboratory of Urban Security and Disaster Engineering, Ministry of Education, Beijing University of Technology, Beijing 100124, China.

Email: Yuyh@emails.bjut.edu.cn

Discontinuous Deformation Analysis for Progressive Toppling of the Rock Slopes with Fractured Slab Structure

Tonglu Li[1,2,*], Sifan Yuan[1,2]

[1] Department of Geological Engineering, Chang'an University, Xi'an 710054,Shaanxi, China
[2] Observation and Research Station of Water Cycle and Geological Environment for the Chinese Loess Plateau, Ministry of Education, Zhengning 745399, Gansu, China
[*] Corresponding author. Tel: +86-29-13992891737; E-mail: dcdgx08@chd.edu.cn

Abstract

Toppling of rock slopes with fractured slab structure is quite common accounted in the slopes of hydro-powers and highway engineering, and it has become one of the potential threatens of engineering construction and safe operation. The field investigation found that toppling mostly occurs on slope with tectonic stress concentration, strong river down erosion and slopes with a set of discontinuities dipping steeply into the free face. From a macroscopic point of view, toppling is neither a collapse nor a slide, but a discontinuous deformation. The traditional limit equilibrium method is not suitable for the slope stability assessment. In this study, taking a toppling slope with metamorphic schist in the Qinling mountains as an case, as shown in figure 1. The whole process of toppling is simulated with the discontinuous deformation analysis (DDA) method considering the effect of mountain uplift and river down erosion, as shown in figure 2. The results show that with the river down erosion, the fractured rock slabs sheared along the inter-layers of the schistose surface under gravity and toppled to the slope face. It is a slow evolution process induced by the river erosion rather than a sudden failure.



Fig. 1 The toppling slope investigated.

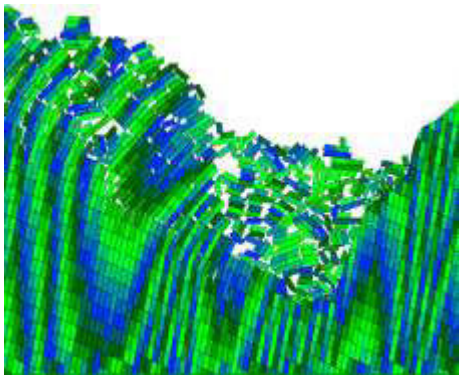


Fig. 2 The simulated results for the toppling.

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From practice to theory: the multihazard case study of Casamicciola Ischia, Italy

Sabatino Cuomo[1,*]
[1] Department of Civil Engineering, University of Salerno, Italy
[*] Corresponding author. Tel: +39 089964231; E-mail: scuomo@unisa.it

Abstract

Shallow deposits of volcanic and pyroclastic materials are widespread in the Campania region, due to the past explosive activity of both Campi Fegrei and Vesusius volcanoes.

In the last centuries several rainfall-induced catastrophic flow-like mass movements occurred in the inland areas (e.g. Sarno and Quindici 1998, Cervinara 1999, and Nocera Inferiore 2005) along the coastline (Atrani 2010, and before Salerno and Vietri sul Mare in 1954) and on the beautiful island of Ischia (Monte Vezzi 2006, up to the latest disaster Nov. 2022).

Significant scientific efforts have been pursued at the University of Salerno, where Prof. L. Cascini established in the latest 90s' what is now GEG (Geotechnical Engineering Group), that has been jointly working with geologists, geomorphologists, hydrologists, hydraulic engineers and experts from other disciplines including social sciences and experts of risk related to natural hazards and also to other human-related risks.

Moving along this path the past and future flow-like mass movements are nowadays conveniently discriminated into four main classes, namely Debris Flows (DF), Debris Avalanches (DA), Hyperconcentrated flows (HF), and Flash Flood (FF).

A first clear distinction among the multiple possible triggering mechanisms was proposed by Cascini et al. (2008). The role of bed entrainment during flows propagation was outlined (Cuomo et al., 2016) in this district and in other similar contexts in many countries. Indeed, the transition from one type of flow to another are currently under investigation (e.g., Cascini et al., 2022).

The Casamicciola case study is another breakthrough due to multiple cascading effects along slopes and piedmont areas, with different geohazards interacting in short time scales. Thus, a robust geomechanical framework is relevant to individuate and thoroughly analyse all the possible slope instability scenarios and the related flow-like mass movements. Such accurate analyses were casted for enhancing the risk zoning, promoting a rationale urban planning and especially for the safeguard of the local populations and tourists.

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Investigation on dynamic impacts of fluid-driven debris cascade on elastic structures via high-fidelity multi-physics tools

Salvatore Capasso[1], Iván Martínez-Estévez[2], Bonaventura Tagliafierro[3], Giacomo Viccione[1*]

[1] Environmental and Maritime Hydraulics Laboratory (LIDAM), University of Salerno, Italy

[2] EPHYSLAB, CIM-Uvigo, Universidade de Vigo, Spain

[3] Universitat Politècnica de Catalunya (UPC) – BarcelonaTech, Barcelona, Spain

[*] Corresponding author; E-mail: gviccion@unisa.it

Abstract

Fluid–debris interaction with structures is a phenomenon that still poses severe challenges for its multifaceted complexity and strong physical coupling, leaving consistent degrees of uncertainty when anticipating the urban fabric resilience. A numerical framework capable of simulating free-surface flows, fluid driven objects, impacts, and structural elasticity, is offered by the fully Lagrangian framework of the open-source DualSPHysics solver. Within its Smoothed Particle Hydrodynamics-based environment, a structural model able to deal with dynamic impacts and hydroelastic deformations is proposed and implemented using the coupling with the core module of the Chrono library [1]. Distributing the structural properties according to a lumped parameters approach [2] and using DEM contact methods, the hydroelastic response of a 3-D box-shaped building under fluid/debris impact is investigated. The structure is, hence, discretized using a set of 3-D elastic super elements capable of six degrees-of-freedom motion, mutually exerting bending moment, shear and normal forces via linear mechanical restrictions. Figure 1 shows the main features of the code in handling a violent impact of a breaking water column on to a deformable U-shaped wall; the model provides a realistic representation of the coupled and interactive fluid-structure deformation. This preliminary work identifies an innovative and powerful high-fidelity approach for more complex applications to come, in which the approaching column contains boulders and debris.

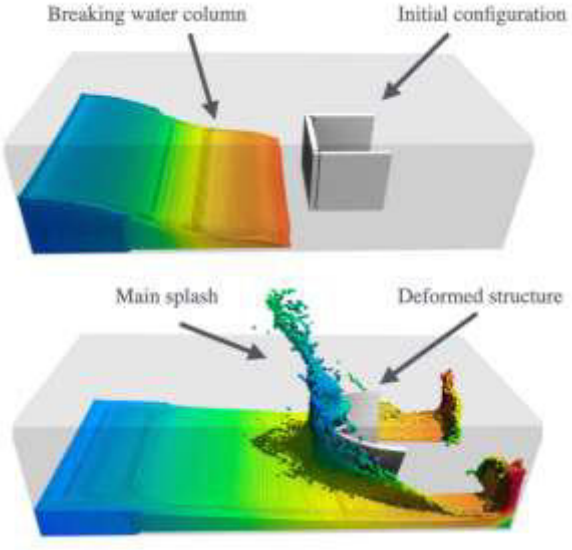


Fig. 1 Snapshot of the evolving dambreak impact on a flexible structure.

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Volcanic hazard assessment in the complex multisource Neapolitan area

Paola Petrosino[1*], Ines Alberico[2]

[1] Department of Earth, Environment and Resource Sciences, University of Napoli Federico II, Italy
[2] Istituto di Scienze Marine (ISMAR), CNR, Napoli, Italy
[*] Corresponding author. Tel: +39-0812538327; E-mail: paola.petrosino@unina.it

Abstract

The Neapolitan Volcanic Area (NVA) hosts three active volcanic sources: Somma Vesuvius and Ischia, currently in a quiescent phase, and Campi Flegrei, which is now experiencing an unrest phase. Although the significant differences in the geological setting and eruptive behavior of the three areas require a specific approach for long-term hazard assessment, we cannot ignore the necessity of a comprehensive view of their potential effects. Here we present a review of our efforts to globally assess volcanic hazard from Neapolitan sources on both a regional and local scale. An application to the Naples megacity, which is endangered by the possible emplacement of pyroclastic fall and flow products from Campi Flegrei and Somma-Vesuvius, is presented.

Furthermore, the complexity of drawing comprehensive volcanic hazard and risk maps will be discussed. These maps could be successfully used as a basic tool for decision-making regarding territorial management and future planning throughout an entire region. Nevertheless, single subsets of the multi-source maps (Fig. 1) can also be drawn on a larger scale with different formats and become effective in managing volcanic crises, establishing preparatory, response, and recovery actions related to volcanic risk.

Finally, we will discuss the key role of hazard indices and hotspots in disaster risk management. We will start from the results of Totaro et al. (2020), in which maps of monothematic and synthetic indices were drawn to describe the hazard status of the Naples and Pozzuoli municipalities. A hazard hotspot map was also elaborated to identify both the areas with high hazard for single dangerous events and those characterized by the concomitance of several hazards.

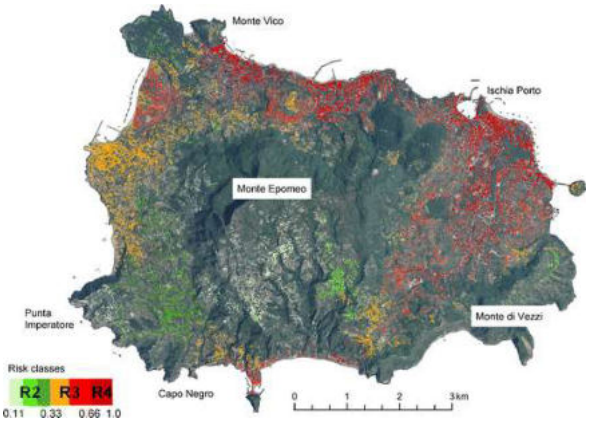


Fig. 1 Risk classes represented on the ortho-photo map for Ischia Island

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Opportunities and challenges for the monitoring of complex archaeological sites: the Pompeii case study

Luigi Petti[1,*], Carmine Lupo[1], Gabriel Zuchtriegel[2] and Vincenzo Calvanese[2]

[1] Department of Civil Engineering, University of Salerno, Via Giovanni Paolo II, Fisciano, SA, 84084, Italy

[2] Archaeological Park of Pompeii, Ministry of Culture (MiC), Via Plinio, 80045, Pompeii, NA, Italy

[*] Corresponding author. E-mail: petti@unisa.it

Abstract

The advent of new technologies has enabled the development of methodologies and tools that facilitate the acquisition of essential information for the protection of human life, buildings and environment. During the last decades, a particular attention has acquired the preservation of cultural heritage, which often lives in high hazardous conditions, such as the Archaeological site of Pompeii (Southern Italy). The ancient city of Pompeii is a complex area extending over more than 40 hectares and it comprises a large number of buildings and monuments of inestimable value. The site is situated in an area where a high volcanic hazard meets a widely recognised active seismotectonic context and a significant hydrogeological hazard, all under the impending influence of climate change. In these conditions, the development of monitoring strategies to assess the evolution of the health state of the heritage became fundamental to let proactive maintenance activities.

This paper is focused on an innovative methodology developed to monitor the Archaeological site of Pompeii (Southern Italy). In particular, the methodology considers multi-scale (data and time resolution) and multi-level (assessment accuracy) approaches and it consists of three screening levels: Local, General and Detailed Assessment.

The paper fully describes the General Assessment Level (GA) also showing examples of its application.



Fig. 1 A view of the archaeological site of Pompeii with Vesuvius in the background.

Acknowledgments

The research was conducted thanks to the willingness to develop an innovative monitoring approach to allow a sustainable and resilient management of a complex archaeological site by the Archaeological Park of Pompeii.

ReNDiS system: The WebGis tool to support the governance on landslide risk reduction.

Valeria Licata[1], Alessandro Fraccica[1], Francesco Menniti and Daniele Spizzichino[1,*]
[1]Institute for Environmental Protection and Research (ISPRA) - Dept. of Geological Survey of Italy.
[*]Corresponding author. Tel: +39-3386192214; E-mail: daniele.spizzichino@isprambiente.it

Abstract

Italy suffers from a high number of landslides which represent a critical issue from both social and economic point of view. According to the Italian Landslide Inventory (IFFI - https://idrogeo.isprambiente.it/app/) the Italian territory is affected by more than 600.000 landslides interacting with urban centers, lifelines, road/rail infrastructures, Cultural Heritage and strategic buildings. In this context, approximately 20 years ago, ISPRA set up a support system to manage measures aimed at mitigating hydro geological risk and in a broader sense to the reduction of geo disasters the support system is called ReNDiS.

ReNDiS is actually a web-GIS based tool, which supports the Italian Environmental Ministry's policies to plan and to fund countermeasures to reduce the landslide risk and related effects and disasters, among others. The system currently collects a significant number of interventions (approximately 25.000) for different typologies of hazards (see Fig. 1). The data referred to the landslides contain more than 8000 mitigation projects. For each project, the database provides the implementation status, the required funds, the geographical information and all the technical elements relevant to the design level adopted. The present work reports the description of ReNDiS system as fundamental tool for territorial and land use planning as well as a useful decision support system able to proactively guide the governance and mitigation policies of geo-disasters facing the landslide risk.



Fig. 1 Distribution of hazards on the Italian territory collected by ReNDiS system.

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Coping with floods and landslides: the May 2023 Emilia-Romagna hydrogeological events and the forecasting and post-event assessment activities of the Italian civil protection system

Andrea Duro[1,*]

[1] Italian Department of Civil Protection, Via Vitorchiano, 4, 00189 Rome, Italy

[*] Corresponding author. Tel: +39-0668202439; E-mail: andrea.duro@protezionecivile.it

Abstract

During May 2023, Emilia-Romagna region (Northern Italy) was hit by two severe hydrogeological events (1-3 May, and 16-17 May 2023), that caused 15 casualties, 36.000 internally displaced people, 5.500 assisted people and 8.500 rescuers.

Ground effects were very relevant: on main watercourses, there were floods affecting floodplains and embankments in the central-eastern sector. In the Romagna basins and tributaries of the Reno, water levels were close to the historical maximum. On the minor river network there were significant increases in hydrometric levels leading to bank erosion and potential flooding of the neighbouring area. 23 rivers flooded, with the interruption of 200 roads. Many landslides occurred along slopes with fragile hydrogeological conditions affecting the road network.

The activities of the National Civil Protection Service covered not only emergency management, but also the forecasting and post-event residual risk assessment phase.

In the forecasting phase, criticality bulletins were issued concerning the expected ground effects. This made it possible to predict which areas would be most affected by the events. Forecasting and monitoring activities were carried out by the National Early Warning System for Hydraulic and Hydrogeological Risk.

Following the events, the National Civil Protection Service carried out technical activities for the post-event residual risk assessment: field surveys, analysis of satellite images, use of optical and LIDAR drones, etc.

Field surveys were very useful for: landslides and flooded areas delineation, event scenario assessment; residual (post-event) risk evaluation, identification of alternative routes, etc. They were carried out by Regional technicians, supported by DPC technical officers and researchers of Research Institutes and Universities as CNR-IRPI, CNR-IGAG, ISPRA, and the University of Florence.

Remote sensing (RS) data provided crucial information of the extension of the disaster in the short aftermath of the event. In flooded areas RS data can be used to quickly assess the degree of the damages and to identify safe and reliable roads for emergency needs. The extensive use of ICT facilitated technical evaluation activities.

The general aim of technical assessment activities is to assess residual risk conditions after hydrogeological events and, based on these, to plan non-structural prevention measures to be taken pending the implementation of structural interventions, which take longer.

Civil Protection Technical volunteering: a fundamental contribution to reduce the emergency cost and lessons learned

P. Angeli[1], C. Ambrosio[1], M. Cagelli[1]

[1] Net.PRO

[*] Corresponding author. E-mail: ing.marco.cagelli@gmail.com

Abstract

Following the seismic events that occurred in Umbria and Marche in 1997, the Civil Protection Department began using the GNDT damage assessment checklists to provide a homogeneous tool that provided a certain judgment on the state of damage to buildings with two purposes: - facilitate the identification of the needs for temporary housing to assist the affected populations; - calculate an estimate of the costs for the reconstruction or repair of the asset. This model was afterwards improved and implemented by further experiences such as the 2009 earthquake in L'Aquila where freelance technicians were involved for the first time. During the 2012 earthquake in Emilia Romagna, technicians qualified to compile Aedes forms were involved, a qualification which was obtained following specific training courses. In fact, with the issuing of the Decree of the President of the Council of Ministers of 5 May 2011, the damage detection, emergency response and usability model for ordinary buildings in post-seismic emergencies and the related compilation manual were approved and it was indicated that specifically trained technicians could be used to support post-earthquake inspection campaigns.

The earthquake swarm of 2016 and 2017 in Central Italy involved a vast area and put the new procedures to the test involving the Civil Protection Department and the National Council of Engineers.

The drafting of hundreds of thousands of checklists and the involvement of thousands of technicians allowed three important aspects to emerge:

- the availability of many technicians to be directly part of the Civil Protection System to give a rapid response to the needs of asses the usability of buildings;

- the need to improve national legislation to allow better numerical modeling of structural units, more consistent with the damage patterns detected in the areas affected by the earthquake; - the effectiveness of the technical volunteering system in reducing the assistance costs to the population and the time required to prepare a certain assessment of the damage following the events.

The desire of many technicians to contribute during an emergency to supporting the affected populations led, in 2019, to the birth of NET.PRO, a national network of civil protection technical volunteer associations which today has more than 300 volunteer technicians and is present in nine regions. The association has so far been mobilized for the flood emergency in Emilia Romagna and, at the time of writing this abstract, is involved in the expeditious survey of the vulnerability of the buildings of the municipalities in the Campi Flegrei area.

Since 2019, specialist training has made it possible to guarantee the availability of specialists for damage assessment:

- for residential buildings via Aedes checklists;

- for structures with large spans via Aedes - GL checklists;

- for buildings listed by the Ministry of Cultural Heritage through A-DC and P-ES checklists; - for buildings subject to geological instability via Ageotec checklists.

NET.Pro technicians of the association ordinarily carry out various activities: - raising awareness of the prevention of natural risks in schools of all levels, to show the correct procedures to follow in the event of seismic, flood and landslide events;

- support the local authorities in drafting emergency response plans;

- participation in National awareness events such as "Io non rischio";

- provide contributions to the national debate on seismic vulnerability of existing buildings; - draw up procedures that comply with the regulations regarding the safety of technicians inside buildings already affected by the earthquake and operating during the assessment damage phase; - participation in various national and local Civil Protection trainings.

- support Local Authorities in defining the priority of intervention between various unsafe buildings. Some evidence emerged from the activities in the earthquake area:

- generally, two generations after the seismic event the population forgets the effects and the drama experienced, effectively canceling the priority of the structural safety of buildings when they want to renovate their building;

- the historical Italian building heritage, made of load-bearing masonry, is the subject of constant modifications over time, generally with reduction of the wall masses for the construction of openings, doors, windows, wardrobes... which reduce the effectiveness of the box behavior of masonry buildings;

- the damage assessment cannot ignore a field analysis of the single structural unit as there are notable differences in the construction methods and the useful life of the buildings is such as to have allowed constant, and sometimes significant, changes in structural behavior. This last point has led to highlighting some necessary precautions in the application of national legislation in the structural field to guarantee safety assessments congruent with the real behavior of the structural unit.

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Post-fire debris flows: an emerging disasters chain

Giuseppe Esposito[1,*]

[1] National Research Council – Research Institute for Geo-Hydrological Protection (CNR-IRPI), Perugia 06128, Italy

[*] Corresponding author. Tel: +39-0755014422; E-mail: giuseppe.esposito@cnr.it

Abstract

Debris flows in recently burned watersheds pose significant hazards to life and property, especially where they have been absent or less common in the past. These flows are able to progressively entrain and transport large amounts of burned and unburned sediments downstream, up to low-gradient areas where human structures are often located. Collection of the event locations and magnitudes, properties of the triggering rainfall, burn severity, runout paths, and other environmental features of the affected sites is of paramount importance to start a reliable analysis.

This contribution provides a summary of the first outcomes related to post-fire debris flows affecting mountainous sectors of the Campania region, southern Italy. Here, wildfires occur mostly between July and September, during the dry season, and are the result of both accidental and intentional human caused ignitions, like in the rest of Italy. Post-fire debris flows are very common in steep watersheds that in the late summer-autumn period are hit by short-duration high-intensity rainfall associated to convective cells. The basic knowledge presented in this work derives from the analysis of predisposing and triggering conditions of some recent events causing also severe damage to human structures (Esposito et al., 2017; Esposito et al., 2019). These events are included in an inventory of 113 post-fire debris flows described in the work of Esposito et al. (2023) together with properties of the triggering rainfall. Fire-induced geotechnical changes in soils properties, that promote enhanced runoff and erosion processes transitioning into debris flows downstream, are discussed in the article of Peduto et al. (2022).

The acquired knowledge represents thus the first step towards reliable hazard and risk assessments for the whole region.

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A first review of coastal hazards in the active volcanic caldera of Campi Flegrei, Italy.

Fabio Matano[1]

Abstract

The coastline of the Campi Flegrei active volcanic district is one of the most densely urbanized rocky coastal zones in Italy. Here, coastal cliffs are formed by volcaniclastic deposits and include remnants of ancient volcanic edifices formed in the last 15 ka. Due to the petrographic, geotechnical and geostructural properties of volcaniclastic deposits, these cliffs have been affected by rapid recession processes since their origin. A review of the initial analysis and monitoring activities, with the objective of achieving a quantitative understanding of the retreat processes that are contributing to the geomorphic evolution of the Campi Flegrei rocky coast, is presented. The research is based on the use of geomatic techniques and focuses on four main coastal cliffs: Coroglio, Punta Epitaffio, Torrefumo and Baia dei Porci, which are located in different sectors of the Campi Flegrei area. The results demonstrated that mass wasting processes are effectively contributing to the dismantling of coastal cliffs in the Campi Flegrei volcanic area. Episodic cliff failures can displace large volumes of rock, resulting in sudden and localized cliff recession. However, despite the occurrence of numerous small-scale failures, which are typically between 10 and 2 m³, the frequency of larger collapses remains higher. The findings of the multi-temporal monitoring of landslides demonstrated that after the main failure, landslide scars can be affected by further collapses. Furthermore, deposits eventually formed at the cliff toe can be rapidly eroded by the sea wave action and by rainfall-induced surface runoff. The short-term retreat rates were found to range from 0.001 to 0.025 m/yr during the period 2013-2016, while the mid-term rates varied between 0.17 and 1.2 m/yr. **Keywords**

coastal cliff, failures, erosion, volcanic caldera

Introduction

Many sea cliffs worldwide are affected by retreat processes, mainly occurring with landslides, surface erosion and weathering (Trenhaile, 1987; Sunamura, 1992; Hampton and Griggs, 2004). Retreat processes of cliffs developed in volcanic rocks (lava, tuff, etc.) can be explained by the mineralogy of the rocks, weathering, gases given off by the volcano and presence of water in the cliff, favouring periodic large rockfall (Douglas et al., 1994; Matsukura et al., 1994). In Italy, nearly 2,850 km of coasts are occupied by rocky shores and cliffs (Iadanza et al., 2009) which are directly in contact with the sea water and are exposed to landslides. This represents a serious risk for liv-ing people, as well as for buildings, roads and railway networks. The coastline of the Campi Flegrei active volcanic district (Figs. 1 and 2) is one of the rocky coastal zones more densely urbanized of Italy. Here, coastal cliffs are made by volcaniclastic deposits and include remnants of ancient volcanic edifices formed in the last 15 ka (Peccerillo, 2005). The cliffed coastlines of the Campi Flegrei area are among the most affected by dismantling processes in Italy. Generally, retreat of volcaniclastic sea cliffs is promoted by weak mechanical properties and low resistance to erosion of the outcropping sediments (loose to welded ash, pumices, scoria), as well as by the local action of natural factors, such as sea waves, wind, rain, salt spray, wetting and drying cycles, changes in temperature and bioerosion. Many landslides affected the Campi Flegrei coastline in the past years. The tuffaceous cliffs are often affected by rock falls that caused sudden and localized reces-sions by displacing hundreds to thousands cubic meters of rocks (Fig. 2). This paper presents a first review of the analyses and monitoring activities aimed at achieving a quantitative knowledge about retreat processes that are contributing to the geomorphic evolution of the Campi Flegrei rocky coast. The research is based on the use of geomatic techniques and focuses on several coastal cliffs (Fig. 2), located in different Campi Flegrei sectors.

Campi Flegrei geological setting

The Campi Flegrei (Fig. 2) is an active volcanic area located in southern Italy, along the Tyr-rhenian coast, and corresponds to a quasi-circular depression extending for about 200 km2, a large part of which develops off the Pozzuoli Bay (e.g., Santacroce et al., 2003; Peccerillo, 2005; D'Auria et al., 2015).

The catastrophic eruption of the Campanian Ignimbrite took place about 39 ka BP, causing the formation of a large caldera that was reshaped by the more recent phreato-plinian eruption of the Neapolitan Yellow Tuff (NYT; Scarpati et al., 1993; Denio et al.,2004), occurred 15 ka BP. The NYT consists mostly of pyroclastic-flow and minor fall deposits that occur as two different diagenetic facies: lithified and non-lithified (Scarpati et al., 1993). The lithification is the result of a diagenetic zeolitisation process (Norin, 1955). Within the central part of Campi Flegrei, the unlithified facies outcrops mainly in the western part of the study area (Baia dei Porci, Torrefumo and Punta epitaffio cliffs; Fig. 2) whereas the lithified facies occur mainly in the eastern part (Coroglio cliff; Fig. 2) (e.g., Scarpati et al., 1993).

Then volcanic activity is subdivided in three main epochs when minor explosive events occurred creating at least 52 monogenic phreato-magmatic vents, interrupted by long periods of quiescence. The latest eruption, known as "Monte Nuovo Eruption", occurred in the 1538 A.D (e.g., Pappalardo & Mastrolorenzo, 2012). Currently, the caldera is characterized by unrest conditions, as demonstrated by widespread fumaroles, thermal springs, earthquakes and ongoing ground deformation. The current geomorphological configuration of the Campi Flegrei was determined by the action of volcanic and geomorphic processes occurred in the last 15 ka, after the NYT eruption. Relics of volcanic edifices partially dismantled by sea erosion and landslides were identified along or near the present coastline and in the submarine part of the caldera. The inland volcanoes were also affected by slope instability processes, and some of them result partially col-lapsed because of volcano-tectonics forming crater lakes. Landslides due to earthquakes, rain-storms, marine erosion, or human actions have repeatedly affected the slopes of ravine and streams, as well as the steep scarps of gullies and coastal cliffs through time.

Methods and results

The geomorphic evolution of the Torrefumo cliff (years 1956 to 2016), located in Monte di Procida municipality, was investigated by Esposito et al. (2018), by comparing digital elevation models (DEMs) derived from the processing of archive aerial images and airborne lidar data (Fig. 3). This activity allowed quantifying volumetric changes and maximum retreat of the cliff top edge. The mid-term average retreat rates (1956-1974 and 1974-2008) were calculated by applying the method of Young and Ashford (2006) that considers the total eroded volume, the average sea cliff height and length. The short-term geomorphic changes that affected part of this cliff were detected by comparing 3D point clouds acquired by Terrestrial Laser Scanning (TLS) in two multi-temporal surveys (years 2013 and 2016) by Esposito et al. (2020). A cliff failures inventory map was developed, and the magnitude-frequency distribution of the events was calculated.

At the Baia dei Porci cliff, a large rockslide occurred in October 2013 threatening a series of buildings located close to the cliff edge, located in Monte di Procida municipality (Fig. 4). The mass movement was monitored until January 2016 to assess its geomorphic evolution (Esposito et al., 2017). In this case, the "Structure-from-Motion" (SfM) photogrammetry technique was used and four surveys (years 2013, 2014, 2015, 2016) were executed from a boat and a Remotely Piloted Aircraft System (RPAS). Moreover, a topographic survey of ground control points was carried out to guarantee the exterior orientation of images. Comparison of the 3D point clouds derived from the processing of photogrammetric data allowed accurate identification of the areas where geomorphic changes occurred, as well as quantifying volumes of material mobilized during and after the 2013 main failure.



Fig. 1 Typical rock fall affecting Campi Flegrei coastal tuff cliffs (Trentaremi, Naples).

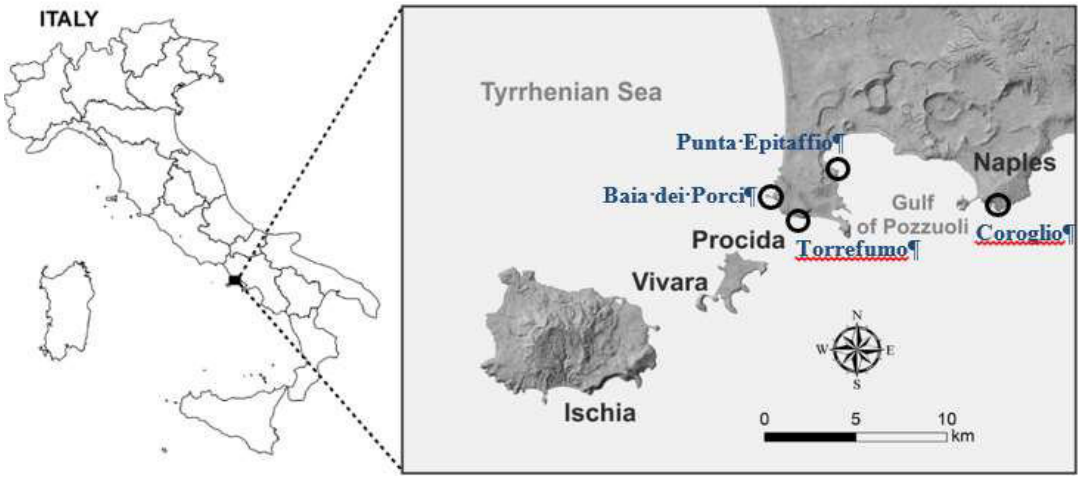


Fig. 2 Location on shaded relief of the Campi Flegrei volcanic area. The circles are the studied cliffs.

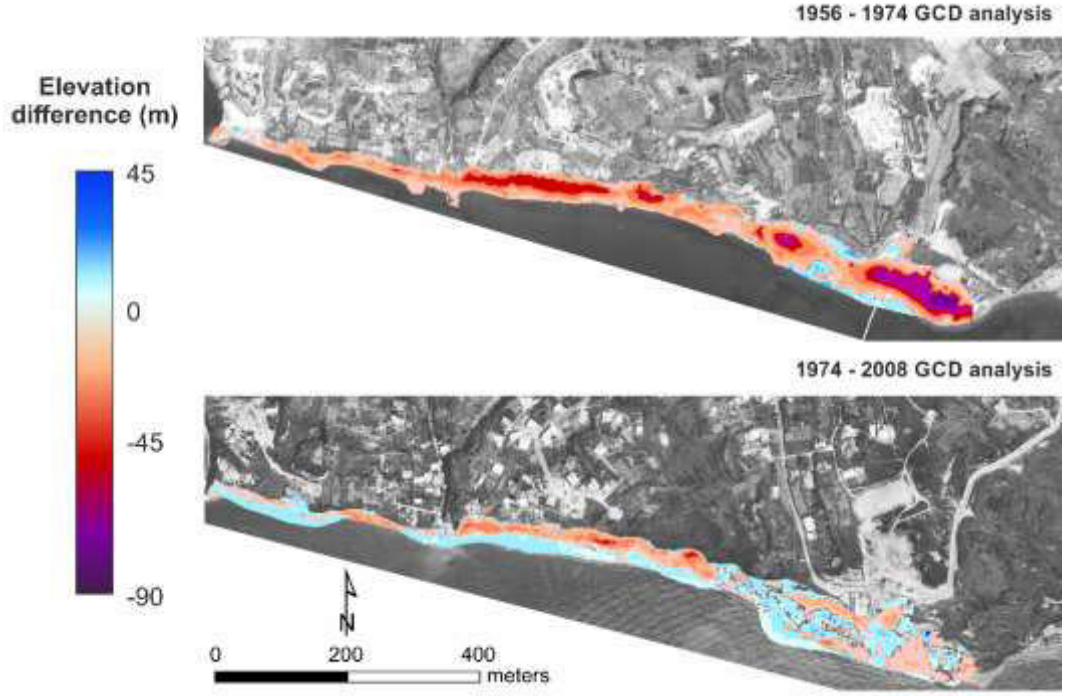


Fig. 3 DEMs of difference (DoDs) related to the two analyzed time intervals; negative elevation changes (reddish colors) highlight erosion, as well as positive changes (bluish colors) highlight deposition. Orthophotos of the year 1956 (above) and 1974 (below) are shown in background (Mod. from Esposito et al., 2018).

Long-range Terrestrial Laser Scanner (TLS) applications were aimed at characterizing the geostructures of the Coroglio cliff, located in Naples, and the structural pattern of Punta Epitaffio tuffaceous coastal cliff, located in the Bacoli municipality (Matano et al., 2015, 2016; Caputo et al., 2018).

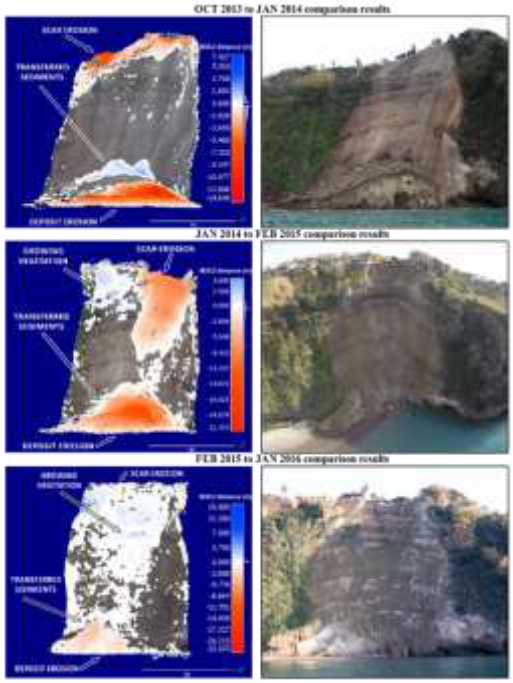


Fig. 4 Change detection and sediment budget by M3C2-derived multitemporal point clouds. On the left, comparison results are showed for 2013-14, 2014-15 and 2015-16. Real images (right) were used for a qualitative validation of the computed changes (Mod. from Esposito et al., 2017).

Regarding the Punta Epitaffio cliff, the geostructural analysis was based on the method of automatic extraction of discontinuity orientation by TLS data, as proposed by Fanti et al. (2013) and modified in Matano et al. (2015). The 3D digital model of the Punta Epitaffio cliff derived from TLS data, provided a base for the classification of rock discontinuities by geostatistical analysis. In detail, the workflow of geostructural data processing included: 1) statistical analysis of spatial orientation of the facets of the 3D mesh derived by the TLS survey point cloud; 2) extraction of the best-fit attitudes (dip and dip direction) of discontinuity sets for each sub-planar patch of the rock face; 3) cluster analysis of best fit structural discontinuities; 4) definition of all the discontinuity sets and geo-structural classification of 3D model facets; 5) kinematic analysis for the definition of possible failure mechanisms.
The procedure adopted in the study of the Coroglio cliff consisted in an integration of geomorphological analysis, structural field survey, Terrestrial Laser Scanner data acquisition and processing, 3D digital model (triangle mesh) development and analysis, geostructural classification of discontinuity orientation data, and 2D vertical cartography elaboration, managed with Geographical Information Systems. The 3D model was derived from a filtered point cloud, and each triangular facet characterized by its spatial orientation (dip and dip direction) and elevation data at 1:500 scale. With an integration of a statistical analysis of mesh facets of the 3D model based on their spatial orientation, structural field data from field surveys in selected sectors and data from literature, we identified the orientation of six main discontinuity sets (F1 to F6). The pattern of mapped discontinuities sets has been utilized for producing the vertical geostructural map of the Coroglio cliff, that is relevant for kinematic analysis of unstable blocks along the cliff surface.
Analysis of Terrestrial Laser Scanning (TLS) datasets referred to multi-temporal (2013-2015) acquisitions, were aimed at evaluating volumetric changes and average retreat rates of the Coroglio tuff cliff (Fig. 5). The multi-temporal analysis was carried out in different steps. The obtained point clouds were aligned and georeferenced using the ICP algorithm. In order to analyse morphological changes on the cliff surface two different approaches were used and combined: a) distance calculation between two 3D point clouds (3D Cloud-to-Cloud comparative analysis) acquired on different dates using Multiscale Model to Model Cloud Comparison (M3C2) algorithm to obtain main morphological changes, b) GIS-based 2.5D processing approach which relies on a rasterization/gridding process of the point clouds to map the limits of the main morphological changes occurred and to calculate the volumetric changes that occurred in recent years with a very high degree of accuracy. Finally, a morphometric analysis was carried out to evaluate both retreat rates and failure indices with GIS tools.

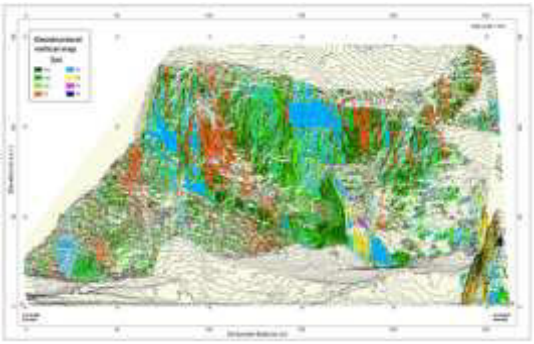


Fig. 5 Geostructural map of Coroglio cliff (Mod. from Matano et al., 2016).

In order to assess the rockfall hazard for the Coroglio coastal cliff, two comprehensive datasets were used. The first one refers to multi-temporal acquisitions of topographic data with a Terrestrial Laser Scanner (TLS) aimed at assessing volumetric changes and

average retreat rates of the cliff (Caputo et al., 2018). This dataset includes volumetric changes related to 247 rockfall events, with magnitude ranging from 0.002 m^3 to 6.81 m^3, over an observation period of 1.85 years (from April 29, 2013, to March 4, 2015). The second dataset comes from an extensive monitoring activity (from December 2014 to October 2018) carried out in the field by Matano et al. (2020). Specifically, five unstable tuff blocks were monitored with electrical crack gauges and clinometers, along with temperature probes. Further data were gathered from a weather station located near the top of the cliff.

Based on the aforementioned datasets, a first part of this work focuses on the elaboration of Frequency-Magnitude (F-M) curves differentiated in relation to the volcanic geo-material involved (Neapolitan Yellow Tuff and Trentaremi Tuff), and to the different portions of the cliff where the same instabilities occurred. This made it possible to enucleate the most active portions, i.e., those in which the frequency values are greatest. The second part of the work focuses on evaluating the role played by two meteorological variables, namely air temperature and precipitation, on the rockfall inception. This revealed that the incidence of rainfall was four times greater than the change in air temperature.

Conclusion

Outcomes presented in this paper highlight that mass wasting processes are effectively contributing to the dismantling of coastal cliffs in the Campi Flegrei volcanic area. Episodic cliff failures, like the one occurred along the Baia dei Porci cliff site, can displace huge masses of rock that induce sudden and localized cliff recessions. Notwithstanding, monitoring of the Torrefumo cliff performed by means of TLS showed that small-scale failures, between 10-2 m3 and 1 m3, are more frequent than larger collapses. The TLS-based monitoring activity also highlighted that those failures are quite superficial and involve an average rock thickness lower than 1 m. This could be related to a general weakness of the surficial material induced by weathering processes, or by the occurrence of incoherent deposits of ash, pumices and scoria that are easy to be affected by grain-by-grain surface erosion.

Findings gained by the multi-temporal monitoring of the Baia dei Porci landslide (Fig. 4) demonstrated that after the main failure, landslide scars can be affected by further collapses, as well as deposits eventually formed at the cliff toe can be rapidly eroded by the sea wave action and, secondly, by rainfall-induced surface runoff. The sea wave action could be one of the most important drivers inducing retreat of cliffs in the Campi Flegrei area, but additional accurate measurements are necessary to ascertain this. An indirect marker of the sea wave action is the massive retreat measured for the Torrefumo cliff (1.2 m/yr), in the time period 1956-1974. Interruption of this ac-tion with the seawall induced a dramatic decrease to 0.17 m/yr of the cliff retreat rate (1974-2008). On the other hand, it did not lead to a stable equilibrium condition of the slope, as testified by short-term retreat rates ranging from 0.001 to 0.025 m/yr during 2013-2016. These rates are comparable with the ones measured along the Coroglio tuff cliff, ranging from 0.05 to 0.10 m/yr, during 2013-2015 interval (Caputo et al., 2018).

Moreover, the contrasting measured retreat rates of Torrefumo and Coroglio cliffs could also be related to the different volcaniclastic facies cohesion. Specifically, the natural retreat rate of the Torrefumo cliff (1,2 m/y), where more unlithified deposits occur, is higher than that of Coroglio cliff (0,05-0,10 m/y), made up mainly of lithified tuff facies.

Geostructural maps and analysis of the tuff coastal cliffs with TLS surveys and processing show accuracy, precision and resolution that are adequate and consistent for the scope of geomechanical analysis of rock failures (Matano et al., 2015, 2016). In summary, the described findings high-light that dismantling of the analyzed volcanoclastic coastal cliffs is due to a complex interaction of marine and subaerial processes. Some processes have been analyzed in this research, but more research is necessary for a complete knowledge of the cliff behavior.

The work also showed the importance of using multi-temporal survey techniques (such as those referable to Terrestrial Laser Scanners), for the construction of the dataset necessary for the generation of F-M curves. In addition, the role of instrumental monitoring – possibly prolonged over the years – was highlighted for the full understanding of the factors controlling the cause-effect relationships and, therefore, for the definition of the most appropriate risk mitigation interventions (whether structural or nonstructural, such as warning systems).

Acknowledgments

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Fabio Matano ()
[1] CNR, ISMAR, Naples 80133, Italy
E-mail: fabio.matano@cnr.it

Seismic performance of earth dams: field data and numerical predictions

Orazio Casablanca[1*]

[1]Department of Engineering – University of Messina Contrada di Dio – Sant'Agata - Messina (Italy)
[*] Corresponding author. E-mail: ocasablanca@unime.it

Abstract

Large dams are essential infrastructures as they provide vital water resources to local communities. Their importance has become even more pronounced in recent times due to the increasing frequency of drought periods caused by pressing climate changes. As a result, ensuring the functionality, structural integrity, and safety of dams has emerged as a top priority, especially in regions prone to seismic activity. Indeed, powerful earthquakes have the capacity to undermine the stability of dams, posing significant risks of triggering flash floods that can pose grave threats to both lives and property. Over recent decades, innovative performance-based approaches have been developed to assess the seismic behaviour of dams and estimate the associated financial and social risks of irreversible damage or failure due to earthquakes. In this framework, it is essential to establish suitable seismic performance index and ensure they remain within specified target values across various seismic scenarios. According to international standards (e.g., FEMA, 2005; USACE, 2004), assessing the seismic performance of dams involves verifying that stability and serviceability conditions are maintained against anticipated ground motions, which can have either a low or high probability of occurrence throughout the dam's lifespan. This strategy helps prevent catastrophic failures and uncontrolled water releases during extreme (low probability) events and ensures the dam's operational conditions (and that of the entire facility) during less severe (high probability) earthquakes. Adhering to these principles, the Italian seismic code for dams, NTD14, defines four limit states with different probabilities of exceedance (PVR) over a specific period (VR). These include two serviceability limit states (Operational Limit State and Damage Limit State) and two ultimate limit states (Life Safety Limit State and Collapse Limit State), based on the extent of damage caused by seismic loading and the occurrence of uncontrolled water release.These limit states must be evaluated through appropriate analyses to verify that the specified conditions are not reached.

Field observations and theoretical studies suggest that accelerations within the dam structure and the magnitude of earthquake-induced permanent displacements are crucial for evaluating the dam's response to significant ground motions. Specifically, comparing the peak acceleration at the dam crest to that in the foundation soils or at the dam base offers a quantitative assessment of the amplification effects during seismic shaking. This comparison can identify a non-linear dam response, characterized by reduced soil shear stiffness and permanent deformations. The settlement of the dam crest resulting from earthquake loading is closely linked to the level of damage caused by the seismic event and can serve as a performance index, reflecting the dam's overall seismic response and the extent of damage caused by seismic loading. Thus, predefined threshold values for crest settlement can be established to determine the occurrence of a specific limit state.

This study presents a comprehensive dataset consisting of both case studies and numerical analyses of seismic-induced damages to dams.The dataset obtained following the classification described above is represented in the following figure where the proposed performance indexes are also represented together with the intervals of settlement ratio (represented with arrows in the plots) suggested by Swaisgood (2003) for each level of damage. This dataset enables the definition of appropriate threshold values for crest settlement, which can be used as reference benchmarks in performance-based seismic analyses of dams.

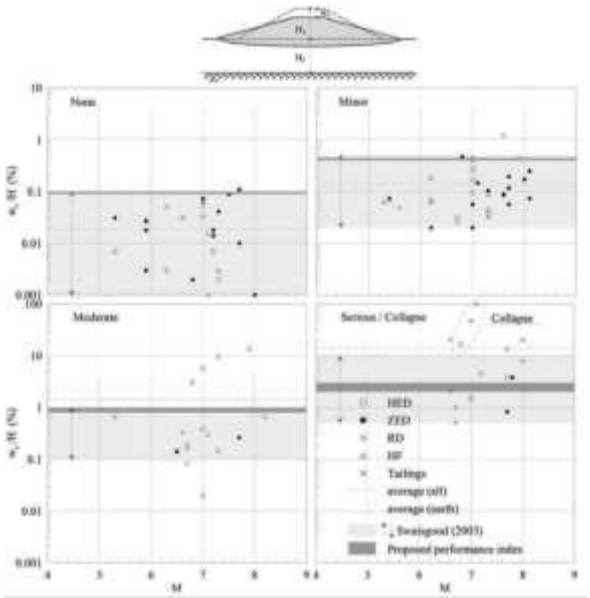


Figure 1: Dataset considered to define threshold values of the non-dimensional crest settlement for different dam type (RD=Concrete/bituminous rockfill, HF= Hydraulic fill, HED= Homogeneous earth, ZED= Zoned earth).

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Study on the attenuation relationship of the earthquake response spectrum in the loess region of China

Yuxin Chen[1], Ping Li[1*],Yaqian Ju[1], jianyuan Xu[1], ganglei Ouyang[1]

[1] Institute of Disaster Prevention, Sanhe City, Hebei Province, China
[*] Corresponding author. Tel: +86-13373168390; E-mail: chinaliping1981@126.com

Abstract

Earthquake-induced building damage and geological disasters have caused incalculable losses to human beings. Geological hazards such as gullies, collapses, landslides and debris flows are especially common in loess areas due to intense surface cutting, and the impact of earthquake damage is severe and extensive. Therefore, it is necessary to conduct special research on the attenuation characteristics of ground motion in loess areas, which provides a very important theoretical basis for the hazard analysis of major earthquakes, improving the level of seismic resistance and seismic zonation in loess areas.

We have collected, screened, and organized 1092 station records from 141 seismic events that occurred in the Loess Plateau region of China between 2001 and 2019. The fitting values were obtained by selecting the high horizontal peak ground acceleration values from the stations and utilizing a two-step regression method to fit the attenuation relationship of horizontal ground motion response spectra in the Loess Plateau region of China. The distribution trends of the residuals with the magnitude and distance were analysed, and the attenuation relationships in this paper were compared with the models of Wang and Yu.

The results show that most of the observed values are within ±1 times the standard deviation of the acceleration attenuation curve, and the model proposed in this paper effectively reflects the trend of response spectrum attenuation with distance in the loess region of China.

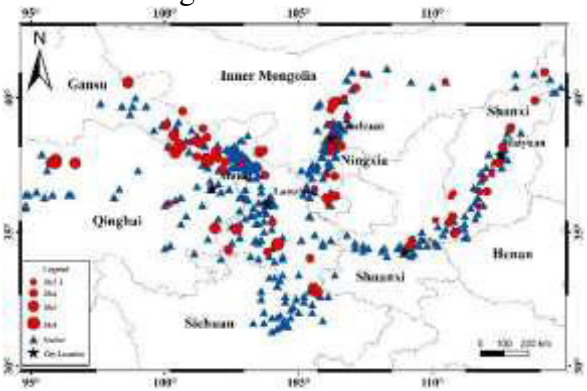


Figure 1. Seismic and station distribution map

Acknowledgments

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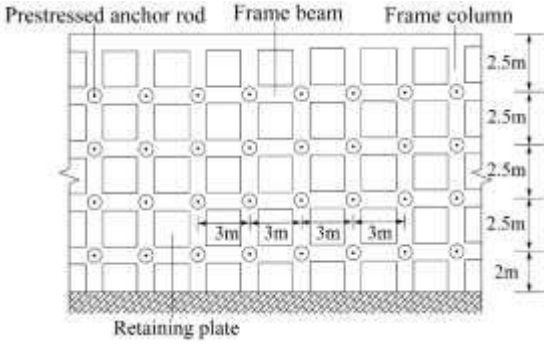
Seismic Displacement Analysis of Slope Reinforced by Frame Prestressed Anchor Rod Structure

Guoliang DAI [1], Zhengzhen WANG [2,*], Rui ZHOU [2]

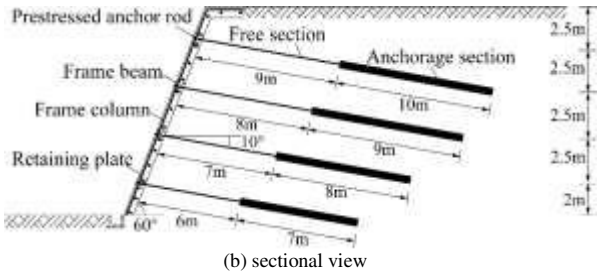
[1] School of Civil Engineering, Southeast University, Nanjing, Jiangsu, 210096, China.
[2] School of Civil Engineering, Lanzhou University of Technology, Lanzhou, Gansu, 730050, China.
[*] Corresponding author. Tel: +86-13611587550; E-mail: wangzz@lut.edu.cn.

Abstract

Instability of slope induced by earthquake seriously affects casualties and property losses and hinders earthquake rescue, so the study of slope displacement under earthquake is of great significance. For the slope reinforced by frame prestressed anchor rod structure under earthquake, the calculation method of slope displacement and axial force of anchor rod under seismic action were derived by combing Newmark method and limit analysis method. This method proposed in this paper comprehensively considers the amplification effect of earthquake acceleration on the slope surface, the inertial force of reinforce structure and the dynamic change of axial force of anchor rod. The solution program of this method was compiled in MATLAB and its rationality was proved by a numerical model in FLAC3D software. The results show the slope displacement calculated by the method proposed is consistent with the variation law of slope displacement obtained by traditional methods. Regardless of whether the anchoring force is considered or not, the slope yield acceleration obtained by this method is very close to the slope yield acceleration obtained by the Bishop method. The critical sliding surfaces obtained by the two methods are very close, and the sliding points are located at the foot of the slope. Compared with the Bishop method, this method can consider the amplification effect of seismic force on the free surface of the slope and the gravity of the supporting structure. Ignoring the change of the axial force of anchor rod will make the slope displacement larger, while not considering the amplification effect of the seismic acceleration on the slope surface and the gravity of the reinforce structure will make the slope displacement smaller. This method proposed in this paper comprehensively considers the above factors and is more accurate. In the process of landslide mass sliding, the axial force of the anchor rods increase gradually, and the growth rate of the axial force of the lower anchor rods are larger than that of the upper anchor rods. The increase of anchor rod prestress and elastic modulus will reduce the seismic displacement of the slope, and the increase of the dip angle of the anchor rod will increase the seismic displacement of the slope. The dip angle of the anchor rod should be minimized without affecting the grouting effect of the bolt.



(a) front view



(b) sectional view

Fig.1. Frame prestressed anchor rod structure

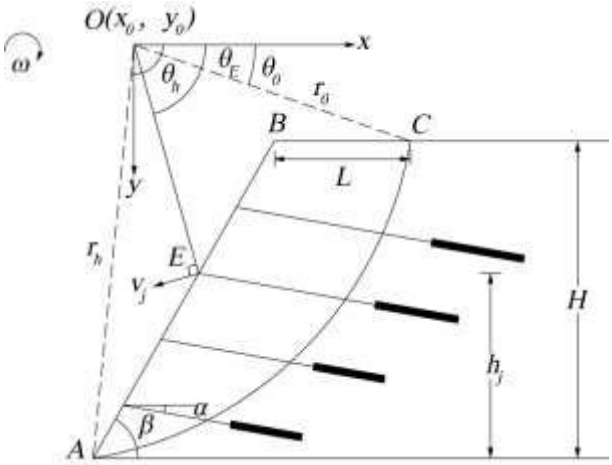


Fig.2. Calculation model of slope reinforced by frame prestressed anchor rod structure

Acknowledgments

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«EARLY WARNING» SYSTEM FOR RISK PREVENTION FROM EXTREME STORMS

Angela Di Leo[1*], Fabio Dentale[1], Anna Sansanelli[1], Eugenio Pugliese Carratelli[1]

[1] Department of Civil Engineering, University of Salerno, 84084 Fisciano, Italy
[*] Corresponding author. Tel: +39-089969088; E-mail: andileo@unisa.it

Abstract

The coastal environment is a highly complex dynamic system influenced by natural and human factors that contribute to its spatial and temporal evolution.

Specifically, the Southern Tyrrhenian Sea, and in particular the Campania coastal area, is heavily affected by human activities and extreme weather/marine events can cause extensive landscape, as well as social and economic damage.

The work presented here is aimed at developing a real time system to forecast and manage the potential risks for citizens and infrastructure deriving from overtopping events of extreme sea storms.

In particular, the wave forecast data on a regional scale, which allow us to see an estimate of the offshore waveform characteristics, acquired from WRF/ NOAA (Weather Research and Forecast) or from ECMWF (European Center for Medium-Range Weather Forecasts) or similar systems, they will constitute the boundary conditions for a propagation model that will allow the identification of the wave characteristics on coastal structure.

The propagation model is necessary because the action of the wave motion near the coast does not only depend on the offshore marine weather climate and the exposure of the area, but also on the effects that small variations in the bathymetry or the presence of obstacles induce on propagation towards the ground (refraction, shoaling, etc).

The daily assessment of the overtopping flow can be done using the wave parameters calculated at multiple points identified in relation to the different coastal structures and compared with the safety limits defined by the Eurotop Manual. This procedure, developed using analytical methodologies or using advanced numerical models, will allow identifying any critical situations in relation to the danger to human life.

To correctly evaluate the wave characteristics in front of the maritime structures, the entire numerical procedure will initially be calibrated using experimental field data detected by suitable instrumentation for wave monitoring.

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Physics-based artificial intelligence regional-scale models for the prediction of liquefaction-induced lateral spreading

Maria Giovanna Durante[1*]
[1]University of Calabria, Arcavacata di Rende, Italy
[*] Corresponding author. E-mail: mgdurante@unical.it

Abstract

The increased availability of high-quality data from post disaster field reconnaissance and their collection in open-source datasets and/or databases enabled the use of modern and innovative tools. An interesting and promising example of such tools is represented by those based on Artificial Intelligence (AI) techniques. Using a large amount of data, AI based tools can be used to develop models to solve problems in the field of geotechnical earthquake engineering. In this connection, the 2010-2011 Canterbury earthquake sequence in New Zealand represents and ideal case study thanks to the amount and quality of information available through the New Zealand Geotechnical Database (NZGD) and to the significant damage due to abundant manifestation of liquefaction-induced lateral spreading (Figure 1, Durante and Rathje, 2021a, b). Such data are used to train and test different types of AI models, based on Random Forest and Neural Network algorithms. The developed models compare favorably against observed data. Explainable AI methods (e.g., tools used in this study to check that results are compatible with physics-based behaviors) are then used to ensure that the models are able to correctly reproduce the physics behind the phenomenon. These computationally expensive analyses were carried out utilizing cloud-based computing capabilities offered by the Texas Advanced Computing Center (TACC) available to the natural hazard community through the cyberinfrastructure DesignSafe (Rathje et al, 2017).

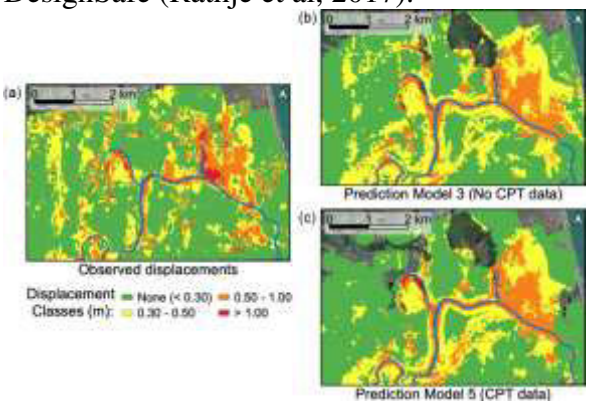


Fig. 1. Regional-scale lateral spreading displacement maps for the 22 February 2011 Christchurch earthquake. (a) Observed and (b, c) predicted displacements using AI models (Durante and Rathje, 2021a, b).

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Initiation and long-runout motion mechanism of an earthquake induced loess landslide

Youqian Feng [1,*], Fawu Wang [1]

[1] College of Civil Engineering, Tongji University, Shanghai 200092, China.
[*] Corresponding author. Tel: +86-18119453657; E-mail: 2310496@tongji.edu.cn

Abstract

On December 18, 2023, an earthquake-induced liquefied loess landslide occurred in Minhe County, Qinghai Province, China. The landslide, which is called Minhe landslide thereafter, quickly mobilized into a flowlike sliding after its initiation, with a horizontal distance of 3200 m, and caused 20 deaths. The source of water in the loess layer is a key issue in clarifying the initiation and movement mechanisms of the Minhe landslide.

During the field investigation, portable dynamic cone penetration tests were conducted to understand the strength of soil in each layer. In order to explore the groundwater distribution and soil permeability, electrical resistivity tomography and in situ permeability tests were conducted. The cyclic loading triaxial tests on intake specimens are intended to replicate the failure process of various soil layers under a PGA of 0.4g, and investigate the seismic liquefaction of loess under varying overlying soil thicknesses. A pore pressure controlled ring shear test was also designed to simulate the liquefaction behavior of loess flow. The cyclic loading triaxial tests and the ring shear test are on going.

The permeability tests showed that geological and topographic factors control the high water content of loess rather than irrigation. The failure process of the Minhe landslide is summarized in Fig. 1. The earthquake triggered the liquefaction of saturated loess in the source area, causing it to flow out of the natural valley and form a flowlike landslide. After that, the overlying alluvium in the source area lost its support, forming a rotationally inclined structure. The liquefied loess swiftly passed through the motion area and gradually slowed down, entering two villages and causing devastating disasters.

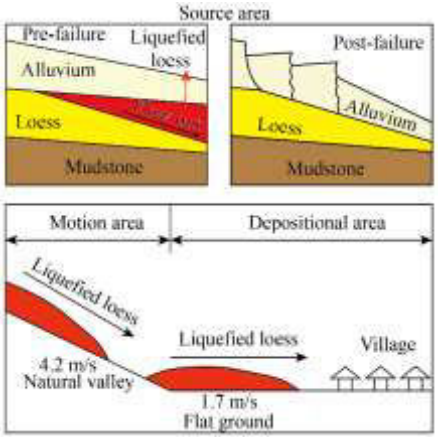


Fig. 1 Failure process of the Minhe landslide

Acknowledgments

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Thanks to Ye Chen, Bo Zhang, Zijin Fu, Hao Ma and Shengzhe Cao for their work in the field investigation.

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The use of soil bio-engineering techniques for landslide risk mitigation in Italy: statistics from ReNDiS database

Alessandro Fraccica[1 *], Valeria Licata[1], Francesco Menniti[1], Daniele Spizzichino[1]

[1] ISPRA – Dept. Geological Survey of Italy

[*]Corresponding author. Tel: +39 3337783279; E-mail: alessandro.fraccica@isprambiente.it

Abstract

ReNDiS is a web-GIS open data developed by ISPRA as part of the technical-scientific support provided to the Italian Ministry of the Environment and Energy Security to monitor the implementation of hydrogeological risk mitigation measures and to manage the evaluation of funding requests from all Italian Regions, since 1999 [Gallozzi et al., 2020]. Information is organised starting from individual mitigation measures. Design phases of the projects, type of solutions adopted and accounting data are collected for all of them. Each measure is subdivided into lots (corresponding to a single project) and these lots may be composed of one or more types of works. The database implements also the types of works that can be recognized as soil bio-engineering techniques: from this detailed classification it is possible to estimate which measures implement some of them, either entirely or in combination with other *'traditional'* works.

So far, about 25.000 mitigation measures financed with a total of more than 17 billion Euro were already collected and monitored in ReNDiS. Among the surveyed measures, detailed and reliable technical information about the types of works and hazard mitigated is available for more than 3.000. Within these measures, 65% include works for landslide risk mitigation and 51% include works for hydraulic risk mitigation. Among landslide risk mitigation, 17% of the types of works fall within soil bio-engineering category, while for hydraulic risk mitigation the percentage is 11.8%. Among the types of works for landslide risk mitigation, selective vegetation cutting is the most commonly used in areas where the overburdening of vegetation has a destabilising action that prevails over root reinforcement. Moreover, it is found that vegetation reinforcement is, to a large extent, combined with structures or reinforcements made of biodegradable materials such as wood, bio-mats, bio-nets and stones. The most adopted traditional engineering works include reinforced concrete piles and walls, cortical reinforcement through steel mesh, and drainage systems. Soil bio-engineering solutions are used to improve soil strength over time, thanks to root growth, and they are used to cover large areas, thus mitigating especially shallow landslides and erosion phenomena over large areas. Given the landscape, cultural, economic and sustainability interests involved, the monitoring of such bio-engineering solutions through the ReNDiS database is a fundamental tool for planning new landslide mitigation works throughout Italy, in order to reduce visual impact with the same efficiency.

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Multiscale simulation of progressive landslides by the adaptive RDFA method

Bin Gong[1,*], Tao Zhao[1]

[1] Department of Civil and Environmental Engineering, Brunel University London, London UB8 3PH, UK

[*] Corresponding author. Tel: +86-13654987937; E-mail: bin.gong@brunel.ac.uk

Abstract

A novel hybrid rock discrete fracture analysis (RDFA) method has been developed through the integration of the rock failure process analysis (RFPA) (Tang et al. 2020) and the discrete element method (DEM) (Cundall 1971). RDFA offers a comprehensive framework for simulating the continuous-discontinuous behaviors within rocks, encompassing fracture and fragmentation phenomena.

Utilizing a newly devised nodal updating scheme, RDFA allows for adaptive node adjustments at critical crack tips while satisfying strength criteria, effectively capturing the initiation and propagation of zero-thickness cracks. Furthermore, the unified global equilibrium equation has been deduced, enabling the simultaneous solution of continuous and discontinuous problems using a unified computational model. Meanwhile, it facilitates the automatic transformation of rock materials from continuum to discontinuum.

Notably, RDFA accommodates the inherent heterogeneity of rock masses, allowing for the synchronized consideration of localized damage and the evolution of fine cracks. RDFA was successfully calibrated via the Brazilian splitting test, demonstrating alignment with the analytical solutions. Subsequently, the uniaxial compression tests were conducted on the rock specimens containing single or double flaws. The results show that as the flaw inclination angle (α) increases from 0° to 60°, the distance between the initiation position of cracks and the flaw tip decreases exponentially. The crack initiation stress follows a non-linear pattern, decreasing initially and then increasing with the growth of α. Besides, it was used to model the progressive landslide and capture multiscale fracturing process and failure characteristics.

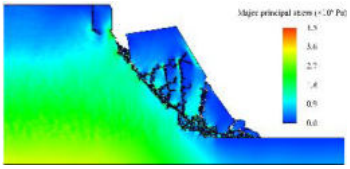


Fig. 1 Landslide modelled by RDFA

Acknowledgments

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Characterization of Groundwater Flow Using Actively Heated Fiber Optics Based Thermal Response Test

Kai Gu[1,*], Bo Zhang[1], Bin Shi[1]

[1] School of Earth Sciences and Engineering, Nanjing University, Nanjing 210023, China
[*] Corresponding author. Tel: +86-13851856337; E-mail: gukai@nju.edu.cn

Abstract

Groundwater flow is considered to be an important triggering factor of many geological disasters, but it is still a challenge to accurately characterize groundwater flow.

Here we present a novel method of characterization of groundwater flow using actively heated fiber optics based thermal response test (ATRT) in grouted boreholes. A series of numerical models are set up to study the effect of the grout and the jacket of an actively heated fiber-optic cable on the simulated thermal response measured along a heated borehole. The findings are utilized to further develop existing groundwater flow rate estimation procedures based on the moving infinite line source model (Diao et al. 2004). The developed approach is demonstrated in a case study in a borehole near a bank collapse site that penetrates different aquifer layers. Accordingly, significant local groundwater flow rates (9×10^{-7}–5×10^{-6} m·s^{-1}) are found that vary with depth. The values derived by the TRT interpretation closely match the expected rates (2×10^{-6}–7×10^{-6} m·s^{-1}), which supports the good applicability of the estimation procedure in the field.

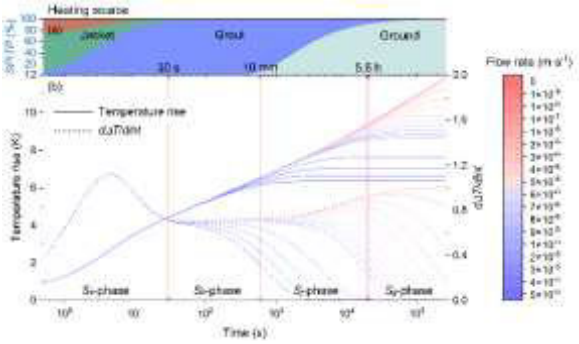


Fig. 1 Temperature response curve of the heat source with different groundwater flow rates.

Acknowledgments

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Study on Deformation Mechanism and Warning Model of Step-like Landslide in Three Gorges Reservoir Area

Fei Guo[1,2*], Xiaohu Huang[1, 2], Maolin Deng[1, 2], Qinglin Yi[1, 2]

[1] Key Laboratory of Geological Hazards on Three Gorges Reservoir Area, Ministry of Education, Yichang 443002, China;

[2] College of Civil Engineering & Architecture, China Three Gorges University, Yichang 443002, China.

[*] Corresponding author. Tel: +86-15926968669; E-mail: ybbnui.2008@163.com

Abstract

Since the impoundment in 2003, more than 5000 landslides or potential landslides have been identified in the Three Gorges Reservoir area, and these hazards seriously threaten the continued operation of the Three Gorges reservoir and the safety of dams, waterways, and residents. By studying the deformation characteristics, inducing triggering factors, and instability mechanism of landslides, it is helpful to evaluate the stability of landslides and construct early warning and prediction models. Taking the Bazimen Landslide in the Three Gorges Reservoir Area as an example, the deformation characteristics and instability mechanism of the landslide are studied by comprehensively analyzing the data of rainfall, reservoir water level, manual and automatic GNSS monitoring, combined with the field macro inspection and exploration data, and the reasonable early warning criterion and threshold are determined. The research shows that: (1) the overall deformation of the Bazimen landslide is obvious and in the creep deformation stage. The landslide deformation is mainly concentrated in May to September every year, and the landslide accumulation curve presents typical "step-like" deformation characteristics. (2) The deformation of landslides is controlled by slope structure, lithology, and other factors. The decline of reservoir water level is the main driving factor of landslide deformation and is positively correlated with the decline rate of reservoir water. In addition, severe rainstorms and continuous rainfall will promote landslide deformation in the stage of water-level decline, reservoir low water level operation, and water level rise, which is the second driving factor of landslide. (3) The threshold displacement rate for "step-like deformation" of Bazimen Landslide is 4.6 mm/d, the threshold of 7-d cumulative rainfall is 60 mm, the threshold of reservoir water level is 159 m, and the threshold of reservoir water level decline rate is 0.4 m/d, obtained by refined data analysis and improved tangent angle method.

Acknowledgments

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Remote Sensing Image Recognition of Loess Earthquake Landsilides Based on GoogLeNet

Ping Li [1,2,*], Liansheng Wang2, Xiaobo Li 2, Zhongyuan Fan 2

[1] Hebei Provincial Key Laboratory of Earthquake Disaster Prevention and Risk Assessment；Sanhe , 065201, Hebei, China.

[2]Institute of Disater Prevention, Sanhe 065201, Hebei, China.

[*] Ping Li. Tel: +86-13373168390; E-mail: chinaliping1981@126.com

Abstract

The identification of regional loess earthquake landslides provides basic data for landslide disaster risk control. In this paper, the method of deep learning is applied to automatically identify and classify typical earthquake landslides in the loess area of China based on remote sensing image data. Firstly, based on the landslide field survey data in Gansu Province and Ningxia Autonomous Region, the Disaster Prevention Research Institute uses visual interpretation-assisted remote sensing images to select landslide and non-landslide samples; Secondly, the GoogLeNet network model algorithm is used to automatically identify and classify earthquake landslide and non-landslide in loess; Finally, the accuracy of the classification and recognition results of the model was evaluated to analyze its application effect on the identification of seismic landslides in loess area. The results show that the identification accuracy and efficiency of loess seismic landslides are high by using this method, and the key areas of landslides can be quickly determined in remote sensing images.Therefore, the method proposed in this paper can quickly evaluate the same type of landslide area, and can provide technical support for its emergency investigation and large-scale landslide disaster investigation.This is an example text template for the Abstract submission to the 21st International Symposium on Geo-disaster Reduction. The Abstract should be concise and self-contained, clearly stating main conclusions of your presentation. Please avoid scientific or engineering symbols, acronyms and bullets.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. U1939209－1).

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Experimental study on turbidity currents transporting microplastics in submarine canyon topographies

Xiaolei Liu[1,2*], Yang Lu[1], Hong Zhang[3], Xiaotian Xie[1]

[1] Shandong Provincial Key Laboratory of Marine Environment and Geological Engineering, Ocean University of China, Qingdao 266100, China

[2] Laboratory for Marine Geology, Qingdao Marine Science and Technology Center, Qingdao 266237, China.

[3] College of Engineering, Ocean University of China, Qingdao 266100, China.

[*] Corresponding author. Tel: +86-18954291073; E-mail: xiaolei@ouc.edu.cn

Abstract

Turbidity currents are commonly known as the key process for delivering terrestrial matter to the deep-water environments and thus inferred to be the main carrier that brings macro and microplastics to the seafloor of submarine canyons, channels, and trenches, etc. Researchers have pointed out that the seafloor of submarine canyons is the preferential accumulation zone of ocean plastics. Effects of canyon topographies on the plastic accumulation after a turbidity current event have not yet been well understood, although it may determine the final fate of plastics and the impact area of plastics on deep-water ecosystems. For this problem, we make attempts through several flume experiments to investigate the difference between microplastic accumulation on two different types of canyon topographies, in which one is the straight-type canyon, and the other is the sinuous-type one. Results show that the microplastics preferentially settle at the canyon heads where wavy sediment deposits simultaneously form due to the turbidity current propagation. The distribution of microplastic-enriched areas is related to specific depositional phenomena caused by the canyon topography. The canyon sinuosity strengthens the ability of the seafloor for trapping more microplastic particles. Due to the lack of direct observation on the process of microplastics transport by turbidity currents, the present study is complementary to the field-related studies to a certain extent and is of great significance in predicting the location of microplastic enrichment in the submarine canyons and the ecological and environmental effects caused by it.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. 42022052 and No. 42277138).

Investigation of the Fuyang shallow landslides triggered by an extreme rainstorm on 22 July 2023 in Zhejiang, China

Qing Lü[1,2], Junyu Wu[1,2,*], Zhongxuan Liao[1,2], Zihao Deng[1,2]

Abstract

An extreme rainstorm hit Fuyang district, Zhejiang province, China, on 22 July 2023. A record-breaking 320 mm of rain fell in less than 48 hours. In the early evening of 22 July, more than 145 shallow landslides emerged in a 20 km2 hilly area of Fuyang amid turbulent flash flooding and claimed 5 lives. Post-event field investigations was urgently carried out to decipher the characteristics and initiation mechanism of the landslides. The main findings are summarized as follows: (1) The week-long rainfall preceding the event and the intense storm on 22 July are primary triggers for the landslides. Besides rainfall, soil heterogeneity and presence of hollows likely contribute to quicker slope destabilizations. (2) The dominant moso bamboo has an extensive and interconnected root system, which could significantly stabilize slopes by enhancing soil strength and restricting cracks.

Keywords

Rainstorm; Shallow landslide; Field survey

Introduction

An extreme rainstorm hit three towns in Fuyang district, Zhejiang, China (Fig. 1), in the early evening at approximately 17:30 local time on July 22, 2023. A total of 320 mm of rain fell in less than 48 hours, which is incredibly rare even in the rainy season for this region, nearly twice the monthly average for July in Fuyang. Hundreds of landslides, dominantly debris flows initiated by debris slides, occurred in the hilly area of Fuyang amid turbulent flash flooding. The catastrophe caused 5 casualties, 3 residents missing, 26 houses crushed or damaged, 21 roads closed, croplands inundated and a four-day standstill for three towns (Wang et al. 2024).

The rainstorm delivered a record amount of precipitation to the hilly area in Fuyang, triggering more than 145 shallow landslides (Fig. 1), most of which initiated the debris flows. As shown in the remote sensing image of Qingong Village (Fig. 2d), one of the most severely affected villages, the majority of these landslides head near the topographic divides, and their toes adjoin the buildings. The converging floodwaters and debris flows caused severe damage to buildings and properties in Qingong Village (Fig. 2).

Post-disaster investigations were carried out to decipher initiation mechanism of the clustered landslides. This article mainly presents the background to this event, the disaster situation in the affected area, and the disaster characteristics identified by the preliminary investigation.

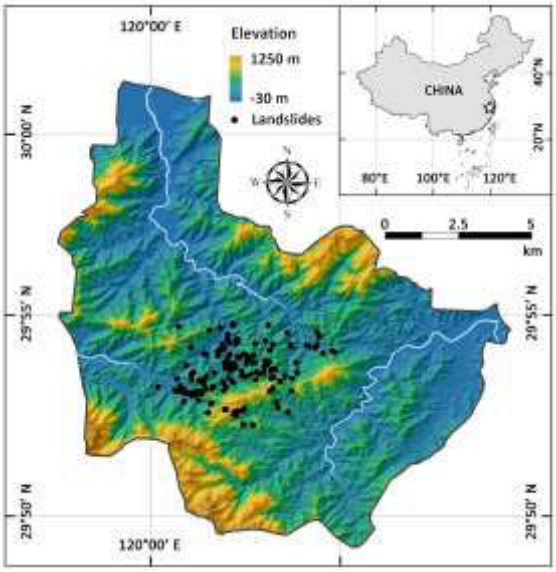


Fig. 1 Landslide distribution map of the study area.

Geological and meteorological conditions

This study focuses on an approximately 250 km2 area which is located in the eastern part of Fuyang district, northwestern Zhejiang Province, China. This area features rugged and overlapping hills.



Fig. 2 Some post-disaster field views in Qingong Village.

The moso bamboo currently occupies a vast majority of the area. Fig. 3 shows the geological map of the study area, located at the southeastern margin of the Yangtze Para-platform. The Proterozoic Jiangshan-Shaoxing Fault Zone runs several kilometers south of the area. The Proterozoic strata are divided into two main groups: the Neoproterozoic, sedimentary Heshangzhen Group, hosting 75% of the Fuyang landslides, and the igneous Shuangxiwu Group, which forms a triangular outcrop area with 22% of the landslides (Fig. 3).

Fuyang has a humid subtropical climate with most precipitation occurring during the monsoon from June to September. The average July rainfall is 178 mm (Zhang 2018), but over 404 mm fell in the week before the landslides. From July 21 to 22, 2023, a record-breaking 320 mm of rain fell, nearly twice the monthly average. Data from the K1690 weather station (Fig. 4b) show a sudden downpour of 77.8 mm in one hour around 18:00, leading to a flood. The 24-hour accumulated rainfall map (Fig. 4a) indicates that the heaviest rainfall and landslide areas do not completely overlap, likely due to geomorphology and lithology influences.

Landslide characteristics

A landslide in Qingong Village was targeted for thorough investigations which included landslide geometries, post-failure topographies, stratigraphic characteristics, erosion features of the propagation path, and debris characteristics in deposition areas. Additionally, an unmanned aerial vehicle was used to reconstruct 3D terrains of the landslide scars for better geomorphological interpretation. The landslide occurred in a Proterozoic sedimentary stratum, mainly composed of the conglomerate. This landslide initiated at an elevation of about 150 m on the slope and rapidly evolved into a debris flow,

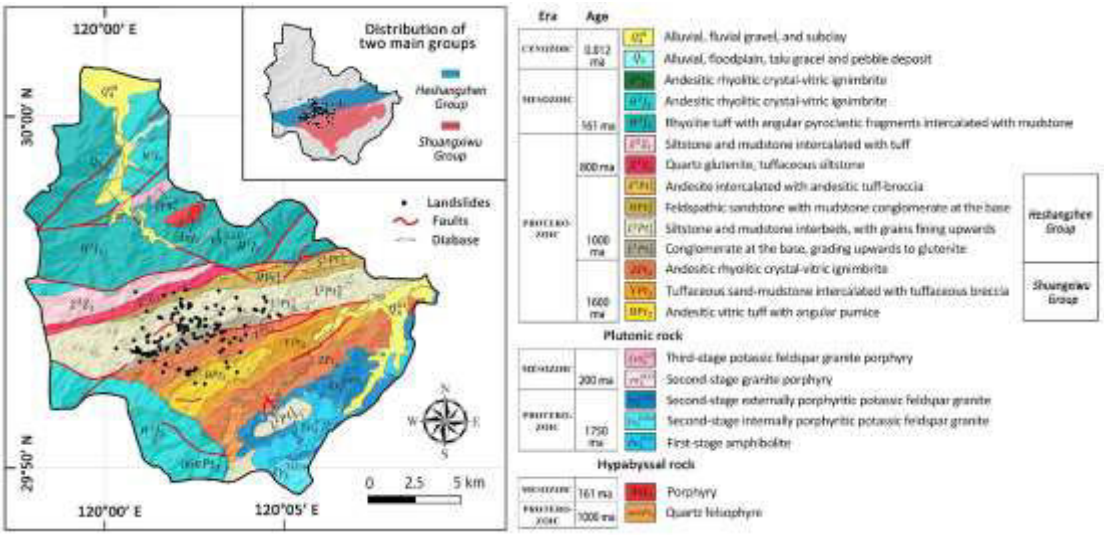


Fig. 3 Geology of the study area. Black dots represent landslides. Red solid lines show faults. Inset shows spread of two main groups of the *Heshangzhen Group* and the *Shuangxiwu Group* (Cai 2001).

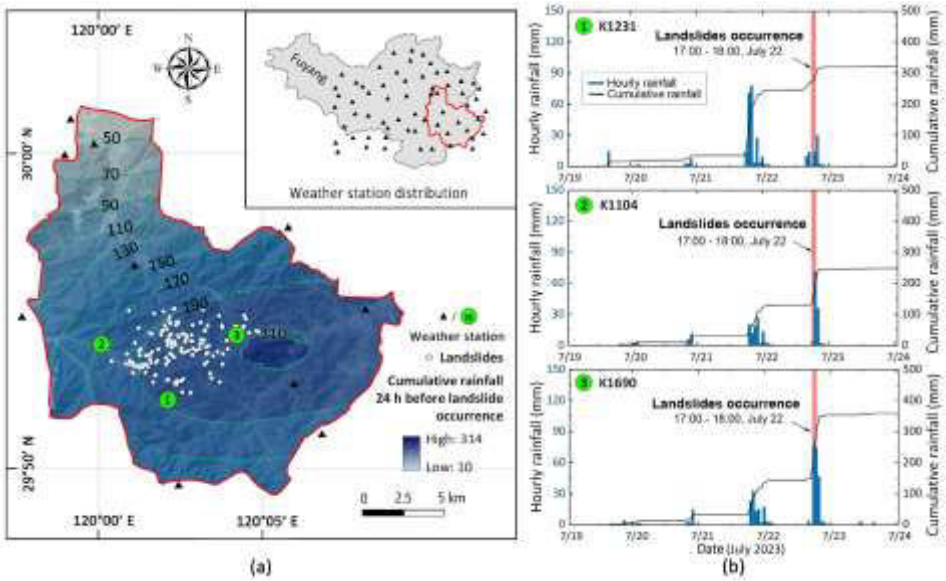


Fig. 4 a. Map of 24-h rainfall amounts before landslides occurred. **b.** Hourly rainfall intensity and accumulated rainfall recorded at three weather stations closest to the struck area

eroding a channel over 300 meters in length. In the following, detailed reconnaissance observations and topographic features of the landslide will be described.

Landslide zones and field observations

Based on field observations and reconnaissance mapping, we delineated the landslide into four distinct zones and several subzones. These zones exhibit different deformation patterns, geometric shapes, topographic features, stratigraphic shapes, topographic features, stratigraphic conditions and material properties (Fig. 6i).

the upper part (zone A1; see Fig. 6a) and partially exposed fractured conglomerate bedrock downslope (zone A2, Fig. 6b). The rough sidewall of zone A2 suggests that the dislocated material is likely colluvium consisting of 5- to 20-cm conglomerate detritus within a loose matrix of nonplastic silt. The original layer in zone A2, less than 1 m thick and characterized by limited sorting

and chaotic structure, is interpreted as old landslide or colluvial deposits. Surface material in zone A1 consists of gravelly soil grading to finer material toward the apex. The soil layer thickens upward to the top of zone A1, where a gentler bare surface was observed. The thin colluvium in zone A2 likely reached saturation and an unstable state earlier due to its steeper incline and higher permeability, leading to retrogressive failure of the residual soil in zone A1.

Zone B, the transport zone, can be subdivided into three subzones (B1 to B3), each with distinct features. Zone B1 is a narrow, relatively flat scour zone with numerous felled, intertwined bamboo oriented downhill (Fig. 6c). The moso bamboo's expansive rhizome system, which robustly connects multiple adjacent bamboo roots (Fig. 6d), is beneficial for reinforcing the soil's lateral mechanical performance. This root system anchors the surficial material, stabilizing the slope and hindering larger destructive mass slides. Zone B2, on the left bank of two converging upland channels (Fig. 6f), is a superelevated area experiencing intense scour from downslope material. This rapid flow stripped vegetation and soil, exposing bare bedrock and creating a high-scoured track. A new gully, formed by post-disaster water flows, separates Zone B2 from Zone B1 (Fig. 6f). These flows come from Zone C, a small gully less than 1 m deep and wide (Fig. 6e), originating in a zero-order basin where a small debris slide occurred. Zone B3 is a narrow, V-shaped valley deeply cut into bedrock (Fig. 6g), showing strong bedrock control. Ancient colluvial talus deposits, 1 to 5 meters thick, with angular gravels at the bottom and finer silt-clay upward, rest on the conglomerate rock. The debris flow eroded and widened the channel, significantly increasing its volume and impact

Zone D is the landslide deposit area (Fig. 6h). Debris, including logs, soil-rock mixtures, and construction waste, filled the area between two concrete houses and a brick cottage. The debris shattered the cottage's side wall but left the concrete houses intact, piling up 3 to 5 meters high and blocking the cottage owners' way home.

Post-failure geometry and topography

The top view of the landslide and its drainage basin was mapped via photogrammetry (Fig. 7, right). The landslide width varies, starting at 10 m at the main scarp, widening to 15 m at the turn, then narrowing to 5 m at the toe. The scar area above the turn is in a convergent, spoon-shaped region, ending in a steep inner gorge. The source and upper transport areas are in gently sloping basins (see A-A' to C-C'). As channels merge below the turn, the basin steepens (D-D' to H-H'), potentially accelerating mass movement but resulting in a short travel distance due to energy dissipation and valley houses, with a travel angle of 26.5°.

The landslide scar shows a slope decrease from top to toe, with a pronounced drop at the turn (Fig. 7, top left). The source area slopes uniformly to SW at 221° and 41.7° steepest angle. After the turn, the flow path orients to WSW (245-251°) with a slope of 25.1°. Outcrop joint measurements at three key sites show consistent attitudes, confirming the underlying bedrock's integrity (Fig. 7, bottom right).

Discussion

Potential factors contributing to the 22 July 2023 landslides

Through meticulous field reconnaissance and laboratory testing, we acquired new insights into the factors contributing to the initiation of the 22 July 2023 landslides, which are summarized as below:

Extreme precipitation preceded the landslides. Starting on July 15, over 404 mm of nearly continuous heavy rainfall fell in the area, including a record-breaking 320 mm storm two days before the event, with a maximum hourly rainfall of 78 mm. This continuous rainfall saturated the slopes and triggered the shallow landslides on July 22. The prolonged rainfall and extreme storms played a major role in triggering the Fuyang landslides. The spatial heterogeneity of soil allowed quicker infiltration through looser regolith, leading to instability in specific slope areas. It is inferred that the thin colluvium in zone A2 reached saturation and became unstable earlier due to its steeper slope and higher permeability compared to zone A1, leading to

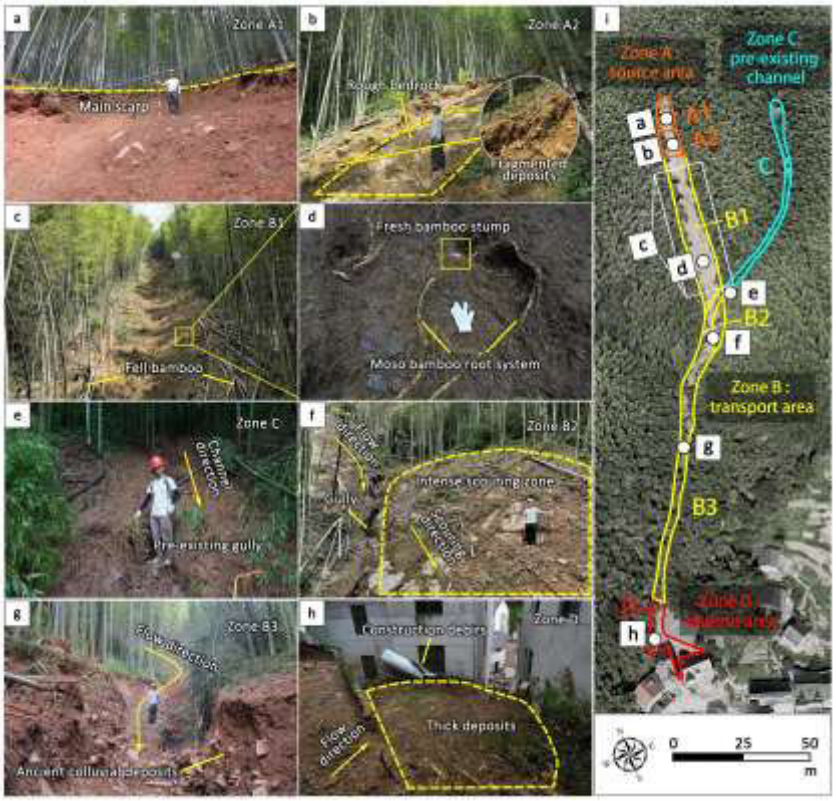


Fig. 6 Different zone views of the landslide.

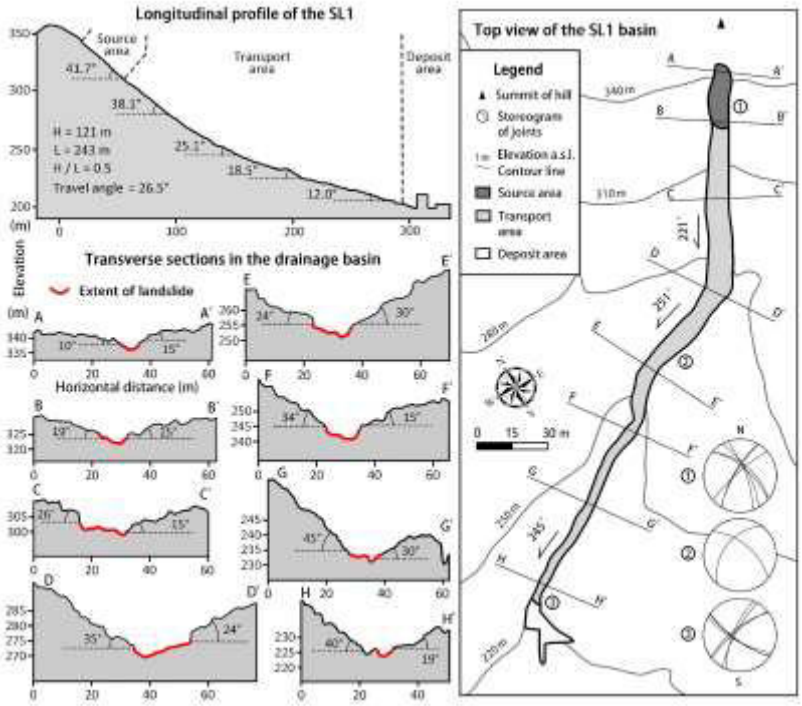


Fig. 7 Top left: longitudinal profile of landslide. Bottom left: transverse sections in the drainage basin; red stripes mark the extents of the landslide. Right: Drainage basin of landslide.

retrogressive failure in zone A1. Field observations also identified small hollows at the landslide sites, which may have facilitated rapid accumulation of runoff and subsurface flow, increasing erosion and pore-water pressure, thereby degrading slope stability.

Bamboo roots limit landslide expansion

Systematic studies indicate that herbaceous plants stabilize slopes better than woody plants (Löbmann et al. 2020). Despite shallow roots, moso bamboo in this region has an extensive root system that significantly stabilizes slopes with thin soil cover. Fresh bamboo stumps found below the SL1 source area, along with intact surrounding soil, confirm the roots' stabilizing role. Additionally, desiccation cracks, typically found on clay-rich slopes, were rare. Research shows abundant vegetation prevents these cracks, enhancing slope stability (Bordoloi et al. 2020). Thus, moso bamboo distribution likely mitigated damage from the event.

Conclusions

In this article, we review the event features, geographic, geologic and hydro-meteorologic conditions of the area, and summarize the observations and data obtained at a lndslide site. Based on these, we gain new evidence and insights into the landslide characteristics and initiation mechanism and. Main conclusions are summarized as follows:

Landslide initiation mechanism: (a) Prolonged rainfall and an extreme storm were the primary factors in the July 22, 2023, Fuyang landslides. (b) Soil heterogeneity allowed quicker infiltration through looser regolith, causing instability. (c) Convergent topography intensified flow concentration and erosion, degrading slope stability.

The dominant indigenous moso bamboo has an extensive and interconnected root system, which could significantly anchor and stabilize the slopes, especially with thin soil cover.

Acknowledgements

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Qing Lü
1 College of Civil Engineering and Architecture, Zhejiang University, Hangzhou 310058, China
2 Center for Balance Architecture, Zhejiang University, Hangzhou 310058, China
E-mail: lvqing@zju.edu.cn

Junyu Wu (✉)
1 College of Civil Engineering and Architecture, Zhejiang University, Hangzhou 310058, China
E-mail: wujunyu@zju.edu.cn

Zhongxuan Liao
1 College of Civil Engineering and Architecture, Zhejiang University, Hangzhou 310058, China
E-mail: liaozhongxuan@zju.edu.cn

Zihao Deng
1 College of Civil Engineering and Architecture, Zhejiang University, Hangzhou 310058, China
E-mail: dengzihao@zju.edu.cn

Clustering mechanisms of landslides triggered by the rainstorm in July 2023 in Beijing, China

Hao Ma[1], Fawu Wang[1]

Abstract

From July 29 to August 2, 2023, Beijing experienced a rare, extremely heavy rainstorm with a peak accumulative rainfall of 745 mm, which broke the historical rainfall record of this area for the past 140 years. The resulting sudden floods and geological disasters destroyed a large number of roads, houses, bridges, and other infrastructure in the mountainous areas of western Beijing. By detailed manual visual inspections, the landslides in the western mountainous areas, which were most seriously affected by the rainstorm, were first identified through multi-period optical remote sensing images. Then, GIS-based spatial analyses were conducted to understand the relationship between landslide distribution and meteorological, geological, and geomorphic factors. The remote sensing interpretation identified 15,383 shallow landslides in a study area of 3,250 km^2. The landslides covered an area of 19.3 km^2 in total and represent 0.59% of the study area. The landslides are characterized by rotational sliding or translational sliding in shallow residual gravel soil, and the depths of landslides are generally less than 3m. The spatial analysis shows that the landslide distribution density has a close relation with the precipitation, except the northwestern area with the maximum density of landslides. The low anti-weathering ability of muddy dolomite is the main cause of the high landslide density in this area. In addition, the slope gradient, slope aspect, topographic position, and elevation also affected the landslide distribution.

Keywords

Extreme precipitation, rainfall-induced landslides, landslide map, Beijing

Introduction

Landslides and related mass movement processes are widespread natural hazards that occur in mountainous areas and frequently cause severe damage to infrastructure, properties, and human life (Zhang, et al., 2023; Froude and Petley, 2018). Rainfall is a primary trigger of landslides, especially shallow ones (Kim et al., 2021). Heavy rainstorm events can trigger up to tens of thousands of landslides simultaneously in rainfall-affected regions (Tsai, et al., 2010). Against the backdrop of global climate change, extreme precipitation events are becoming increasingly common and are hitting various parts of the world (Westra, et al., 2013), even in many traditionally arid and low-rainfall areas (Donat, et al., 2016). These rare extreme precipitation events in typically less rainy areas are likelier to trigger unexpected landslides (Tian, et al., 2022).

From July 29, 2023 to August 2, 2023, a rare extreme rainfall event occurred in Beijing City, which is located in an area of northern China that typically does not receive much rainfall. The rainstorm lasted about 83 hours from July 29 to August 2. During this period, the average cumulative rainfall in the whole Beijing city reached 331 mm, accounting for ~60% of the annual average rainfall (599.5 mm). The maximum cumulative rainfall observed by the meteorological monitoring station reached 745mm, which broke the record in Beijing since the first meteorological monitoring data was available 140 years ago (State Council Information Office of the People's Republic of China, 2023). The rainfall center is located in the western mountainous areas (Fig. 1), and the extreme rainfall has caused a large number of landslides, debris flows, and debris floods in these areas. As of August 9th, the statistical data indicates 33 deaths and 19 people missing, 59,000 houses and 119 bridge collapsed, 147,000 houses severely damaged, and 225,000 acres of crops affected (Securities Times, 2023).

In order to investigate the geological hazard effects of this extreme precipitation event, this paper establishes an inventory to catalog the landslides in

the western mountainous areas (WMAs) of Beijing, which were most severely affected by this extreme precipitation event. Statistical analyses were also conducted to identify the primary controlling factors for the spatial distribution of landslides.



Fig. 1 Location map of the area of interest with the spatial arrangement of the cumulated rainfall. (Source: Beijing Meteorological Service; http://bj.cma.gov.cn)

Study area

The WMAs are located at the northern end of the Taihang Mountains, accounting for approximately 31.2% of the mountainous area of Beijing. The tectonic activity in this area is active. The NE compartmentalized fold structure composed of Tiaojiashan syncline, Jiulongshan-Xiangyudaliang syncline, Beiling anticline, and their intervening narrow anticlines, shaped the basic tectonic framework of this region. From the late Paleozoic to the Mesozoic, the crust in WMAs exhibits continuous subsidence and depression, accumulating thick sedimentary strata with a thickness of 6000-7000 meters. Later, affected by the early Cretaceous Yanshan Movement, the WMAs began to uplift into mountains. Complex crustal activities formed the multilayered strata in WMAs. The lithological types of the WMAs mainly include Cambrian and Ordovician limestone, dolomite and coal; Jurassic sandstone, shale, and conglomerate; Jurassic and Cretaceous volcanic rocks and granite.

The main types of landforms include medium-sized mountains, low mountains, hills, and mountain valleys. Erosion landforms are developed. The ridges generally trend northeastward, consistent with the regional structural direction. The elevation in the mountainous areas generally ranges from 1000 to 1500 meters. The highest altitude is 2303 m a.s.l. Soil erosion in WMAs is severe, and approximately half of the land has a soil layer thickness of less than 30 cm. The WMAs belong to a semi-humid monsoon continental climate, characterized by hot and rainy summers and cold and dry winters. Additionally, due to its significant relative elevation differences and complex terrain, the climate shows clear vertical zonality.

Data and methods

Remote sensing interpretation

Landslide interpretation based on optical satellite images is an effective method for establishing regional landslide inventories. Visual manual interpretation is still a commonly used method for landslide cataloging (Sun et al., 2024). Compared with emerging AI-based automatic mapping algorithms, visual interpretation takes longer but has higher accuracy (Huang et al., 2023). In this study, we fully utilized manual visual interpretation to identify and catalog the landslides, resulting in a rich and accurate landslide inventory.

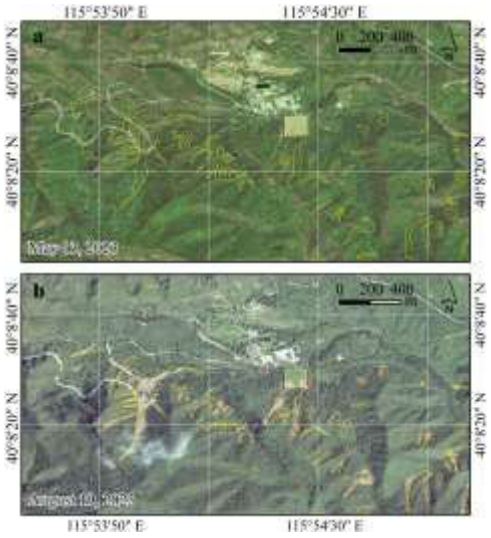


Fig. 2 Comparison of images before and after the studied extreme precipitation event.

The data used to produce and verify the landslide inventory is mainly sourced from high-resolution optical satellite images provided by SIWEIearth (a map service platform led by China Siwei Surveying and Mapping Technology Co., Ltd). The SIWEIearth provides high-quality satellite images with a resolution ranging from 0.8 to 2 meters covering the entirety of China from January 2018 to the present. These images have been precisely merged with an overall cloud cover of less than 20%. To

target our research areas of interest, we mainly utilized images taken on May 13, 2023 (before the studied event, Fig. 2a) and August 19, 2023 (after the studied event, Fig. 2b). It can be observed that there is high vegetation cover in the pre-event image (Fig. 2a) and low or no vegetation cover in the post-event image (Fig. 2b), and there are clear signs of sliding.

The delimiting and mapping procedure for the landslide areas was carried out using Omap software (developed by Beijing Ovital Software Co., Ltd.), which provides both 2D and 3D visualization perspectives of the required optical satellite imagery. The 3D view produced by Omap allows for fully flexible navigation over a virtual mosaic of periodically updated satellite images, including features such as free zoom, rotation, and angle of view. Furthermore, Omap directly links the historical imagery catalog of the SIWEIearth database and supports commonly used graphic vectorization operations, including marking points, lines, and polygons. Therefore, the identification, delimiting, and labeling of landslides can be completed in one sitting through the Omap software. All identified landslides were marked as points and polygons. The point feature indicates the initial position of registered landslides, and the polygon covers the entire identifiable area affected by a landslide.

Environmental factors

Precipitation, geology, and geomorphology are important factors that induce rainfall-induced landslides. Rainfall data comes from publicly accessed data by the Beijing Meteorological Bureau. Geologic factors mainly include lithology, stratigraphic age. The lithology of the study area is based on the 1:1,000,000 geological map. Based on the characteristics and research objectives of the WMAs, four geomorphological factors were selected, including elevation, slope, aspect, and position. These geomorphological factors were extracted from the DEM data from "ALOS PALSAR 12.5 m DEM" using GIS platforms. Position refers to the geomorphic features that describe relative location and specific slope surface shapes, and is divided into six categories: ridge, upper slope, middle slope, lower slope, valley, and flat slope, according to the rules proposed by Weiss (2001). Utilizing visually interpreted landslide point and polygon data, we extracted attribute values such as quantity and area.

Results and analysis

Landslide inventory

After careful comparison of satellite images pre- and post- the rainfall event, a total of 15,383 landslides were identified. An average of 4.7 landslides per square kilometer were identified within the study area. Figure 3a presents the distribution of these landslides, and Figure 3b depicts the quantity density of landslides within the study area. The statistics indicate that there are three main landslide concentration areas within the study area, distributed respectively in the eastern, northeastern, and northwestern parts. The landslide concentration area in the northwestern part of the study area has the highest density of landslide distribution, reaching a maximum of 109 landslides per square kilometer (Fig. 3b). In addition, there are some local landslide concentration areas in the central and southwestern parts of the study area.

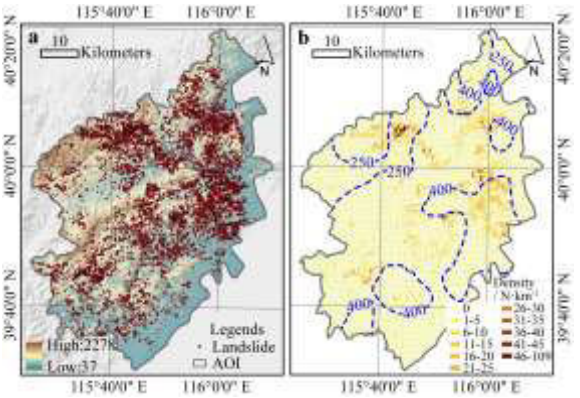


Fig. 3 Landslide distribution map. a) distribution map; b) landslide density map.

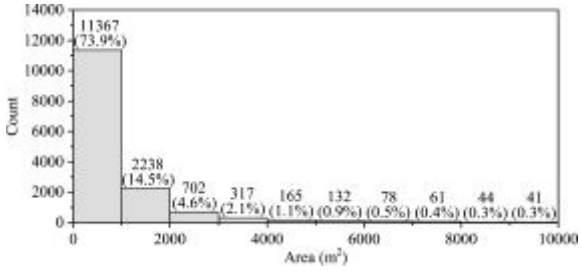


Fig. 4 Number and proportion of landslides.

The total area of identified landslides is 19.3 km^2, which accounts for 0.59% of the area of interest. An individual landslide ranges ~40 m^2 < A_L < 2.1 × 10^5 m^2 in the landslide area (A_L in m^2). Figure 4 presents the statistics of landslide quantity within different area ranges. The majority of landslides caused by this rainfall event have an area of less than 3000 m^2, accounting for 93% (Fig. 4). Among them, 70% of landslides have an area of less than 1000 m^2. A total of 521 landslides has an area greater than 3000 square meters, among which there are 85 landslides with an

area greater than 8000 square meters. The larger landslides commonly exhibit long-distance fluidization movement.

Spatial distribution

Rainfall

The rainfall is the main trigger for the landslides in this event. From Fig. 3b, it can be observed that the distribution of landslides is closely related to the amount of rainfall. The location of the landslide concentration area in the eastern part of the study area is basically consistent with the position of the rainfall center in the eastern part of the study area. In addition, the statistics in Fig. 5 also indicates that the region with the highest rainfall exhibits the highest density of landslides. Therefore, the distribution of rainfall is an important factor affecting the distribution of landslides.

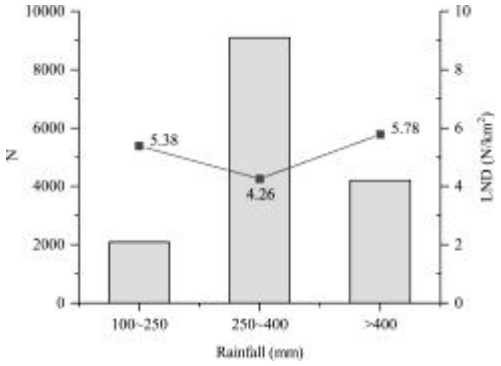


Fig. 5 Statistics of affecting factors: rainfall.

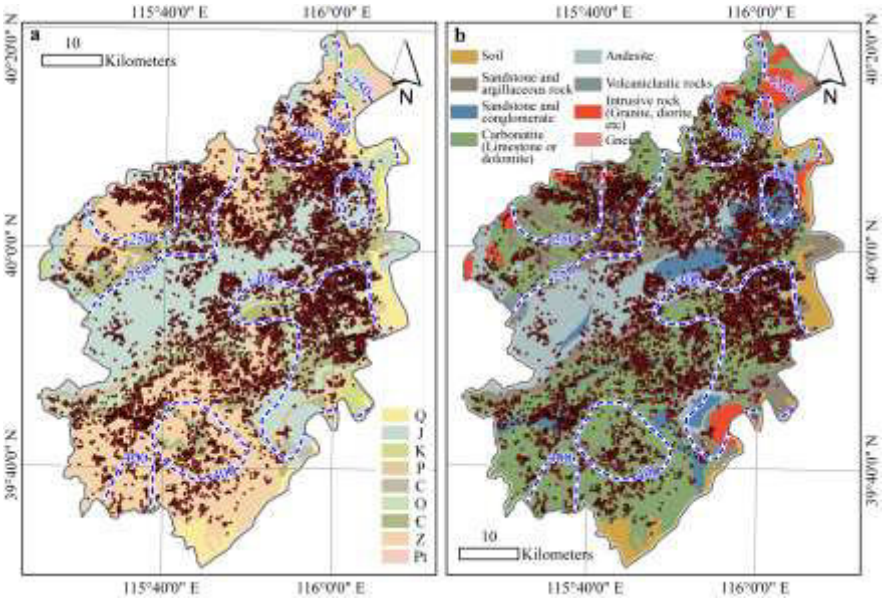


Fig. 6 Geological map of the study area with landslide distribution. a) stratigraphic age; b) main lithology.

Exceptionally, in contrast, the northwestern region, which has the densest distribution of landslides within the study area, experiences the lowest rainfall. In the southwestern part of the study area, where the rainfall center (with rainfall exceeding 400mm) is located, there is only a small number of landslides. Figure 5 also indicates that the density of landslides in the area with the lowest rainfall (5.38 per square kilometer) is second only to the area with the highest rainfall (5.78 per square kilometer). Therefore, it can be expected that besides rainfall, there are other factors controlling the distribution of landslides.

Lithology

Rock lithology is closely related to rock strength and slope stability, and commonly plays a crucial role in the distribution of landslides in many regional landslide trigger events (Sun et al., 2024; Huang et al., 2023). For the WMAs, complex lithology types extracted from a 1:1,000,000 geological map were simplified and categorized in Fig. 6, based on stratigraphic ages (Fig. 6a) and predominant rock types (Fig. 6b). According to the distribution of landslides, it can be observed from Fig. 6a that the most densely distributed area of landslides in the northwest and northeast parts of the study area are

both characterized by Sinian carbonate rocks. The lithologies in the landslide-prone area in the eastern part of the study area are more complex, mainly involving Ordovician, Carboniferous, and Permian carbonate rocks, sandstones, conglomerates, and argillaceous rocks.

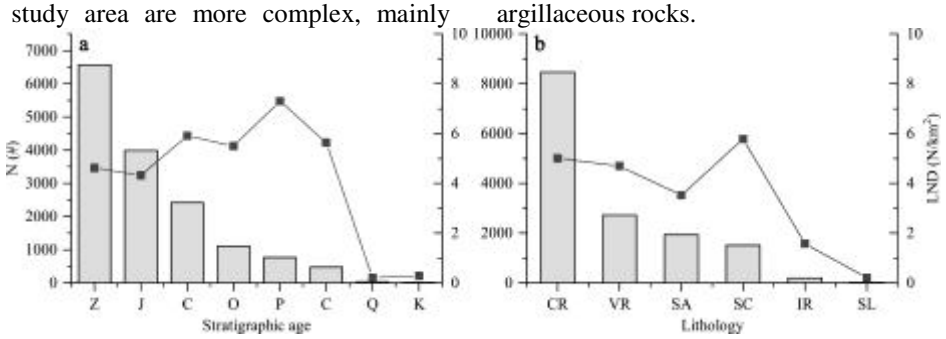


Fig. 7 Statistics of affecting factors: geology. a) stratigraphic age; b) lithology. Note: CR-Carbonatite, VR-Volcaniclastic rocks, SA-Sandstone and argillaceous rocks, SC-Sandstone and conglomerate, IR-Intrusive rock, SL-Soil

Figure 7 provides a statistical breakdown of the number of landslides developed within geological formations of various ages and rock types. In terms of geological ages, the highest number of landslides occurred in the Sinian formations (42%), followed by the Jurassic (26%) and Cambrian (16%) formations. The highest number of landslides occurred within carbonate rock formations (55%), significantly surpassing other rock types such as volcanic rocks, sandstones, and conglomerates. However, when the number of landslides in each geological unit is divided by the area of the formation to obtain landslide density for comparison, a noticeable change in the statistical pattern emerges. For the stratigraphic ages, the formations with the highest landslide density are from the Permian period, followed by the Cambrian, Carboniferous, and Ordovician periods. For the main lithologies, the formations with the highest landslide density are a combination of sandstone and conglomerate, followed by carbonate rocks and volcaniclastic rocks.

Topographic factors

Topographic factors can also significantly affect the formation and distribution of landslides. This study primarily focuses on the influence of elevation, position, slope aspect, and slope gradient (Fig. 8). The elevation was divided into 15 intervals, excluding an interval above 1400 m (Fig. 8a). The remaining 14 intervals were spaced 100 m apart. The statistical results show that the number of landslides increases initially with elevation and then decreases. The number of landslides reaches its peak value of 2400 in the elevation range of 400 m–500 m. 78% of landslides are distributed within the elevation range of 200 m to 800 m.

Slope positions are divided into 6 categories (Fig. 8b), according to the topographic position index. The statistics indicate that the highest number of landslides, reaching 4750 occurrences, is developed at ridge positions, constituting 31% of the total. In addition, landslides formed on middle slopes and valley slopes also account for a significant proportion, with 21% and 23% respectively.

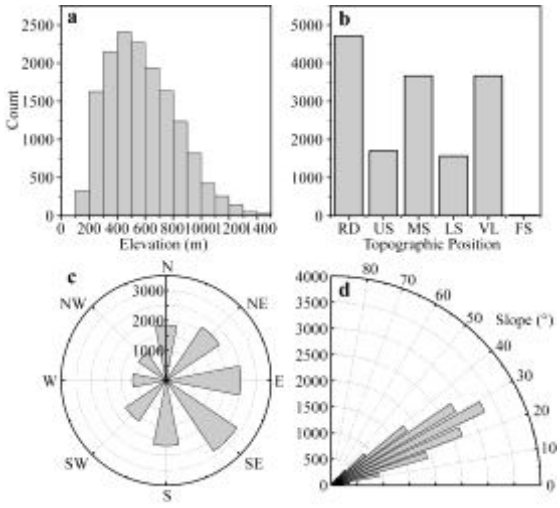


Fig. 8 Statistics of affecting factors: geomorphology. a) elevation, b) topographic position, c) slope aspect, d) slope. Note: RD-ridge, US-upper slope, MS-middle slope, LS-low slope, VL-valley, FS-flat slope

The slope aspect is divided into eight categories (Fig. 8c). The highest number of landslides occurs in the southeast direction, accounting for 18% of the total. The number of landslides in the east and south directions follows as the next highest, accounting for 16% and 14%, respectively. The landslides occurred in the slope aspect from east to south account for 48% of the total number.

The slope gradient in the study area is from 0° to 72.69°. Each 5° interval is divided into a slope interval. As shown in Fig. 8d, the number of landslides increases initially with slope gradient and then decreases. Approximately 55% of the landslides are located within the slope gradient range of 20° to 35°. The highest number of landslides falls within the 25° to 30° range, constituting 21% of the total.

Conclusions

To study the geological hazard effects of extreme rainfall in July 2023 in Beijing, this research identified 15,383 landslides through visual interpretation of satellite images in the western mountainous areas of Beijing. The clustering effect of geological hazards caused by this extreme precipitation event was analyzed through GIS spatial analysis and factor statistics. The results indicate that there are three main landslide cluster areas in the eastern, northeastern, and northwestern parts of the study area. Among them, the landslide cluster area in the east has the largest area, and its location is essentially consistent with the center of the rainfall. The landslide cluster area in the northwest has the highest landslide distribution density, which is related to the specific rock types in the area. For the entire study area, the highest number of landslides develops in carbonate rock formations, while the highest landslide density occurs in formations composed of sandstone and conglomerate. In terms of stratigraphic age, the highest number of landslides occurs in the Sinian strata, while the highest landslide density is observed in the Permian strata. In terms of terrain factors, landslides are most likely to occur at elevations of 400 to 500 meters, ridge positions, southeast-facing slopes, and slopes with gradients of 25 to 35 degrees.

Acknowledgments

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Hao Ma (✉)
[1] College of Civil Engineering, Tongji University, Shanghai 200092, China
E-mail: mah@tongji.edu.cn

Fawu Wang
[1] College of Civil Engineering, Tongji University, Shanghai 200092, China
E-mail: wangfw@tongji.edu.cn

Fluid-solid coupling simulation of landslide disaster chain and its application

Tingkai Nian[1,*], Dongyang Li[1], Hao Wu[2], Yanjun Zhang[3], Defeng Zheng[4,**]

[1] School of Infrastructure Engineering, Dalian University of Technology, Dalian 116024, China

[2] Department of Geotechnical Engineering, Nanjing Hydraulic Research Institute , Nanjing 210024, China

[3] Wuhan Center, China Geological Survey (Central South China Innovation Center for Geosciences), Wuhan 430205, China

[4] Liaoning Provincial Key Laboratory of Physical Geography and Geomatics, Liaoning Normal University, Dalian 116029, China.

[*] Corresponding author. Tel: +86-411-84708189; E-mail: tknian@dlut.edu.cn

[**] Corresponding author. E-mail: defengzheng@lnnu.edu.cn

Abstract

The disaster chain evolution of the landslide-river blockage-impulse wave involves complex dynamic processes and multi-phase coupling which impose a great challenge in developing accurate simulation models. In this research, an extended coupled discrete element method (DEM) and computational fluid dynamics (CFD) numerical method is proposed. The VOF model is introduced to describe the river surface evolution and track the formation and propagation of landslide induced impulse waves. Additionally, a virtual sphere model is suggested to address numerical instability resulting from particle size nearing mesh size. The accuracy and validity of the extended coupled method are verified using a series of test cases involving three-phase interaction and free surface evolution. Based on this, the disaster chain simulations of the landslide-river blockage-impulse wave are carried out and further applied to reproduce the river blockage and impulse wave process of the 10.11 Baige landslide event .The disaster dynamic evolution and the landslide-river interaction mechanism are investigated comprehensively. The results show that the coupled effects of landslide-river-terrain drive the disaster evolution, The landslide mass drives impulse wave propagation, while rivers increase kinetic energy dissipation and have opposing effects on landslide deposition along flow directions; the simulated migration path, deposit morphology and wave erosion area of the Baige landslide are in high agreement with the field survey results about the Baige landslide. The extended coupled DEM-CFD method proposed in this research provides a robust tool for the understanding of the river blockage and impulse wave evolution mechanism. It is of reference value for the disaster prediction and disaster mitigation strategies.

Acknowledgments

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An integrated multi-source data analysis for slow-moving landslide risk assessment on the exposed built-up environment

Gianfranco Nicodemo[1,*], Dario Peduto[1], Settimio Ferlisi[1]

[1] Department of Civil Engineering, University of Salerno, Fisciano (SA)
[*] Corresponding author. Tel: +39-089963440; E-mail: gnicodemo@unisa.it

Abstract

The interaction between slow-moving landslides and human settlements is challenging for both scientific and technical communities involved in identifying the most suitable strategies for land-use planning and urban management that take into account the landslide dynamics and their potential variation under the effect of ongoing climate change.

From this point of view, landslide characterization (in terms of both geometrical and kinematical issues) and the estimation of the vulnerability of structures (e.g., buildings) and/or infrastructure (e.g., roads) interacting with them represent key steps for quantifying the risk as well as in the setting up of forecasting models provided that reliable input data are available. The latter profitably include information gathered from non-invasive spaceborne remote monitoring (e.g., Differential Interferometric techniques) and virtual surveying (e.g., Google Street view imagery) to be integrated with ancillary multi-source data acquired by adopting geological, geomorphological and geotechnical criteria.

In this regards, the methodological frameworks (Peduto et al., 2018; Ferlisi et al., 2019) developed by the authors on the topic in the last few years highlighted that the use of conventional and innovative multi-source monitoring/survey data, if properly integrated according to a multidisciplinary approach, can enhance the studies aimed at addressing the two abovementioned key steps of quantitative risk analyses.

The feasibility of the adopted methods is here presented with reference to well-documented case studies in southern Italy, wherein a rich sample of multi-temporal slow-moving landslide-induced damage to buildings and roads was collected.

Once further validated, the proposed approach could be part of a general procedure to prioritize building/road (extraordinary) maintenance activities and scheduling/implementing risk mitigation strategies by way of structural and/or non-structural measures.

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Enhancing the performance of regional warning models for landslides using local monitoring data

Gaetano Pecoraro[1,*], Rosa Menichini[1], Michele Calvello[1]

[1] Department of Civil Engineering (DICIV), University of Salerno, Fisciano 84084, Italy.

[*] Corresponding author. Tel: +39-3404004413; E-mail: gpecoraro@unisa.it

Abstract

Weather-induced landslides pose significant threats to human lives and properties worldwide. Territorial landslide early warning systems (Te-LEWS) have proven to be a cost-efficient tool to inform the public about the imminent landslide risk (Piciullo et al., 2018). Figure 1 shows a scheme of the main components of a Te-LEWS based on a clear distinction among landslide models, warning models and warning systems (Calvello, 2017).

Warning models employed within Te-LEWS are typically based on rainfall thresholds, which empirically relate the occurrence of landslides to rainfall event characteristics. However, meteorological monitoring does not allow to take into account critical soil properties controlling the initiation of the triggering process. Depending on these conditions, landslides may be triggered in response to a large variety of rainfall combinations.

This research aims at developing and testing a methodology for improving the performance of regional warning models by means of local monitoring data. The proposed procedure has been already applied in some selected test areas (Pecoraro and Calvello, 2021). Currently, the research activities are focused on the development of a diffuse low-cost network of sensors and a proper real-time data transmission, analysis and processing to improve the warning procedures.

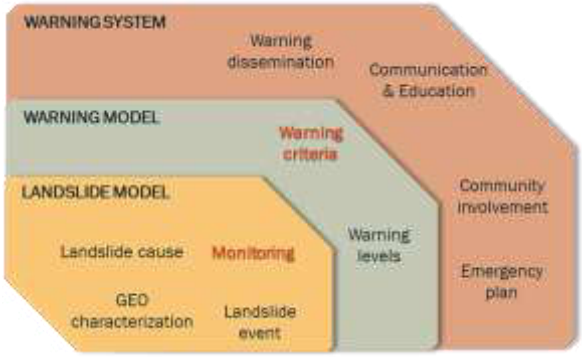


Fig. 1 Framework identifying the main modules of Te-LEWS (modified from Calvello 2017).

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Simulation of landslide dam breach using a 3D-SPH approach

Shuang Li, Ming Peng*

Abstract

Landslide dam breaches pose a significant risk of catastrophic flooding, leading to extensive downstream economic and property losses. Accurately simulating the process requires modeling erosion and slope instability. Traditional mesh-based methods struggle to capture the large deformation and instability of soil slopes directly. Therefore, we introduce a numerical simulation framework based on mesh-free methods. This framework incorporates an erosion rate model to simulate gradual erosion and utilizes the Drucker-Prager plasticity model to capture soil slope deformation. Validation of the model was conducted through simulations of three benchmarks: column collapse, bottom flushing induced by dam-break flows, and landslide dam breach. Comparison with experimental data confirmed the accuracy of the proposed algorithm. Compared to traditional mesh-based methods, our approach enables direct simulation of large deformations in side slopes. The landslide dam breach process unfolds in two stages: headward erosion followed by overall erosion. The onset of overall erosion coincides with the occurrence of slope instability. We employed the proposed method to examine the impact of the internal friction angle ϕ on the evolution of dam slope morphology. Increasing ϕ reduces slope instability and deformation, thus delaying the lateral expansion of the breach.

Keywords

Landslide dam, SPH, Erosion, Large deformation

Introduction

Under overtopping conditions, dams can experience a rapid breach, resulting in substantial flooding and causing severe downstream hazards (Peng and Zhang, 2012; Shen et al., 2020) The precise 3D simulation of dam breach processes, particularly those equipped with spill-ways, presents a significant research challenge (Hu et al., 2024) This challenge involves accurately representing significant deformations in both fluid and soil, erosion phenomena, and the integration of erosion with slope instability.

Currently, the latest 3D fine-scale simulations of landslide dam breaches primarily rely on Flow-3D software using the finite volume method (FVM) (Hu et al., 2024; Mei et al., 2022; Zhang et al., 2023; J. Zhang et al., 2024) However, due to the limitations of grid-based methods, they cannot accurately capture the large deformation movements associated with slope instability of the soil. The Smoothed Particle Hydrodynamics (SPH) method, as a mesh-free approach, offers advantages in handling large deformations and fluid-soil coupling (Bui and Nguyen, 2021), making it suitable for simulating the process of overtopping-induced erosion and slope instability in dams. Zhou et al. (2024) and Chen et al. (2023) have conducted pioneering work in SPH simulations of dam breaches. However, their validations were limited to 2D scenarios. The development of a mesh-free model capable of simulating 3D conditions remains to be explored.

We propose a novel framework based on Riemann-SPH for simulating the 3D dam breach process. The algorithm can directly couple erosion-induced downcutting and slope instability within the spillway. In essence, downcutting leads to an increase in the overhanging face, triggering slope instability. The de-stabilized soil, in turn, affects the fluid dynamics, influencing the erosion process. The proposed method enables the 3D simulation of the evolution of dam slope morphology, which is difficult to achieve with existing grid-based models (Hu et al., 2024).

Methodology

At a certain inflow rate, a dam undergoes breaching due to soil deformation and erosion, as depicted in Fig. 1(a). To simulate this process, we propose a dam breaching model based on the meshless method, involving four different types of SPH particles and three interfaces, as illustrated in Fig. 1(b).

For different SPH particles, fluid particles simulate fluid based on the Newtonian fluid model; soil particles simulate soil based on the elastoplastic solid model; fluidized soil particles simulate soil particles activated by erosion, with similar governing equations to fluid particles but different density and viscosity;

boundary particles are used to simulate non-slip boundaries.

Different types of particles form various interfaces: the mixture interface (Rezavand et al., 2020) defines the mixture between fluid particles and fluidized soil particles; the Adami boundary (Adami et al., 2012) defines the impermeability of boundaries. The erosion interface defines the erosion process of particles, which will be elaborated on next section.

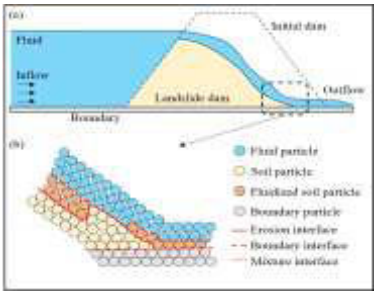


Fig. 1. Illustration of (a) overtopping breach of a dam and (b) various types of SPH particles and interfaces involved in the proposed method.

Governing equation

In the Lagrangian framework, both the soil and fluid must satisfy the conservation equations for mass and momentum. In this study, the fluid is treated as a Newtonian fluid, while the soil is considered an elastoplastic solid. Although both adhere to the same mass conservation equation, their momentum conservation equations differ. The motion of the water body is governed by pressure p, whereas the soil is controlled by Cauchy stress σ, specifically (Bui and Nguyen, 2021; Liu and Liu, 2010):

$$\frac{\mathrm{d}\rho^{f|s}}{\mathrm{d}t} = -\rho\nabla\cdot\mathbf{v}_{f|s} \tag{1}$$

$$\rho^{f}\frac{\mathrm{d}\mathbf{v}^{f}}{\mathrm{d}t} = -\nabla p + \eta\nabla^{2}\mathbf{v}^{f} + \rho^{f}\mathbf{g} \tag{2}$$

$$\rho^{s}\frac{\mathrm{d}\mathbf{v}^{s}}{\mathrm{d}t} = \nabla\boldsymbol{\sigma} + \rho^{s}\mathbf{g} \tag{3}$$

where superscript *f* and *s* represent fluid and soli, superscript *f*|*s* denotes fluid or soil. **v** is the velocity vector, ρ is the density, p is the pressure of the fluid, η is the fluid dynamic viscosity, **g** is the gravity, and $\frac{\mathrm{d}}{\mathrm{d}t} = \frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla$ is the material derivative.

The SPH kernel function W is utilized to discretize the governing equations, and the Riemann dissipation terms proposed by Zhang et al. (2017) and S. Zhang et al. (2024) are employed to address pressure and stress oscillations:

$$\frac{\mathrm{d}\rho_i^{f|s}}{\mathrm{d}t} = -2\rho_i^{f|s}\sum_j\left(\mathbf{v}_i^{f|s} - (\mathbf{v}^{*})^{f|s}\right)\cdot\nabla W_{ij}V_j \tag{4}$$

$$\rho_i^{f}\frac{\mathrm{d}\mathbf{v}_i^{f}}{\mathrm{d}t} = -2\sum_j P^{*}\nabla W_{ij}V_j + 2\sum_j \eta\mathbf{v}_{ij}^{f}V_j\frac{\partial W}{\partial r_{ij}} + \rho_i^{f}\mathbf{g}_i \tag{5}$$

$$\rho_i^{s}\frac{D\mathbf{v}_i^{s}}{Dt} = 2\sum_j \boldsymbol{\sigma}^{*}\nabla_i W_{ij}V_j + \rho_i^{s}\mathbf{g}_i \tag{6}$$

where $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$ is the relative velocity vector, $\mathbf{v}^{*}$, P^{*}, $\boldsymbol{\sigma}^{*}$ are the inter-particle average velocity, pressure, and stress. The detailed definition of the inter-particle average variable can be found in Zhang et al. (2017) and S. Zhang et al. (2024).

Constitutive model

Both fluid and fluidized soil are treated as weakly compressible fluids, where the equation of state (EOS) is used to define the one-to-one correspondence between pressure and density:

$$p = c_0^2(\rho^{f} - \rho_0^{f}) \tag{7}$$

where ρ_0 is the initial fluid density, and c_0 is the artificial speed of sound. $c_0 = 10\ U_{\max}$ is adopted to ensure that pressure fluctuations are less than 10% (Morris et al., 1997) and $U_{\max}$ is the anticipated maximum fluid speed.

The non-eroded soil is treated as an elastoplastic solid defined by the Drucker-Prager (DP) criterion, where the yield criterion is given by (Bui et al., 2008):

$$f(I_1, J_2) = \alpha_\phi I_1 + \sqrt{J_2} - k_c \tag{8}$$

where $\boldsymbol{I}_1 = \mathrm{tr}(\boldsymbol{\sigma})$ is the first invariant of the stress tensor and $J_2 = \mathbf{s}:\mathbf{s}\ /\ 2$ is the second invariant of the deviatoric stress tensor. α_ϕ and k_c are defined as :

$$\alpha_\phi = \frac{\tan\phi}{\sqrt{9+12\tan^2\phi}}, k_c = \frac{3c}{9+12\tan^2\phi} \tag{9}$$

where ϕ is the friction angle and c is the cohesion. More details on the SPH implementation of the DP model can be found in (Bui et al., 2008) and the stress diffusion term (Feng et al., 2021) is also adopted.

Erosion model

To simulate the erosion of dam soil under overtopping conditions, an erosion rate model is used to define the gradual loss of soil material due to erosion. For a soil particle a, its height h gradually decreases over time at a rate defined as (Chang and Zhang, 2010; Huang et al., 2021):

$$\frac{dh_a}{dt} = K_d\left(\tau_a - \tau^c\right) \tag{10}$$

where K_d is the coefficient of erodibility, τ_a is the shear force induced by fluid and τ^c is the critical shear force. Note that, according to Zhou et al. (2024), it is assumed that the shear force and critical shear force are parallel to the surface of the dam, hence both are scalars. τ_a is calculated using the logarithmic velocity distribution (Wang et al., 2016):

$$\frac{\tau_a}{\rho^f} = (\kappa D)^2\left(\frac{u_a^\tau}{D}\right)^2 \tag{11}$$

where u_a^τ represents the average velocity of fluid particles and fluidized soil particles near soil particle a, and it is computed based on the SPH kernel function:

$$u_a^\tau = \frac{\sum_i u_i W_{ai} V_i + \sum_m u_m W_{am} V_m}{\sum_i W_{ai} V_i + \sum_m W_{am} V_m} \tag{12}$$

where indices i and m refer to fluid particles and fluidized soil particles, respectively. The critical shear stress τ^c is defined based on the critical Shields parameter, and its calculation method can be referred to in Wang et al. (2016).

Numerical results

2D column collapse

The benchmark of column collapse is used to validate the Riemann-SPH method for simulating the elastoplastic behavior of soil particles. The laboratory experiment conducted by Nguyen et al. (2017) is adopted. In this experiment, a column consisting of granular materials with a length of 0.2 m and a height of 0.1 m undergoes gravitational collapse after the sudden removal of a barrier, as is shown in Fig. 2.

Fig. 3 presents the simulated results compared with the experimental outcomes. It is observed that under the influence of self-weight, large deformation occurs at the slope angle of the granular material, followed by forward movement. Conversely, at the rear of the column, where the velocity is zero, no deformation or disturbance occurs. As the motion progresses, the velocity gradually decreases until it eventually comes to a halt, forming a deposition body. Overall, the morphology of the column during motion obtained through the Riemann-SPH simulation aligns with the experimental observations, forming a smooth velocity field.

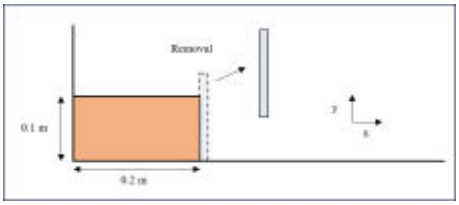


Fig. 2. Schematic of the initial state of column collapse.

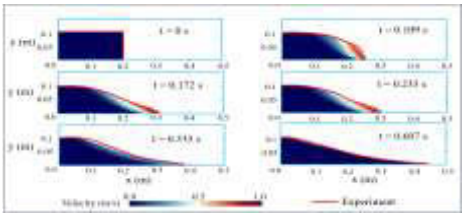


Fig. 3. Comparison between the Riemann-SPH results and the experimental results for the flow progress of the granular column at different moments.

Bottom flushing by dam-break flows

To validate the two-phase model used in this study for simulating the mixture of fluid and fluidized soil, laboratory experiments of bottom flushing by dam-break flows (Fraccarollo and Capart, 2002) are used as benchmarks. In a tank measuring 2.5 m in length, 0.1 m in width, and 0.35 m in height, a sediment measuring 2.5 m in length and 0.06 m in thickness is installed, as is shown in Fig.4. The sediment, comprising PVC spheres with a diameter of 0.0035 m, is placed at the bottom of the tank. A dam-break flow with a length of 1m and a height of 0.1 m is released suddenly from one side of the tank, resulting in the movement of the dam-break flow along the upper surface of the sediment and the formation of a two-phase mixed fluid at the interface.

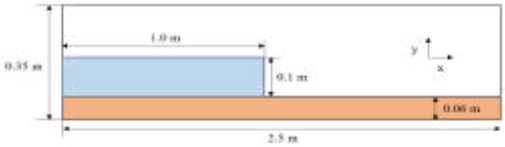


Fig. 4. Schematic of the initial state of the bottom flushing by dam-break flow

Fig. 5 illustrates the simulated progression of the dam-break flow over the sediment at different moments, juxtaposed with experimental data. Following the release of the dam-break flow, localized scouring forms pits (t = 0.25 s). Subsequently, the water flow transports sediment along the interface (t = 0.5 s, 0.75 s). Finally, the leading edge of the fluid gradually thins out, accompanied by a decrease in velocity (t = 1.0 s). The numerical simulation depicts the motion of the flow and entrainment consistent with the experimental findings. However, in the later stages

of the experiment (t = 1.0 s), the height of the sediment on the right side exceeds the experimental results, attributed to the assumption of a two-phase mixture problem. Similar treatments and outcomes can be found in Wang et al.(2016).

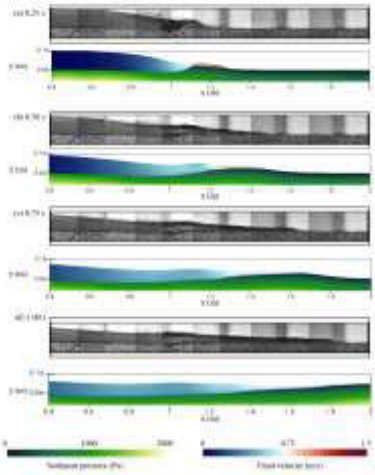


Fig. 5. Comparison between SPH simulation results and experimental data for the case of bottom flushing by dam-break flow at different time points.

Landslide dam breach

A laboratory experiment on the breach process of dams conducted by Zhang et al., (2023) was utilized to validate the algorithm proposed in this study, which couples erosion and slope instability. The dam has a height of 0.3 m and a width of 0.4 m, with an upstream slope angle of 40° and a downstream slope angle of 30°. A trapezoidal region is excavated as the spillway, with widths of 6 cm and 2 cm at the top and bottom respectively, and a height of 4 cm. The dam is placed on a slope with an inclination of 6°, and the upstream inflow rate is 0.5 L/s.

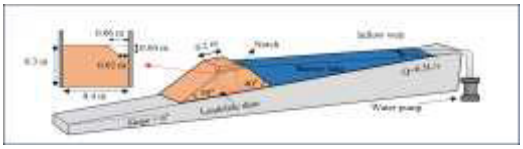


Fig. 6. Schematic of the laboratory experiment of landslide dam breach.

Fig. 7 depicts snapshots at different moments obtained from the SPH simulation. Under the overtopping action in the upstream, initial erosion occurs primarily at the spillway's crest (t = 19 s). At t = 38 s, there is overall scouring at the spillway crest, and slight instability occurs on the slopes. Subsequently, there is a general erosion, and the slopes gradually collapse (t = 60 s), ultimately resulting in the complete removal of the spillway area (t = 80 s).

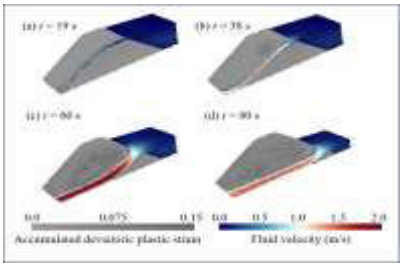


Fig. 7. Snapshots of the breach process of the dam at different moments

Fig. 8(a) further demonstrates the comparison between the simulated and experimental longitudinal profiles, showing good agreement between the two. Figure 8(b) illustrates the comparison between the simulated and experimental dam-breaching discharge, accurately predicting the timing and magnitude of peak flow. In conclusion, the dam breach process can be divided into two phases: headward erosion and overall scouring. During the headward erosion phase, erosion mainly occurs on the downstream slope of the dam, while the top of the dam has not yet shifted overall. At this stage, there is minimal occurrence of slope instability. During the overall scouring phase, both the downstream slope and the top of the dam experience erosion simultaneously, leading to widespread slope instability and a significant increase in breach discharge. The onset of the overall scouring of the dam body coincides with the occurrence of slope instability. Similar categorization was also observed in Zhang et al. (2023), but the occurrence of slope instability was not analyzed.

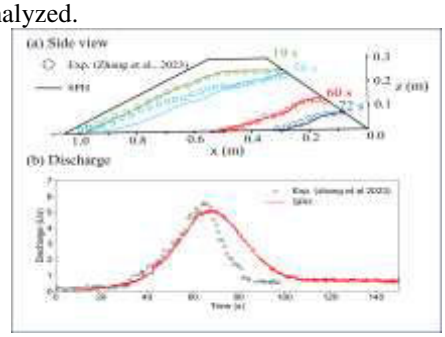


Fig. 8. Comparison of simulated results with experimental data: (a) longitudinal profiles, and (b) discharge.

Discussion

The friction angle ϕ is a crucial parameter controlling the strength of the soil, which directly influences the slope instability process. Building upon existing experiments, this study compares the differences in the breach morphology of dams under two friction angle

conditions: ϕ = 19° and ϕ = 57°. Following Pickert et al. (2011), the width w of the breach top is selected as a quantitative descriptor for the evolution of the breach. The variation of w over time under different ϕ conditions is depicted in Fig. 9. With the passage of time, under different friction angle ϕ conditions, the width w exhibits a trend of growth, eventually expanding to the entire width of the flume, which is 40 cm. However, as the friction angle increases, the rate of expansion of w gradually slows down. This is attributed to the higher friction angle leading to increased soil strength, making slope instability less likely to occur and thus delaying the process of slope expansion. Fig. 10 further illustrates the slope morphology at different time points. In the early stages (t = 20 s, 40 s), the results under different friction angles ϕ are generally consistent. However, as time progresses, larger friction angles tend to maintain slope stability more effectively (t ≥ 60 s), resulting in steeper slope angles and slowing down the lateral expansion of the breach.

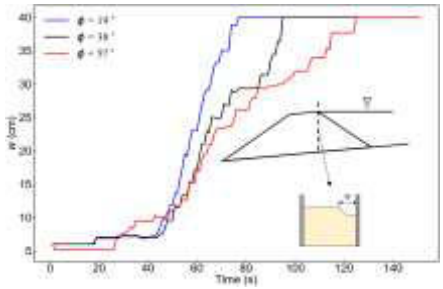


Fig. 9. Evolution of breach width under different friction angles ϕ.

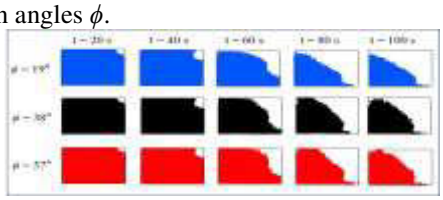


Fig. 10. Side slope morphology at different time intervals under varying friction angles ϕ.

Conclusion

This paper proposes a novel meshless method for simulating the breach process of dams, enabling the simulation of 3D landslide dam breach processes. The algorithm presented in this paper is validated through three cases, and the influence of soil strength on the expansion of the breach is discussed using this algorithm. The key concluding remarks are drawn as follows:

(1) The proposed method can directly simulate the gradual erosion of the dam body and the process of slope instability, thereby achieving the simulation of the 3D evolution of the dam body's shape.

(2) The landslide dam failure process can be divided into two stages: headward erosion and overall erosion. During the headward erosion stage, erosion primarily occurs on the downstream slope of the dam. In the overall erosion stage, the erosion process accelerates rapidly, affecting both the top and downstream slopes of the dam simultaneously. The occurrence of slope instability coincides with the onset of overall erosion.

(3) The friction angle ϕ affects the evolution of the breach by influencing the strength of the soil. A higher ϕ results in better stability of the slope. With a higher ϕ, fewer soil particles experience instability, thus slowing down the evolution of the breach.

Future improvements to the algorithm should include consideration of seepage and sedimentation effects on the breach process of dams. Additionally, implementing GPU acceleration will enable the simulation of field-scale breach processes.

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Shuang Li
Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, 1239 Siping Road, Shanghai, China
E-mail: 2111042@tongji.edu.cn

Ming Peng (✉)
Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai 200092, China
E-mail: pengming@tongji.edu.cn

Experimental and numerical studies on granular flows

Luca Sarno[1,*], Maria Nicolina Papa[1]

[1]Laboratory of Environmental and Maritime Hydraulics, Department of Civil Engineering, University of Salerno, Fisciano 84084, Italy

[*] Corresponding author. Tel: +39-089963409; E-mail: lsarno@unisa.it

Abstract

Granular materials are involved in various geophysical flows, like avalanches, debris flows and pyroclastic density currents. A comprehensive understanding of the granular mechanics is necessary to correctly model these hazardous phenomena and, hence, to better assess and mitigate the associated risks in mountainous areas. Due to the high complexity of the granular rheological behavior, several aspects of the granular dynamics are yet to be fully understood (e.g., Sarno *et al.*, 2018a). Our work is based on the integration of laboratory measurements, theoretical approaches, and numerical modeling. Through laboratory activities we measure the local values of the significant quantities, velocity and volume fraction. These measures are used to develop and test suitable rheological models. Numerical modeling, then, allows simulations under various conditions (e.g., unsteady flows).

The experimental setup is composed of a Plexiglas flume equipped with high-speed cameras, no-flicker LED lamps and a load cell at the outlet, while the investigated material is made of resin beads. A particle image velocimetry (PIV) approach is employed for the velocity measurements (Sarno *et al.*, 2018b). For the estimation of the volume fraction, the stochastic-optical method SOM (Sarno *et al.*, 2016) is adopted. This latter method allows an indirect estimation of the near wall volume fraction under controlled illumination conditions by leveraging a measurable quantity (denoted as *2D concentration*) and a transfer function by Monte Carlo simulations. The measurements are, then, successfully compared with a compressible multilayer model, based on a depth-averaged approach and on a slightly modified $\mu(I)$ rheology (Sarno *et al.*, 2022).

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Point Cloud Alignment in Slope Terrain with Quadruped Robot Assistance

Xiaohang Shao[1], Chun Liu[1,2], Yanyi Li[1]

Abstract

Robots are coming to play important role in the traditional field of engineering for their exceptional mechanical capabilities and flexible movement abilities. The quadruped robots equipped with professional surveying instruments will significantly contribute in field investigations and mapping in complex slope terrains. However, challenges persist in terms of integrating point cloud alignment of surveying observations with robotic motions. To address these limitations, this paper proposes a novel system and an alignment assistant module focusing on introducing an optimal trade-off between accuracy and efficiency in point cloud processing of complex terrain environments. In contrast to rough alignment method between point cloud sets, it utilizes the transformation results based on SLAM techniques, which sets up initial values of gradient directions in the translation and rotation components. Experimental results show that, compared to the direct ICP registration method, an average error of 3.16 centimetres is achieved costing 4.11 seconds between two station-based point cloud sets.

Keywords

Slope Investigation, LiDAR, quadruped robot, SLAM

Introduction

With the rapid development of autonomous unmanned systems, mobile robot plat-forms, such as quadruped robots, are assuming an increasingly pivotal role in traditional surveying and observation domains. Quadruped robots possess exceptional mechanical capabilities and flexible movement abilities, enabling them to effectively navigate complex terrains. Consequently, in our point of view, they can make a substantial contribution to field investigation of surveying and mapping.

In order to achieve continuous and stable field motion, quadruped robots require real-time and robust localization and mapping capabilities. The multi-sensor fusion SLAM technology, which incorporates Light Detection And Ranging (LiDAR), im-ages, and Inertial Measurement Unit (IMU), becomes the mainstream solution in this field. For example, VINS-Mono is a typical representative of low-cost methods that primarily couple visual cameras with IMUs (Qin and Shen, 2018); LIO-SAM integrates LiDAR and IMU measurements to achieve robust localization, while simultaneously generating point cloud maps of large-scale environments (Shan *et al.*, 2020); In contrast to the aforementioned graph-based backend optimization solutions, FAST-LIO employs an efficient and tightly-coupled iterative Kalman filtering method, which exhibits superior performance in scenarios with high dynamic motion conditions (Xu and Zhang, 2021). However, in the practice of integrating surveying with quadruped robot, field observations often generate a large amount of point cloud data. It is subsequently a topic worth of study to leverage the aforementioned techniques to enhance the efficiency and accuracy of point cloud processing in complex terrain environments.

This study proposes a method for point cloud alignment assisted by a quadruped robot, leveraging the terrain adaptation capabilities of a multi-degree-of-freedom robotic platform. By incorporating SLAM techniques and automated registration methods, this approach achieves an optimal trade-off between accuracy and efficiency in complex terrain environments.

Method

System Overview

In contrast to rough alignment method between point cloud sets, we proposed an alignment assistant module utilizing the mobility and flexibility of quadruped robots. Point cloud observations in complex terrain scenarios are usually performed by station-based laser scanner. In this context, the proposed alignment assistant contributes to point cloud aligning in translation, rotation and fine registration. The primary procedure of the technology is as follows.

1) The quadruped robot collects point cloud data at the initial station, and conducts path planning to the next station based on the so-far observations. 2) During

its' traveling to the next station, a SLAM technique is employed to continuously obtains its transformations in translation and rotation. 3) Before performing station-based laser scanning at a new location, the quadruped robot triggers an auto levelling by controlling its related joints of legs in order to keep horizontal balance. These aforementioned processes are then repeated until all observation tasks are finished to execute.

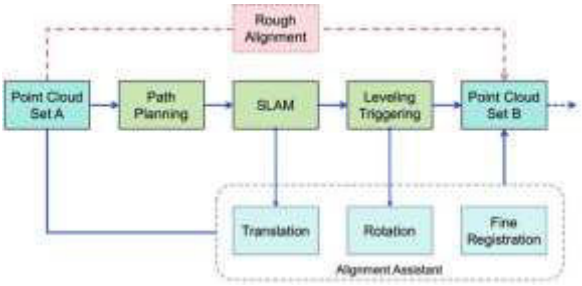


Fig. 1. Flowchart of Alignment Assistant of Quadruped Robot.

Transformation Constraints in Alignment Assistant

The point cloud alignment is to find the transformation matrix to transform coordinates of points from two different sets based on points' feature descriptors. For a pair of point cloud sets with co-observed region P and Q, P = {p_1, p_2, …, p_n}, pi∈R3, and Q = {q_1, q_2, …, q_m}, $q_i \in R^3$ are aligned and referenced point cloud sets respectively. In the alignment process, the optimized transformation matrix T* = [R | t] is estimated based on points sharing the same feature. It can be presented as a nonlinearized least square problem as shown in following equation:

$$\frac{1}{N}\sum_{i=1}^{N} \|q_i - (R \cdot p_i + t)\|_2 \qquad (1)$$

where {p_i, q_i} is the nearest point pair in P and Q sets, N is the number of pairs with-in a pre-defined distance threshold, R is the rotation matrix of 3×3, and t is the translation vector of 3×1.

In assistance of the SLAM technique, the transformation matrix of the robot from two stations can be used as the initial estimation T_0 = [R_0 | t_0], and the rotation component R_0 can be decomposed to three rotations around x-y-z axis as follows:

$$R_0 = R_x \cdot R_y \cdot R_z \qquad (2)$$

As shown in Fig. 2, the quadruped robot can control the related leg joints after triggering auto levelling in slope terrain, so the station-based laser scanner on the workbench of the robot maintains horizontal balance while acquiring point cloud data.

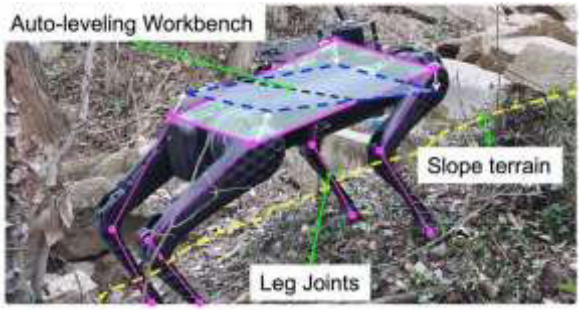


Fig. 2. Auto-levelling of the quadruped robot on a complex slope terrain.

In this context, the rotation component can be assumed to degrade to a Z-axis-only rotation matrix as shown in following equation:

$$R_0 = R_z = [cos\ cos\ \square\ \ -sin\ sin\ \square\ 0 \quad \square\square\square\ sin\ \square \quad \square\square\square\ cos\ \square\ 0\ 0\ 0\ 1\ ] \qquad (3)$$

where θ is the rotational angle around Z-axis. Then, the optimal alignment transformation matrix can be expressed as:

$$T^* = T_0 + \Delta T = [\Delta R \cdot R_0 | \Delta t + t_0] \qquad (4)$$

where ΔR and Δt are optimal increment values. Eq. (1) can be following presented as:

$$\frac{1}{N}\sum_{i=1}^{N} \|q_i - [\Delta R \cdot (R_0 p_i + t_0) + \Delta t]\|_2 \qquad (5)$$

Experiment

Datasets in Slope Terrain

The point cloud datasets were acquired in a challenging field environment with intricate terrain variations, as depicted in the Fig. 3. Six black-white check-boards were strategically placed within this scenario to facilitate accurate estimation of results. During the experiment, the quadruped robot employs auto-levelling control to maintain its back within a specific range of inclination at station A. Subsequently, the laser scanner positioned on the workbench atop its back initiates the collection of point cloud data. After completing the scanning task, the quadruped robot makes route planning and navigates to next station where it will replicate the aforementioned scanning procedure.

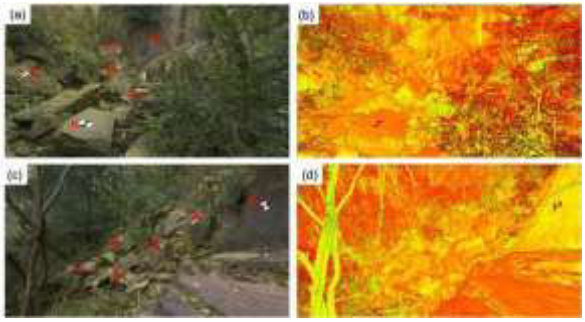


Fig. 3. Scenario description and black-white checkboard set up for result estimation. (a) and (c) are RGB colour images for station A and B. (b) and (c) are the corresponding intensity images acquired from the station-based laser scanner.

We have acquired point cloud data from two stations within this slope scenario as shown in Fig. 4, exhibiting a relative descent of approximately 19 meters. The location of Station A is situated at the lower section of the slope, while Station B is positioned at the higher portion. The vertical separation between these two stations amounts to approximately 10 meters. Considering the scene coverage, we conducted a preliminary screening of the point cloud within a 15 meters range, with the scanner as the central reference point. Due to the proximity of station B to a section of nearly vertical cliff, some parts in the selected circle range lack point cloud data.

Additionally, during the robot's movement from point A to point B, a Simultaneous Localization and Mapping (SLAM) technique is employed based on multi-sensor observations including LiDAR, images, and integrated IMU to ensure continuous and robust localization. The transformation component derived from the inter-station transformations is subsequently utilized for alignment assistance in practical applications.

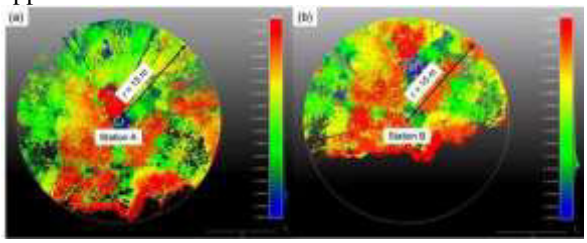


Fig. 4. Point cloud datasets coloured in relative height of axis-Z. (a) and (b) are acquired from two corresponding stations and pre-processed with a scanning range of 15 meters.

Datasets in Slope Terrain

The point cloud alignment assistant provides support in three aspects: translation, rotation, and fine registration. The translation component is derived from the three-dimensional translation vector within the transformation matrix obtained from SLAM results. Meanwhile, the rotation component arises from a degenerate two-dimensional rotation matrix present in the transformation matrix, as auto-levelling effectively eliminates two degrees of freedom other than the vertical Z-axis.

The original two sets of point cloud are referenced in a station-centred independent coordinate system (Fig. 5 a). After applying the aforementioned translation and rotation, the two sets are initially aligned as depicted in Fig. 5 b, where the trajectory of the quadruped robot from station A to station B is represented by the yellow line. Then, the initial alignment results are followed by the utilization of an iterative closest point (ICP) method for fine registration (Besl and McKay, 1992). This unsupervised automatic technique for aligning point clouds produces outcomes as illustrated in Fig. 5 c. As contrast, a rough ICP result is shown in Fig. 5 d without alignment assistance in translation and rotation for a comparison in the following section.

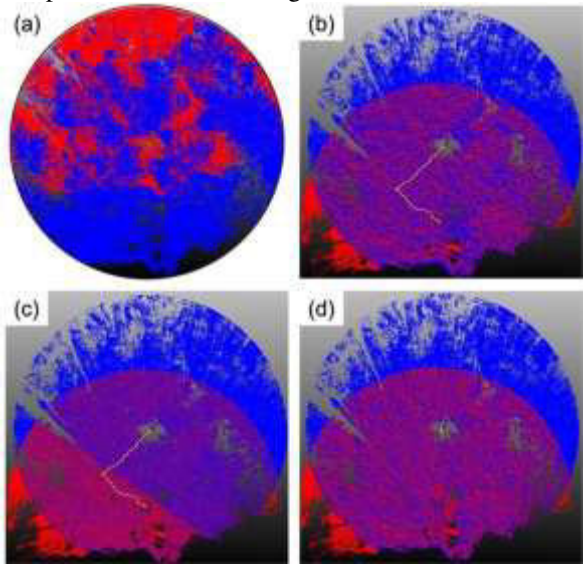


Fig. 5. Results of point cloud alignment between station A (blue) and B (red). (a) the original station-centred point clouds, (b) initial alignment after transformation assistance, (c) the final fine registration result, (d) roughly automatic alignment result based on the ICP technique.

Comparisons of Results

In order to make comprehensive comparisons on alignment results between the prosed method and the rough ICP, we conduct four tests by changing number of sample points, as shown in Table 1. It is worth noting that this parameter has a decisive impact on

accuracy due to the characteristics of ICP method in randomness and automation. A large number of sampling points also implies time-consuming iterations to find optimal gradient directions.

Table 1. Comparison of fine registration between the rough ICP and the proposed technique

	Rough ICP		Ours	
	Test-1	Test-2	Test-3	Test-4
Points	50000	100000	100000	**215440**
RMSE(cm)	33.98	30.56	30.67	**27.56**
Time(s)	21.17	42.74	**3.46**	4.11

In practice of test 1 and test 2, achieving a 10% improvement in accuracy requires more than double the time cost. However, our proposed method exhibits remarkable efficiency advantages, for example, more than 90% drop in time cost from test 2 to test 4. In test 4, despite doubling the number of points, only an additional 19% execution time is incurred while achieving better performance in RMSE than in test 3. Additionally, to be clarified, the computational platform used for these experiments was an Intel (R) Core (TM) i9-1094 X CPU @3.36HZ with a maximum of 28 threads available for parallel computation.

A detailed comparison between the proposed method and rough ICP results on alignment accuracy is performed based on the six involved black-white check-boards. Three dimensional distances of paired checkerboard's centres are calculated between two aligned point cloud sets, as illustrated in Table 2.

The distance error of the proposed method is notably lower than that of rough ICP, with an average decrease of almost 60% to 3.16 cm as illustrated. However, it should be noted that the distribution of these extreme values across the six checkboards lacks consistency as they appear on different checkboards. This observation suggests a potential correlation between this phenomenon and the inherent randomness associated with the ICP method.

Conclusion

This paper introduces a technique for point cloud alignment in assistant with a quadruped robot in complex terrains. It utilizes the position and rotation from the robot's SLAM results to establish initial registration values, meanwhile reduces the complexity of rotation solving through horizontal balancing control of the robot. Experimental results in the typical slope scenario with changing terrains shows that the proposed method enhances final registration accuracy as well as concurrently mitigating computational costs.

Table 2. Evaluation and comparison on the checkboard alignment results (cm)

	Rough ICP	Proposed
1	8.02	**4.03**
2	**8.91**	2.99
3	**7.01**	3.71
4	7.39	2.60
5	8.28	3.13
6	7.71	**2.48**
Avg	7.88	**3.16**

Acknowledgments

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Xiaohang Shao
[1] College of Surveying and Geo-Informatics, Tongji University, Shanghai, 200092, China.
E-mail: 1533336@tongji.edu.cn
Chun Liu
[1] College of Surveying and Geo-Informatics, Tongji University, Shanghai, 200092, China
[2] National College of Elite Engineers, Tongji University, Shanghai, 200092, China
Email: liuchun@tongji.edu.cn
Yanyi Li
[1] College of Surveying and Geo-Informatics, Tongji University, Shanghai, 200092, China.
Email: 2131939@tongji.edu.cn

Runout process and failure mechanism of the Shaziba landslide in Enshi, China

Kun Song[1,*], Jianxiang Chen[1], Fawu Wang[2], Yisheng Cai[1]

[1] Hubei Key Laboratory of Disaster Prevention and Mitigation, China Three Gorges University, Yichang 443002, China.

[2] Department of Geotechnical Engineering, Tongji University, Shanghai 200092, China

[*] Corresponding author. Tel: +86-15997617617; E-mail: songkun@ctgu.edu.cn

Abstract

Shaziba landslide is a major geological disaster caused by heavy rainfall in 2020 in Enshi Hubei Province, China. The landslide is located at the upper reaches of the Qingjiang River, a tributary of the Yangtze River, with coordinates of 30.36° N and 109.30° E. The length of the landslide is about 1500 m, the maximum transverse width is 320 m, and the volume is estimated to be 1.0×10^7 m^3.

On 17 July 2020, the first fresh crack in the Shaziba landslide was found, ittle sliding mass entered Qingjiang River until 18 July. owever, on 19 July, about 3×10^5 m^3 of debris on the west side slipped into the gully along a slope of about 270°. Some of the sliding mass entered the Qingjiang River in the form of debris brought by the water flow into the gully. On 20 July, the deformation scope expanded, and about 2.5×10^6 m^3 of debris in the west side slipped into the gully. At about 5:30 am on 21 July, more debris in the west gully slipped into the Qingjiang River. On 21 July, by 6:00 am, the volume of mass that flowed into the Qingjiang River reached about 1.5×10^6 m^3.

Based on the detailed engineering geological survey of the landslide, the regional geological environment and triggered factors are analyzed. We conducted some ring shear tests of sliding zone soil to investigate the cohesion, internal friction angle and friction coefficient (Fig.1).

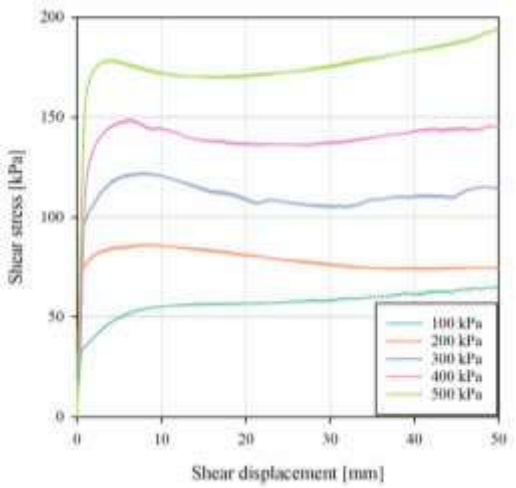


Fig. 1 Ring shear result of the sliding zone soil.

The Shaziba landslide runout process was simulated for the spatial and temporal variation of thickness, running speed of the landslide. The rate-and-state dependent friction law of the sliding zone soil, and soil particles mixed to result shear strengthen in the sliding process were the possible explanation for the failure mechanism.

Acknowledgments

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Landslides triggered by the 18 December 2023 Ms 6.2 Jishishan earthquake, Gansu Province, China: A field reconnaissance

Yingying Tian*, Siyuan Ma, Renmao Yuan

Key Laboratory of Seismic and Volcanic Hazards, Institute of Geology, China Earthquake Administration, Beijing, 100029, China

* Corresponding author. Tel: +86-13141327377; E-mail: tian_yingying@foxmail.com

Abstract

On December 18, 2023, an Ms 6.2 earthquake struck Jishishan Bao'an, Dongxiang, and Salar Nationality Autonomous County, Linxia Hui Autonomous Prefecture, Gansu Province. The earthquake had an epicenter depth of 10 km and a maximum intensity of VIII. The study provides an intuitive and detailed co-seismic landsliding scene in the meizoseismal area based on our comprehensive field surveys immediately after the earthquake. This moderate-to-strong earthquake triggered more severe co-seismic landslides than expected, including the Zhongchuan Town mudflow, which resulted in the loss of >20 lives. Most of the co-seismic landslides are small-to-middle scale shallow ones and concentrated in the loess area in Zhongchuan, Guanting towns of Qinghai Province and Dahejia, Shiyuan, Liugou towns in Gansu Province. Loess collapses are the most common co-seismic landslide type and are usually distributed on steep cut-slopes of roads and loess terraces where human activities are intense. The second-most common loess slides tend to occur on valley slopes covered by thick loose loess. Landslides in the bedrock area are less frequent and the majority are compound geo-disasters involving rockfalls and debris avalanches that originated in the upper part of the slopes and tend to cluster at the northern end of Jishishan Mountain. Generally, the distribution and occurrence mechanism of the co-seismic landslides are closely related to the lithology, terrain, hydrogeology, and seismic response of the areas where they originated. Lots of discontinuities (i.e., cracks or holes) and suspended loose debris on the slopes pose potential threats of re-sliding and should be given more attention.



Fig. 1 Field survey routes, investigation points, and the detailed study cases of the co-seismic landslides.

Acknowledgments

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Coastal hazard in tuffaceous cliff under the influence of climate change: The case of Capo Miseno (Gulf of Naples)

Maria Francesca Tursi[1], Fabio Matano[2], Annarita Casaburi[2], Alberto Fortelli[2], Marco Sacchi[2], Pietro P.C. Aucelli[1]

Abstract

In Italy, there are many volcanic areas in coastal domains. Scientific interest in these areas is primarily focused on volcanic and seismic hazards, with relatively little attention paid to processes such as landslides and coastal erosion. This research aims to identify the main predisposing factors to rockfall/rockslide phenomena in the tuffaceous coastal cliffs of the Campi Flegrei volcanic area (Southern Italy). In addition, it attempts to assess possible relations between the different predisposing factors and the triggering factors. In Campi Flegrei area, significant erosion processes affect a large part of the coastal cliffs, which are mainly composed of fractured volcanic tuff and pyroclastic deposits. The large landslide that affected the Capo Miseno cliff on March 25, 2015, was used as an example. The event, a rockslide phenomenon, occurred on the right side of Miseno promontory causing the detachment of approximately 90.000 m^3 of material from the slope. This reshaped the morphology of the coastline. In order to ascertain the factors that may have contributed to the occurrence of this landslide, a multistep analysis was conducted, incorporating a range of datasets, including field surveys, geomechanical analyses, UAV surveys, and meteo analyses. The example of Capo Miseno serves to illustrate the evolution of tuffaceous cliffs, which is strictly related to both predisposing and triggering factors. The findings of this study represent a significant advancement in the understanding of coastal hazards in volcanic high coasts in the Campi Flegrei area. These have led to an accurate zonation of these coasts through the ranking of coastal sectors according to the degree of hazard of coastal landslide occurrence. Furthermore, an analysis of the correlations between the different influencing factors may prove highly valuable for early warning actions and policies, given the urbanised nature of this area.

Keywords

Coastal hazards, Volcanic cliff, Campi Flegrei, Italy

Introduction

Campi Flegrei area is located on the Tyrrhenian side of southern Italy, near Naples. It is an active volcanic caldera considered among those with the highest volcanic risk in the world and is used as a reference to understand mechanisms controlling volcanic activity.
Landslides in this area have been recognized since the Roman imperial period as reported in a comprehensive landslide geodatabase of the area referring to the last two centuries (Esposito & Matano, 2023).
According to the latter, more than 100 landslides have occurred in the coastal sector in the considered time span. So, it is important to understand the main causes that lead to the triggering of landslide phenomena, as they represent a high risk both for coastal navigation and for urban settlements. In fact, despite the high coastal instability, the area registers strong urbanization and it is a popular tourist destination during the summer season.
The large landslide that affected the Capo Miseno cliff on March 25, 2015, was used as an example. The event was a rockslide phenomenon that occurred on the right side of the Miseno promontory which caused the detachment of about 87.000 m^3 of material from the slope, reshaping the morphology of the coastline.
This research aims to identify the main predisposing factors to rockfall/rockslide phenomena in the tuffaceous coastal cliffs of the Campi Flegrei volcanic area (Southern Italy), attempting to assess possible relations between the different predisposing factors and the triggering factors.

Study area

Campi Flegrei area is an active volcanic area located in the central part of the Campanian Plain, southern Italy. It corresponds to a quasi-circular structure extending for about 200 km2, a large part of which develops off Pozzuoli Bay (Sacchi et al., 2011, 2020) while the islands of Ischia, Procida, and Vivara represent the insular sector (Aucelli et al., 2020). Campi Flegrei corresponds to a complex volcanic system setting within a poly-caldera structure (Fig. 1) formed as a consequence of two explosive super-eruptions (the Campanian Ignimbrite (39 ky BP, Isaia et al., 2016) and the Neapolitan Yellow Tuff (15 ky BP Deino et al., 2004) followed by three epochs of volcanic activity (15

- 10.6 ky BP, 9.6 −9.1 ky BP, and 5.5 - 3.5 ky BP, Smith et al., 2011). The youngest phreatomagmatic eruption occurred in Monte Nuovo in 1538 AD (Di Vito et al., 1987). This area has been characterized since the Early Holocene by sudden vertical ground movements (bradyseismic crisis) that have induced great modifications in inland areas and led to abrupt coastal changes (Aucelli et al., 2017).

Despite these volcano-tectonic activities, intense urbanization occurred in this area since historical times. In addition, the study area is inhabited by approximately 500,000 people, and it is visited each year by thousands of tourists due to the wide range of cultural and environmental attractions (Mattei et al., 2023).

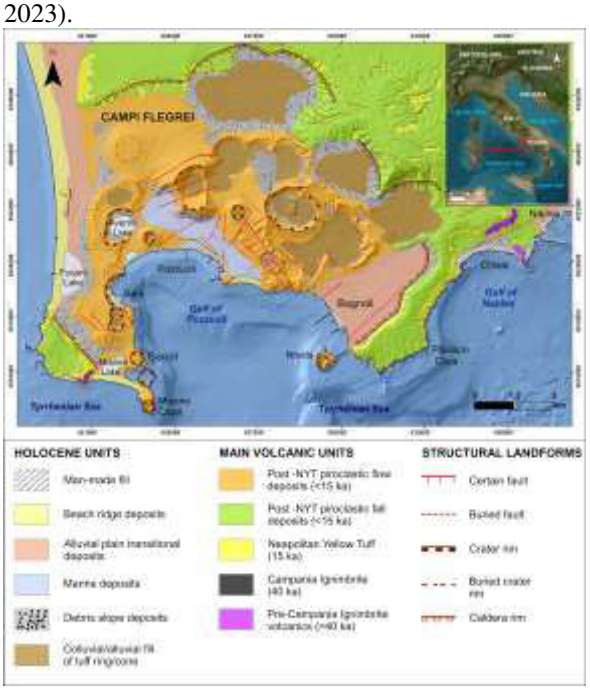


Fig. 1 Geological map of Campi Flegrei area

Capo Miseno Landslide

The analyzed landslide event was a rock slide type that occurred on the right side of Miseno promontory, which caused the detachment of about 87.000 m^3 of material from the slope, reshaping the morphology of the coastline.

The detachment area interested about 5000 m^2 with a horizontal thickness of collapsed material between 5 and 30 m, a vertical thickness between 5 and 20 m and a volume of deposed material of about 100.000 m^3 (Fig. 2). This event was selected because it is one of the most important landslide events occurred in the last decades in a sector where other evidence of ancient events can be observed. In addition, it is oriented transversally to the direction of occurrence of the predominant wave events (WSW-WNW) as demonstrated by the Ponza buoy analysis (Aucelli et al., 2022). Despite the transversal wave action characterizing this sector, even this area was subject to extreme events, with less intensity than the opposite sector, coming from E-SE.



Fig. 2 Main characteristics of Miseno Landslide

Methods

In order to determine the predisposing factors involved in this large landslide event, a multistep analysis considering different information was carried out and reported in the following sections.

Literature review and geomorphological analysis of ancient maps

The analysis started with an intense literature review of previous work to obtain information regarding the number of events that previously occurred in the area.

In addition, the historical morpho-evolution that occurred in the area in the last 200 years (since the end of 1700) was considered by using ancient maps derived from different sources.

The original dataset included additional historical pictures and maps. They were discarded due to distortions of the images and georeferencing issues.

Geological and geomorphological surveys

Careful geological and geomorphological surveys were carried out along the whole promontory to recognize the main geomorphological features characterizing the cliff and map the main indicators of slope instability.

UAV Survey

In a third step, the analysis provided the comparison with a survey carried out in 2014 on the contiguous cliff known as 'Spiaggetta Verde', which shares broad characteristics with Miseno cliff to understand the geomorphological characteristics of the cliff before the landslide occurrence. The drone used for high-resolution image acquisition was a DJI™ Phantom 4 RTK quadrotor UAV platform equipped with a 4 K video camera with a 1/2.3" CMOS sensor, 94-degree field of view, and 12.4 MP images.

Flights were performed at a height of 100 m, defining a resolution (Ground Sample Distance) of 3 cm. The images were acquired with an overlap of 80% and were successively elaborated with Structure from Motion (SfM) techniques (AgiSoft Metashape 2.0.3 software, Over et al. 2021, Tursi et al., 2023).

In addition, to define the failure mechanism of the tuffaceous cliff, field data from two structural stations (S1 and S2) were selected and analyzed, whose locations are reported in Fig. 3. The two stations were selected according to the procedures proposed by the International Society for Rock Mechanics (ISRM) and are both located in the central part of the cliff where the most representative geostructural conditions can be observed.

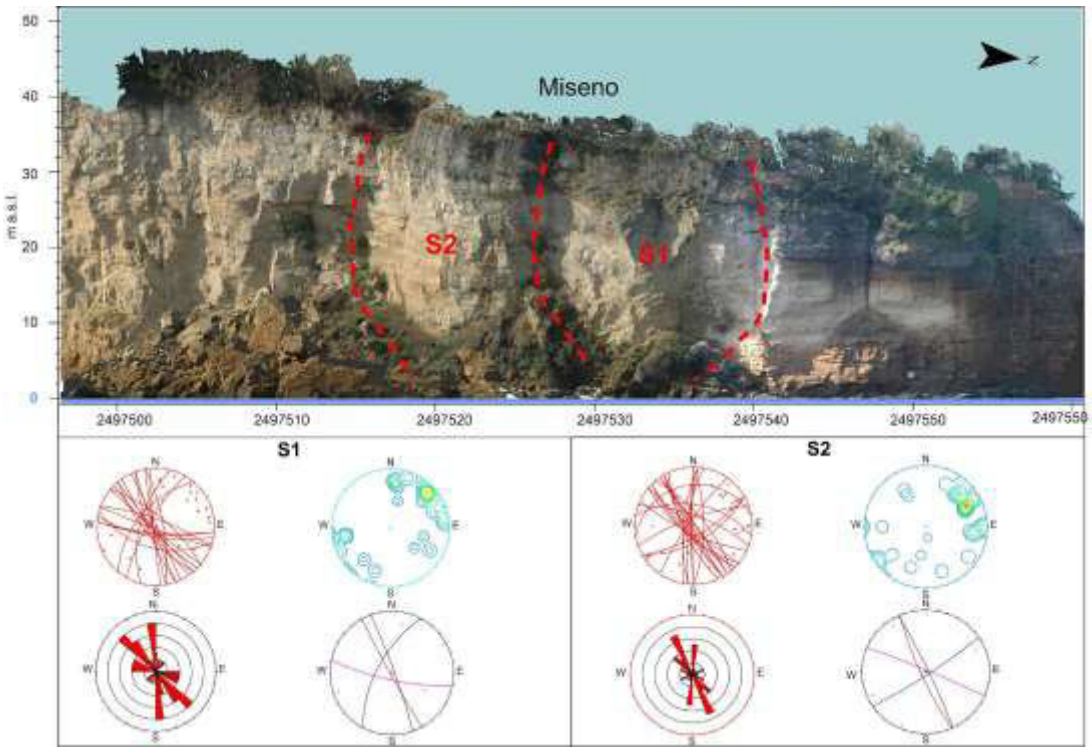


Fig. 3 Wide shot showing the studied sea-cliff and the identified homogeneous geo-structural sectors. The red lines mark the area of the geo-structural stations. Stereographic projections of the joint sets affecting the two sectors

Meteorological analysis

The above-mentioned analyses were integrated with the analyses of a database of meteo parameters measurements derived from a weather station, measuring rain, temperature, wind and atmospheric pressure. This database (Fortelli et al., 2019) encompasses 262944 measurements acquired with a 10 min frequency for 1826 days spanning from January 2014 to December 2018 (Matano et al., 2020).

On March 25, 2015, the meteorological situation was characterized by the presence of a barometric deep low over the central-western Mediterranean, associated with the typical frontal surfaces of mid-latitude barometric depressions.

Since there was no data availability concerning sea conditions, the wave height reached as a consequence of this event was evaluated through a deductive procedure that reproduces the energy transferring from winds to the sea surface.

In the considered coastal sector, two wind datasets were considered, derived by Bacoli and Denza Capo Posillipo weather stations. Bacoli weather station is

located close to the landslide site but on the leeward side respect to Capo Miseno, while Denza-Capo Posillipo weather station is located far away from Capo Miseno but it is well exposed to South-East winds since there are no significant obstacles in this direction.
In order to evaluate information regarding Hs (significant wave height), the two wind series were used separately. The analysis of the wind speed diagrams highlights a division of the whole event into two sub-events, the first of lesser intensity, preparing the wave base conditions on which the second sub-event acted. The reconstruction of the wave motion (offshore conditions) through the Groen-Dorrestein diagram, allows estimating the characteristics of wave height.
Capo Miseno eastern cliff has a fetch equal to 30 km in the South-East direction, and this value was used to enter in Dorrestein nomogram. For the waves entering the Gulf of Naples the same wind characteristics were considered, aggregating the values of Bacoli and Capo Posillipo through an averaging operation. In this case, the event was considered completely developed (not limited for fetch). To evaluate the height of diffracted waves, the Irribarren equation was applied.

Results and Discussion

Miseno cliff has been affected by large landslide phenomena at least since the end of the 7th century as reported by ancient geomorphological maps underlying that these events cyclically affected the promontory since a long time.
Based on the collected data through UAV analysis is possible to observe that the studied cliff was characterized by steep, up to overhanging walls affected by a fracture network, which locally isolates blocks in precarious equilibrium. The geotechnical analysis allows defining the main characteristics of the rock forming the cliff (Table 1) and to evaluate the final value of Rock Mass Rating (RMR) equal to 38 and 59 respectively for S1 and S2 stations. On this basis the rock composing this cliff can be classified as a poor rock and fair rock, as suggested by Beniawski (1989) and Romana (1985) classifications. In addition, the accurate detection of structural discontinuities resulted in understanding the possible failure mechanism and their kinematic analysis demonstrating that the possible failure kinematics are toppling, planar and wedge sliding.

Table 1. Main geomechanical characteristics of the S1 and S2 stations.

Parameter	**S1**	**S2**
UCS	5 MPa	5 MPa
RQD	71%	91%
JRC	6	6
SOD	435 mm	970 mm
Jv	13.39 l/m	7.33 l/m
VRU	0.60 m^3	1.09 m^3
c	293 kPa	363 kPa
ϕ	34°	41°
E	17 GPa	45 GPa
RMR	38	59
SMR	31	49

After the evaluation of the main predisposing factor, the analysis provided the study of possible triggering factors. Among them, attention was paid to sea conditions determined through an analysis of meteorological parameter datasets.
The anemometric event that affected the Campania coast on 25 March 2015 began on the evening of the 24th, when around 10:00 PM-CET (9:00.PM-UTC) the wind suddenly increased its speed from a modest value up to values with numerous peaks over 10.3 m/s (max 13.4 m/s), with a prevalent direction between E and SE. The analysis of wind and rain intensity in the previous 10 days gives evidence of a long-lasting light winds condition before 25 March, so underline an almost calm sea condition before March 24. Rainfall has also been very scarce, with amounts scarcely infiltrating, limiting their action to surface top soils. Following the procedure described in the methods section by using the wind condition, it was possible to estimate the wave height on the day of landslide occurrence which is equal to Hs=2.51.
However, even though these large landslide events occur frequently and cyclically, it is important to note that the landslide body is dismantled very quickly due to wave action (Fig. 4). In fact, a total percentage of 78% of the emerged landslide body has been eroded in the last 9 years, leaving only the crown zone as clear evidence of the event (Table 2).



Fig. 4 Landslide body variation for time ranges measured on five longitudinal sections (t0-t4).

Table 2. Meters of landslide body variation

Year	t0	t1	t2	t3	t4
2015	56.9	83.2	109	103	78
2016	35.6	71.6	87.9	89.3	48.5
2017	29.7	57	75.2	79.3	42
2019	27.6	47.1	58.4	64.7	33.8
2021	25.9	45.3	48.5	51.7	43.9
2024	22.8	37.3	34	39	26

Conclusions

Miseno cliff shares broad characteristics with several tuff cliffs from the Campi Flegrei coastal area affected by severe retreat processes. In this research, we focused the attention both on predisposing and triggering factors of a landslide event that occurred on March 25, 2015. However, these are not the only aspects that need to be evaluated, since other stressful conditions should be better assessed. Among others the seismic component should not be neglected this area suffers recurrent seismic swarms due to bradyseism crises that may contribute to the occurrence of these events.

In general, the results of this research provide important advances in the knowledge of coastal hazards in volcanic high coasts in Campi Flegrei area, leading to an accurate zonation of these coasts *i.e.* the division of the coastal area into sectors, and the relative ranking according to degrees of actual or potential hazard of coastal landslide occurrence.

In addition, the analysis of the correlations between the different influencing factors could prove highly valuable for early warning actions and policies, especially considering that it is a densely urbanized area.

Acknowledgements

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Maria Francesca Tursi (✉), Pietro P. C.Aucelli
[1] Department of Science and Technology, University of Naples "Parthenope", Centro Direzionale Isola C4 80143 Naples, Italy
E-mail:
mariafrancesca.tursi@assegnista.uniparthenope.it
pietro.aucelli@uniparthenope.it

Fabio Matano, Alberto Fortelli, Marco Sacchi
[2] National Research Council, Institute of Marine Sciences (CNR-ISMAR), Calata Porta Di Massa 80, 80133 Naples, Italy
E-mail: fabio.matano@cnr.it;
annarcasaburi@gmail.com
afortelli@hotmail.com
marco.sacchi@cnr.it

Modeling the vegetation effect on shallow landslides' initiation using CRITERIA-1D

Giada Sannino[1], Fausto Tomei[2], Marco Bittelli[3], Massimiliano Bordoni[4], Claudia Meisina[4], Roberto Valentino[1,*]

[1] Department of Chemistry, Life Sciences and Environmental Sustainability, University of Parma, Italy

[2] Regional Agency for Prevention, Environment and Energy of Emilia-Romagna, Bologna, Italy

[3] Department of Agricultural and Food Sciences, University of Bologna, Italy

[4] Department of Earth and Environmental Sciences, University of Pavia, Italy

[*] Corresponding author. Tel: +39-0521 906112; E-mail: roberto.valentino@unipr.it

Abstract

This note presents the implementation of a slope stability method for rainfall-induced shallow landslides in CRITERIA-1D, which is an agro-hydrological model based on Richards' equation for transient infiltration and redistribution processes (Bittelli et al., 2010).

CRITERIA-1D can simulate the presence and development of roots and canopies over time, the regulation of transpiration activity based on real meteorological data and the delay effect of vegetation on rainfall infiltration.

Besides assuming the initial position of the water table, CRITERIA-1D can consider different soil layers, characterized by different hydraulic conductivities and soil water retention curves. The validation of the proposed slope stability model is conducted using field data of water content and water potential of a monitored slope, located in Montuè (Oltrepò Pavese - northern Apennines), where a shallow landslide occurred in 2014.

The results show that the model is able to capture the landslide initiation with less overestimation when vegetation is considered with respect to the bare soil condition, which is an issue for early warning systems related to rainfall-induced shallow landslides.

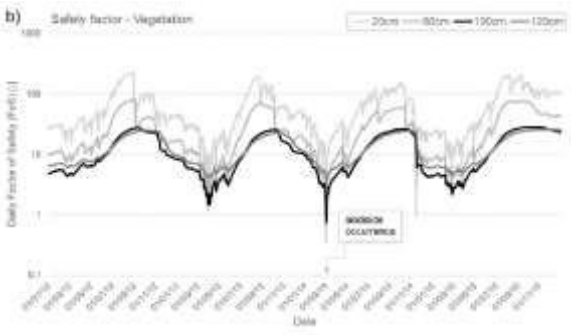


Fig. 1 Trend of Factor of Safety in time.

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Typical Geo-disasters Caused by Heavy Rainfall in Zhejiang Province, China in Recent Years

Fawu Wang[1,*], Ye Chen[1], Bo Zhang[1], Xingliang Peng[1], Kongming Yan[1]
[1] Department of Geotechnical Engineering, Tongji University, Shanghai 200092, China
[*] Corresponding author. Tel: +86-19121765044; E-mail: wangfw@tongji.edu.cn

Abstract

Zhejiang Province plays an important role in Chinese economic development. In recent years, because of extremely heavy rainfall and typhoon event, a number of geo-disasters, including landslide, debris flow and flooding, have occurred and made strong impacts to the local community and infrastructures. We have carried out field investigations on several typical geo-disasters, and tried to figure out their mechanisms.

Wuxie debris flow was triggered by a record-breaking rainfall on 9 June 2021. It occurred in a rhyolite porphyry area that was not recognised as an area prone to debris flow. It was demonstrated that an extreme rainfall event, complex topographic profiles and abundant diluvium in the gullies were the main contributing factors to the disastrous damage (Wang et al., 2022a).

On 20 May 2021, a fault-controlled landslide occurred during heavy rainfall near Chengtian village. The landslide destroyed a provincial highway in front of the slope and rushed into a river, resulting in a barrier lake, which blocked local traffic for 17 days. It is found that the geological structures of fault zones control the occurrence of landslides and determine their characteristics and motion styles, especially in igneous rock areas with high rock strength (Wang et al. 2022b).

On 25 July 2021, a landslide caused by Super Typhoon In-Fa struck the village of Hongchi in Hangzhou City. It blocked a county road for 150 m and damaged six houses. It is found that the landslide had been affected by a fault crossing the main scarp, creating a seepage channel for the rainwater to infiltrate into the sliding mass such that pore-water pressure could be formed at the back of the sliding mass (Wang et al., 2022c).

On 22 July 2023, an unprecedented rainfall event struck Fuyang District, resulting in five deaths and three missing persons by severe geo-hydrological hazards, including 32 small-scale mountain floods and over 55 shallow landslides and debris flows. Wang et al (2024) reported the bamboo collapses obstructed the village's drainage channels and debris flow barriers, causing a delay and intensification in the peak of the mountain flood.

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Research on the Mechanism of Large Deformation Flow of Rock and Soil Particles

Su-ran Wang[1*], Yao Lu[1], Yu Huang[2]
[1] **University of Shanghai for Science and Technology**
2 Tongji University
wangsuran@usst.edu.cn
ORCID:0000-0002-4704-0451

Abstract:

We investigate the evolution of shear behavior in quartz sand subjected to high temperatures in long run-out landslides. Quantitative heat treatment is applied to qua rtz sand samples within a specific peak temperature range. Following the irreversibility principle of mechanical properties in soil materials post-heating, diverse shear experiments on quartz sand properties are conducted, and analyses include sample density and X-ray powder diffraction patterns. Evaluation encompasses shear properties, shear strength, residual shear stress, sample density, particle size, and mineral composition. The study explores the impacts of temperature, particle size distribution, normal stress, and shear velocity on the structure. Results indicate an increase in shear stress fluctuation amplitude with rising shear velocity. Building on prior research, this paper further elucidates the influencing mechanism. It is established that smaller particle sizes correspond to greater residual shear strength. Temperature elevation affects quartz sand, with its shear characteristics, such as shear stress, exhibiting a lower degree of deterioration. The mechanism of sample dilatation compaction under significant shear displacement is further explored. Analyzing the large displacement shear behavior of samples at 800□, we posit that temperature, particle breakage, and normal stress influence the samples, leading to strain softening during large displacement shear after heat treatment, with an upper limit of approximately 600 kPa. Simultaneously, elevated temperatures influence the shear characteristics of the landslide base phase through their impact on rock and soil fractures. Furthermore, we delve into the interlocking force variations resulting from particle shape and fragmentation, along with the consequent fluctuations in shear stress.

Keywords:

ring-shear test, shear characteristics, high temperature

Materials and methods

Materials

To investigate the influence of changes in sand particles subjected to different normal stresses and temperatures, pure quartz sand particles from Fujian Province (China) were selected for the experiments conducted in this study. The content of SiO_2 of the quartz sand exceeded 99.5%. The particle size of all samples was 1 mm. The dry density of the samples was tested in accordance with the standards for soil test methods (GB/T50123, 1999). Details of the dry density of the samples before the ring shear tests were conducted are shown in Table 1. Additionally, we investigated how alterations in individual minerals impact the shear characteristics of soil at elevated temperatures. The minimum dry density and maximum dry density of each group of samples before ring shear test are listed in Table 2 and Table 3, respectively.

Table 1 Maximum and minimum dry density of the samples (kg/m3)

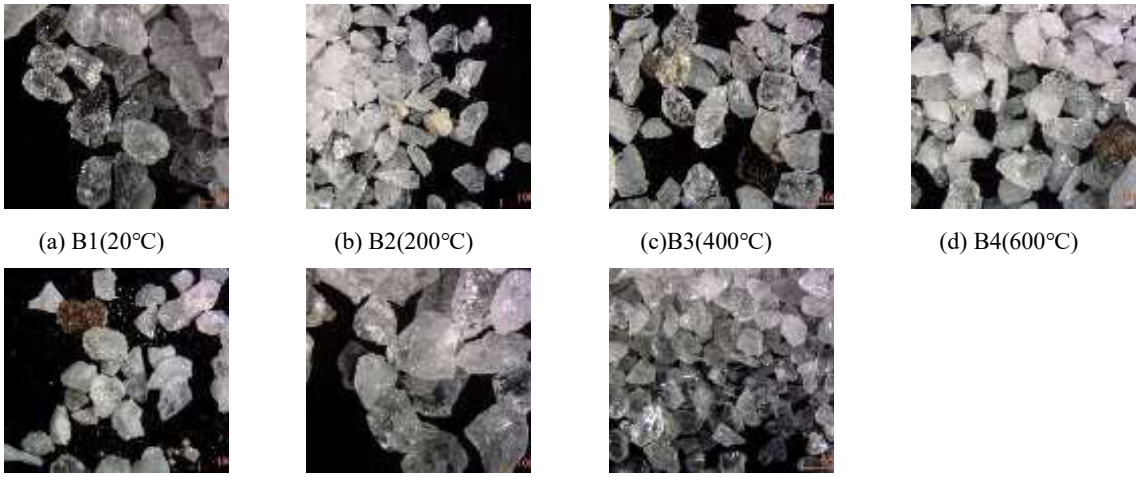
	20℃	800℃
Maximum dry density	1482.9	1468.4
Minimum dry density	1414.3	1407.1

Table 2 Minimum dry density of the samples (kg/m^3)

	20°C	200°C	400°C	600°C	800°C
1mm	1414.3	1428.6	1400.1	1393.1	1407.
0.5mm	1308.4	1346.2	1386.1	1359.4	1341
0.25mm	1171.6	1295.1	1296.3	1308.5	1318.3

Table 3 Maximum dry density of the samples (kg/m^3)

	20°C	200°C	400°C	600°C	800°C
1mm	1482.9	1462.1	1482.6	1475.4	1468.
0.5mm	1466.2	1464.2	1471.4	1472.9	1472.1
0.25mm	1470.6	1446.8	1453.9	1440.9	1458.



(a) B1(20℃) (b) B2(200℃) (c)B3(400℃) (d) B4(600℃)

(e) B5(800℃) (f) PS1-200(200℃) (g) PS3-200(200℃)

Fig. 1 Quartz sand samples before and after heat treatment

By examining SEM images of sand particles, we posit the existence of fluid inclusions within quartz sand. Figure 1 illustrates the samples exposed to varying temperatures.

Experimental procedure



(a) SMF 1900-50 box resistance furnace



(b) GCTS SRS-150 ring shear apparatus

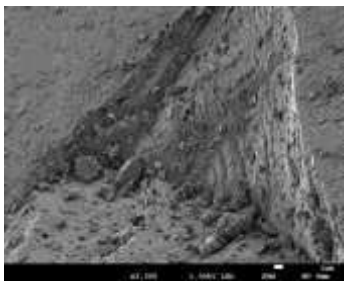
Fig. 2 Test equipment

An SMF1900-50 box resistance furnace at Shanghai University of Technology was used for the heat treatments (Fig. 2a). An SRS-150 ring-shear apparatus (Geotechnical Consulting and Testing Systems, Tempe, AZ, USA) at the Key Laboratory of the Ministry of Geotechnical and Underground Engineering, Tongji University, was used for the ring-shear experiments (Fig. 2b). This instrument can perform high-precision dynamic ring shear tests, used to determine the shear strength and residual shear parameters of unsaturated soil. During the experiment, the shear plane is kept constant and the displacement is continuous. The performance of the SRS-150 ring-shear apparatus: Axial displacement: 50mm, displacement accuracy: 0.001mm, normal stress: 1000 kPa, maximum shear stress: 1300kPa, stress accuracy: 0.1 kPa, shear speed: 0.001-360°/min, sample external diameter: 150mm, inner diameter: 100mm, maximum height: 25 mm.

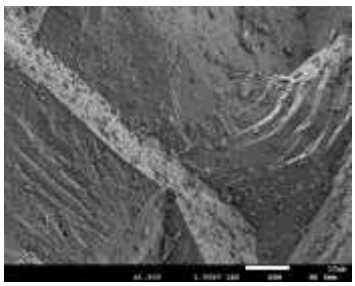
Results

Variable-speed ring shears

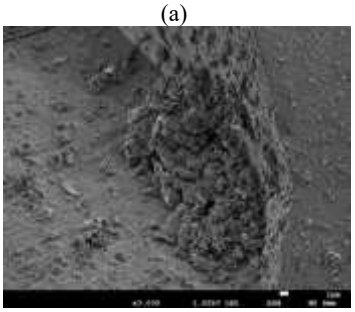
The SEM micrograph reveals that, prior to heat treatment, the quartz boundary exhibits only a few bulges (Fig. 3(a) and (b)). However, after 800□ heat treatment, large amount of bulges accumulate on the boundary of quartz mineral (Fig. 3(c) and (d)). This is mainly observed as the grain boundary bulging from low dislocation density to high dislocation density at low temperatures or high strain rates to form new, independent, and fine-grained crystals. Bulges and recrystallized grains can be found at the micro scale along grain boundaries and, to a lesser extent, along microcracks. Furthermore, grain shear scratches occurred on the mineral surface in Fig. 3(d), showing that this area was shear as a shear plane during the sample shear process.



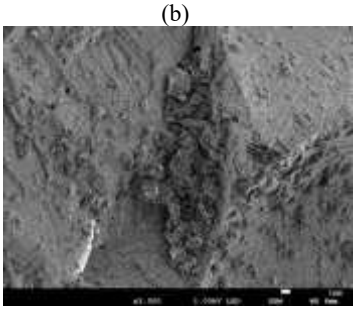
(a)



(b)



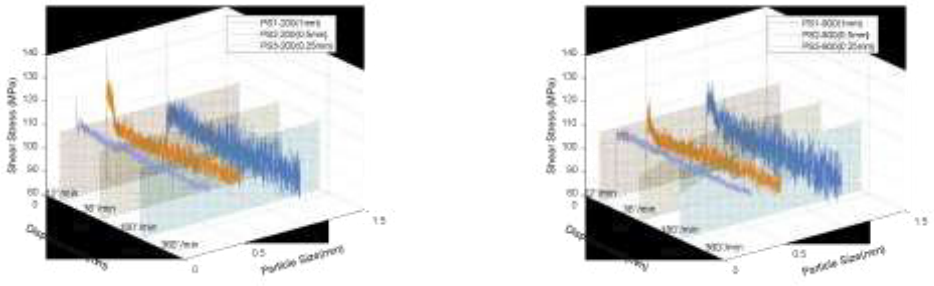
(c)



(d)

Fig. 3 SEM micrographs of bulging recrystallization (low temperature): Along grain boundaries and to a lesser extent along microcracks, bulges and recrystallized grains can be found: (a) 0.25mm particle of B1(20℃) sample; (b) 0.5mm particle of B1(20℃) sample; (c) 0.25mm particle of B5(800℃) sample; (d) 0.1mm particle of B5(800℃) sample

Particle size distribution



(a) (b)

Fig. 4 Shear behavior of the ring-shear tests with different particle size showing the shear stress versus displacement: (a) 200 °C; (b) 800 °C

The relationship between the shear stress and displacement of the quartz sand samples after high-temperature treatment is taken as an example. Figure 4 shows the shear stress behavior of the quartz sand samples treated at 200 and 800 °C. The shear stress is shown to significantly increase with decreasing particle size. The average residual shear values of the quartz sand samples after 800 °C heat treatment is 104.06, 103.54, and 103.69 kPa at shear velocities of 36, 180, and 360°/min, respectively, which are slightly lower than those of the quartz sand samples after the 200 °C heat treatment (105.18, 104.57, 105.08 kPa). The results indicate that the average residual shear stress of quartz sand is only slightly affected by high-temperature treatment.

The density variation of the 0.25-mm samples shows a different trend from those of the other particle sizes. The results indicate that the sample density of the 0.25-mm samples subjected to different temperatures increases during the entire shear process, and the initial density decreases insignificantly when the velocity increases. This shows that the dilatancy phenomenon is not apparent when the 0.25-mm sample is sheared. Differing from the other two size samples, the density of the 0.25-mm sample does not indicate a notable peak during the shear process and continues to increase, and the effect of temperature is not distinct. The density of the quartz sand samples subjected to 800 °C after shearing is 1738 kg/m^3, which is substantially higher than the densities of the 1- and 0.5-mm samples. According to the dry density of the unsheared sample , smaller particle sizes of the same mass sample are associated with smaller maximum and minimum dry densities. Yan (2006) and Wang et al. (2018) showed that the maximum dry density of dry sand does not continue to increase with decreasing particle size, although the maximum dry density of a 0.075-mm sample in that study was higher than that of a 1-mm sample. After shearing, the particles in the sample are broken, thus the PSD tends to be more consistent. It is therefore not a test error that the density of the 0.25-mm sample rapidly increases and exceeds its own maximum dry density, which may represent an important problem of multi-scale phenomena.

Several conclusions can thus be obtained. (1) Under the same treatment conditions, larger particle sizes are more easily crushed. (2) Temperature has a certain influence on particle breakage, but it is not significant. (3) Smaller particle sizes are more difficult to crush.

The experimental results reveal a significant difference in the microstructure between samples before and after heat treatment. Compared with samples without heat treatment , a large number of fragmented minerals are attached to the quartz surface after heat treatment and shearing . This also indicates that the sample's crushing degree is higher after exposure to high temperatures. These

micro variations, on the other hand, are less obvious in macro shear properties and only have a little impact on them.

Variation in normal stress

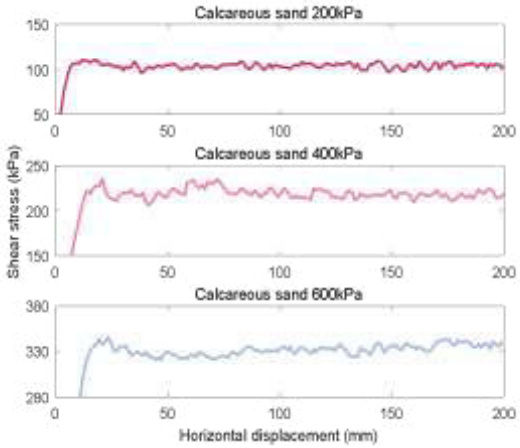


Fig. 5 Shear stress of calcareous sand under different normal stress

Based on the principle of the mechanical properties of minerals being irreversible after heating, ring-shear tests with different normal stress were conducted on quartz sand subjected to 800□ heat treatment. The variation laws of the shear characteristics and breakage of the quartz sand were investigated. Based on the analysis of the experimental results, the following conclusions were drawn.

First, under identical normal stress conditions, the relative breakage of samples that undergo high-temperature treatment is greater than that of samples not treated at high temperature. When the normal stress exceeds 200 kPa, the relative breakage between the sample prior to and after high-temperature treatment is increasingly apparent. Second, the relative breakage is >0.6 in samples for which the shear stress increases with shear displacement. Additionally, the magnitude of the fluctuation of the displacement shear stress of the samples is proportional to the relative breakage. In conjunction with previous research findings, it can be confirmed that the shear breakage of particles within this shear velocity range influences the increase in shear stress, and the bigger the relative breakage, the greater the increase in shear stress. Because current shear equipment is incapable of observing the movement of particle rearrangement in a sample after the internal particles are broken, there is no direct evidence to prove whether the broken particles enter the sample directly as part of the rearrangement process or accumulate in the shear zone.

Conclusions

1. Quartz sand particles that have been exposed to high temperatures are more susceptible to crushing. Under the same experimental conditions, the residual shear stress of the sample with 0.25mm particle size is around 105kPa, which is higher than that of the sample with 1mm particle size 100kPa. Meanwhile, the sample with a smaller particle size has a smaller shear stress fluctuation range.
2. Heat treatment has an influence on the microstructure of quartz sand, mostly in terms of particle breakage, but have little effect on the residual shear stress. After high-temperature treatment, the shear stress fluctuation amplitude of the quartz sand samples decreases with increasing temperature. The shear stress fluctuation amplitude of the 1- and 0.5-mm samples increases with increasing shear velocity, whereas that of the 0.25-mm samples

decreases.

3. The relative breakage of 1mm particle size is roughly 0.5 with the same test conditions. It of 0.25mm and 0.1mm particle size are less than 0.2. The degree of breakage appears to have a size effect.The densities of the 1- and 0.5-mm samples before and after the ring-shear tests were 0.5% higher than their initial maximum dry densities, whereas the density of the 0.25-mm samples after the test was 20% higher than the initial maximum dry density. This characteristic is not caused only by particle breakage, but also related to particle size effect.
4. In the large-displacement shear process of quartz sand, the vertical displacement data verify that the samples follow the compression-dilatancy law of non-broken particle material to a certain extent. However, these changes are sensitive to the material characteristics.
5. Normal stress causes the absolute value and amplitude of fluctuation of shear stress to rise. Exposure to high temperature causes the quartz sand particles to become fragmented more easily, alters the particle size composition of the sample, and ultimately impacts the large-displacement shear characteristics of the material. This characteristic is highly probable to influence the transport morphology of landslides with high temperatures.
6. Strain softening of the specimen during large-displacement shearing is related to the normal stress and particle breakage, and particle breakage reduces the magnitude of the strain-softening phenomenon. It suggests that a significant stress drop phenomenon may not exist on fragile rock and soil slopes.

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Earthquake-triggered landslide mapping based on remote sensing change detection

Xin Wang*, Xuanmei Fan

State Key Laboratory of Geohazard Prevention and Geoenvironment Protection, Chengdu University of Technology, Chengdu 610059, China

* Corresponding author. Tel: +86-15062262523; E-mail: wangxin@cdut.edu.cn

Abstract

A powerful earthquake has the potential to trigger tens of thousands of landslides, posing an immediate danger to the lives and properties of local residents and exerting a lasting impact on the surrounding ecological environment. Remote sensing technology offers a valuable tool for detecting and monitoring large-scale landslides.

We propose a novel approach for identifying earthquake-triggered landslides based on remote sensing change detection. Initially, we obtain high-resolution images before and after the earthquake and apply relative radiometric and geometric corrections to ensure accurate matching of radiometric and positional information at corresponding locations. Subsequently, we extract the multi-scale morphological features of bi-temporal images to comprehensively characterize the spatial features of land surface before and after the earthquake, particularly those related to landslides. Next, we utilize the change vector analysis to preliminarily screen out the pixels with significant and insignificant surface reflectance, which are adopted as training samples for identifying landslides and non-landslides. Finally, we train a machine learning algorithm based on the integration of multiple classifiers to construct a model for accurately recognizing earthquake-triggered landslides.

Three experiments were carried out and the results demonstrated the effectiveness and superiority of the proposed approach with the overall accuracy of 97.13% (2017 Jiuzhaigou event, China), 97.26% (2017 Mainling event, China), and 97.16% (2021 Nippes event, Haiti), providing an effective solution for complicated co-seismic landslide mapping task in the future.

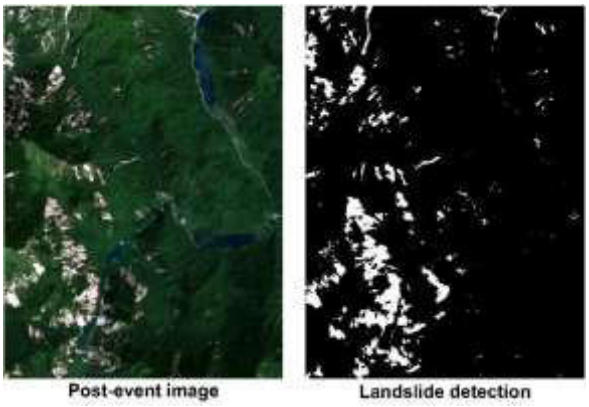


Fig. 1 2017 Jiuzhaigou earthquake-triggered landslide detection.

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Hybrid MPM+SPH analysis of Wangjiashan landslide-induced wave in Baihetan Reservoir, China

Hao Wu[1,*]

[1] Department of Geotechnical Engineering, Nanjing Hydraulic Research Institute, Nanjing 210014, China

[*] Corresponding author. Tel: +86-13913039798; E-mail: wuhao@nhri.cn

Abstract

Impulse waves caused by landslides pose a significant threat to human life, property, and infrastructure. Following the construction of the Baihetan Hydropower Station in China in 2022, numerous unstable slopes were identified in the reservoir area, with the slope of Wangjiashan being the most notable. This paper investigates the potential impact of the Wangjiashan landslide-induced wave in the Baihetan Reservoir by using a novel hybrid numerical method. The material point method analyzes the deformation and sliding velocity of the potential Wangjiashan landslide, while the smoothed particle hydrodynamic method simulates the impulse wave propagating in reservoir. Our results reveal that an 80% reduction in the landslide's initial strength will result in a maximum runout distance of approximately 188 m and a maximum sliding speed of 16.17 m/s. The impulse wave reaches the residential area at t = 75 s with a velocity of approximately 2.5 m/s after the Wangjiashan landslide occurs. The impulse wave has two crests at the residential area, with maximum wave height of 4.238 m and 2.915 m, respectively. Moreover, particular topographic features of the residential area amplify the risk of landslide-induced wave. Our simulation indicates that the residual strength of the landslide material has a significant impact on wave height. A 28% increase in residual strength results in a 62% reduction in the maximum wave height. Therefore, detailed geotechnical investigation and testing are crucial for assessing the risk of landslide-induced waves. Our study provides important information for assessing this risk in the Wangjiashan area and is critical for ensuring safety in the Baihetan Reservoir of China.

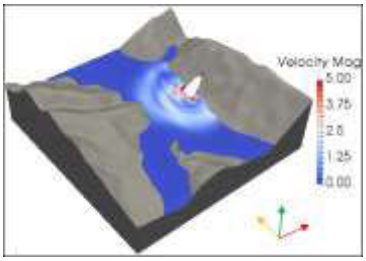


Fig. 1 Simulation results of the Wangjiashan landslide-induced wave propagation

Acknowledgments

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Estimation of sediment yield in Wushe Reservoir by conceptual model combined with environmental indicators

Shao Wei Wu [1] Chao Yuan Lin [1*], Tzu Yu Chung [1] Fu Jun Tu [12]

[1] Department of Soil and Water Conservation, National Chung Hsing University, Taiwan

[2] Forest Ecology Division, Taiwan Forestry Research Institute, Taiwan, R.O.C.

[*] Corresponding author. Tel: +886-52318556; E-mail: yuanlin@dragon.nchu.edu.tw

Abstract

Assessing soil loss in watersheds is a crucial task in soil and water conservation. However, traditional methods like the Universal Soil Loss Equation (USLE) have limitations, particularly in watersheds with complex terrain and those affected by climate change. These limitations hinder the immediate reflection of current soil loss conditions, impeding effective planning and design.

In this study, we introduce a novel approach by integrating the concept of risk assessment with the USLE method. We use environmental indicators to better capture the current soil loss environment. By incorporating annual rainfall (Pn), the Greenery Deterioration Index (GDI), and the Topographic Wetness Index (TWI) as indicators of Hazard, Vulnerability, and Exposure respectively, we develop a watershed-scale sediment yield estimation formula.

Our model utilizes big data from relevant departments to immediately estimate spatial information within the watershed. This enables us to identify sediment yield hotspots in reservoir watersheds. This approach is innovative and crucial, addressing both academic methodologies and practical needs in soil and water conservation.

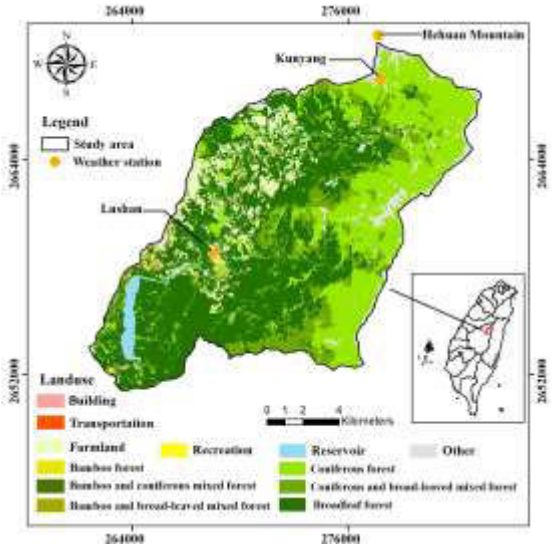


Fig. 1 Study area and landuse

Acknowledgments

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Response of fault-crossing pipelines considering the directionality of subgrade reaction

Yue Pan[1*], Yasuko Kuwata[1]

[1] Dept. of Civil Eng., Graduate School of Engineering, Kobe University, Kobe 657-8501, Japan
[*] Corresponding author. Tel: +81-70-3815-4816; E-mail: 209t136t@stu.kobe-u.ac.jp

Abstract

When an earthquake occurs, the pipeline crossing a fault may be caused different degrees of damage due to the magnitude and direction of fault displacement. More than that, buried pipelines are subjected to subgrade reaction due to ground deformation and fault movement during earthquakes. However, for pipelines buried near the surface, the subgrade reaction is dependent on the direction of movement. Therefore, the subgrade reaction acting on a pipeline is different when comparing a strike-slip fault and a reverse fault, even if the same amount of fault displacement is given, and the response of the pipeline is also considered to be different.

Therefore, in this study, a three-dimensional FE model and its parameters to explain the directionality of the subgrade reaction were developed to analyze the seismic behavior of a fault-crossing pipeline. First, the pipeline with a ground spring was modeled using a conventional shell-spring model, and the behavior of the pipeline with changes in the ground springs was confirmed. Next, the ground was modeled with solid elements without ground springs, and the model and its response were clarified to be able to consider the directionality of the subgrade reaction.

Acknowledgments

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The impact of horizontal well layout on the stability of submarine hydrate-bearing slope

Hao Zhang[1], Tingkai Nian[1,*]

[1] State Key Laboratory of Coastal and Offshore Engineering, Dalian University of Technology, Dalian 116024, China.

[*] Corresponding author. Tel: +86-411-84708189; E-mail: tknian@dlut.edu.cn

Abstract

Natural gas hydrates have attracted widespread global attention as a new type of alternative energy in recent years. Extensive seismic surveys and field drilling data both indicate that submarine hydrates are typically found in seabed formations with certain slopes. Extraction can easily trigger submarine landslides. The stability of submarine slopes containing hydrates is primarily constrained by geological conditions such as reservoir characteristics, as well as engineering factors such as well placement during extraction. This paper developed a numerical simulation program considering the coupled process of hydrate extraction with thermal-hydraulic-chemical (THC) effects, investigating the influence of different horizontal well positions on the stability of submarine slopes. The hydrate thermal stimulation method during extraction was simulated using COMSOL. The stability coefficients of submarine slopes were calculated using the limit equilibrium method for three scenarios, ie slope crest, middle, and toe. The results show that thermal stimulation extraction significantly reduces the stability of submarine slopes. The stability is most affected when the well is positioned in the slope toe. The research findings can offer theoretical guidance for the extraction of natural gas hydrates.

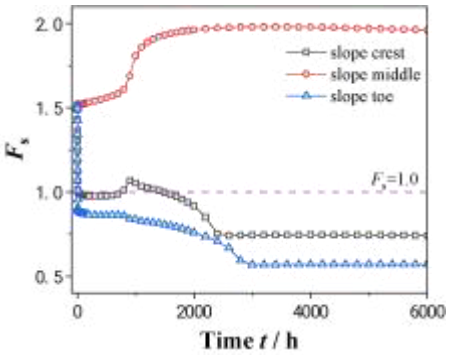


Fig. 1 F_s curves for shallow sliding surfaces considering different wellbore locations

Acknowledgments

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Catastrophic flow-like loess landslides in Northwest China

Shuai Zhang[1,2,*], Ping Sun[1,2], Ran Li[1,2], Fawu Wang[3,4,5]

[1] Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing 100081, China.

[2] Observation and Research Station of Geological Disaster in Baoji, Shaanxi Province, Ministry of Natural Resources, Baoji 721001, China.

[3] College of Civil Engineering, Tongji University, Shanghai 20092, China.

[4] Key Laboratory of Geotechnical and Underground Engineering (Tongji University), Ministry of Education, Shanghai 200092, China.

[5] Professor Emeritus, Shimane University, Japan.

[*] Corresponding author. Tel: +86-13269675138; E-mail: 13269675138@163.com

Abstract

Catastrophic rainfall-induced loess landslide has long been a major concern in Northwest China, and this situation has been exacerbated recently due to the warming and humidity change along with the westward and northward movement of the 400 mm isohyet in China. In the past decade, rainfall-induced catastrophic landslides frequently occurred in Northwest China. In July 2013, three successions of rainstorms with daily precipitation higher than 100 mm struck Tianshui, Gansu, Northwest China and the ensuing thousands of shallow loess landslides resulted in 25 fatalities and the economic loss of 8.275 billion RMB. Generally, clustered rainfall-induced loess landslides are characterized by small scale, wide occurrence, unascertainable distribution, imperceptible pre-failure deformation, and frequent occurrence with severe destruction. However, large (even giant) loess landslides could be triggered in these events as well and the disintegrated loess can rapidly transform to flow-like loess landslides or even loess mud flows. Different from cascades of shallow loess landslides induced by heavy precipitation, large loess landslides induced heavy rainstorms are generally characterized by relatively deep sliding surface, large volume, and amplified disaster effect. Favorable infiltration channels such as tensile cracks and sinkholes are the prerequisite for large loess landslide occurrence as the infiltration depth of loess is limited. In addition, liquefaction sometimes appears during long and rapid motion in large loess landslides, rather than static liquefaction at the initiation stage in shallow loess landslides induced by rainfall.

Acknowledgments

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Research on engineering geological zoning and evaluation of plateau mountains traffic corridors

Bowen Zheng[1,2*], Shengwen Qi[1,2], Songfeng Guo[1,2], Yu Zou[1,2], Ning Liang[1,2], Shuaihua Song[1,2], Yue Yang[1,2], Gao'ang Wei[1,2]

[1] Key Laboratory of Shale Gas and Geoengineering, Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing 100029, China

[2] College of Earth and Planetary Sciences, University of Chinese Academy of Sciences, Beijing 100049, China

[*] Corresponding author. Tel: +86-15801295295; E-mail: zhengbowen@mail.iggcas.ac.cn

Abstract

With the continuous extension of major engineering construction to high-altitude mountainous areas, especially large-scale linear projects spanning different geological structural units, the difficult engineering geological problems faced are becoming increasingly prominent. However, there is currently a lack of universal technical methods for multi-scale engineering geological zoning and engineering geological evaluation in high-altitude mountainous areas. In response to the above issues, this study constructed a systematic technical approach.

Carry out engineering geological zoning and evaluation around the needs of different stages such as pre-feasibility study and feasibility study, and divide the engineering geological zoning and evaluation scale into regional scale and key section scale. Determine the elements of engineering geological zoning and engineering geological evaluation indicators based on different zoning and evaluation scales, and establish an engineering geological zoning and engineering geological evaluation indicator system. Regional scale zoning/evaluation and key section scale zoning/evaluation are based on regional crustal stability theory and engineering geological stability theory, respectively. Qualitative semi quantitative methods are used for zoning and evaluation, and the results of engineering geological zoning and evaluation are divided into four levels: good, medium, poor, and poor engineering geological conditions. For sections where both planar and sectional engineering geological zoning and engineering geological evaluation are carried out simultaneously, it is advisable to integrate the planar results of different elevations or the sectional results of different distances from linear engineering to obtain the corresponding three-dimensional engineering geological zoning and engineering geological evaluation results of the sections. On this basis, major engineering geological problems are predicted and evaluated in zones and segments for different types of engineering, including their likelihood, spatial location, activity intensity, impact range, and potential risks.

The above technical methods have achieved good application results in the transportation corridor engineering in high-altitude mountainous areas.

Acknowledgments

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Seismic and multi-hazard system reliability of distributed geotechnical infrastructure systems

Paolo Zimmaro[1,2*]

[1] Department of Environmental Engineering, University of Calabria, Arcavacata di Rende, Italy

[2] The J. Garrick Institute for the Risk Sciences, University of California, Los Angeles, USA.

[*] Corresponding author. Tel: +39-0984496652; E-mail: paolo.zimmaro@unical.it

Abstract

The global population living in urban areas is growing at an unprecedented rate. As a result, regionally-distributed infrastructure such as pipelines, levees, and electricity distribution systems are becoming of paramount importance to the resilience of such areas in case of natural and anthropogenic disasters. Traditional approaches to analyze geotechnical or structural systems focused on site-specific and/or structure-specific analyses. Such models neglected or underestimated the importance of proper regional approaches able to capture spatial distributions and correlations of demand and capacity. Nowadays, efficient and pragmatic approaches are available to identify the risk to which such distributed systems are subjected to in case of earthquakes and other natural disasters. Zimmaro et al. (2019, 2021) presented novel frameworks to evaluate system reliability (which can be viewed as a probabilistic measure of risk for practical purposes) of river-protection levees and gas pipelines. These approaches (and related studies) rely upon three main pillars: (1) there is need to move from probabilistic seismic hazard maps to scenario events, (2) the strength of these systems and the demand to which they are subjected to are characterized by statistical distributions, and (3) strength and demand have variability in space which can be described by spatial distribution models using correlation structures (Figure 1). Once these pillars are established and modeled, vulnerability can be analyzed by means of fragility functions which relate the probability of a certain damage state being exceeded to a demand parameter (e.g., the peak ground acceleration for earthquakes). This presentation shows how such framework can be implemented, providing a powerful tool to risk identification and potential mitigation strategies at the territorial scale.

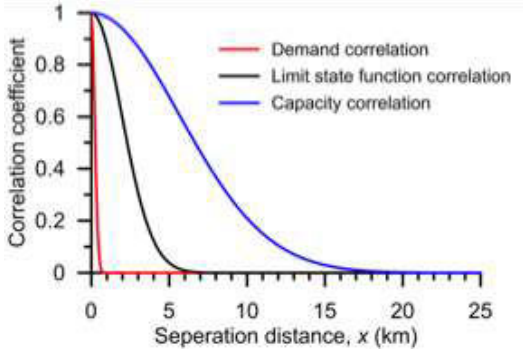


Fig. 1 Generic spatial correlation models.

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Improving Rockfall Safety Measures: Analyzing the Role of Compressible Cushion Layers in Rock-Shed Tunnel

Riya Bhowmik[1,*], Utkarsh Gautam[1], Shreya Maheshwari[1]

[1] Department of Civil Engineering, Indian Institute of Technology Jammu, Jammu and Kashmir-181221, India.

[*] Corresponding author. Tel: +91-01912741170; E-mail: riya.bhowmik@iitjammu.ac.in

Abstract

Rockfall incidents pose significant challenges in the Himalayan regions of Jammu and Kashmir due to the rugged terrain and geological conditions. Despite various infrastructure initiatives, the region's highway frequently faces closures during the rainy season due to rockfalls. To combat this issue, efforts are underway to transform the highway into an all-weather road, requiring protective measures against rockfall impacts. Rock-shed tunnels and galleries are deemed effective solutions to safeguard transportation routes from mid- to high-energy rockfall impacts. Understanding the impact energy of falling rocks is crucial for designing these protective structures.

Researchers have extensively studied rock-shed structures to mitigate rockfall impacts in steep valleys. Conventional structures use concrete or steel slabs with soil-based cushion layers to dissipate impact energy. However, these cushion layers may not always provide sufficient buffering, potentially increasing construction costs and maintenance efforts. To optimize cushioning layers and reduce expenses, researchers have explored geosynthetic materials like expanded polystyrene (EPS) and expanded polyethylene (EPE) geofoams and geogrids. Very limited studies have demonstrated that combined sand-EPS cushion layers are the most effective in impact resistance
. The present study focuses on evaluating the optimization and effectiveness of an EPS-sand cushioning system of rockshed structure proposed for a rockfall-prone road-cut slope along the National Highway (NH-44) in the Union Territory of Jammu and Kashmir, India. By conducting comprehensive site investigations and laboratory tests on rock, soil, and EPS samples, the study establishes key parameters related to slope geometry and cushioning materials. Utilizing RocFall2 software, rockfall trajectory analyses are conducted, simulating and analyzing rockfall trajectories based on motion and collision theory principles.

The data obtained from trajectory analysis are then utilized to estimate rockfall impact forces. Dynamic finite element analyses are subsequently employed to evaluate the performance of an EPS-sand cushioning system, focusing on stress transmissibility and deformation patterns of the cushioning systems. The findings reveal that the integration of a sand cushion with an EPS geofoam provides a lightweight, less stiff, and efficient solution for rock-shed cushioning.

Acknowledgments

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Deformation of unsaturated clayey soils under climatic influences

Qing Cheng[1,*], Chao-Sheng Tang[1]

[1] School of Earth Sciences and Engineering, Nanjing University, Nanjing 210023, China
[*] Corresponding author. Tel: +86-13236589389; E-mail: chengqing@nju.edu.cn

Abstract

Under the influence of global climate change, severe droughts, super-rainstorms, heat waves and other extreme climate events occur frequently and will intensify in the future. The frequent occurrence of extreme weather events brings a wide range of geotechnical problems and disasters such as landslides, dam failures, land subsidence and surface cracks. These geotechnical engineering problems are intrinsically related to soil deformation.

Cracking is a typical soil deformation problem in arid climates. For complex crack networks, fast recognition techniques based on machine learning have been developed for a variety of soils and environmental conditions. Using the newly developed technique, we identified and quantitatively analyzed the desiccation cracking process in compacted soil samples with different microstructures. It was revealed that as the water content of the compacted soil increases, the soil changes from an aggregated to a dispersed structure, and desiccation cracks develop from disordered to ordered (Cheng et al., 2020). It has also been found that cracks occur not only from the surface of the soil, but also from the bottom due to interlayer friction.

The presence of cracks provides a channel for rapid infiltration of rainwater. Experimental studies have revealed a three-stage infiltration law for cracked soils (Cheng et al., 2021). With the infiltration of rainwater, the existing cracks may heal. The healing of desiccation cracks in clayey soils is mainly attributed by two factors: wetting-induced swelling and disintegration. The healing of desiccation cracks with a lower wetting rate is mainly due to the wetting-induced swelling of soils. However, that with a higher wetting rate is dominated by both swelling and disintegration (Tian et al., 2023).

In terms of soil deformation caused by heating and cooling, a temperature-controlled triaxial apparatus was developed. By using the newly developed equipment, the thermally induced stress softening of structural soil and the soil deformation subjected to a heating and cooling cycle were revealed. It was found that for normal consolidated soils, the stress softening caused by the rise in temperature leads to the rearrangement of soil particles, after which the soil undergoes plastic shrinkage and deformation. During cooling, soil deformation in the early stages is an elastic contraction caused by the cooling of soil particles, and plastic contraction deformation may occur with further cooling. Based on this phenomenon, a yield limit of "temperature reduction" was proposed.

Acknowledgments

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Microscopic Initiation of the Granular Columns Collapse

Guowei Dai[1], Ruihao Li[1], Hu Zheng[1,2*]

[1] Department of Geotechnical Engineering, Tongji University, Shanghai 200092, China
[2] Key Laboratory of Geotechnical and Underground Engineering, Ministry of Education, Tongji University, Shanghai 200092, China.
[*] Corresponding author. Tel: +86-13524608289; E-mail: zhenghu@tongji.edu.cn

Abstract

The macroscopic behavior and dynamics of the granular columns collapse have been extensively studied through numerical and experimental research over an extended period. However, the microscopic initiation of granular collapse remains elusive due to a limited understanding of the internal stresses within the system. To reveal the onset mechanism of granular columns, a two-dimensional flume apparatus has been devised utilizing the photoelastic technique. This experimental setup enables us to observe the collapse behavior and the force chain network properties, providing a microscopic comprehension of the initiation process of granular materials.

The experimental findings demonstrate that a near-wedge shape block slides along the failure line, with the resulting failure angle measuring approximately 59.6°. In addition to this sliding collapse phenomenon, we also observe significant particle motion occurring away from the front of the granular collapse. This observation suggests the presence of a nonlocal effect.

Fig.1 illustrates the distribution of force chains during the granular collapse. It is evident that the particles forming strong force chains (SFC) are distributed in a homogeneous and dispersed manner. Moreover, the quantitative disparity in the number of SFC particles and their proportions indicates the occurrence of force transition in three regions. This force transition is believed to be the underlying cause of the nonlocal particle motion. These valuable insights offer a deeper understanding of the initiation mechanism behind geo-granular failure.

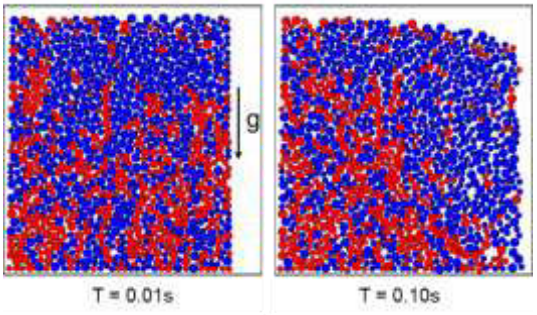


Fig. 1 Strong force chain particles on granular collapse.

Acknowledgments

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Specificity of tectonic environment, earthquakes, and earthquake-triggered landslides in slope displacement

Danny Love Wamba Djukem [1,*], Xuanmei Fan [1], Hans-Balder Havenith [2]

[1] State Key Laboratory of Geohazard Prevention and Geoenvironment Protection, Chengdu University of Technology, 610059 Chengdu, China.

[2] Geology Department - B18, Liege University, B-4000 Sart Tilman, Liege, Belgium.

[*] Corresponding author. Tel: +86-1888 4676 269; E-mail: wdjukem@gmail.com

Abstract

The use of the Geographic Information Systems (GIS) has significantly advanced the analysis of slope stability under seismic loading at regional scales. GIS facilitates the application of well-known slope stability calculation techniques like the Newmark Displacement (ND) method (Newmark, 1965) by integrating geospatial and seismic data. The ND approach has long been effective in assessing earthquake-induced slope deformation. Currently, there is no universally accepted ND equation for predicting earthquake-triggered landslides (ETLs) at regional scale. This study investigates why certain coefficient values work better in some regions compared to others. We propose a versatile approach that can be adapted to various tectonic and mountainous environments, at different scales. This approach enables the incorporation of region-specific conditions, such as the type of the activated fault, hanging/footwall and related fault mechanism effects, site and topographic amplification, terrain roughness, and wetness coefficients. These 'adjustments' are applied to the spatial analysis of six ETLs' distributions (Fig. 1). Slope failures appear to start when ND values fluctuate between 2 and 10cm in Haiti, Taiwan, Lushan, and Luding (Fig. 1). The more efficient the ND model, the more landslides fall into high ND categories as also highlighted by Djukem et al. (2024). This site-adaptable ND approach can be implemented as soon as the activated fault segment, focal mechanism, and epicenter of the earthquake are available.

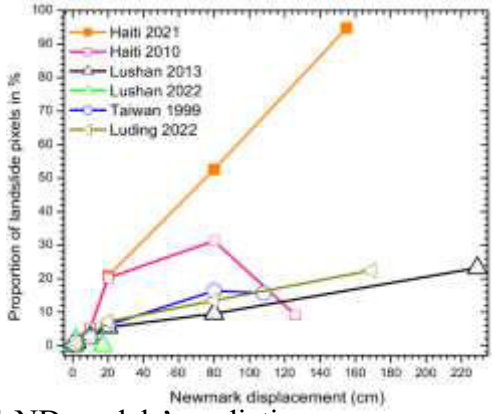


Fig. 1 ND models' predictive power curves.

Acknowledgments

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Research on the impact of atmospheric pressure on the flow characteristics of geotechnical granular materials

Yuxiang Hu[1], Hu Zheng[1,2,*]

[1] Department of Geotechnical Engineering, Tongji University, Shanghai 200092, China
[2] Key Laboratory of Geotechnical and Underground Engineering, Ministry of Education, Tongji University, Shanghai 200092, China.
[*] Corresponding author. Tel: +86-13524608289; E-mail: zhenghu@tongji.edu.cn

Abstract

The interstitial gaseous medium plays a pivotal role in influencing the flow dynamics of geotechnical granular materials. To delve deeper into this phenomenon, the present study employs a bespoke low-pressure, horizontal rotating drum apparatus coupled with digital image processing technology. This approach enables the precise quantification of various flow characteristic parameters under different atmospheric pressures, thereby elucidating the effect of atmospheric pressure on the granular flow behavior.

The findings reveal that at reduced pressure conditions, both the upper and lower repose angles of the granular avalanche flow diminish to a certain extent. This reduction in repose angles undermines the stability of the particle accumulation within the drum, thereby enhancing flowability. In the case of the horizontal drum particle flow collapse system, the diminished air molecule-induced disturbances (airflow) at low pressures culminate in a more orderly and stable motion state; notably, the collapse angle and collapse duration exhibit minimal sensitivity to atmospheric pressure variations.

These insights underscore the profound influence of atmospheric pressure changes on the flow dynamics of geotechnical granular materials, offering a robust scientific framework for advancing our comprehension of the dynamic mechanisms underpinning disasters triggered by geotechnical granular material flows.

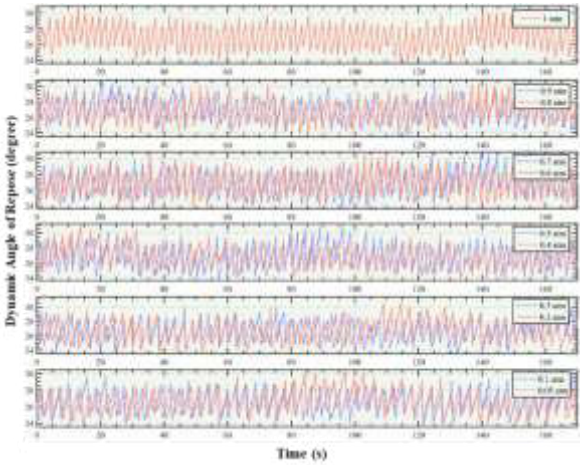


Fig. 1 The variation of the repose angle over time under different atmospheric pressures.

Acknowledgments

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3D improved bond–base Smooth Particle Hydrodynamics method for modeling large-scale rockslide considering the cracking and contact behavior

Bo Li[12*], Zhenming Shi[12], Chengzhi Xia[12]

[1] Department of Geotechnical Engineering, Tongji University, Shanghai 200092, China
[2] Key Laboratory of Geotechnical and Underground Engineering, Ministry of Education, Tongji University, Shanghai 200092, China.
[*] Corresponding author. Tel: +86-17769735586; E-mail: libotj@tongji.edu.cn

Abstract

A 3D improved bond–base Smooth Particle Hydrodynamics (IBB–SPH) framework for modelling the cracking and contact behaviour of rock–like material is proposed, with its application to rock slope simulation of complex discontinuous structures. An improved kernel function integrates the momentum equation formulation to represent the interactions between particles and to capture the crack propagation modes during brittle failure of rock–like material based on virtual bond. A new contact algorithm coupled the cracking and fracture surface is proposed to simulate frictional slip and separation/contraction behaviour between intact particles and damage particles. Moreover, particle domain searching (PDS) algorithm is developed to search and locate particles in intact matrix and discontinuous surfaces. Following this way, high computational efficiency and natural coupling of cracking and contact in IBB–SPH is achieved. The present 3D IBB–SPH model is validated by a number of theoretical and computational examples and reproduction of the progressive failure of Tangjiashan rockslide's valley-blocking event considering different rock layer thickness, dip angles and rock strengths. The findings indicate that the IBB–SPH method accurately captures the entire process of crack initiation and propagation, formation of contacts, frictional slip, and a time–dependent gravitational motion of rock slopes.

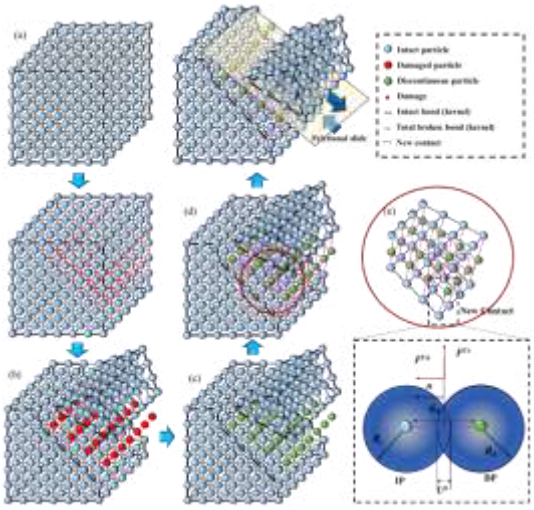


Fig. 1 Schematic view of the procedure for simulating the cracking and contact in IBB–SPH

Acknowledgments

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Field test study on the internal force distribution of prestressed anchor cable lattice beam on a high slope

LI-Jingbang[1,*]

[1] College of Civil Engineering, Lanzhou Institute of Technology, Lanzhou 730050, China.
[*] Corresponding author. Tel: +86-13519676159; E-mail: lst_123143567@163.com

Abstract

Prestressed anchor cable lattice beams are the common supporting form in high slope support engineering. At present, theoretical calculations or numerical simulations are used to study the internal force distribution of lattice beams, but the related field tests are few. In this paper, considering the lattice beam's double-reinforcement section and the materials' nonuniformity, a new calculation method of the measured moment was introduced in the field test conducted in a high slope support project in northwest China. Comparisons between the measured moment and the theoretical moment were performed. The results showed that the stress variation of the lattice beam section agreed with the stress change state of the slope rock and soil mass after the anchor cable tension, which indicated that the test was practical. The distribution trend of the measured moment calculated was consistent with the theoretical moment, which verified the reliability of the proposed calculation method. Due to the different relative stiffness, the difference in slope rock and soil mass significantly affected the lattice beam's internal force distribution. In a specific range, the loading of adjacent nodes had a non-negligible effect on the internal forces of lattice beams. The measured moment obtained by the test was smaller than the theoretical moment, which indicated that the lattice beam was safe at the present stage. For some monitoring points, the torque effect between the cross and vertical beams was also a reason for the more significant difference between the measured moment and the theoretical value.

Acknowledgments

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Effects of sliding-control structures on the seepage processes and failure modes of mudstone slopes: an experimental study

Kun Li[1,2], Ping Sun[1,2], Haojie Wang[1,2]

[1] Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing 100081, China

[2] Observation and Research Station of Geological Disaster in Baoji, Shannxi Province, Ministry of Natural Resources, Baoji 721001, China.

[*] Ping Sun. Tel: +86-15810865191; E-mail: sunpingcgs@foxmail.com

Abstract

Neogene mudstone is a widely distributed stratum in the Loess Plateau of China, which is generally deposited beneath the Quaternary loess, acting as an important medium for most landslides. Previous studies have demonstrated that the scale, failure process, movement type, and deposit characteristic of the mudstone landslides are closely related to the form and properties of the structural plane (Sun et al., 2017; Peng et al., 2019; Zhang et al., 2020).

In this research, the effects of sliding-control structures on the failure modes and hydrological conditions of the mudstone slope are investigated via model experiment. The results show that the seepage process and failure mode vary with the structural conditions of the experimental slope. The vertical joints serve as preferential seepage paths, which accelerate rainfall infiltration into their surrounding areas, resulting in earlier responses of volumetric water content and pore water pressure. With the participation of vertical joints, the failure mode tends to transform from shallow failure to deep-seated failure. On the other hand, the presence of a weak interlayer leads to significant increases in the velocity and runout of the sliding mass. The variation in the failure extent and deformation characteristics with varying structures further changes the distributions of volumetric water content and pore water pressure in time and space. The variation in failure mode indicates the difference in the sliding-control mechanism, which is dominated by the type of structural plane and its interaction with hydrological process. In these mechanisms, the pore water pressure and seepage force play significant roles for the reduction in effective stress and shear strength.

Acknowledgments

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Preliminary investigation on the failure mode and kinematic feature of the catastrophic Jiujiawan landslide occurred on September 2022 in Qinghai, Northwest China

Ran Li[1,2*], Ping Sun[1,2], Shuai Zhang[1,2], Fawu Wang[3,4,5]

[1] Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing 100081, China.

[2] Observation and Research Station of Geological Disaster in Baoji, Shaanxi Province, Ministry of Natural Resources, Baoji 721001, China.

[3] College of Civil Engineering, Tongji University, Shanghai 20092, China.

[4] Key Laboratory of Geotechnical and Underground Engineering (Tongji University), Ministry of Education, Shanghai 200092, China.

[5] Professor Emeritus, Shimane University, Japan.

[*] Corresponding author. Tel: +86-13261553306; E-mail: 13261553306@163.com

Abstract

At approximately 15:30 (UTC + 8) on 15 September 2022, a catastrophic mudstone landslide occurred in Jiujiawan Village, Xining City, Qinghai Province, Northwest China and resulted in the outage of the Lanzhou-Urumqi High-speed Railway for nearly 10 months. To investigate the failure mechanism and motion feature of the catastrophic Jiujiawan landslide, detailed field investigation, unmanned aerial vehicle (UAV) survey, trench observation, engineering geological drilling, and numerical simulation were conducted. In this work, the geometric feature, potential triggering factors, failure mechanism, failure process, and runout prediction were preliminary discussed. Based on the field reconnaissance, it is confirmed that the 2022 Jiujiawan landslide occurred at a counter-tilt slope comprising the interbedding of thick red mudstone and thin gypsum, and it is a typical reactivated thrust load-caused rockslide. Mudstone and gypsum can swell and disintegrate strongly under the cyclic drying-wetting effect, leading to the continuous development and expansion of fractures. Dynamic infiltration of overland flow from virescence irrigation and torrential precipitation in August through cracks of original landslides may be the foremost factor responsible for the occurrence of the catastrophic Jiujiawan landslide. In addition, numerical simulation was conducted with the aid of Massflow software to reveal the motion feature of the catastrophic Jiujiawan and the potential runout when suffering further slope failure in future considering that large portions of the sliding mass are rested at the slope at present.

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Effects of typical loess landslides induced by the 1920 Haiyuan earthquake in China

Xiaobo Li[1,2*], Junjie Duan[1], Yaokuo Liu[1], Xinghao Zhou[1], Jingshan Bo[1,2]

[1] Department of Geological Engineering, Institute of Disaster Prevention, Sanhe 065201, China
[2] Heibei Key Laboratory of Earthquake Disaster Prevention and Risk Assessment, Sanhe 065201, China.
[*] Corresponding author. Tel: +86-15832609679; E-mail: lixiaobo@cidp.edu.cn

Abstract

Based on a field survey, data on 568 typical loess seismic landslides (Fig. 1) induced by the 1920 Haiyuan earthquake were acquired. The relationships between the landslide distribution, seismic factors, and terrain factors were analyzed to better understand the geometrical, directional, and kinematic characteristics of loess seismic landslides. The results indicate that: (1) The dense area of loess landslides triggered by the 1920 Haiyuan earthquake corresponds to the IX-degree seismic intensity zone, situated within epicentral distances ranging from 70 to 110 km and fault distances between 45 and 70 km. (2) The original slope angle of seismic landslides is generally moderate, and the dominant angle ranges from 19° to 27°. (3) The main sliding directions of the landslides are concentrated in the ranges of 60° to 90° and 270° to 300°. The correlation between the main sliding direction of the landslides, the original slope aspect, and the relative epicenter azimuth demonstrates that the propagation direction of seismic waves and the original slope aspect influence the main sliding direction of landslides. (4) The equivalent friction coefficient (μ) of these landslides is between 0.03 and 0.28, with an average value of 0.15. The empirical relationships of kinematic characteristics suggest that these landslides are characterized by significant scaling effects, low slope angles, and long sliding distances. The findings of our study are worthy for use in engineering applications and will pave the way for additional studies into the mechanism and risk assessment of loess seismic landslides.



Fig. 1 Photos of typical loess seismic landslides.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. U1939209, 51808118).

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Probabilistic Seismic Landslide Hazard Zonation Based on Slope Units

Jiamei Liu[1,2*], Mengtan GAO [3], Tao Wang[1,2]

[1] Institute of Geomechanics, Chinese Academy of Geologi-cal Sciences, Beijing 100081, China

[2] Observation and Research Station of Geological Disaster in Baoji, Shaanxi Province, Ministry of Natural Resources, Baoji 721001, China.

[3] Institute of Geophysics, China Earthquake Administration, Beijing 100081, China.

[*] Corresponding author. Tel: +86-15501066098; E-mail: **liujiamei@mail.cgs.gov.cn**

Abstract

Recent landslide hazard analyses were carried out based on ground motions with different probability of exceedance, representing a conditional probability of earthquake induced landslides. This paper presents a method for potential seismic landslide hazard zonation at regional scale in terms of probability based on the probabilistic seismic hazard analysis and Newmark's displacement model. The conditional probability of seismic landslides under a certain ground motion is given by the Newmark model. The probability distribution of the ground motion in a given time interval is obtained through the probabilistic seismic hazard analysis. The seismic landslide hazard zonation gives the likelihood of earthquake induced landslides. We produce a regional seismic landslide probability hazard map in future 50 years for the Ganquan town of Gansu province. The landslide probability of each grid cells is estimated based on the infinite slope model assumption basic of Newmark's model. Then, the maximum probability of slope unit is extracted from the grid cells. The results show that the high hazard zones and can provide a reference for land planning and seismic landslide risk assessment and management.

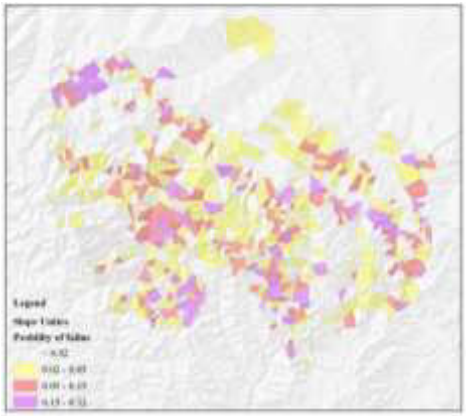


Fig. 1 Probability of seismic induced landslide occurrence in future 50 years

Acknowledgments

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Fully Probabilistic Analysis of Dikes Against Inner Slope Stability: A Practical Guide

Elham Mahmoudi[1], Mark van der Krogt[2]

Abstract

Central to a flood defence strategy in the Netherlands is the safe design and maintenance of geotechnical structures like dikes, which serve as critical barriers against flood. However, the inherent complexities of geotechnical systems introduce uncertainties that necessitate a probabilistic approach to design and assessment. By integrating uncertainty quantification into the design process and adopting probabilistic frameworks for assessing existing structures, engineers can enhance the resilience and longevity of flood defence infrastructure. This paper presents a comprehensive guide to conducting fully probabilistic failure analyses for the inner slope stability of dikes. It offers practical insights alongside theoretical underpinnings to facilitate their application in engineering practice. Building upon the advancements in failure probability analysis methods and their integration into accessible software tools such as LEM analysis and the Probabilistic Toolkit, this work explains the process of assessing failure probabilities for the inner slope stability of the embankment. This fully probabilistic analysis also considers the uncertainties in the water level, for example, represented by extreme value distributions. A detailed step-by-step plan for executing the fully probabilistic analysis using Monte Carlo Importance sampling and FORM, complemented by a fragility curve approach, is presented.

Keywords

flood defence reliability, dike stability, probabilistic analysis, fragility curve

Introduction

Each year, floods rank among the most prevalent natural disasters. For instance, in 2019, flooding accounted for 27% of reported disasters, impacting 69 countries, resulting in 1586 fatalities, and displacing over 10 million individuals. The World Disasters Report[1] indicates a consistent upward trend in reported flood occurrences, with 151 floods documented from 1960 to 1969, compared to 1499 floods between 2000 and 2009. Predictions of rising sea levels and increased rainfall intensity due to climate change suggest that these numbers will likely escalate further in the coming years (Van, M.A. et al., 2022).

The Netherlands has a long history of flood protection with dykes, dams, dunes and storm surge barriers. Develop and implement realistic, innovative techniques and solutions to improve the (assessment of) flood defences, enable sustainable construction and allow for integral spatial implementation. While a flood defence's resistance parameters are typically deterministic, practical considerations introduce uncertainties. Furthermore, predictive models employed to anticipate critical combinations of parameter values, which could trigger failure, may yield outcomes diverging from the unknown truth. Failure probability analyses for flood defences have developed for a long time in the Netherlands, where probabilistic calculations are transferred from safety factors to failure probability indicators. In general, a sharper assessment is arguably possible with the probabilistic method. This is mainly because the semi-probabilistic analysis usually uses a (too) conservative relationship between the allowable safety factor and the allowable failure probability. This impression that the strength related to macro stability was underestimated initiated a research program named Actual Strength (Actuele Sterkte in Dutch) (E. Rosenbrand and A. Rozing, 2020). This is done using targeted ground investigation, monitoring, and advanced calculation techniques, such as failure probability analysis and updating. This program suggested performing a four-stage assessment on each levee segment. The first two stages concern primitive assessment based on available data and semi-probabilistic methods to determine if the segment can benefit from further investigation. In the third stage, fully probabilistic failure analyses take place. The current work reports the key aspects of an updated practical guide, which the authors have prepared for the third stage in the framework of this program (Mark van der Krogt, 2023; Schweckendiek et al., 2017a). Also, the last stage of the assessment involves a failure probability updating process, in which additional information from survived loads is considered in the assessments, which has not been covered in the current work.

This guide describes how failure probability analyses can be carried out for the inner slope stability of dikes. Moreover, we have introduced proprietary software and libraries enabling practitioners to conduct assessments. The procedures outlined in this document can be executed within these platforms, which offer free access and have undergone extensive testing by water authorities, consultants, and engineering

[1] World Disasters Report 2020

firms. The main workflow to perform a fully probabilistic analysis is presented after highlighting the utilised methods and software in the following two sections. The last section presents a practical example to showcase the introduced workflow.

Theoretical background

The failure definition for inward inner slope stability concerns the occurrence of shearing of the inner slope of the dike, as shown schematically in Figure 1The assessment concerns the likelihood of an inward shearing of the dike slope. Embankment stability can be quantified using several stability analysis methods. The major methods include limit equilibrium methods (LEM) and finite element methods. LEM methods (e.g., Bishop, Uplift-Van, or Spencer) are by far the most popular stability analysis approach available due to their simplicity.

D-Stability software (Deltares, 2024), is a LEM software commonly used in the Netherlands for design, assessment, and renovation purposes by flood defence managers (water boards and Rijkswaterstaat) and consultancy and engineering firms that carry out the assessment or design for primary flood defences on behalf of the flood defence managers. This software has also implemented methods for failure probability analysis, such as the first-order reliability method (FORM) and Monte Carlo importance sampling (MCIS). The GEOLib[2] Python library is also dedicated to D-Stability software. It automates different parts of analysis in D-Stability and connects its inputs and outputs to other probabilistic software and platforms.

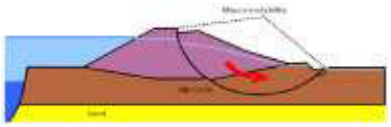


Figure 1 Macrostability of a dike (Robert 't Hart et al., 2016)

By definition, with perfectly known input and a perfectly accurate model, failure would occur at a stability factor SF < 1.0. However, because the model is imperfect, model uncertainty must be considered in a probabilistic analysis. In this aspect, the failure probability is defined as the probability of occurrence of an inner slope slide, considering all uncertainties in the model and other parameters that influence inner slope stability.

Probabilistic analysis

Failure probability analyses (or reliability analyses) for the inner slope stability of dikes have essentially the same theoretical background as failure probability analyses for other failure mechanisms. The probability of failure (P_f) equals the probability that load (S) exceeds resistance/strength (R):

$$P_f = P(R < S) \rightarrow P_f = P(Z < 0)$$

$With\ Z = R - S$, where Z is the limit state function. The probability of failure P_f is also often expressed in terms of the reliability index β, which is defined as:

$$\beta = \Phi^{-1}(1 - P_f)$$

where Φ is the cumulative standard normal probability distribution. Various methods exist for performing failure probability analyses, each with distinct advantages and disadvantages regarding accuracy and computational efficiency. These methods are typically categorised into three levels. However, this guide focuses solely on level 2 or 3 calculations in the future.

At Level 3, failure probabilities and reliability indices are calculated over a specific period, like a design lifespan or one year, often using Monte Carlo simulation (MC). While accurate, classic MC can be time-consuming and is best suited for higher failure probabilities exceeding approximately 1%.

Level 2 methods offer significant computational efficiency by utilising approximations. Techniques include FORM, MCIS, Directional Sampling (DS), and Subset Simulation (SS). FORM reveals parameter uncertainties, while MCIS focuses on failure-prone areas. DS and SS offer alternative methods for exploring multiple failure modes. Refer to (Hasofer and Lind, 1974; Baecher and Christian, 2003; Rob Brinkman, 2023) for theoretical details.

At Level 1, or semi-probabilistic, analyses, explicit failure probability estimates are not provided. Instead, checks using calculated values assess whether a structure meets the required failure probability. For inner slope stability, these checks typically involve evaluating the stability factor against a specified quantile, adjusted by factors considering damage, model, and schematisation. The Probabilistic Toolkit (PTK) (Deltares, 2023) can be used to calculate the probability of failure. It enables probabilistic analyses across various models, including geotechnics and hydrodynamics. It offers a range of reliability methods, from numerical integration to advanced techniques like SS and DS.

Fragility curves

A failure probability analysis for macro stability can account for all uncertainties in strength and load. However, integrating stochastic variables like outside water level may cause convergence issues or prolonged computation due to nonlinear effects, such as bursting and saturation from overtopping or critical slip surfaces at varying water levels. Fragility curves offer a solution by depicting the probability of failure given a load, typically using outside water level (h) for inner slope stability as

[2] https://github.com/Deltares/GEOLib

$P(Z < 0|h)$ or (β), as shown in Figure 3. While a limited number of points can suffice to determine a fragility curve, interpolation between points assumes linear reliability index behaviour. In cases of pronounced nonlinear behaviour, generating additional points around inflexion points may be necessary for accuracy. In some failure analyses like overtopping and uplift, the strength parameters, such as strength reduction, may follow different scenarios in different water levels. In such cases, the fragility curve of each scenario must be drawn up and combined to evaluate the realistic failure probability (for more details, the reader is referred to (Mark van der Krogt, 2023).

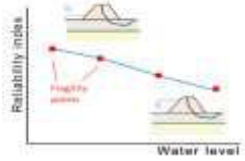


Figure 2 Fragility curve illustration

The total failure probability is ultimately determined by combining the conditional failure. This step is often called "out integrating" because the following integral must be solved:

$$P_f = \int P(h) f_h(h) dh$$

where $f_h(h)$is the probability density function of the water level (or other load variables). The "total" failure probability P_f is determined by combining the fragility curve ($P(Z < 0|h) = F_R(h)$) with the water level statistics ($f_h(h)$), see Figure 3. Usually, solving this integral is facilitated in failure probability software, such as PTK. Alternatively, the result can be determined with a manual Riemann integration in a spreadsheet. The reader may use the Python notebook prepared for this topic in the GEOLib-Plus[3] platform.

For each fragility point, influence coefficients ($\alpha_i \mid h$) are determined based on the parameter values at the corresponding design point.

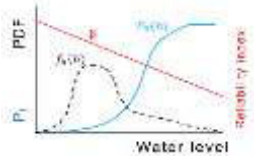


Figure 3 Integrating fragility curve over probability density of water level

The influence coefficient of the water level is then determined by:

$$\alpha_h = \frac{u^*}{-\beta} = \frac{\Phi^{-1}(Fh(h^*))}{\Phi^{-1}(P_f)}$$

Where $\alpha_h$is the influence coefficient of the water level and h^* is the water level's design point. $u*$ is the design point (normalised water level). β and P_f are the "total" reliability index and "total" failure probability. F_h represents the cumulative distribution function of the water level (load). To determine the "total" influence coefficients after "out-integration", including strength and load contributions, we can use the fact that the total sum of the squared influence coefficients must be equal to 1:

$$\Sigma_{i=1}^{n} (h^*)^2 . (1 + \alpha_h^2) = 1$$

Given the conditional influence coefficients at the design point water level ($\alpha_i \mid h^*$), we can normalise by:

$$\alpha'^2_i = (h^*)^2 \cdot (1 - \alpha_h^2)$$

where α'_i is the transformed influence coefficient(s) of strength variable i. $\alpha_i \mid h^*$ represents influence coefficient(s) of strength variable i directly from FORM at the design point water level (e.g., based on interpolation between fragility points).

Failure probability analyses must be calculated for multiple water levels to draw fragility curves. Calculations for multiple water levels can be automated using the GEOLib-plus Python package to reduce manual work.

Failure probability analysis for macrostability

This chapter aims to provide the reader with a workflow for efficiently performing a failure probability analysis for macro stability. It is essential to make well-founded and sensible choices about the uncertainties to be modelled so that realistic failure probabilities are calculated and the calculation time is not unnecessarily long. A failure probability analysis for macrostability essentially contains the following steps, shown in Figure 4.

In short, one needs to take further steps: **Step 1**: Establishing a (semi-probabilistic) baseline as the starting point for the failure probability analysis, providing an estimate of expected reliability. It also serves as the basis for sensitivity analyses. In **step 2**, sensitivity analysis identifies influential parameters and uncertainties affecting stability. It also involves studying the sliding slopes and the predefined areas to search for them. It helps determine discontinuity points and fragility curve inclusions, aiding result interpretation. **Step 3** involves defining stochastic input (probability distribution, scenarios, …) for the relevant uncertainties in the input parameters. The first three steps are sometimes an iterative process.

[3] github.com/Deltares/GEOLib-Plus

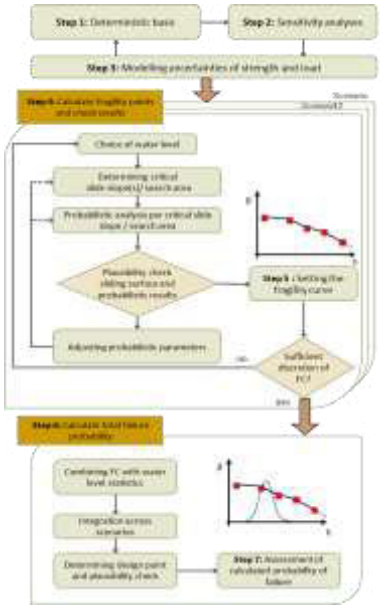


Figure 4 Roadmap to perform failure probability analysis in macro stability.

Step 4 calculates the failure probabilities for fragility points at different water levels, also known as scenarios. It uses stability models that consider water level and pore pressure schematisation, with the outside water level representing peak river discharge. Moreover, results are verified for plausibility through checks and verifications at the design point, including identifying relevant slip surfaces. **Step 5** involves constructing the fragility curve by combining fragility points from **Step 4**. The situation's complexity determines the number of fragility points needed, especially around water levels where discontinuities occur. **Step 6** entails calculating the total failure probability per year by integrating the fragility curve with the load statistics, as detailed in the last section. **Step 7** involves assessing, interpreting, and verifying the results.

In short, the results of probabilistic analysis in **Steps 4** and **6** can be cross-checked and verified considering the following aspects:

- Convergence of probability calculation
- Representative influence coefficients for parameters and the summation
- Review stability analysis with design points as an outcome of probability analysis
- Consistency with semi-probabilistic sensitivity analysis results
- Comparison with calibrated WBI 'damage factor' line (Kanning et al., 2017)

Based on these results, subsequent steps can be determined, providing insights into further reducing uncertainties through additional investigations or failure probability updating (Schweckendiek et al., 2017b).

Hands-on practical examples

This section gives an example of developing a failure probability analysis for inner slope stability using the probabilistic functionality of D-Stability and the PTK. The example considers one representative cross-section in a uniform dike section. Figure 5 shows that the dike crest is at NAP +12.6 m, with an 11.5 m thick cohesive layer of peat and clay beneath the water barrier. A 2.5 m wide traffic load near the inner crestline is accounted for. Sandy layers use drained strength parameters, while cohesive layers use undrained parameters using the SHANSEP model. The pre-overburden pressure (POP) is divided into areas under the crest, the foreland, and the hinterland, with the boundary at the mid-slope.

Two construction phases are considered: daily conditions and a high-water situation. The daily water level is NAP+5.50 m, and the high water level for a return period of 10,000 years is NAP+12, calculated with a software for safety analysis of Dutch primary flood defences ("Riskeer," 2023).

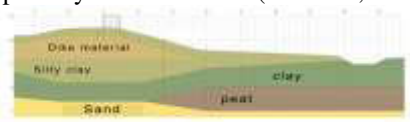


Figure 5 Geometry and soil structure of flood defence

This example evaluates failure probability against inner slope stability for a flood defence in the river area. The maximum probability of flooding of the dike section is 1/10,000 per year. The failure probability requirement for a design for inner slope stability at the cross-section, $P_{norm-CS}$ has been derived based on (Kanning et al., 2017). With a failure probability budget of $w = 0.04$ and length effect factor $N = 16$ as a starting point.

$$P_{norm-CS} = \frac{P_{norm} \times \omega}{N} = 2{,}52 \times 10^{-7}\, 1/year$$

Where P_{norm} is failure probability per dyke section on an annual basis.

The associated reliability requirement is $\beta_{norm-CS}\ \Phi^{-1}(1 - 2{,}5 \cdot 10^{-7}) = 5{,}02$. Checking the damage line in Figure 10, the SF=1,16. Assuming a schematisation factor of 1.0 and a model factor of 1.06 (Uplift-Van), the minimum required stability factor in a semi-probabilistic calculation is 1.23.

Following the basis calculation of **Step 1**, a stability factor of 0.96 has been calculated for the model; see Figure 6. This Figure also indicates other slip planes with higher SF values. Given the substantial deviation between the calculated stability factor (0.96) and the requirement (1.23), it is unlikely that the failure probability analysis will meet the failure probability requirement. However, the analysis may provide a better estimation of failure probability and offer insight into the critical uncertainties affecting failure probability, facilitating more precise determination of new (soil) investigations or monitoring valuable for reinforcement design.

In **Step 2**, a sensitivity analysis was carried out on different loading scenarios (including changes in

daily and high water and traffic load). Different sets of material parameters (namely, mean and characteristics) are considered in different analyses. The sensitivity results are important for the soil strength parameter values and the water level behind the dike. The sensitivity analysis also revealed that the deep slip surfaces are critical for the failure probability analysis. This is important because, with FORM, the sliding surface is fixed, and with the MCIS method, it is impossible to find all sliding surfaces with one search grid efficiently.

Table 1 Probability distribution of soil parameters

Parameter	Average	Standard deviation	Characteristic value
$\gamma_{dry}/\gamma_{sat}$ [kN/m3]	16/16	-	16/16
S [-]	0,35	0,06	0,26
m [-]	0,95	0,025	0,91
POP_{crown} [kPa]	23,0	9,0	11,5
POP_{Toe} [kPa]	26,0	11,0	12,3

Following the workflow presented in the last sections, in **Step 3**, we defined the uncertainties. The ground parameters, model uncertainty, and open water level are modelled with a probability distribution because it is expected that these variables will have a significant impact on the result. The probability distributions are driven using the method presented in the GEOLib-plus Python notebook. As an example, the properties of clay material are presented in Tabel 1. For model uncertainty, the recommendations of (Van Duinen, A., 2015). The values follow a lognormal distribution.



Figure 6 Daily water scenario, traffic load = 13.3 kPa, SF = 0.96

Following **Steps 4 and 5**, a fragility curve represented in Figure 6 is calculated for four water levels with return periods mentioned in Tabel 2. The table represents each input parameter's fragility points and influence coefficient in different scenarios. The influence coefficients in Tabel reveal that the uncertainty of SHANSEP strength parameters (S) and (m) of the cohesive layers contributes the most to the calculated failure probability. Notably, the contribution of sand and dike material's internal friction angle (φ) has a very low influence. A low influence coefficient for dike material is expected because the coefficient of variation of φ is relatively small, and only a small part of the slip surface passes through the dike material. Similarly, a low influence coefficient for sand is mainly because the governing slip surface typically does not pass through the sand layer.

The validity of MCIS results is confirmed by comparing them with a basic MC model employing PTK. While the final P_f outcomes exhibit no more than 2% difference, the time required for convergence with MC amounted to 80 hours, sharply contrasting with MCIS, which achieved convergence within a mere 20 minutes. Additionally, the hierarchy of influential parameters remains consistent across both methodologies.

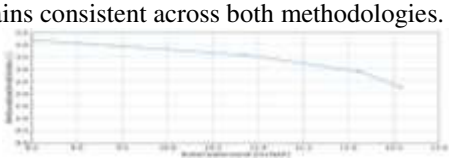


Figure 7 Created fragility curve (blue line) from fragility points (crosses).

Determining the total failure probability involves integrating the fragility curve with the probability distribution of water levels, performed in **Step 6**. This can be performed using spreadsheets or scripts; here, PTK was utilised. Firstly, the fragility curve exported from D-Stability (as a .JSON file) can be directly imported into the PTK as an external fragility curve.

Tabel 2 MCIS results at different water levels

Water level [m+NAP]	**8,50**	**10,84**	**12,12**	**12,58**
Return period (year)	-	100	10.000	100.000
		Probability analysis results		
Reliability index	4,20	3,59	2,92	2,27
		Influence coefficient α		
Silty clay (S,m)	0,40	0,40	0,52	0,55
clay (S,m)	0,43	0,39	0,46	0,40
Peat (S,m)	0,65	0,64	0,44	0,45
Sand (φ)	-0,02	0,03	0,07	0,05
Dike material (φ)	0,12	0,06	0,01	-0,03*
POP toe	0,37	0,38	0,35	0,38
POP crown	0,17	0,22	0,26	0,28
Model uncertainty	-0,24	-0,30	-0,36	-0,33

In the PTK, users can specify any possible probability distribution for water levels. The user may perform it by *i)* directly entering the data on the water level in various return periods, where PTK handle the data via extrapolation, or *ii)* using a parametrised probability distribution for the annual maximum water level, e.g., a Gumbel distribution. The calculated fragility curve can be integrated into the following probability distribution of water levels. After integration, the reliability index of the failure probability analysis and influence coefficients are calculated; see Figure *9*.

Through numerical integration, the total failure probability is calculated as $\beta_{norm-CS} = 4.1$,

corresponding to a failure probability of 1/56,800 years. The design point of the water level is NAP +9.53 m, with an influence coefficient of -0.32. Squaring this yields an influence of 10%. Instead of using PTK, one may perform the out-integration process manually using the online Python scripts presented in the GeoLib-Plus platform. A verification step is also required to complete the assessment. Integrating the fragility curve over the probability distribution of the water level results in a reliability index $\beta = 4.1$. The position of the example is shown in the point cloud of the WBI calibration (see Figure 10).

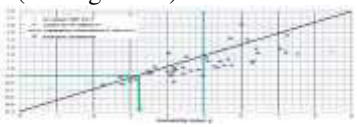


Figure 10 Point cloud of the WBI calibration (Kanning et al., 2017), including example

Based on the safety factor-reliability index relationship, the calculated reliability index is higher than expected (green arrow). However, the calculated probability of failure does not meet the specified maximum probability of failure at the cross-section level (blue dotted line).

Conclusion

The guide aims to facilitate failure probability analysis for the macrostability of flood defences in practice by highlighting practical examples in conjunction with the theoretical backgrounds. Failure probability analysis can be used to assess an existing situation and verify design situations. Using the suggested workflow, the assessment is optimised to avoid underestimating the system's safety. Moreover, an insight into the possible modifications will be gained.



Figure 9 Results after integration.

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Elham Mahmoudi ()
[1] Department of Geotechnical Engineering, Deltares, The Netherlands
E-mail: elham.mahmoudi@deltares.nl

Mark van der Krogt
[2] Department of Geotechnical Engineering, Deltares, The Netherlands
E-mail: mark.vanderkrogt@deltares.nl

Rapid Identification of Rainfall-induced Shallow Landslides Based on Machine Learning

MENG Huajun[1,2*], QI Lining[1,6], GUO Chaoxu[3,4], CHU Kelei[3,4,5], WANG Yuanyuan[1,6]

[1] Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing 100081, China.

[2] Observation and Research Station of Geological Disaster in Baoji, Shaanxi Province, Ministry of Natural Resources, Baoji 721001, China.

[3] Key Laboratory of Geohazard Prevention of Hilly Mountains, Ministry of Natural Resources of China, Fuzhou, Fujian Province, 350002

[4] Key Laboratory of Geohazard of Fujian Province, Fuzhou, 350002

[5] Chinese Academy of Geological Sciences, Beijing, 100037

[6] School of Earth Sciences and Resources, China University of Geosciences In Beijing, Beijing, 100083

[*] Corresponding author. Tel: +86-13381142663; E-mail: mhjun-521@qq.com / mhjun521@cags.ac.cn

Abstract

In recent years, affected by global warming, China's southeast coastal provinces have frequently suffered from super typhoon rainstorms and Meiyu, which have induced a large number of cluster shallow landslides, seriously threatening the lives and property safety of local people. In order to obtain more accurate information about disaster spatial distribution, disaster size and other data to support the scientific decision-making of emergency relief, aiming at the cluster shallow landslides induced by heavy rainfall and formed by granite strong weathering layer, a fast identification model of shallow landslides based on decision tree Light GBM algorithm was proposed by using machine learning method. The high-precision aerial image and DOS data of Wuping County, Fujian Province were used to select slope, elevation, plane curvature (SOA), sectional curvature (SOS) and R, G, B three bands as the training features of the model, and the model was trained and tested. The results show that:(1)the recognition accuracy of multi-feature combination is better than that of single feature recognition;(2)the recognition model composed of slope, elevation, plane curvature, sectional curvature and R,G,B three bands has the highest recognition accuracy, and the recall value is greater than 0.95;(3)the F1score of the model is as high as 0.97,indicating that the model has strong generalization ability, and can accurately realize the rapid automatic identification of landslide disasters induced by similar geological environment. It can provide a basis and scientific and technological support for the risk prevention and control of typhoon rainstorm landslide geological disasters and disaster relief decision-making.

Acknowledgments

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Estimation of Ultimate Bearing Capacity of Ground–Log Composite Ground

Takumi Murata[1,*], Atsunori Numata[2], Shuhei Sasaki[3], Atsushi Kawasaki[4]

[1] Civil Engineering Headquarters, Tobishima Corporation, Minato-ku, Tokyo 108-0075, Japan

[2] Director, Soilwood, Koshigaya-shi, Saitama 343-0041, Japan

[3] Technology and Product Development Department Housing Division, SUMITOMO FORESTRY CO., LTD., Chiyoda-ku, Tokyo 100-8270, Japan

[4] Product and Technology Development Department, Misawa Homes Co., Ltd., Shinjuku-ku, Tokyo 163-0833, Japan

[*]Corresponding author. Tel: +81-80-8775-5978; E-mail: takumi_murata@tobishima.co.jp

Abstract

This study investigated the bearing capacity of composite ground consisting of logs inserted into the ground for soft-ground countermeasures and the ground between those logs with the aim of rationalizing the design by clarifying this characteristic.

Although the bearing capacity of the ground between piles tends to be neglected, it is not zero. It is believed that a reasonable design can be achieved that accounts for the bearing capacity of the ground.

This study examined whether the ultimate bearing capacity of composite ground as determined by large-scale plate loading tests can be estimated from ground investigations. For comparison, the ultimate bearing capacity was calculated by adding the ultimate bearing capacity of the log section and that of the ground.

Figure 1 shows the relationship between the ultimate bearing capacity determined by the large-scale plate loading test and that estimated from the ground investigation. The results of both comparisons were in general agreement, regardless of log length or pitch.

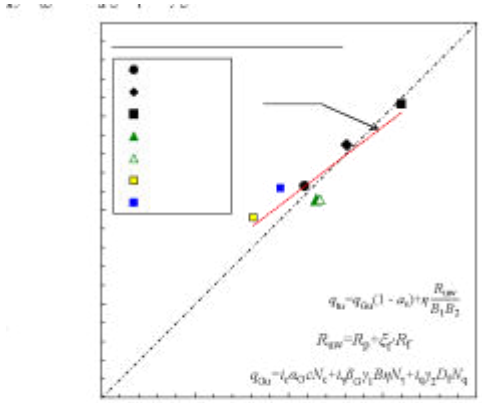


Fig. 1 Relationship between ultimate bearing capacity determined by the large-scale plate loading test and that estimated from the ground investigation

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Harnessing elastic metasurfaces composed of in-filled pipes for surface wave attenuation in layered half-space

Anchen Ni [1], Zhifei Shi [1], Ioannis A. Antoniadis [2]

Abstract

In this work, we further investigate the surface wave attenuation performance of elastic metasurfaces composed of in-filled pipes in a layered half-space, focusing on the dispersion relations and transmission properties of Rayleigh waves. The introduction of soil layers will reduce the width of attenuation zones. Additionally, transmission simulations reveal complex propagation patterns for elastic metasurfaces in a layered half-space, including wave reflection, wave resonance, and higher-order wave modes, which will hinder the penetration of converted shear waves into the half-space. In contrast, in reference cases, only surface-shear wave mode conversion is observed. Moreover, the attenuation performance of elastic metasurfaces is also diminished in layered soils in the frequency domain, and nonuniform displacement distribution behind the elastic metasurface is also found. Last but not least, the feasibility of elastic metasurfaces to train-induced ground-borne vibration mitigation is numerically verified in the time domain. Although performance of elastic metasurfaces in layered soils is inferior to that in the homogeneous soils, they are better than traditional trenches within the main frequency range. Thanks to the excellent surface wave attenuation performance and easy realization, these novel elastic metasurfaces hold great potentials in ambient vibration mitigation.

Keywords

Elastic metasurfaces, ambient vibration, surface waves, layered half-space

Introduction

Surface waves and bulk waves are generated by the train-induced vibration in half-space. Surface waves predominate, comprising 67% of the energy, while shear waves and compressional waves contribute 26% and 7%, respectively (Murillo et al,. 2009). Due to the higher energy and slower decay of surface waves compared to compressional and shear waves, their mitigation is a primary focus in controlling ambient vibration.

Installing wave barriers is an effective strategy to interrupt the propagation of surface waves. This method is common, given its simplicity and cost-effectiveness. Various barrier types have been proposed and implemented, such as open/in-filled trenches, pile barriers, wave impeding blocks (WIBs) and surface walls (Chen et al., 2022). These methods have proven effective in isolating ground vibration through analytical, numerical, and experimental studies. However, their limitations become apparent with the expansion of urban rail systems. The predominant limitations include: (1) design often targets a solitary frequency, resulting in limited operational bandwidth without consideration for specific scenarios. (2) The working mechanism of barriers is limited (wave reflection, refraction, and scattering), which relies on the strong impedance ratio (IR) between soil and barriers. (3) the geometric size significantly influences isolation performance. In light of these points, the pursuit of innovative, high-efficiency wave barriers with novel mechanisms and practical applicability continues to be a priority for ambient vibration mitigation.

The development of a novel class of artificial composites, termed elastic metamaterials (EMs), holds promise for innovative wave barrier design. The local resonance mechanism in EMs facilitates the formation of subwavelength AZs, representing one of the most intriguing properties where elastic waves are strongly attenuated. In our previous works (Ni et al., 2024; Ni et al., 2023), we introduced a novel type of elastic metasurface composed of in-filled pipes to effectively attenuate surface waves. Importantly, our previous works extensively investigated the effects of geometry/material parameters, layouts, the water table, and the viscosity of the soil and rubber materials on the performance of vibration mitigation. Furthermore, through laboratory-scale experiments, we have demonstrated the exceptional effectiveness of the proposed elastic metasurfaces. Specifically, in this study, we take a step further by investigating the influence of layered soil on the performance of elastic metasurfaces. In general, the foundation soil is formed

through geological processes, resulting in well-defined stratification characteristics (Fig.1(a)).

Model and method

Governing equations and dispersion model

As shown in Fig.1 (b), the In-filled pipes are buried underground with the periodic constant $a=0.2$ m. The buried depth $d=0.5a$, and the total height of the unit cell model $H=30$m. Specifically, the elastic parameters of layered soil can be found in Table 1. Details of in-filled pipes are shown in the inset of Fig.1 (b). The outer radius is $R=0.4a$, while the inner radius is $r=0.3a$. The thickness of rubber coating is $t=R-r=0.1a$, and the inclusion is made of concrete. The elastic parameters of in-filled pipes can be found in Table 2. For simplicity, the layered soil and in-filled pipes are assumed to be linear elasticity, due to the comparatively small strain on the ground induced by train. Hysteretic, slip and nonlinear phenomenons between the interfaces will not to be considered in this work. Ignoring the damping and body force, the decoupled governing equation of elastic waves propagating within the 2D plane-strain model can be described as:

$$(\lambda+\mu)\nabla(\nabla\cdot\boldsymbol{u})+\mu\nabla^2\boldsymbol{u}=\rho\frac{\partial^2\boldsymbol{u}}{\partial t^2}, \quad (1)$$

where λ and μ are the Lamé constant, ρ is the mass density, $\boldsymbol{u}=[u_x,u_y]$ is the in-plane displacement vector, and $\nabla=\left(\frac{\partial}{\partial x},\frac{\partial}{\partial y}\right)$ is the gradient operator. Next, we present the procedure of in-plane elastic wave propagation. It is assumed that soil layers interface are bonded perfectly. While surface of layered-half is set to be traction-free, and the bottom is set to be fixed to avoid undesired rigid motions.

According to the periodic structure theory, the typical unit cell can be extracted for dispersion analysis shown in Fig. 1(b). At the same time, the displacement of this periodic system can be expressed as:

$$\mathbf{u}(\mathbf{r})=e^{i(\mathbf{k}\cdot\mathbf{r})}\mathbf{u}_{\mathbf{k}}(\mathbf{r}), \quad (2)$$

where k is the wave vector, and $r=(x,y)$ is the position vector; $u_k(r)$ is the modulation function of the displacement vector. Besides, the modulation function is a periodic function with the same periodicity as the periodic system:

$$\boldsymbol{u}_k(\boldsymbol{r})=\boldsymbol{u}_k(\boldsymbol{r}+\boldsymbol{A}), \quad (3)$$

where A is the constant vector. Subsequently, substituting the Eq.(3) into Eq.(2), the Bloch periodic boundary conditions (BPBC) can be described as:

$$\boldsymbol{u}(\boldsymbol{r}+\boldsymbol{A},t)=e^{i\boldsymbol{k}\cdot\boldsymbol{A}}\boldsymbol{u}(\boldsymbol{r},t). \quad (4)$$

Combining the BPBC with Eq.(1), the dispersion relation of the infinite periodic system can be calculated and converted into an eigenvalue problem of the typical unit cell as follows:

$$(\boldsymbol{K}-\omega^2\boldsymbol{M})\boldsymbol{U}=0, \quad (5)$$

where K is the stiffness matrices, M is the mass matrices, and U is the displacement vector. Note that BPBCs are applied at the left and right boundaries of the unit cell. Next, the dispersion relation can be obtained by letting k sweep the first Brillouin zone of the lattice shown in Fig.1(c). To identify the surface wave modes from dispersion relations, we adopted our previously proposed strain energy method (Pu & Shi 2020a), which can be expressed as:

$$P=\frac{\int_{S(2\lambda)}W_\varepsilon ds}{\int_{S(H)}W_\varepsilon ds}, \quad (6)$$

where W_ε is the elastic strain energy, while the λ is the wavelength of the considered surface wave. Consquently, the surface wave mode can be identified when $P\geq 0.8$.

Transmission model

The transmission model of elastic metasurfaces is plotted in Fig. 2, where $l_1=150a$, $l_2=200a$, $l_3=25a$, $l_4=75a$, $l_5=10a$, $l_6=25a$, and the number of in-filled pipes is $N=20$. Besides, the low-reflecting boundary conditions (LRBCs) are applied at the right, left and bottom boundary to avoid the undesired wave reflection. The harmonic vertical/shear excitation with 10^{-5} m is applied to the black arrows, and the receiver line is imposed behind the elastic metasurfaces. Note that the distance between the excitation point and elastic metasurfaces is long enough to eliminate the interference from bulk waves. Quantitively, the amplitude reduction factor A_R is adopted to assess the surface wave attenuation performance of elastic metasurfaces, respectively, which are expressed in Eq.7, respectively.

$$A_R=\frac{1}{l_5}\int_{l_5}u_{y1}dl\Big/\int_{l_5}u_{y0}dl, \quad (7)$$

where u_{y1} is the vertical displacement with elastic metasurfaces, while u_{y0} is the vertical displacement without elastic metasurfaces. l_s is the length of the receiver line.

Dispersion and transmission analysis of Rayleigh waves in layered half-space

In-plane dynamic characteristics of elastic metasurfaces on layered half-space will be discussed in this section, including the dispersion and transmission characteristics. For comparsion, the homogeneous half-space is taken as a reference. Note that the homogeneous half-space is assumed to be made of fill soils presented in Table 1.

Dispersion relation and mode shapes

Dispersion relations of Rayleigh waves on homogeneous and layered half-space are shown in Fig. 3(a) and Fig. 3(b), respectively. We observe that classic "avoided-crossing" behavior which results in a Rayleigh wave attenuation zone (RWAZ) in both cases, noting that RWAZ is denoted by the shaded area. It can be found that the RWAZ is within 63.9-83.7 Hz in homogeneous half-space, while the RWAZ is within 63.9-71.1Hz in layered half-space. Noteworthy, the periodic constant a is about 1/6 of the max Rayleigh wavelength in the frequency range of interest. With respect to layered soils, the lower bound of RWAZ remains constant, while the upper bound shifts to the lower frequency, leading to the reduction of RWAZ width. Note that the dispersion curve within the RWAZ shows horizontal local resonance, which cannot fully interact with Rayleigh waves and affect the existence of RWAZ. Additionally, we plot the mode shapes of bounds of RWAZs in Fig. 3(c). As shown in point A and point C, we can notice that the RWAZ is opened by the strong interaction between the vertical local resonance of in-filled pipes and Rayleigh waves. Note that surface soil and in-filled pipes have the same in-phase displacement shown by the red arrow and blue arrow, respectively. Therefore, the fundamental dispersion curve bends off to approach the vertical resonance frequency of in-filled pipes asymptotically. In other words, the lower bound frequency of RWAZ is determined by the vertical resonance frequency of in-filled pipes. More details of design of resonant unit cells can be found in (Mitchell et al., 2014). In contrast, both mode shapes of upper bound of RWAZ exhibit different characteristics, although the same downward vertical local resonance is observed. Specifically, a high-order Rayleigh wave mode is found between the in-filled pipes and the interface 1 at point D. The interaction of vertical local resonance of in-filled pipes with the high-order Rayleigh wave is also observed.

Quantitively, the profile of the vertical displacement along the depth is presented in Fig. 4, which is extracted from the pink and green dashed line in Fig. 3(c). For comparison, the displacement is normalized by their corresponding displacement at the surface. As shown in Fig. 4(a), both displacement profiles of lower bounds of RWAZ show a high consistency, which can explain why lower bounds of RWAZ remain constant in both cases. However, the interaction between Rayleigh waves and vertical resonance of in-filled pipes is weakened a lot in layered soils due to the existence of high-order Rayleigh wave modes. In fact, the high-order Rayleigh wave mode is induced by the interaction of P and S waves at the contact interface of soil layers, and contributes a little to the surface vibration (Pu et al., 2020b). Thus, the reduction of width of RWAZ can be attributed to the weakening of interaction. From the perspective of dispersion relations, the presence of layered soils will weaken the performance of elastic metasurfaces for Rayleigh wave attenuation to some extent.

Rayleigh wave attenuation performance of elastic metasurfaces

Rayleigh wave attenuation performance of elastic metasurfaces is shown in Fig. 5. As expected, we can notice that a significant wave attenuation within the RWAZ in both cases, implying the accuracy of the predicted dispersion properties. Although the width of RWAZ is reduced in layered soils compared to the homogeneous soil, a similar Rayleigh wave attenuation within the overlapping RWAZ is observed within the overlapping range. However, the weakening of Rayleigh wave attenuation is found outside the RWAZ of layered soils and within the RWAZ of homogeneous soils, which can be explained by the existence of high-order Rayleigh waves as mentioned. Nonetheless, the effect of high-order Rayleigh waves is limited, which weakens performance of elastic metasurfaces smaller than 10%.

To gain a further insight, the displacement field of the first three layers is extracted as shown in Fig. 6, including the layered soil and homogeneous soil. An extremely complex propagation pattern is found in layered half-space when compared to the homogeneous half-space, especially within the shallow layers. Specifically, two marked features can be summarized as follows: (1) The strong wave reflection at the layer interface, which will interact with fundamental Rayleigh waves and prevent the mode conversion by elastic metasurfaces. As a result, shear wave front will return to the free surface, inducing an

amplification of surface displacement (resonance); (2) Although the remarkable phenomenon of Rayleigh-to-shear wave conversion occurs due to the hybridization between Rayleigh waves and metasurface resonances. It should be noted that the converted shear waves excite the high-order Rayleigh waves below the metasurfaces, where high-order Rayleigh waves propagate along the layer interface. Comparatively, only mode conversion from Rayleigh waves to shear waves is observed in the homogeneous soil. The existence of shallow soil layer will induce nonuniform distribution of dynamic response behind the elastic metasurface within/outside the RWAZ. Therefore, we can conclude that the consideration of soil layers is necessary for the design of elastic metasurfaces, especially for ambient vibration mitigation. Due to the nonuniform displacement distribution behind the elastic metasurface, i.e., the vibration strengthening zone and vibration mitigation zone coexist, which will affect the selection of the location of the protected area.

Feasibility of elastic metasurfaces for train-induced ground vibration mitigation

The transmission model in Fig. 2 is used, and material parameters remain the same as above mentioned in Table 1 and Table 2. In fact, the 2D plain-strain model is the simplest, and the train-loading is regarded as the point excitation. The length of elastic metasurfaces is set to $N=20$. To highlight the performance of elastic metasurfaces for surface wave attenuation, the commonly used concrete trench barrier is taken for comparison (width $w=0.3$m, depth $D=3$m).

First of all, we conducted a field test of train-induced ground vibrations of Beijing subway line 13, noting that the measured subway line is next to the university and dense residential areas. The picture of field test and equipment is shown in Fig. 7(a), including accelerometers (LC0106), signal acquisition (DEWE-43-A) and laptops. The measuring point is located about 5 m from the nearest railway center. The time history of train-induced ground vibration from 0-12 s is presented in Fig. 7(b). Next, the frequency-domain results are obtained by the fast Fourier transform (FFT) shown in Fig. 7(c). It can be found the main frequency range of ground vibration is between 55-80 Hz, which is set to be the target range of elastic metasurfaces. Next, we put the measured time history into our transmission models, and extract the displacement at the receiver line. The time history and corresponding Fourier spectrum are presented in the Fig. 8(a) and Fig. 8(b), respectively. We can notice that elastic metasurfaces perform better than commonly used trenches in layered soils within the RWAZ. Quantitively, the average insertion loss ($IL=20\lg(A_R)$) of elastic metasurfaces within the RWAZ is 4.4 dB larger than that of trenches. At the same time, elastic metasurfaces in homogeneous soils perform better than those in layered soils, which supports our findings in the frequency domain. Quantitively, the average IL of elastic metasurfaces in homogeneous soils is 2.7 dB larger than that of metasurfaces in layered soils within the overlapped RWAZ. However, all cases exhibit a similar vibration mitigation performance outside the RWAZ. To obtain a further insight of the effect of soil layers, the displacement field from 1.232 – 1.412 s is plotted in Fig. 9. The displacement is well confined within the first three soil layers. The wave reflection at the lower interfaces prevents the converted shear waves further infiltrating into the half-space. In addition, the reflected waves returned to the surface, and interacted with the propagative Rayleigh waves, causing the wave resonance and nonuniform displacemtn distribution behind the elastic metasurface. In short, the wave reflection at the lower soil interface will weaken the performance of elastic metasurfaces to some extent. Nonetheless, the elastic metasurface shows more potentials in ground vibration mitigation than the trench, including the better performance and lower cost. It is worth noting that the total area of elastic metasurfaces (0.8 m^2) is smaller than the total area of in-filled trenches (0.9 m^2). Therefore, elastic metasurfaces can be regarded as an effective alternative to train-induced ambient vibration mitigation.

Conclusion

This work further investigates the performance of elastic metasurfaces in attenuating surface waves in a layered half-space, focusing particularly on how the soil layer interface affects the dispersion relations and transmission characteristics of elastic metasurfaces. Some remarks are summarized as follows:

Regarding dispersion relations, the introduction of soil layers will narrow the attenuation zones for surface waves. Besides, in terms of transmission characteristics, wave reflection, wave resonance, and higher-order modes will hinder the penetration of converted shear waves into the half-space. This ultimately diminishes their effectiveness and leads to nonuniform displacement distribution behind the elastic metasurface, including amplified surface displacement in the protected area.

For application of elastic metasurfaces, we first conduct a field test to record the primary frequency range of train-induced vibrations on Beijing subway

line 13, which is within 55-80 Hz. It is set as the target frequency range for the metasurfaces. Subsequently, we input the measured acceleration time history into our transmission model, a strong wave attenuation is observed within the RWAZ. Specifically, the insertion loss of elastic metasurfaces exceeds that of trenches in a layered half-space by 4.4 dB. Furthermore, within the overlapping RWAZ, the insertion loss of elastic metasurfaces in homogeneous soils exceeds that in layered soils by 2.7 dB. Snapshots of the transient simulation confirm these findings in the frequency domain again.

Figures and figure captions

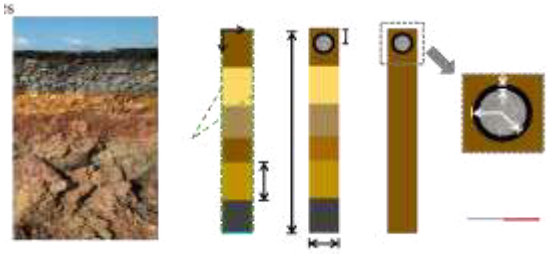


Fig. 1 Elastic metasurfaces in layered half-space. (a) Layered characteristic of soils; (b) unit cell model and details of in-filled pipes; (c) the first irreducible Brillouin zone.

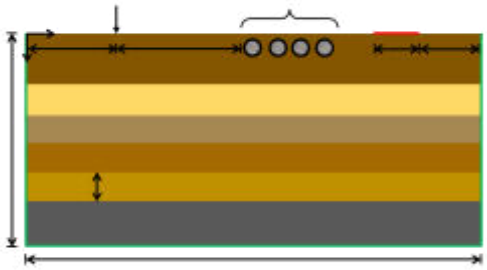


Fig. 2 Transmission model of elastic metasurfaces in layered half-space.

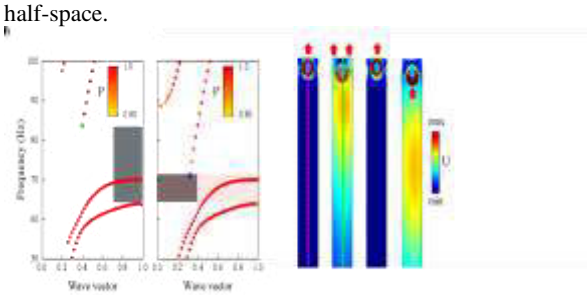


Fig. 3 Dispersion relation of Rayleigh waves and mode shapes. (a) Dispersion relation of Rayleigh modes in homogeneous soils and (b) in layered soils; (c) mode shapes of bounds of RWAZ.

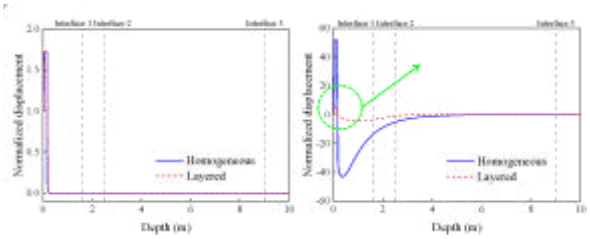


Fig. 4 Normalized displacement profile of the bound modes of Rayleigh waves. (a) Displacement profile of lower bound of RWAZ and of upper bound of RWAZ. Note that the buried location of in-filled pipes is denoted by the yellow shaded area.

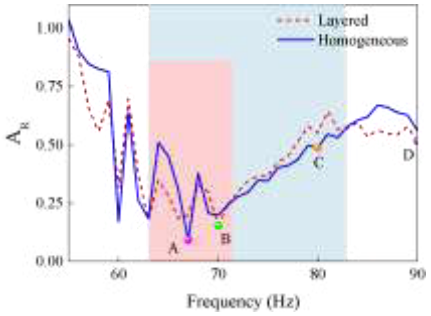


Fig. 5 Transmission spectra of Rayleigh wave attenuation by elastic metasurfaces in half-space.

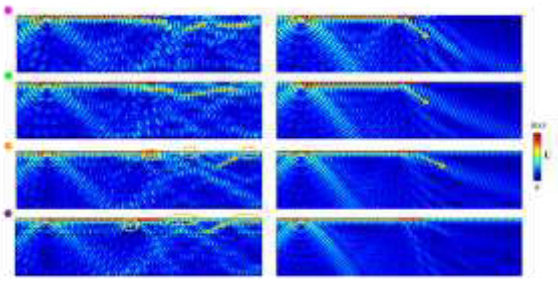


Fig. 6 Displacement fields of Rayleigh wave attenuation by elastic metasurfaces in half-space. (a) In layered half-space; (b) in homogeneous half-space.

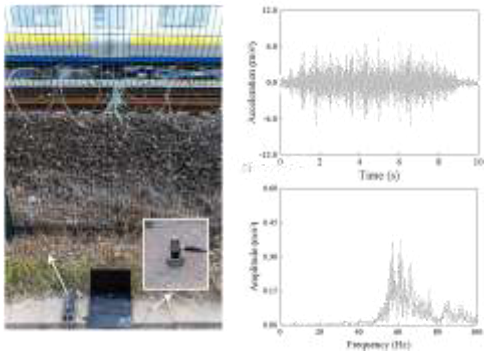


Fig. 7 Field test of train-induced ground vibration in Beijing subway line 13. (a) Measuring equipments; (b) vertical acceleration data of train-induced ground vibrations; and (c) corresponding Fourier spectra.

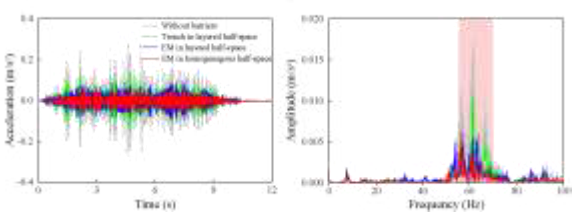


Fig. 8 Train-induced ground vibration mitigation by elastic metasurfaces in time domain. (a) Average vertical acceleration response and (b) corresponding Fourier spectra.

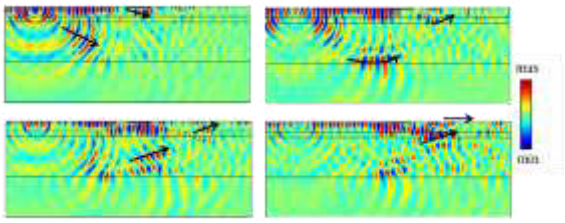


Fig. 9 Snapshot of the vertical displacement field of elastic metasurfaces in layered half-space from 1.232-1.412 s.

Tables

Table 1. Material parameters of layered soils (Gao et al., 2017).

Soil type (No.)	Depth (m)	Density (kg/m³)	Young modulus (MPa)	Poisson ratio
Fill (1)	0.0-1.6	1830	37	0.4
Clayey silt (2)	1.6-2.5	1939	44	0.4
Sandy silt (3)	2.5-9.0	1888	53	0.4
Silt (4)	9.0-15.5	1898	141	0.45
Sandy silt (5)	15.5-25.5	1837	90	0.45
Clay (6)	25.5-30.0	1847	155	0.45

Table 2. Parameters of the materials of in-filled pipes.

Materials	Density (kg/m³)	Young modulus (MPa)	Poisson ratio
Rubber	1300	0.102	0.47
Concrete	2500	30e3	0.2

Acknowledgments

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Anchen Ni (✉)
[1] Institute of Smart Materials and Structures, School of Civil Engineering, Beijing Jiaotong University, Beijing 100044, China
E-mail: anchenni@bjtu.edu.cn

Zhifei Shi
1 Institute of Smart Materials and Structures, School of Civil Engineering, Beijing Jiaotong University, Beijing 100044, China
E-mail: zfshi178@bjtu.edu.cn.

Ioannis A. Antoniadis
2 Dynamics and Structures Laboratory, School of Mechanical Engineering, National Technical University of Athens, Zografou Campus, GR-157 80 Athens, Greece
E-mail: antogian@central.ntua.gr

Reactivation mechanism and run-out processes of the Wangqi landslide

Haojie Wang [1,2,3], Ping Sun [1,2,3 *], Jian Ren [1], Jiarui Mao [1,3,4], Chaoying Ke [1]

[1] Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing 100081, China;
[2] Observation and Research Station of Geological Disaster in Baoji, Shaanxi Province, Ministry of Natural Resources, Baoji 721001, China;
[3] Key Laboratory of Active Tectonics and Geological Safety, Ministry of Natural Resources, Beijing 100081, China;
[4] Observation and Research Station of Crustal Stress and Strain in Beijing, Ministry of Natural Resources, Beijing 100081, China
[*] Corresponding author. Tel: +86-18811326840; E-mail: sunpingcgs@foxmail.com

Abstract

On April 30, 2022, the old Wangqi landslide was reactivated in Weinan Town, Tianshui City, Gansu Province, China. The landslide with a volume of approximately 547,500 m3 destroyed 30,000 m2 farmland and 100 m of the rural road. Based on field investigation, unmanned sub-aerial photography and numerical calculation, the reactivation mechanism, run-out processes and failure mode of the landslide were analyzed. The results indicate that the new Wangqi landslide is a typical reactivated thrust-load-caused loess landslide affected by the leakage of water storage wells, and two water storage wells mainly control its boundary. The new Wangqi landslide has apparent characteristics of deformation transferring from the trailing edge to the leading edge, and its failure pattern is creep-cracking type. The duration of the landslide movement is approximately 25 s, the maximum velocity of the sliding mass is about 2.5 m/s, and about 9 % of the potential energy is converted into kinetic energy during the landslide movement. The landslide undergoes four stages from deformation to failure: deep creep, tension cracking, penetration of sliding zone and sliding stages. Our results may be used as a reference for the risk management of landslides induced by human engineering activities in rural areas.

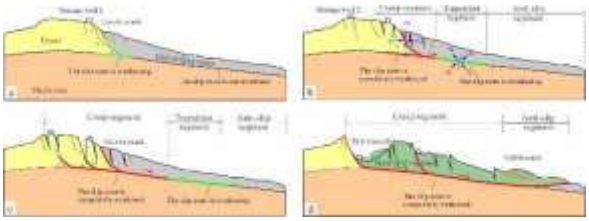


Fig. 1 Failure mode of the landslide.

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Particle flow simulation of rapid loess landslides with frontal plowing effect

Sun Xinglai[1] , Shen Wei[1*] , Tonglu Li[1]

[1] Department of Engineering Geology, Chang'an University, Xi'an 710064, China

[*] Corresponding author. Tel: +86-18729540814; E-mail: wei.shen@chd.edu.cn

Abstract

During the run-out process of some rapid loess landslides, the loose terrace materials at the slope toe can be plowed and pushed forward by the loess sliding mass, resulting in large upheaval deformation and horizontal transport of the terrace materials and the enlargement of the landslide impact area. This phenomenon is called the frontal plowing effect in literature. To investigate the dynamic evolution mechanism of the frontal plowing effect of rapid loess landslides, we conducted a case study on the Ximiaodian landslide occurred on the northern margin slopes of the loess tableland in Xianyang City, Shaanxi Province. Field investigation was carried out to study the depositional geomorphology of the landslide and the deformation characteristics of the terrace material. Then we built up a particle flow model to analyze the frontal plowing effect formed during the run-out process of this landslide, and finally, the influence of pre-failure slope angle on the frontal plowing mode of the landslide was discussed. The results show that the rapid movement of the loess landslide provides the kinematic condition for the formation of the frontal plowing effect. The upheaval deformation and horizontal transport of the front terrace material are caused by the squeezing of the loess landslide, and in this process the loess sliding mass basically does not mix with the terrace materials. According to the simulated run-out characteristics of the landslide, the frontal plowing process can be divided into four stages. The squeezing-shearing stage plays an importance role in the formation of the multiple thrust-nappe structure in the disturbed terrace area. Additionally, it also indicates that the pre-failure slope angle determines the frontal plowing mode of the landslide. The research results of this paper could provide useful reference to the prevention and mitigation of this type of landslides in loess area.

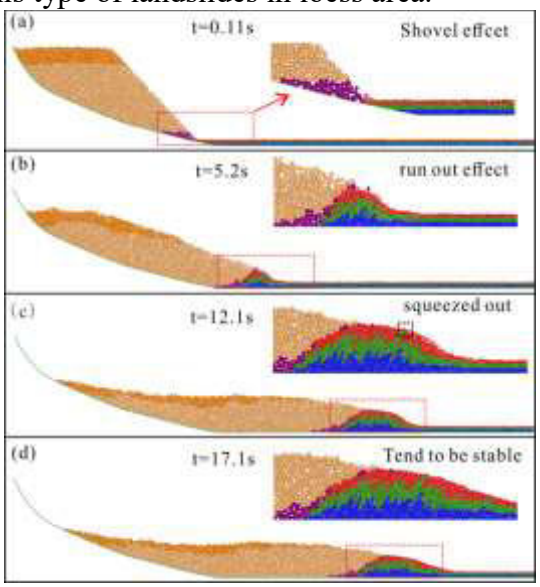


Fig. 1 Characteristics of loess landslide and frontal terrace at different times

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Microseismic observation of seabed geological processes: Internal waves intensify seabed methane release

Zhaoyang Tian[1], Yonggang Jia[1*], Hongxian Shan[1]

[1] Shandong Provincial Key Laboratory of Marine Environment and Geological Engineering, Ocean University of China, Qingdao, 266100, China
[*] Corresponding author. Tel: +86-13905324116; E-mail: yongang@ouc.edu.cn

Abstract

Internal waves are an important natural phenomenon that controls the deep-sea seabed environment. They transfer energy and material from the upper water layers to the deep sea, causing movements in the deep-sea water, altering seafloor topography, and resuspending sediments. However, due to the lack of effective observation methods, our understanding of the impact of internal waves on deep-sea methane release is limited.

We conducted studies in the northern South China Sea, specifically in the depths of 655 meters and 1450 meters in the Pearl River Canyon area. Through in-situ observations of seafloor microseisms and internal waves, we identified and confirmed the characteristics of seafloor microseismic signals associated with internal waves and methane release events. Analyzing these records allowed us to establish a connection between internal waves and seafloor gas release, confirming their correlation. We discovered active methane release phenomena on the seafloor in this area. Internal waves were found to increase seafloor methane release intensity by 4.2 times and 2.4 times on varying time scales, intensifying the methane emissions.

In global marginal seas, the areas where internal waves develop overlap significantly with regions containing seabed natural gas hydrates. This implies that when studying issues related to geological disasters and climate change caused by seafloor methane release in these areas, the influence of internal waves must be thoroughly considered.

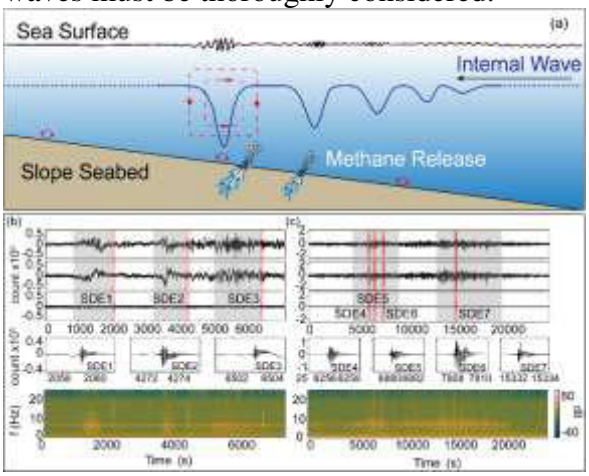


Fig. 1 Microseismic evidence of Internal waves intensify seabed methane release

Acknowledgments

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Identifying critical landslide events: a novel approach based on equivalent displacements

Alessandro Valletta[1], Marco Conciatori[1], Andrea Segalini[1]

Abstract

The growing frequency and intensity of natural hazards has resulted in increased efforts towards their study and control. As a consequence, monitoring systems have emerged as pivotal components of landslide Early Warning Systems (EWS), facilitating comprehension of process evolution and assessment of mitigation needs. This progress stems from technological advancements, yielding more precise sensors and automated data acquisition procedures. Nevertheless, challenges persist in interpreting data, particularly in identifying critical events and forecasting failures. This paper presents the application of a methodology developed by the authors to identify significant deviations in potentially critical events from previously sampled data, with the objective of separating them from false alarms. The core of this approach relies on establishing a threshold value based on antecedent landslide behavior. Specifically, the reference value is derived from assessing equivalent displacements, defined as displacements observed within a time interval equal to that of the critical event. The paper presents various case studies highlighting the efficacy of the proposed model as a valuable tool for near-real-time assessment of landslide behavior.

Keywords

Landslide, Monitoring, Displacements, Alert threshold, Early Warning System

Introduction

Within the framework of landslide risk management, Early Warning Systems (EWS) are a realistic and efficient solution, particularly when structural interventions are impractical for financial or practical reasons. (Londoño 2011). The advantages of newly available technologies for monitoring activities, the decreased economic costs and environmental impact, and the increased availability of reliable databases to calibrate the warning models are some of the reasons for the popularity of these approaches (Pecoraro et al. 2019). According to the United Nations Office for Disaster Risk Reduction, complete and effective EWS should include four components: risk knowledge, monitoring and warning service, dissemination and communication, and response capability (UNISDR 2006). Each of these components is critical, and a flaw in any one of them could lead to substantial issues in the system functionality.

Methodologies for defining alert levels and thresholds are among the most challenging to implement. Because of their particular characteristics, these should be determined with the goal to represent a critical event in the context of the researched phenomenon, i.e., a condition that can trigger a landslide when exceeded. (Guzzetti et al. 2007). For what concerns the landslide monitoring framework, these occurrences are usually associated with slope collapses generated by a situation of irreversible instability of the studied element.

In many circumstances, the threshold evaluation procedure is entirely empirical, based on expert judgment and available monitoring data, and produces values appropriate only for the specific landslide for which they are first developed (Intrieri et al. 2019). Numerical modelling is another possible approach to assess warning levels for a specific landslide. These techniques seek to establish one or more thresholds that correlate to various stages of the slope evolution over time by comparing displacements recorded by monitoring instruments with values derived from a reference model (Zhao et al. 2020; López-Vinielles et al. 2021; Dai et al. 2022).

More recently, technological improvements regarding the computational ability of modern computers, together with the increased availability of powerful software able to process advanced algorithms, has greatly expanded research effort in this area. Consequently, many authors have published new findings and techniques based on an extensive variety of methods, including algorithms powered by Artificial Intelligence and Machine Learning (Guardiani et al.

2021; Meena et al. 2022; Pota et al. 2022; Nocentini et al. 2023).

Alternative approaches have been created over time with the goal of developing a procedure that is more generic in nature and not exclusively dependent on a particular case study. In these cases, failure predicting techniques are usually the foundation of the design process (Manconi and Giordan 2016; Carlà et al. 2017; Valletta et al. 2020; Wu et al. 2021). Several slope-scale EWS can incorporate these techniques because of their relevant exportability. However, they should not be employed in isolation or with a "closed box" approach, since they tend to share the same issues affecting the methodologies from which they originate. Consequently, integrating multiple methods to obtain a more comprehensive description of the phenomenon is the most dependable approach.

Materials and Methods

The introduction of innovative technologies in the geotechnical monitoring field has notably increased the systems reliability and sampling rate, thus making them able to provide consistently a considerable amount of information regarding the phenomenon evolution. Starting from this consideration, the methodology here presented was developed by the authors to exploit the availability of a large amount of information regarding the past movements of the monitored landslide as a comparison to determine the impact of newly recorded displacements on the slope stability conditions (Valletta et al. 2023).

The main objective of this method is the identification of false alarms generated by events that present a data trend geometrically compatible with an accelerating pattern, while featuring a displacement magnitude which does not correspond to a critical occurrence if compared to previously observed events. The process to define the alert threshold value can be summarized as follows:

- When the software detects a potentially critical event, it extracts the corresponding dataset and evaluates the displacement generated S^* and its duration d^*. In this phase it is expected that the event is following an upward trend in the displacement vs. time plot, assuming that other processes (such as the one presented in Valletta et al. 2021) have already elaborated and validated the dataset.
- Taking as a reference the date of the first point x included in the dataset, the software retrieves all monitored data sampled by the same sensor during the 30 days preceding the event. These values are going to serve as a term of comparison for the event identified at the previous step.
- Equivalent displacements are evaluated by considering the available dataset selected in the previous set and taking d^* as a reference for each displacement. By doing this, the algorithm produces a series of displacements S_n generated in the same time interval d^*. Table 1 reports an example of this procedure for a dataset composed of six displacement values.

Table 1 Equivalent displacement values for a potentially critical event composed of six data

Equivalent displacements (dataset = 6 points)	
Event	$S^* = d_x - d_{x-5}$
Eq. Disp. 1	$S_1 = d_{x-1} - d_{x-6}$
Eq. Disp. 2	$S_2 = d_{x-2} - d_{x-7}$
Eq. Disp. 3	$S_3 = d_{x-3} - d_{x-8}$
…	…
Eq. Disp. n	$S_n = d_{x-n} - d_{x-5-n}$

- In this step, it is possible to assess an alert threshold S_{th} based on the values of mean μ_S and standard deviation σ_S calculated from the equivalent displacement values:

$$S_{th} = \mu_S + 3\sigma_S$$

- Finally, it is possible to compare this outcome with the displacement S^*, in order to verify if the event generated a displacement with a magnitude similar to values previously observed during the considered time period, or if the resulting values overcomes the alert threshold, thus indicating a potentially critical phenomenon.

Case studies

The purpose of the following case studies is to demonstrate how the suggested technique might be applied in real-world situations to elaborate on data collected by automatic instrumentation. In particular, examples here discussed involves the analysis of displacement datasets sampled by Vertical Array automatic inclinometers. Based on MUMS (Modular Underground Monitoring System) technology developed and produced by ASE S.r.l., the instrumentation is an array composed of different nodes (named Links) connected by a quadrupole electrical and an aramid fiber cables in order to form an arbitrary long chain of sensors (Segalini et al. 2014). It can be equipped with 3D MEMS, electrolytic cell, piezometer, thermometer, and other typologies of

sensors, while a dedicated datalogger connected to the Array automatically queries each different Link. Data are transmitted to the elaboration server, where they are processed to return the physical monitored quantities, and are finally displayed on an interactive web-based platform.

The on-site location and composition of each array should be carefully considered in the project phase with the aim of providing a comprehensive description of the landslide, focusing on different sectors of the monitored slope. This aspect is essential for any monitoring activity intended for early warning purposes.

Case study #1

The first case study here reported involves the monitoring activity of a slope, located in Southern Italy, crossed by a high-speed railway tunnel currently under construction. After the identification of a quiescent landslide in the area, an automatic monitoring system was installed in order to control the deformations potentially induced by the excavation works. It included four Vertical Array automatic inclinometers, featuring different lengths (between 30 and 80 metres) and number of sensors (from 31 to 81 Links) depending on the specific device. Interspace between nodes also varied depending on their vertical position, with a distance of two metres between each node in the supposedly stable area, reducing to 0.5 metres in proximity of the sliding surface to increase the detail of measurements. The system also included four Piezo Arrays, each one composed of two piezometers, to control the water level variations over time at different depths.

During the entire period of the monitoring activity, all Vertical Arrays evidenced a relevant degree of activity of the entire monitored site, with several datasets triggering the algorithms integrated in the elaboration software. The event here presented was detected on 08 July 2021, when a 6-point dataset sampled by Vertical Array DT0113 – Tilt Link 52 activated the OOA identification process, which defined the onset-of-acceleration for the event at 21:51 of that same day (Fig. 1).

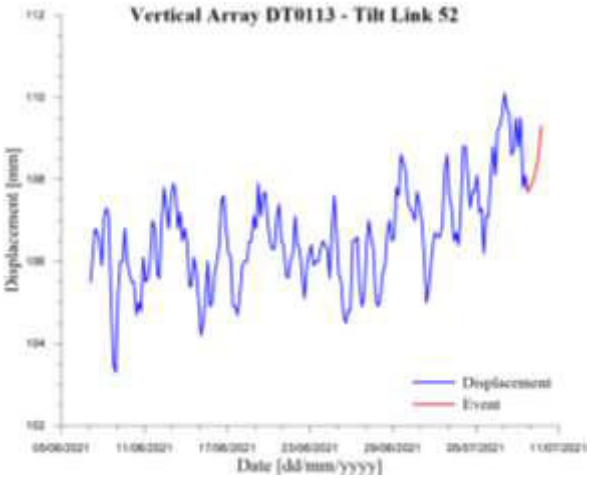


Figure 1 Displacement values measured by Vertical Array DT0113 - Tilt Link 52, referred to the event with OOA on 08/07/2021 21:51, and to the 30 days preceding the event itself

As previously detailed, the first step following the alert level assessment involves the determination of the displacement generated by the event itself. This corresponds to a value of 1.7 mm over a time period of approximately 20 hours. The subsequent operation consists of retrieving monitoring data from a time period of 30 days preceding the event of interest, in order to assess the equivalent displacements and the correlated threshold for each dataset involved in the analysis. Each displacement value is computed by considering a time interval equal to the one obtained from the event, which in this case equals to 20 hours (corresponding to six monitoring values under the hypothesis of constant sampling frequency). The threshold values derived from the mean and standard deviation values of the equivalent displacement dataset, comprising a total of 190 entries, is equal to 3.28 mm.

Fig. 2 presents a graphical comparison between equivalent displacements and the threshold computed at the previous step. It is possible to observe how the resulting value referred to the event of interest is comparable to previously recorded displacements, not overcoming the reference value defined with the procedure previously described.

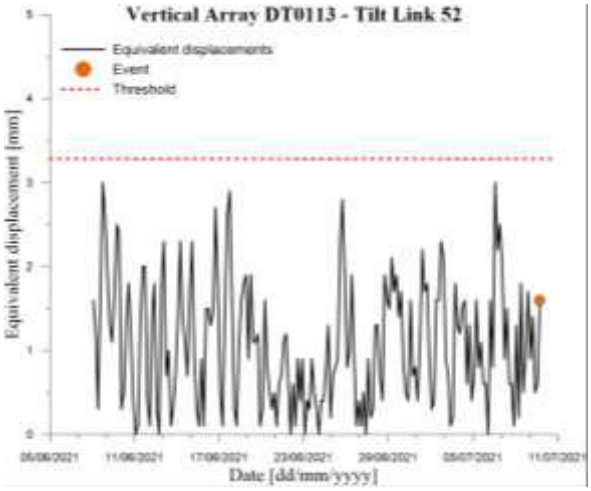


Figure 2 Graphical representation of equivalent displacements referred to the event of interest, the time period of 30 days preceding the event itself, and the threshold evaluated from these values for DT0113 – Tilt Link 52

Case study #2

The second case study refers to the monitoring of a landslide, located in Central Italy, which persists on the construction site of a state road connecting the Adriatic and Tyrrhenian Seas, through Abruzzo and Molise regions. The site is affected by the presence of several landslides in the western sector of the area of interest, showing fast kinematics and sliding surfaces at a depth between 8 and 10 metres, with other instabilities appearing in the first meters of material in the Eastern areas. Following a series of preliminary surveys evidencing further problems related to settlements and damages to pre-existing instrumentation, a MUMS-based monitoring system was designed, with a total of nine Vertical Arrays installed on site over approximately four years. Each Array featured a different number of Links, equipped with 3D MEMS and electrolytic tilt sensors, and customized interspace between nodes:

- One 35-meter long Vertical Array was installed in November 2016
- One 35-meter long Vertical Array was installed in September 2017
- Four Vertical Arrays, with length ranging from 66 to 111 meters, were installed in September 2018
- Three 100-meter long Vertical Arrays were installed in November 2020

It should be noted that the first Vertical Array installed on site was irreversibly damaged due to excessive deformation caused by a superficial landslide between February and March 2017. Despite being compromised by the event, the device was still able to identify the occurrence of the event, disseminating a series of alert messages to authorities responsible of the monitoring activity (Segalini et al. 2019).

On 16 June 2023, the elaboration software reported the presence of an accelerating pattern detected by Vertical Array DT0120 in correspondence of Tilt Link 81, located at a depth of 16 metres. The automatic routine for the determination of the onset-of-acceleration identified the beginning of the event at 12:41. After the definition of the dataset of interest, the elaboration software evaluated the displacement generated by the event itself and the corresponding time interval. In this case, since this device was set to sample new data every hour, the monitoring data showed a displacement of 2.2 mm over a time period of approximately 6 hours. Taking the OOA as a reference, displacements recorded since 17 May 2023 12:41 were retrieved in this phase (Fig. 3)

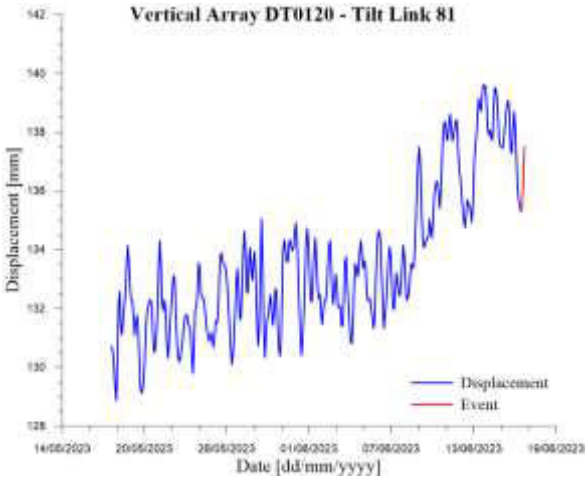


Figure 3 Displacement values measured by Vertical Array DT0120 - Tilt Link 81, referred to the event with OOA on 16/06/2023 12:41, and to the 30 days preceding the event itself

Each equivalent displacement value is computed by considering a time interval equal to the one obtained from the event, resulting in 721 equivalent displacements. By exploiting the mean and standard deviation values calculated on this dataset, the alert threshold based on equivalent displacements is 3.76 mm (Fig. 4). As a result, it is possible to state that the identified event does not display a concerning behaviour if compared to previously sampled monitoring data, which presented comparable values in terms of magnitude without influencing the stability conditions.

It should be pointed out that the presence of other values overcoming the threshold are not indicative of a critical event, considering that the process to evaluate the threshold value is strictly related to a specific event and should not be taken out of context.

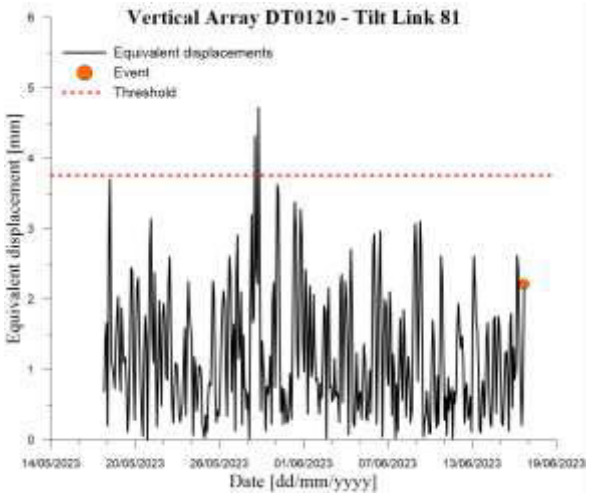


Figure 4 Graphical representation of equivalent displacements referred to the event of interest, the time period of 30 days preceding the event itself, and the threshold evaluated from these values for DT0120 – Tilt Link 81

Conclusions

In a landslide Early Warning System (EWS), failure forecasting and assessment of alert thresholds are crucial elements for identifying slope instabilities and potential collapses. In particular, the correct definition of alert levels should aim to minimize the occurrence of both false positive and missed alerts.

This paper proposes a procedure to evaluate alert thresholds using equivalent displacements, defined as the displacements generated in a time interval equal to the one showed by a specific event identified by the elaboration software When compared to pre-event data, equivalent displacements offer insight into the past behaviour of the monitored element, providing a term of comparison in order to understand if the recorded event generated a displacement which does not correspond to a critical occurrence if compared to the entity of previously observed events, despite being geometrically compatible with an accelerating pattern.

Case studies involving on-site implementation of automatic monitoring instrumentation are presented, where the methodology successfully provided an effective and easily applicable procedure for the analysis of potentially critical events and the identification of false alarms.

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Alessandro Valletta (✉)
[1] Department of Engineering and Architecture, University of Parma, Italy
E-mail: alessandro.valletta@unipr.it

Marco Conciatori
[1] Department of Engineering and Architecture, University of Parma, Italy
E-mail: marco.conciatori@unipr.it

Andrea Segalini
[1] Department of Engineering and Architecture, University of Parma, Italy
E-mail: andrea.segalini@unipr.it

Characteristics, Mechanism and Susceptibility of Seismic Landslides Induced by the Gulang M8.0 Earthquake of 1927, China

Tao Wang[1,2,3*], Jiamei Liu [1,2,3], Haojie Wang [1,2,3], Shuai Zhang [1,2,3], Shuai Liu [1,2,3], Jianjun Du [1,2,3]

[1] Institute of Geomechanics, Chinese Academy of Geological Sciences, Minzu University South Road, Haidian District, Beijing 100081, China;

[2] Observation and Research Station of Geological Disaster in Baoji, Shaanxi Province, Ministry of Natural Resources, Baoji 721001, China;

[3] Key Laboratory of Active Tectonics and Geological Safety, Ministry of Natural Resources, Beijing 100081, China;

[*] Corresponding author. Tel: +86-10-8881 5190; E-mail: wangtaoig@cags.ac.cn

Abstract

On May 23, 1927, a magnitude 8 earthquake struck Gulang County, Gansu Province, China, with an epicenter intensity of 11 degrees. It was one of the most catastrophic earthquakes on the northeastern margin of the Tibetan Plateau, causing nearly 100,000 deaths and triggering a large number of coseismic landslides. Unfortunately, there are few reports on the research of this landslide disaster. In this study, through detailed field survey and high-precision remote sensing interpretation, a more reliable seismic landslide catalog database was established in the high seismic intensity area (X-XI degrees), the main slope structure types and coseismic landslide types in the Gulang earthquake area were identified, the regional seismic landslide distribution law was initially revealed, and the geometric characteristics of landslides were statistically analyzed. The empirical formula of landslide sliding distance fitting is presented. By using engineering geological drilling, the original slope structure and multistage sliding characteristics of Dengshan village landslide, which caused hundreds of deaths in extreme earthquake area, are revealed. By using the numerical simulation method, the seismic cracking and shear sliding instability model of landslide is inverted. The quantitative correlation analysis between seismic landslide and various influencing factors is carried out by using random forest model, and the landslide susceptibility assessment is completed. The relevant research results can provide reference for the quantitative assessment and genetic mechanism research of seismic landslides in the northeast margin of Qinghai-Tibet Plateau.

Acknowledgments

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Study On The Mechanism Of Water Potential Response of Loess And Plant Roots Improved by Water-retention Agents

Lida Wei[1,2], Wenqian Gong[1,2], Ping Li[1,2,*], Kai Yang[1,2]

1 School of Geological Engineering and Geomatics, Chang'an University, Xi'an 710054, China;
2 Observation and Research Station of Water Cycle and Geological Environment on the Chinese Loess Plateau, Ministry of Education, Zhengning 745300, China

[*] Corresponding author. Ping Li: +86-13891936410; E-mail:dcdgx07@chd.eddu.cn

Abstract

This study is dedicated to exploring the mechanism of a water-retention agent (WRA) in improving loess's response to water potential. By applying the filter paper method and axis translation technique, we measured the soil-water retention curves (SWRC) of pure WRA, plant root, pure loess, loess with 2%WRA, as shown in Figs. 1, 2, 3, 4. The detailed characteristics of their water potential changes with water content were revealed. Comparison between Fig. 3 and Fig. 4 indicate that, under sufficient soil moisture conditions, water-retention agents can significantly enhance the water-retention capacity of loess, thereby aiding in the extension of the plant growth cycle. However, facing long-time very drought conditions, once the soil moisture level approaches the plant's wilting point (point 1 in Fig. 2) - the suction corresponding to the wilting point might inversely extract water from the plant roots, accelerating the plant demise. This finding suggests that soil amended with water-retention agents is particularly suited to plants with a higher wilting point; for those with lower wilting point, the moisture in the amended soil might not be effectively absorbed, and sometimes might even exacerbate plant demise. Therefore, the application of water-retention agents in soil improvement and plant cultivation has both positive and negative effects, it suggests that we have to use it carefully in the positive aspect.

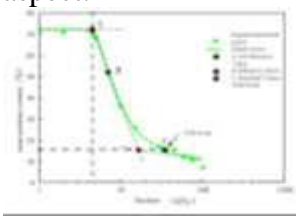


Fig. 1 SWRC of the pure WRA

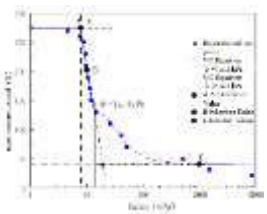


Fig. 2 SWRC of Plant Root

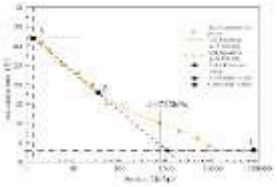


Fig. 3 SWRC of pure loess

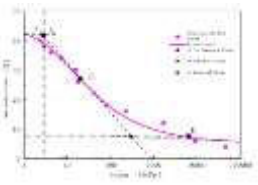


Fig. 4 SWRC of loess with 2% WRA

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A Transformer-based slope stability agent model for shallow landslide forecast

Shih-Chao Wei[1,*], Song-Yue Yang[2], Hsin-Chi Li[3], Eric Tan Chen Hee [2]

[1] Department of Soil and Water Conservation, National Chung Hsing University, Taiwan (R.O.C.)
[2] Department of Urban Planning and Spatial Information, Feng Chia University, Taiwan (R.O.C.)
[3]National Science and Technology Center for Disaster Reduction, Taiwan (R.O.C.)
[*] Corresponding author. Tel: +886-4-22840381 #217; E-mail: scwei@nchu.edu.tw

Abstract

Real-time landslide prediction, a complex challenge, requires integrating various methodologies. Traditional physical models, though robust, often fall short in real-time applications due to their high computational needs. This study introduces a new approach using the Transformer model (Dosovitskiy et al., 2021) to predict shallow landslides in real-time, focusing on Taiwan's Xindian River basin.

Utilizing the validated TRIGRS model (Wei et al., 2018) and the projection rainfall scenarios, we created a comprehensive landslide dataset for developing a Transformer-based slope stability model, which was rigorously evaluated for its accuracy and efficiency in predicting landslides.

Results show that the Transformer model's predictions closely match those from the TRIGRS model, especially where the Factor of Safety (FS) is below 2, as illustrated in Figure 1. The performance comparison revealed a Root Mean Square Error (RMSE) of 0.13, with most prediction errors falling below 10%. Notably, the Transformer model predicted outcomes for a 48-hour rainfall event in just 7.68 seconds—far faster than the TRIGRS simulations, highlighting its potential for effective real-time landslide forecasting.

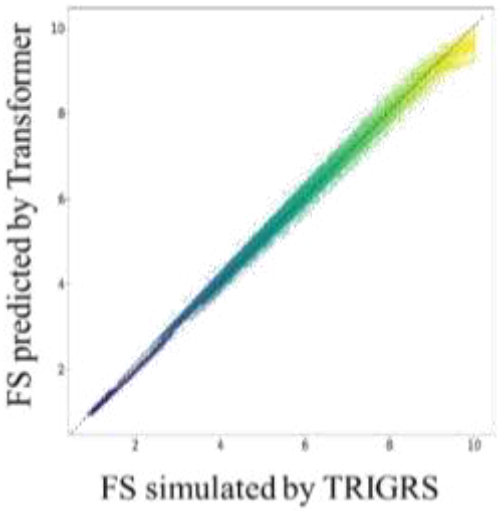


Fig. 1 Comparison of Factor of Safety: Transformer vs. TRIGRS

Acknowledgments

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An improved meshless method for simulating Baige landslide event considering fluid-solid interaction

Chengzhi Xia[12], Zhenming Shi[12], Bo Li[12*]

[1] Department of Geotechnical Engineering, Tongji University, Shanghai 200092, China

[2] Key Laboratory of Geotechnical and Underground Engineering, Ministry of Education, Tongji University, Shanghai 200092, China.

[*] Corresponding author. Tel: +86-17769735586; E-mail: libotj@tongji.edu.cn

Abstract

An improved Smoothed Particle Hydrodynamics method was employed to simulate the surge event of the Baige landslide. In this method, rock cracking was realized by an enhanced kernel function of SPH, while contact and sliding were modeled using a point-to-point contact algorithm. The simulation of fluid-solid interaction was achieved through a two-phase neighborhood coupling method. Three benchmarks were utilized to validate the fracture, contact sliding, and fluid-structure coupling methods, respectively. Subsequently, the Baige landslide surge event was recreated, with discussions on the influence of sliding surface friction coefficients and river height on the landslide. The results revealed that the improved SPH method effectively captures the entire process of the Baige landslide, from rock fracture and contact sliding to wave generation and propagation, culminating in the formation of the landslide dam, consistent with field observations. Both the rockslide velocity and water flow propagation velocity exhibit a linear decrease with increasing surface friction coefficient of bed rock. The shape of the rockslide dam transitions from a fan shape to a wedge shape when the water level is below 20 m.

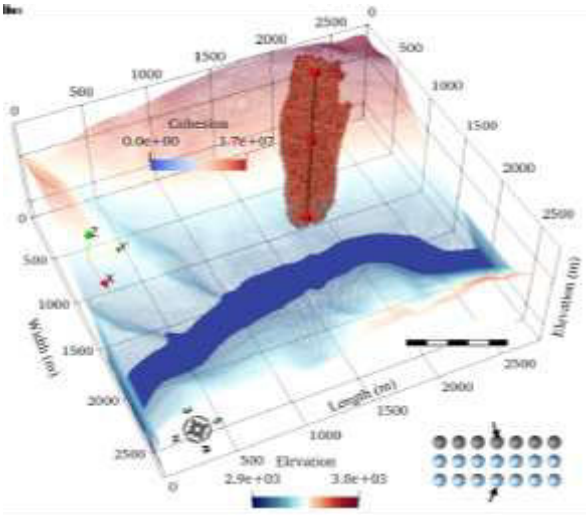


Fig. 1 Establishment of Baige rockslide model and adaptive boundary conditions.

Acknowledgments

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Evaluation on the Rationality of Spatial Distribution of Landslide Susceptibility Assessment Results

Shuo Yang[1,2], Deying Li[1,*], Mariagiovanna Moscariello[2], Sabatino Cuomo[2]

[1] Faculty of Engineering, China University of Geosciences, Wuhan 430074, China
[2] Department of Civil Engineering, University of Salerno, Fisciano 84084, Italy
[*] Corresponding author. Tel: +86-13659892519; E-mail: lideying@cug.edu.cn

Abstract

The selection of landslide susceptibility mapping (LSM) usually depends on the prediction accuracy of the landslide susceptibility model. However, the spatial distribution of landslide susceptibility evaluation results is ignored, which may cause limitations in the practical application of the LSM. In order to optimize the selection of the LSM, an evaluation method that combines prediction accuracy analysis and spatial distribution analysis of landslide susceptibility assessment results was proposed. The Mayang County in northwest of Hunan Province, China was selected to study area, and 4 different machine learning models were introduced in landslide susceptibility assessment. The proportion of susceptibility level partitions and spatial autocorrelation of the evaluation results of 2 different basemap units (grid units and slope units) were analyzed. The results demonstrated that: (i) The average AUROC value of Random Forest (RF), Logistic Regression (LR), Support Vector Machine (SVM) and Multi-Layer Perceptron Model (MLP) were 0.83, 0.77, 0.74 and 0.80, and the RF had the best predictive performance. (ii) All of the model results had weak spatial heterogeneity and strong spatial autocorrelation based on the grid and slope unit. At the grid unit scale, the evaluation result of RF was the most rationality, and at the slope unit scale, MLP was the most rationality. The spatial distribution pattern of the landslide susceptibility evaluation results is influenced by base map units. Evaluation from the model prediction accuracy and the rationality of the spatial distribution can make the selection of LSM more consistent with practical applications.

Acknowledgments

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Study on the particle breakage of coral sand in the ring shear experiment

Weilong Zhang[1], Hu Zheng[1,2,*]

[1] Department of Geotechnical Engineering, Tongji University, Shanghai 200092, China
[2] Key Laboratory of Geotechnical and Underground Engineering, Ministry of Education, Tongji University, Shanghai 200092, China.
[*] Corresponding author. Tel: +86-13524608289; E-mail: zhenghu@tongji.edu.cn

Abstract

Coral sand, a geotechnical material commonly encountered in geotechnical engineering practice, is susceptible to particle breakage under external loads. Particle breakage has always been a hot topic in geotechnical engineering. This study focuses on the detailed evolution of particle breakage in granular materials under large shear displacements by conducting ring shear tests on three graded dyed coral sands. In this study, image processing technology was utilized to quantitatively identify particles of different colors, and the breakage of particles in each particle size range within different graded sand samples was presented and analyzed. The results reveal significant differences between the actual breakage amount of particles and the relative breakage amount in most particle size ranges. Additionally, a linear relationship was observed between the absolute particle breakage rate in each particle size range and the logarithmic value of the normal stress. Finally, a new particle breakage index was proposed based on the detailed particle breakage analysis, which determined the breakage mode of coral sand particles under large shear displacement. These findings contribute valuable insights to understanding the particle breakage behavior of granular materials in engineering practice.

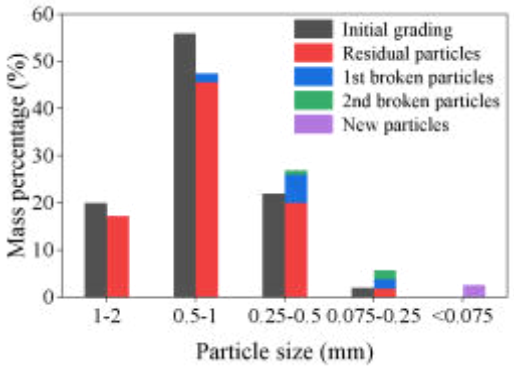


Fig. 1 Detailed particle size distribution of RS1 under the normal stress of 100 kPa.

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Study on the initiation of liquefaction of loose systems under monotonic and cyclic loads with DEM

Minyi Zhu[1], Hu Zheng[1,2,*]

[1] Department of Geotechnical Engineering, Tongji University, Shanghai 200092, China

[2] Key Laboratory of Geotechnical and Underground Engineering, Ministry of Education, Tongji University, Shanghai 200092, China.

[*] Corresponding author. Tel: +86-13524608289; E-mail: zhenghu@tongji.edu.cn

Abstract

Factors such as earthquakes, waves, and abrupt loads predominantly trigger soil liquefaction, during which even non-cohesive soils might lack adequate time for drainage, resulting in an escalation in pore pressure and a notable strength reduction. This underscores the necessity for an exhaustive investigation into the liquefaction behavior of granular media under different stress states. However, traditional lab works or theory are capable of capturing only the macroscopic undrained behaviors. As an alternative, the discrete element method is used in this study, offering an insight view into the microscopic scale.

This study seeks to scrutinize the incipient phase of liquefaction in loosely consolidated sand, subjected to various loading methods: 1. monotonic; 2. strain-controlled cyclic; 3. stress-controlled cyclic. The undrained state is executed maintaining a constant volume. Based on the gathered data, both the macro- and micro-scale behaviors are assessed. The quasi steady state is noted in samples subjected to monotonic loading. In the case of strain-controlled cyclic loading, there's a gradual reduction in deviator stress amplitude until it eventually approaches zero. When it comes to stress-controlled cyclic loading, the deviator stress amplitude remains constant, and liquefaction takes place when the stress becomes too feeble to reach the intended amplitude. A modified redundancy index is defined, which includes constraint loss attributed by sliding. The susceptibility of liquefaction can be well captured by the phase transition line - a transition from solid-like manner to liquid-like manner.

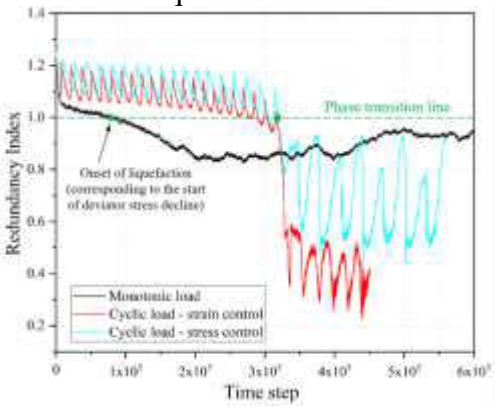


Fig. 1 Redundancy index against time step for tests with different loads

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Factors Controlling the Spatial Distribution of Coseismic Landslides Triggered by Ludian Earthquake

Yu Zou[1,2], Shengwen Qi[1,2,*], Yongchao Li[1,2], Songfeng Guo[1,2], Bowen Zheng[1,2]

[1] Key Laboratory of Shale Gas and Geoengineering (KLSGG), Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing 100029, China;

[2] Innovation Academy for Earth Science, Chinese Academy of Sciences, Beijing 100029, China;

[*] Corresponding author. Tel: +86-01082998055; E-mail: shengwenqi@mail.iggcas.ac.cn

Abstract

Understanding the spatial distribution regularity of coseismic landslides is important for mitigation in future earthquakes. Unveiling the controlling factors of the distribution of coseismic landslides is a long-standing unsolved problem. On 3 August 2014, a *Mw* 6.1 earthquake occurred in Ludian County, Yunnan Province, China, and resulted in a great number of landslides and substantial losses. This work conducted a spatial analysis to reveal the controlling factors in the Ludian earthquake. A total of 1,414 landslides were identified after a combination of image interpretation and field investigation in an area of 704.7 km^2. A geographic information system (GIS) was used to analyse the spatial characterization of the landslides. Correlations between the occurrences of landslides and factors were characterized with a spatial analysis in GIS. It can be concluded that the slope gradient and the distance to the coseismic fault are the two most important factors for determining the spatial distribution of coseismic disasters. Some other factors play a role by influencing the slope gradient to a certain extent, and the movement of the coseismic fault is the prone direction of coseismic landslides. Slope cutting fractures the rock mass, increases the slope gradient, and significantly increases the occurrence of coseismic landslides.

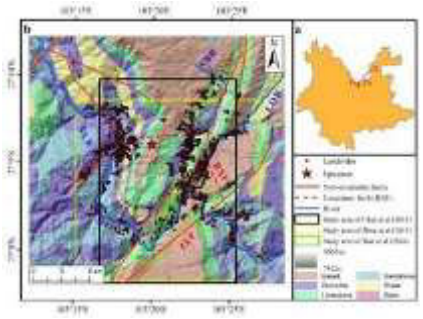


Fig. 1 Geological background and seismic landslide distribution map.

Acknowledgments

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Experimental evidence of fine long rooted grass grown in pyroclastic soils involved in debris avalanches

Mariagiovanna Moscariello, Rita Ciaglia, Sabatino Cuomo

Geotechnical Engineering Group (GEG), University of Salerno, Italy

Abstract

Te potential of using long-root grass to reduce or prevent debris avalanches has been investigated through an experimental field setup was established in 2020 and further refined in 2021. Three different species of long-root grass were considered. Regular measurements of soil suction and water content were taken from both the vegetated and original soils. Additionally, soil samples were collected for laboratory geotechnical testing to determine changes in water retention and strength properties. The results showed that all species of long-root grass increased soil suction and shear strength, despite some variations among the species. These findings suggest that planting long-root grass could potentially reduce the volume of soil mobilized in the initiation zones of debris avalanches.

Keywords

Field, laboratory, landslides, debris avalanche, root, grass

Introduction

Researchers and practitioners must face relevant issues for modelling flow-like landslides due to the different kinematic features of failure, post-failure, and propagation stages. In particular, Hungr et al. (2001) defined the Debris Avalanches (DA) as "very rapid to extremely rapid shallow flows of partially or fully saturated debris on a steep slope, without confinement in an established channel". Both in northern Italy, in the debris deposits of Valtellina (Chen et al., 2006) and in the Vesuvian pyroclastic deposits of southern Italy (Cascini et al., 2008; Guadagno et al., 2005), debris avalanches typically occur along slopes steeper than 30°, in cohesionless soil deposits that are usually 1-2 meters thick.

Despite the efforts, two challenges remain: i) reducing the volume of soil mobilized during the inception of DAs through eco-friendly solutions, and ii) evaluating the performance of engineering control measures.

This paper outlines some experimental evidence for long-root grass to be used to possibly mitigate the debris avalanche volumes in the inception zone. Various types of grasses and roots have been tested for different purposes in the literature (Wu et al., 1979, 2013; Ng et al., 2013a,b, 2016a,b). Recently, long-root grasses have also been tested in coarse-grained soils prone to shallow landslides (Capobianco, 2018) and in fine-grained soils affected by slow-moving landslides (Tagarelli, 2019).

In this paper, the scientific literature on debris avalanches and soil-root interactions will be briefly reviewed. At last, a theoretical framework for a new eco-friendly technique to mitigate debris avalanche inception will be discussed. The results of a laboratory and in-situ experimental campaign on artificially seeded soils are also presented.

Literature review

Debris avalanches have been examined from various perspectives, revealing the combined mechanisms of impact loading and bed entrainment during propagation, which significantly increase the volume of initial unstable soil.

Wang et al. (2002) and Sassa et al. (2003) analyzed the impact loading mechanism through ring shear tests, showing that soil contraction and excess pore pressure generation lead to a substantial drop in shear resistance. The bed entrainment mechanism also occurs during the propagation stage, where soil is mobilized due to overload and shearing from the flow over stable ground. Cuomo et al. (2014) investigated the role of the propagating mass's rheological features, slope geometry, and triggered volume characteristics, concluding that lateral enlargement along a slope is influenced mainly by bed shear resistance and slope

steepness, with soil thickness in the landslide source area also playing a role. Cuomo et al. (2016) extended this work, finding that the pattern of soil thickness erosion along a mountain catchment is complex and interconnected with slope morphology and landslide dynamics.

To mitigate the severe effects of debris avalanches, active/passive interventions aim to reduce landslide volume within source areas. However, practical difficulties in reaching steep areas with standard construction machinery limit the feasible of many available techniques. A novel intervention could involve soil reinforcement through artificially grown roots.

Literature extensively reports the hydrological and mechanical effects of roots at different scales (Boldrin et al., 2017; Leung et al., 2015; Ng et al., 2014, 2016; Leung et al., 2017). Some roots primarily affect soil suction and water content (Leung et al., 2015), while others provide additional shear strength (Stokes et al., 2009). Generally, soil-root interaction encompasses both hydrological and mechanical effects, as noted by Capobianco et al. (2020) and Foresta et al. (2020). However, the use of artificially seeded roots to mitigate flow-like landslides has yet to be explored.

The roots to mitigate the Debris Avalanches

A new mitigation technique has been proposed for geo-environmental contexts characterized by steep slopes with shallow, unsaturated soil deposits by the researchers of the University of Salerno. Vegetated soils, being multiphase materials, undergo changes in soil void ratio due to root growth (Fig.1).
The presence of roots significantly modifies the soil water retention curve, which represents the change in saturation degree with soil suction changes.

The effects of roots within the soil are as follows:

- Density effect (D): roots typically reduce soil void ratio (e), making the soil denser.
- Fine effect (F): roots alter the Grain Size Distribution (GSD), introducing a finer material into the soil. When the same amount of external water is added to both natural and rooted soil, the volume of voids available for seepage is slightly lower in the rooted soil. However, the saturation degree remains almost the same. Therefore, due to the combined D and F effects, soil suction increases in vegetated soil (VEG) compared to the original non-vegetated soil (NV).
- Mechanical effect (M): in vegetated soils an increase in soil shear strength can be observed, due the presence of long fine vertical roots.
- Hydrological effect (H): the presence of roots can increase the soil suction; such increment has been widely documented in the literature, as previously discussed. In a context prone to landslide triggering, the increase in soil suction causes the increase in effective Bishop stress, hence the increase in soil shear strength.

A global measure of that is provided by the experimental determination of the shear strength envelope of the rooted soil, which always result in larger strength for a rooted (VEG) compared to a non-vegetated soil (NV), as depicted in Fig. 2a-b.

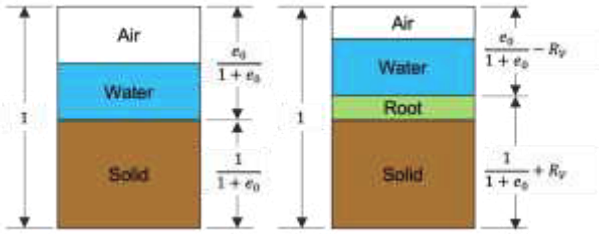


Figure 1. Volumetric phase diagram of (a) bare (non-vegetated, NV) soil and (b) a vegetated soil (VEG).

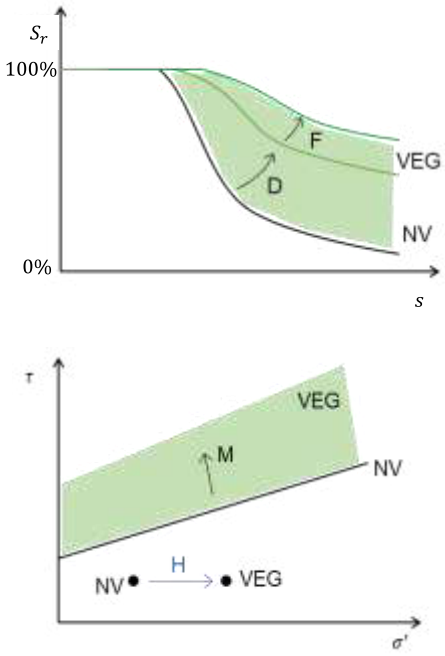


Figure 2. a) Density (D) and Fine (F) effects on saturation degree (S_r) and soil suction (s), b) Hydrological (H) and Mechanical (M) effect of roots grown in unsaturated soil (VEG) versus NV.

An experimental campaign for artificially rooted soils

The experimental campaign involved both in situ and laboratory investigations. The test site was chosen within the Nocera Inferiore territory in the Campania region of Southern Italy. The site was divided into nine square plots, each measuring 1m x 1m (Fig. 2a).
The soil tested was a pyroclastic, non-plastic silty sand with a median diameter of 0.425 mm, high porosity (0.55 - 0.58), and low density (11.04 kN/m3 - 12.82 kN/m3).

From November 2020 to May 2021, soil water content and suction were monitored for six months prior to seeding. During this time, the soil was bare, as the existing vegetation had been removed to allow for the addition of new plants. The original vegetation was cleared because its high density prevented the introduction of other plants.

The number of seeds per square meter was calculated based on an assumed germination rate of 85% to 95%, with one seed planted every 1 cm². The specific weight of the seeds was known for each of the species used (VEG 1, VEG 2, VEG 3). These species were fine, long-rooted perennial grasses, known for their high water consumption.



Figure 3. Test site: a) before seeding (Nocera Inferiore mounts are visible), b) three months after seeding, c) same as b) with the Pizzo d'Alvano massif (affected by huge flow-landslides in 1998) as background.

After seeding, soil matric suction was measured weekly at depths of 0.2 m and 0.6 m, while gravimetric water content (the difference between wet and dry soil mass divided by dry soil mass) was measured at 0.2 m. Suction measurements were taken using a Quickdraw portable tensiometer, which provides results within 15-30 seconds. Volumetric water content (the ratio of water volume to soil volume, dependent on soil porosity) was estimated using three porosity values representing the minimum, average, and maximum values from the samples collected in the nine plots over time.
Field data on suction and water content were interpolated using the van Genuchten equation (1959).

Cumulative daily rainfall and daily minimum and maximum atmospheric temperatures were recorded from a nearby weather station in the Campania Region. These data helped define the boundary conditions for the test site. Temperature data were used to estimate potential evapotranspiration (ET0) and the expected evapotranspiration (ET)

Grass growth was monitored monthly by collecting leaf samples from the three species and measuring their length, similar to the method used by Capobianco et al. (2020), which found a correlation between root and leaf length and grass age. More than 40 soil samples, both vegetated (VEG) and non-vegetated (NV), were manually collected up to a depth of 0.30 m. The laboratory testing program aimed to investigate changes in soil shear strength induced by the long fine roots of the three species (VEG 1, VEG 2, and VEG 3).
Direct shear tests on saturated and unsaturated soils were performed at constant water content or suction.

Root reinforcement was expressed through an increased intercept cohesion (c_r), using a semi-empirical expression proposed by Wu et al. (1979) and later modified by Preti and Schwarz.

Experimental results

The long-root grasses were planted in late spring (May 21, 2021), and the first leaves appeared 15 days later. Figure 6a shows the height of the foliage acquired during a year of observation, compared to those reported by Capobianco et al. (2020). The growth rate at the field test site was higher than that observed in the twin columns (0.084 m/month), and after several months of consistent growth, the foliage height stabilized before the grasses entered a dormancy stage. VEG 2 exhibited the highest growth rate at 0.73 m/month during the first 50 days, while VEG 1 had the lowest growth rate at 0.11 m/month in

the same period. VEG 2 reached the maximum foliage height, and even VEG 1 attained a greater height than noted by Capobianco et al. (2020).

Root diameters were measured and across all species, the median diameter was approximately 0.5 mm, consistent with findings by Foresta et al. (2020). VEG 2 had the maximum median diameter of 0.50 mm and the largest range between the 25th and 75th percentiles.

It is worth noting that the three perennial graminae grass species developed distinct root morphologies:

1. VEG 1 presented vertical, oblique, and deep tap roots that are thin and originate from the grass stem, with main and secondary roots having similar diameters.
2. VEG 2 has a root structure similar to VEG 1, but the roots are coated with a jelly-like substance, and the main and secondary roots have slightly different diameters.
3. VEG 3 features a tap root system with one large central fibrous root and finer lateral roots.

All species achieved a rooting depth of about 1 meter within one year (Fig. 4).

In vegetated soils, consistently lower water content was observed compared to non-vegetated soils. Among the vegetation species, VEG 1 showed the highest water content values. Additionally, higher suction levels were detected in the vegetated soils relative to the non-vegetated soils. Significant differences in suction at 0.2 m depth between vegetated and non-vegetated soils were evident shortly after seeding. At 0.6 m depth, the divergence in suction between these soil types appeared after 9 months, aligning with the growth depth of the long-root grass species, which reached 1 m depth within approximately one year.

The soil water retention curves (SWRC) derived from interpolating field data at 0.2 m depth are illustrated in Figure 5, with three void ratio values considered for each soil type. Root presence led to (i) reduced porosity and saturated water content, (ii) increased air entry value, and (iii) a slight decrease in residual water content. These findings are consistent with observations by Leung et al. (2015), highlighting roots' significant influence on SWRCs.

Shear strength parameters were determined through laboratory tests (direct shear tests). Root-soil interaction increased shear strength, although each grass type affected the failure criterion differently (Fig. 6).

While non-vegetated soil exhibited very low intercept cohesion (6.44 kPa), VEG 2 showed the highest cohesion (c' = 42.43 kPa). VEG 1 and VEG 3 also demonstrated substantial increases in cohesion, reaching 20 kPa and 29 kPa, respectively.

The introduction of fine roots slightly reduced the soil friction angle in vegetated soils compared to original non-vegetated soil conditions. Despite this drawback, in practical applications where in-situ stresses are low (e.g., less than 100 kPa), the shear resistance of vegetated soils consistently surpassed that of non-vegetated soil across all three investigated species (Fig. 6).

The R2 values in the figure indicate coefficient of determination, emphasizing that data for VEG 1 and VEG 2 are less scattered compared to VEG 3.

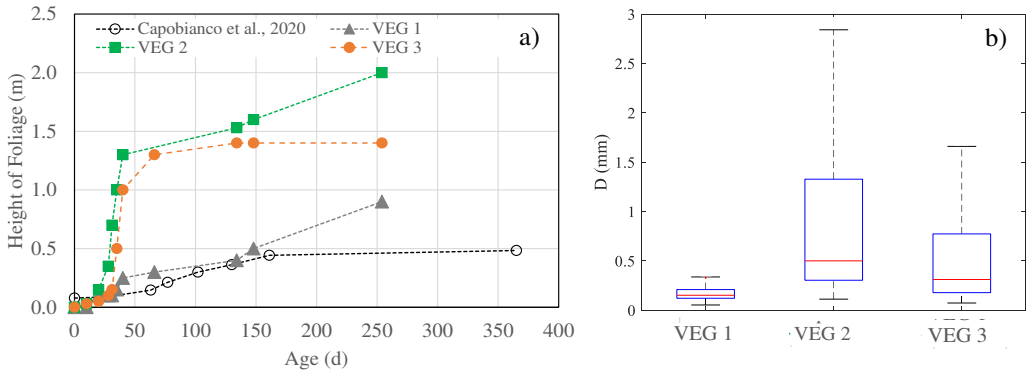


Figure 4. a) Growth rate for the three species b) Summary statistics on root measured diameter. The total number of measurements is 905, i.e., VEG 1 (n=244), VEG 2 (n=249), VEG 3 (n=412).

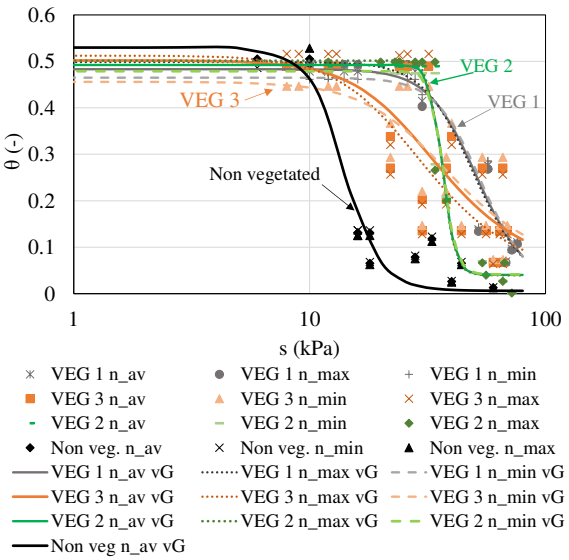


Figure 5. Soil Water Retention Curves of non-vegetated soil (NV) and vegetated soils (VEG 1, VEG 2 and VEG 3). In the legend, the term "n" means porosity, while "av", "min", "max" means that the volumetric water content was estimated considering the average porosity, minimum porosity, and maximum porosity respectively. The label "vG" is referred to the interpolation curve of experimental data (markers) obtained using the van Genucthen equation.

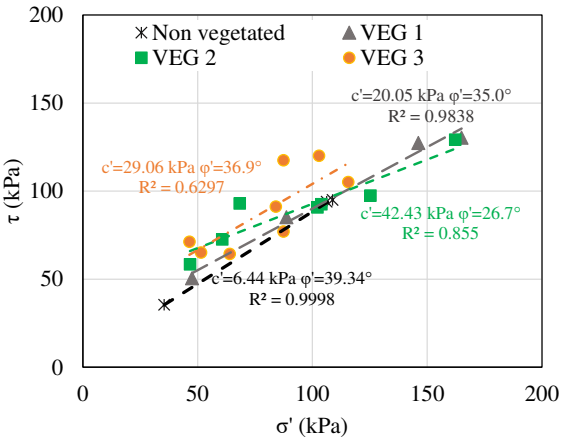


Figure 6. Shear failure envelopes of the non-vegetated soil (NV) and the vegetated soils (VEG 1, VEG 2, VEG 3), with R2 indicating the coefficient of determination.

Conclusions

This study explored the effectiveness of long-root grasses as a potential mitigation strategy for debris avalanches. Three different vegetation species (VEG 1, VEG 2, VEG 3) were planted in a field test area located near slopes previously affected by significant flow-like landslides, including debris avalanches.

The experimental setup consisted of nine square plots, each measuring 1m x 1m, where the species were carefully seeded under controlled conditions. Over subsequent months, the growth of roots was closely monitored.

Samples of the rooted soils were then collected for geotechnical laboratory testing, focusing on assessing changes in soil water retention curves and shear strength. Additionally, field measurements were conducted to analyze variations in soil suction within the vegetated areas.

Preliminary results have shown promise, suggesting that vegetation could indeed offer a viable approach for mitigating landslide hazards, further investigation is needed.

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Mariagiovanna Moscariello (✉)
Geotechnical Engineering Group (GEG), University of Salerno, Italy
E-mail: mamoscariello@unisa.it

Rita Ciaglia
Geotechnical Engineering Group (GEG), University of Salerno, Italy
E-mail: r.ciaglia@studenti.unisa.it

Sabatino Cuomo
Geotechnical Engineering Group (GEG), University of Salerno, Italy
E-mail: scuomo@unisa.it

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